Projecting (your) voice:
A theory of inversion and defective circumvention

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Abstract of the Dissertation

Projecting (your) voice:
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This doctoral dissertation revolves around colloquial agreement alternations in many constructions in several languages involving postverbal thematic subjects. These alternations are productive and well-attested, and are exemplified by the following examples from English.

(1) a. There {were/was} no seats left
b. Off the plank {walk/walks} several bad pirates
c. "Moo!" {go/goes} the cows
d. What I love most about you {are/is} your outfits

This agreement alternation is notably only possible in the inverted form. When the postverbal elements in (1) appear preverbally, the optionality disappears.

(2) a. No seats {were/*was} left
b. Several bad pirates {walk/*walks} off the plank
c. The cows {go/*goes} "Moo!"
d. Your outfits {are/*is} what I love most about you

This colloquial agreement alternation has widespread consequences for theories of Agree, inversion, argument structure, and the connections among the three. Using a maximally simple formulation of Agree that effectively reduces to Minimal Search/closest Match, I show that the aforementioned agreement alternations are derivable from a single set of syntactic operations through an optional process which I dub *defective circumvention*, in which a probe can conditionally Agree past a featurally deficient goal and undergo sequential multiple Agree with more than one goal, given that the features of the two goals are compatible with one another. I motivate this theory primarily with data from English and Spanish, among other languages. Defective circumvention explains the agreement optionality shown in (1) as well as PCC effects which I demonstrate the existence of in various multiple subject constructions.

Given the minimal theory of Agree presented in this thesis, along with a minimal formulation of argument structure in which thematic arguments are uniformly and hierarchically externally merged, a conundrum emerges. The sentences in (1) involve A-movement (as evidenced by the effects on agreement) of an internal argument over an argument that was externally merged in a higher position. How is it that A-movement possible without the higher argument serving as an intervener? Given the understood uniformity of the projection of arguments, as well as the locality and strict minimality of this formulation of Agree, a mechanism is needed to affect the accessibility of arguments for A-operations.

As such, I demonstrate that these constructions necessarily involve inversion *via smuggling*, in which the internal argument is smuggled to a position above the external argument via phrasal movement of a verbal projection containing the internal argument, which makes it a viable goal for A-movement that would otherwise be blocked by the higher argument. I show that this VP smuggling to make internal arguments accessible for A-operations is characteristic of voice constructions. I present a definition of voice alternations in which voice is characterized by mediating between the mapping of thematic arguments from the positions in which they are externally merged and the potential realization of arguments at LF. Voice, then, is what allows for inversion to take place and ultimately what allows for the agreement alternation facilitated by defective circumvention. I demonstrate this analysis of inversion via detailed derivations of various inversion constructions in English, including locative inversion, quotative inversion, predicate inversion, and existential ‘there’ sentences, which I show are derived via inversion as well.

Dedication

To Mads. You always knew I could do it.


HE WRITES THE
WAY PEOPLE TALK!

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List of Abbreviations

#	Number
ACC	Accusative
COP	Copula
DAT	Dative
EI	Existential inversion
EPP	Extended Projection Principle
HNPS	Heavy-NP Shift
INV	Inverse
LI	Locative inversion
LOC	LOCATION
MLC	Minimal Link Condition
NOM	Nominative
NonPart	Non-participant
OBJ	Object
Part	Participant
PCC	Person-case constraint
Pers/π	Person
PI	Predicate inversion
PL	Plural
PLC	Person licensing condition
PRED	Predicate
QI	Quotative inversion
SG	Singular
SMT	Strong Minimalist Thesis
Spkr	Speaker
TC	Transitivity Constraint
UTAH	Uniformity of Theta-Assignment Hypothesis
WCO	Weak crossover
γ	Gender
ϕ	Phi(-features)

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It really does take a village. This is true not only of language acquisition and other aspects of childrearing, but also of completing a doctoral dissertation. I could not have done this (or even come close to doing so) without the invaluable contributions of many people. The list I present here is by no means exhaustive, so, if you think you belong here and do not see your name mentioned, you're probably right. Just know that I am thinking of you as well.

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1 Introduction

While languages are descriptively categorized as having tendencies toward certain word order patterns (i.e., SVO, SVO, etc.), there are numerous cases where those tendencies do not hold in a given language. For example, English – a canonically rigid SVO language – has a multitude of OVS constructions which I show in this thesis are derived via A-movement of the fronted object. Among these constructions are those known as *inversion* constructions, in which the thematic subject, which is typically the most external argument to the verb, appears in a postverbal position, and some sort of object-like material (a direct object, a locative, a predicate, etc.) comes to precede the verb. These alternations are as follows.

(1) a. Alex walked into the room (locative inversion)
b. Into the room walked Alex

(2) a. Madeline shouted "Hey!" (quotative inversion)
b. "Hey!" shouted Madeline

(3) a. Stephen is my brother (predicate inversion)
b. My brother is Stephen

Similarly, we have English existential sentences with the structural subject 'there', in which the thematic subject obligatorily follows the verb. If English existential sentences are OVS, the SVO version of existential sentences are not felicitous, which I explain in Chapter 4, though these existential sentences share many properties with the inversion constructions mentioned above.

(4) a. #A problem was there (existentials)
b. There was a problem

The (b) examples here all involve a clear and marked departure from canonical English word order, especially since English is not a language which allows subject-object inversion with most transitive verbs (unlike, for instance, Russian (Bailyn 2004), Algonquian (Oxford 2019; 2023), and many Bantu languages (Shlonsky 2024)). As such, it is an immediate issue how the combinatoric constraints of a language such as English allow for such inverted word orders.

While the English inversion constructions presented in (1) through (3) involve a limited set of syntactic predicates that allow for such alternations, there are other canonical word order inversions in English that have a similar output for surface form, such as passivization.

Both inversion and passivization involve a departure from a canonical SVO "kernel" sentence form in which the external argument appears preverbally and the internal argument appears postverbally to a more contextually-restricted form in which the position of the external and internal arguments are switched (creating an OVS linear form). See the kernel derivations in (5) and the passivized/inverted derivations in (6).

(5) a. [Drew] read [the book]
S V O
b. [Paula] walked [into the room]

(6) a. [The book] was read [by Drew]
O V S
b. [Into the room] walked [Paula]

Despite some marked differences between these constructions, such as passives involving dedicated verbal morphology ('be' + participial verb form) and the famous inclusion of a *by*-phrase that introduces the external argument, which are notably both lacking from the inversion derivations, the linear output of these alternations is fundamentally the same. Furthermore, both alternations create a pragmatically-marked variant of the underlying derivation which is information-structurally distinct from the non-inverted kernel sentences from which they are derived.

Passives necessarily involve A-movement of the internal argument; this is clear from (at minimum) the fact that verbal agreement in passives must track the structural subject that began as the internal argument. I argue that these inversion constructions involve A-movement as well, as evidenced by the fact that inversion of this kind creates new agreement possibilities. See the following examples which are grammatical in colloquial English.

(7) a. Into the room {walk/walks} three scary clowns
b. "Quack!" {go/goes} the ducks
c. My favorite thing about you {are/is} your outfits
d. There {were/was} problems with his analysis

Furthermore, these inversion constructions involve new scope and binding relations that are different from their non-inverted counterparts (Culicover & Levine 2001; Rizzi & Shlonsky 2006), just as passives do (Jackendoff 1972). See the following examples involving locative inversion and passive sentences obviating weak crossover.

(8) a. Into every dog$_i$'s cage peered its$_i$ owner (Culicover & Levine 2001:289)
b. *Its$_i$ owner peered into every dog$_i$'s cage

(9) a. Every dog$_i$ was fed by its$_i$ owner
b. *Its$_i$ owner fed every dog$_i$

Based on facts such as these concerning agreement and A-movement, I develop a theory of inversion in which these sentences are derived by A-movement facilitated by the same operation that allows for passives (smuggling (Collins 2005)), as well as a theory of Agree that accounts for the complex facts concerning agreement with postverbal subjects in inversion constructions.

In order to accomplish this goal, I systematically survey and analyze the possibilities in these inversion constructions regarding agreement and Case assignment, informed by naturally-occurring data across several different inversion types. I show that my theory of Agree works for multiple constructions in other languages as well, including dative subject constructions in Icelandic and inversion constructions in Spanish. Once the theory of Agree has been established, I present derivations of locative inversion, quotative inversion, predicate inversion. I additionally show that sentences involving existential 'there' in English are also inversion constructions of the same variety of these other ones. As part of these analyses I explain several interesting properties of these inversion constructions including: the transitivity constraint, which is a restriction from locative inversion constructions taking a direct object (Diercks 2017) and quotative inversion verbs taking an indirect object (Collins & Branigan 1997); the fact that inversion constructions display different combinatorics when there is so-called Heavy-NP Shift (Ross 1967); the information-structural properties of these constructions; and the syntactic and semantic distribution of so-called expletive subjects in English inversion.

1.1 Hypothesis and results

The core questions of this dissertation revolve around the following colloquial agreement alternations shown in a variety of English sentences involving postverbal subjects.

(10) a. There {were/was} problems in the paper (existentials)
b. Off the plank {walk/walks} several bad pirates (locative inversion)
c. "Moo!" {go/goes} the cows (quotative inversion)
c. What I love most about her {are/is} her cats (predicate inversion)

These alternations are notably not present outside of the inverted constructions. That is, when the postverbal subjects in (10) are preverbal, there is no such alternation (Greenberg 1963).

(11) a. Problems {were/*was} in the paper
b. Several bad pirates {walk/*walk} off the plank
c. The cows {go/*goes} "Moo!"
d. Her cats {are/*is} what I love most about her

I demonstrate that these alternations are real and productive with convincing amounts of Internet data, and I also show that such alternations occur in other languages in some constructions involving postverbal thematic subjects. These data reveal several interesting, closely related questions concerning Agree, A-movement, and the nature of inversion and postverbal subjects.

The analysis of inversion constructions and their agreement alternations that I present in this dissertation has widespread consequences and implications for the theories of Agree and inversion, and how they are related. Central to these classes of inversion is the agreement optionality in which the verb may alternate between tracking agreement with the postverbal thematic subject or surfacing as a third person singular so-called "default" value. I take this agreement alternation – that extends across four classes of inversion constructions in English (and other things in other languages as well) – as a signature of a kind of voice alternation involving smuggling (Belletti & Collins 2021) and what I dub *defective circumvention*, which is Agree that takes place through a defective goal to a non-defective one. The main goal of this work is to present a systematic and comprehensive analysis of these signatures and show how they fit together to inform not only a theory of inversion constructions, but a theory of Agree, argument structure, voice, and how these phenomena are all related.

Crucial to the theory of Agree informed by these data is that Agree reduces to Minimal Search/closest Match. This formulation of Agree, however, does not preclude the possibility of a probe undergoing Agree with more than one goal as long as doing so is compatible with the featural specifications of the probe. I demonstrate throughout this thesis that this formulation of Agree sufficiently derives the complex facts concerning agreement in these inversion constructions involving number agreement optionality and PCC effects with postverbal subjects, even across multiple languages in which the facts look different from those of English.

These constructions that show agreement optionality necessarily involve a configuration in which a phi-defective goal stands as an intervener between a phi-probe and a non-defective phi-goal. This is illustrated as follows.

(12) $[P_{[uF\ uG]} \ldots [XP_{1[F]} \ldots [XP_{2[F\ G]}]]]$

In this configuration, the probe undergoes Agree with the closest goal XP_1 for feature [F], at which point the probe still has an unvalued feature [*u*G] for which it has not undergone Agree. Depending on the exact featural makeup of the second closest accessible goal XP_2, the probe may probe further and undergo Agree with XP_2, so long as the features of XP_2 are not incompatible with the features that the probe has already acquired from XP_1. This process of conditionally Agreeing with XP_2 after already having undergone Agree with XP_1 is defective circumvention.

(13) *Defective Circumvention*
A probe P undergoes Agree with a goal XP_2 that has the full set of features that P is specified to search for after first entering into an Agree relation with a deficient goal XP_1 that stands as an intervening goal between P and XP_2 and bears a proper subset of the features that P is specified to find.

As an example of this, take the following alternation in English.

(14) There {was/were} problems

The (simplified) setup for Agree in this example looks as follows. I remain agnostic as to the exact positions that the two arguments 'there' and 'problems' occupy, the crucial claim here being that 'there' (which is phi-deficient (Chomsky 1981; 2001; Béjar 2003:62; Deal 2009; Tsiakmakis & Espinal 2022)) is in a position in which it is closer to the phi-probe on T^0 (as defined by asymmetrical c-command (Chomsky 1995)) than 'problems' is. This structure and its derivation will be expounded upon significantly in Chapter 4.

(15) $[T^0_{[uPers\ uNum]} \ldots [DP_1\ \textit{there}_{[Pers]} \ldots [DP_2\ \textit{problems}_{[Pers\ Num]}]]]$

Here, the phi-probe on T^0 undergoes Agree with the closest visible DP bearing a relevant feature, which is *there*. At this point, the probe may either stop searching altogether and cause verbal agreement to track only the deficient DP 'there' – which results in third person singular 'was' – and leave the postverbal DP_2 untouched. Alternatively, the probe may initiate a second search for its unvalued number feature and Agree with the further argument.

Thus, Agree without defective circumvention results in the so-called default agreement on the verb, whereas Agree with defective circumvention results in verbal agreement tracking the postverbal subject. Defective circumvention is optional in (14) due to the nature of number probing, which I explain in detail in Chapter 2, though it may be obligatory or impossible crosslinguistically and crossconstructionally depending on the featural specifications of the probe and the DP_2 goal and how those features are compatible with one another and with those of the DP_1 goal. The possibilities for defective circumvention also vary parametrically according to how different phi-values are represented featurally within the DP. The exact details of this formulation of Agree are the focus of Chapter 2, and the facts concerning agreement with inversion in this theory shall inform the analysis of the inversion constructions in the subsequent chapters of this thesis.

The general setup that allows for defective circumvention – that is, a deficient goal appearing as a potential intervener between a phi-probe and a non-deficient goal in which the features of both goals are compatible with one another – is present in all of the other inversion constructions mentioned here: predicate inversion, locative inversion, and quotative inversion.

(16) a. Who I love most of all {is/are} my parents
b. [$T^0_{[u\text{Pers } u\text{Num}]}$ … [DP_1 *who I love(...)* $_{[\text{Pers}]}$ … [DP_2 *my parents* $_{[\text{Pers Num}]}$]]]

(17) a. Into my mind {rushes/rush} all these ideas
b. [$T^0_{[u\text{Pers } u\text{Num}]}$ … [PP *into my mind* $_{[\text{Pers}]}$ … [DP *all these ideas* $_{[\text{Pers Num}]}$]]]

(18) a. "Don't kick it!" {shouts/shout} the adults
b. [$T^0_{[u\text{Pers } u\text{Num}]}$ … [XP_1 *"Don't kick it!"* $_{[\text{Pers}]}$ … [DP_2 *the adults* $_{[\text{Pers Num}]}$]]]

At this point, a question emerges. In these English inversion constructions, how does the phi-deficient structural subject come to be an intervener and circumventee between P and XP_2? Especially given what we know about the uniformity of the projection of thematic arguments (Chomsky 1981; Baker 1997; Collins 2024; a.m.o), how is it the case that constituents that are normally considered "object" material (e.g. predicative DPs, locative PPs, theme XPs) not only come to be in the subject search domain of T^0, but also appear in a position structurally higher than the external argument DP in the inversion constructions which show this agreement alternation?

The analysis of these inversion constructions that accounts for these facts necessarily involves *smuggling*, which makes a thematic argument visible to processes such as Agree that would otherwise be blocked by an intervening argument, via movement of a larger constituent that contains the moved argument (Collins 2005; Belletti & Collins 2021). Collins (2005) defines smuggling as follows.

(19) *Smuggling* (Collins 2005:97)
Suppose a constituent YP contains XP. Furthermore, suppose that XP is inaccessible to Z because of the presence of W (a barrier, phase boundary, or an intervener for the Minimal Link Condition and/or Relativized Minimality), which blocks a syntactic relation between Z and XP (e.g., movement, Case checking, agreement, binding). If YP moves to a position c-commanding W, we say that YP smuggles XP past W.

As I show in section 1.2.3 and throughout this dissertation, smuggling is necessary to create the A-dependences seen in these inversion constructions (and other kinds of voice alternations) if one assumes a maximally minimalist formulation of Agree and argument structure that does not allow for stipulative technology such as equidistance and a totally unconstrained means of projecting thematic arguments. Devices such as equidistance in Agree and a non-uniform projection of arguments (i.e., moving away from something like the Theta-Criterion (Chomsky 1987)) are complications to the architecture of grammar that increase the number of computational principles needed to derive the types of data we find in natural language, and, as such, they reduce conditions of computational efficiency needed to derive said data. Additionally, I show in this thesis that we can do without such complications. Smuggling, on the other hand, is two applications of internal merge, which is already assumed in any standard minimalist syntactic framework. Smuggling, then, is not a complication.

Smuggling is visualized as follows in (20), with (a) representing the pre-smuggling derivation in which XP is inaccessible to Z (a probe), and (b) representing the smuggling derivation in which XP is accessible to Z. Without equidistance and multiple specifiers, such a movement is necessary to derive any structure in which XP is able to undergo A-movement to a position above W without incurring a violation of minimality or the MLC.

(20) a. $[\mathrm{Z}_{[u\mathrm{F}]} \ldots [_{\mathrm{BP}}\ \mathrm{W}_{[\mathrm{F}]} \ldots [_{\mathrm{YP}}\ \mathrm{XP}_{[\mathrm{F}]}]]]$ ✗

b. $[\mathrm{Z}_{[u\mathrm{F}]} \ldots [[_{\mathrm{YP}}\ \mathrm{XP}_{[\mathrm{F}]}]_i \ldots [_{\mathrm{BP}}\ \mathrm{W}_{[\mathrm{F}]} \ldots [t_i]]]]$ ✓

This operation is responsible for the word order and phrasal movement facts that are hallmark of these inversion constructions, and it also sets the stage for defective circumvention to occur. See the following example of '"Beans!" said John', keeping in mind that the quote here is deficient for number (see the analysis of the syntax of quotes in Chapter 3).

(21) a. $[\mathrm{T}^0{}_{[u\mathrm{Pers}\ u\mathrm{Num}]} \ldots [_{\mathrm{vP}}\ [_{\mathrm{DP}}\ \textit{John}_{[\mathrm{Pers\ Num}]}] \ldots [_{\mathrm{VP}}\ [_{\mathrm{DP}}\ \textit{“Beans!”}_{[\mathrm{Pers}]}]]]]$ ✗

b. $[\mathrm{T}^0{}_{[u\mathrm{Pers}\ u\mathrm{Num}]} \ldots [_{\mathrm{XP}}\ [_{\mathrm{VP}}\ [_{\mathrm{DP}}\ \textit{“Beans!”}_{[\mathrm{Pers}]}]]_i \ldots [_{\mathrm{vP}}\ [_{\mathrm{DP}}\ \textit{John}_{[\mathrm{Pers\ Num}]}] \ldots [t_i]]]]$ ✓

Here, the entire VP containing the object quote "Beans!" is smuggled over the external argument subject 'John' into a position where it is the first visible goal to the phi-probe on T^0. After entering into an Agree relation with this smuggled object, the probe can then optionally perform defective circumvention and undergo Agree with the (postverbal) external argument as well.

(22) $[\mathrm{T}^0{}_{[u\mathrm{Pers}\ u\mathrm{Num}]} \ldots [_{\mathrm{XP}}\ [_{\mathrm{VP}}\ [_{\mathrm{DP}}\ \textit{“Beans!”}_{[\mathrm{Pers}]}]]_i \ldots [_{\mathrm{vP}}\ [_{\mathrm{DP}}\ \textit{John}_{[\mathrm{Pers\ Num}]}] \ldots [t_i]]]]$

( )

Smuggling in these derivations involves the presence of a functional projection (which I have called XP in (21) and (22)) that is externally merged above the projection that introduces the external argument (which I call vP) and attracts the VP for the smuggling movement. As shown in (21), this smuggler XP is not present when there is no need for smuggling. Crucially, smuggling allows for the manipulation of thematic arguments from the positions in which they are externally merged into new configurations which create new possibilities for A-dependencies using only internal Merge.

Given that smuggling is used to derive voice constructions such as passives (Collins 2005; 2024) and middles (Gotah 2024), which involve the same VP phrasal movement, I refer to this XP henceforth as VoiceP, whose role is to smuggle thematic arguments over one another, thus creating new possibilities for A-movement of arguments and all that comes along with A-movement, e.g. new possibilities for Agree, Case assignment, scope, binding, and helping to facilitate new relations with arguments to Ā-positions such as those in the left periphery (Cinque 1990; 1999; Rizzi 1990; 1997; 2004). The use of VoiceP in this way additionally follows from the discussion of the nature and definition of syntactic voice alternations in the following section.

Thus, the central goal of this thesis is to present an analysis of English inversion constructions which display the agreement alternations (the properties of which are discussed in detail in Chapter 2), which are a hallmark of inversion constructions involving defective circumvention facilitated by smuggling via VoiceP. As a result of this analysis, I argue that English inversion constructions (quotative, locative, predicate, and existential) are all derived by the same smuggling-via-VoiceP operation and thus all show the same agreement optionality made possible by defective circumvention. While the technology underlying these inversion constructions is consistent across them, these constructions vary according to factors such as whether or not the structural subject XP is a locative and whether or not the external argument is obligatory in a focus position. As such, I present the following two-way binary classification of English inversion constructions involving defective circumvention.

(23) English inversion quadrilateral

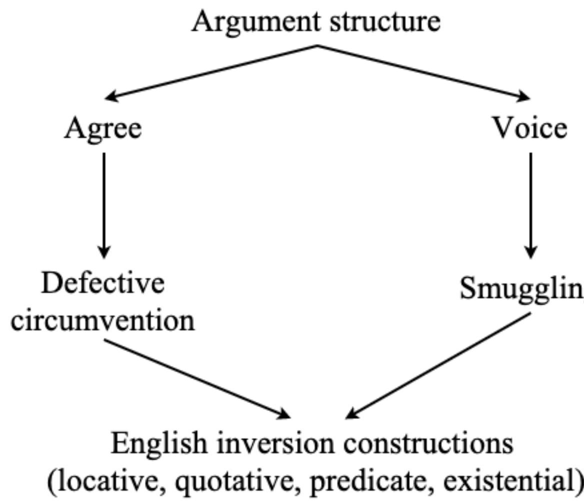
Necessary FocP

Inverted LOC		-	+
	-	Quotative	Predicate
	+	Locative	Existential

As an example, locative inversion involves the smuggling of a locative constituent to the preverbal position, and, as I show in Chapter 3, movement of the external argument into a focus position is not a necessary condition of all locative inversion derivations. On the other hand, predicate inversion does not necessarily involve the smuggling of a locative constituent to a preverbal position, and it does necessarily involve focusing of the postverbal external argument, which I detail in Chapter 4.

Outside of English inversion constructions, the theories of defective circumvention in Agree and smuggling via VoiceP also bear on a variety of constructions in other languages and, ultimately, show the connectedness of Agree, argument structure, and syntactic voice under a unified theory of all these not-so-disparate aspects of the syntax of human language. See the following concept map showing the relation of these ideas.

(24) Dissertation concept map



What this map shows is that argument structure constrains possibilities for Agree. Argument structure also affected by grammatical voice, as voice alternations affect the A-positions of thematic arguments. Agree drives defective circumvention, and voice involves smuggling (see the following section). Defective circumvention and smuggling are the signatures of the English inversion constructions analyzed in this dissertation.

The colloquial agreement alternations observed with English inversion constructions, which I systematically present in this thesis showing their attestation and grammaticality, reveal complex interactions between agreement and voice alternations, which ultimately bear on the way in which thematic arguments are projected.

It is now necessary to discuss some theoretical assumptions that underly the work in this thesis presented thus far and in the remainder of the work. These concern syntactic theory overall, as well as some considerations on argument structure, Agree, and voice.

1.2 Theoretical considerations

The work of this thesis takes place within the theoretical tradition of Minimalist syntax (Chomsky 1995). Specifically, I adopt the spirit and assumptions of the Strong Minimalist Thesis (Berwick & Chomsky 2016; Chomsky & Gallego & Ott 2019; Chomsky 2021; Chomsky et al. 2023) and the Miracle Creed (Chomsky 2024), by which I mean the following.

(25) *Strong Minimalist Thesis* (Berwick & Chomsky 2016:94)
The optimal situation would be that UG reduces to the simplest computational principles which operate in accord with conditions of computational efficiency. This conjecture is sometimes called the Strong Minimalist Thesis (SMT).

In line with SMT, I seek to pursue genuine explanation, about which Chomsky (2021) says the following.

(26) *Genuine explanation* (Chomsky 2021:12)
Conformity to SMT — hence genuine explanation — is achieved when an account relies on "third factor" principles such as computational efficiency, understood in this context as natural law. These can be taken off the shelf when needed, with no cost. SMT sets as an ideal that all linguistic phenomena can receive genuine explanations in this broader sense. A critical task for inquiry is to see how closely this ideal can be approached.

Informed by this notion of simplicity, I seek to pursue full explanatory adequacy (Chomsky 1965) and genuine explanation in my theoretical analyses here. In pursuit of genuine explanation, there are three goals that UG must meet.

(27) *Goals of genuine explanation of UG* (Chomsky 2021:7)
(i) It must be rich enough to overcome the problem of poverty of stimulus (POS), the fact that what is acquired demonstrably lies far beyond the evidence available.
(ii) It must be simple enough to have evolved under the conditions of human evolution.
(iii) It must be the same for all possible languages, given commonality of UG.

These goals are known as learnability, evolvability, and universality, respectively (Chomsky 2021:8). As it pertains to goals (ii) and (iii), I seek explanations of syntactic phenomena in this dissertation that are maximally simple and in accordance with principles of computational efficiency in order to arrive at explanations of syntactic structures that are simple enough to have evolved naturally in humans and to be generalizable across all possible languages. In other words, I seek explanations of argument structure, Agree, and grammatical voice that can explain complex inversion phenomena, voice alternations, and agreement facts in as many contexts in as many languages as possible while maintaining accordance with computational efficiency. I expound on some of these considerations in the following section.

For example, if we can pursue an explanation of complex inversion phenomena using minimal technology (that already exists in other domains) such as phrasal movement (internal Merge), A-movement (internal Merge), and a simple formulation of Agree, then this is preferable to an explanation that must stipulate conditions such as equidistance and technology such as multiple specifiers and a formulation of Agree that must cross multiple transfer domains. This is the same logic underlying the motivation to analyze nuanced and sensitive agreement alternations as the licit output of a single set of syntactic rules and operations, which I lay out in Chapter 2.

As another example of a theoretical assumption informed by SMT, let us see the radically simple formulation of the projection of thematic arguments (argument structure) that is assumed in this thesis, from Collins (2024).

(28) *Argument Criterion* (Collins 2024:5)
a. Each argument is introduced by a single argument-introducing head.
b. Each argument-introducing head introduces a single argument.

This Argument Criterion is an updated version of the theta-criterion.

(29) *Theta-Criterion* (Chomsky 1986:97)
Each argument α appears in a chain containing a unique visible theta-position P, and each theta-position P is visible in a chain containing a unique argument α.

These formulations of argument structure conform to SMT as they are radically simple, computationally efficient, and (as I show throughout this dissertation) sufficient to derive the data, even those with apparent minimality violations with respect to A-movement of arguments from these positions. In Chapter 3, I discuss the motivations for having such a formulation of argument structure, as well as some motivations to adopt assumptions such as argument introducing heads being projected uniformly across different clause types (e.g. UTAH (Baker 1997)).

As mentioned above, this theory of argument structure is intimately connected with the theory of Agree – as well as the theory of voice alternations (i.e., inversion) – presented in this thesis, and, as such, I outline some background theoretical assumptions and motivations on the theories of Agree and voice alternations used here, which I significantly develop in chapter 2 and 3, respectively.

1.2.1 Theory of Agree

I adopt a formulation of Agree which is maximally local and operates on syntactic features. The formulation of Agree, which detail in Chapter 2, builds on theories of Agree from Chomsky (2000; 2001), Béjar (2003), Miyagawa (2010), and Preminger (2014). The definition of Agree adopted in this dissertation is as follows.

(30) *Agree*
a. *Search*
Probe P on head H° initiates the operation Find(F) for a given unvalued feature [_F] present on H°.
b. *Match*
Probe P bearing an unvalued feature [_F] identifies the closest (under c-command) instance of [F] in its search domain
c. *Copy*
The feature geometry containing [F] is copied onto the probe, replacing (and thereby deleting) any instances of [_F]

I assume that the entirety of the operation Agree takes place in the narrow syntax, as this theory of Agree operates strictly on syntactic objects, on features. Under this formulation of Agree, there is no need to appeal to the spellout domains of LF or PF.

Under this definition, Agree is a tripartite operation consisting of Search, Match, and Copy, and the nonobligatory successful execution of these steps leads to the visible redundancies of movement (internal Merge), Case assignment, and, ultimately, the presence of agreement morphology. This is to say that Agree derivationally precedes internal Merge and Case assignment, contrary to such derivational timing considerations presented elsewhere (Bobaljik 2008; Preminger 2014). This is confirmed by data in which verbal agreement may track an accusative postverbal subject, which I present throughout this work. These data are a complication for any theory of Agree in which Agree is fed by morphological Case assignment and suggest that formulations such as the Revised Moravscik Hierarchy (Bobaljik 2008), in which morphological case "delineates an accessibility…for morphological agreement" (Bobaljik 2008:303), are not universal.

(31) *The Revised Moravcsik Hierarchy (M-Case)* (Bobaljik 2008:303)
Unmarked Case > Dependent Case > Lexical/Oblique Case

The theory of Agree adopted in this thesis, as well as the data that inform the theory, present a complication for any theory of agreement that is driven by morphological case. Additionally, it informs the definition of a subject that I use throughout this work, which is any viable goal in the search domain of the phi-probe on T^0, irrespective of case/Case. As such, all of the inversion constructions mentioned thus far are multiple subject constructions involving a preverbal structural subject (with which undergoing Agree is obligatory for T^0) and a postverbal thematic subject (that can conditionally be Agreed with by T^0).

The most important aspect of this syntactic approach to Agree as it relates to the other theoretical assumptions and implications of this thesis is that it is maximally constrained and therefore maximally local. Agree effectively reduces to Minimal Search (as this is what drives the whole operation): what is the closest element to a probe bearing the relevant feature that the probe is specified to search for? The relevant features here are syntactic privative features such as category features (i.e., D, V) and phi (ϕ), which encodes person, number, and gender (though gender will not be as relevant as person and number for the purposes of this dissertation)[1].

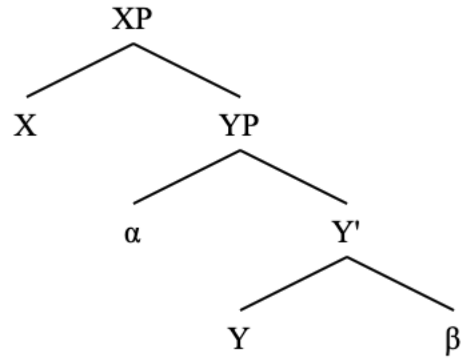
I define closeness in terms of asymmetrical c-command (Chomsky 1995; Ura 2000b) and the Minimal Link Condition (MLC) (Chomsky 1995:286). I assume the following definition of closeness, which does not consider any notion of equidistance.

(32) Closeness (Chomsky 1995:330)
β is closer to the target K than α if β [asymmetrically] c-commands α.

In the following structure, α and β are both asymmetrically c-commanded by X, with α being closer to X than β since α asymmetrically c-commands β

[1] A goal of this dissertation is to reduce the number of abstract features required to derive complex agreement patterns. As such, sticking to syntactically tangible privative features is a key of keeping this system as constrained and local as possible.

(33) Closeness under c-command



If X is a probe searching for feature F, and that α and β both bear an instance of feature F, then this maximally local formulation of Agree ensures that X finds α as a goal before it finds β, which is in accordance with the formulation of closeness presented above. Any notion of equidistance in Agree carries an additional set of stipulations that must work around this definition of closeness (and, indeed, I show in this work that a stipulation such as equidistance is not necessary at all to derive complex agreement alternations).

Phi-agreement probes are designated to target the closest element(s) bearing relevant phi-features, and nothing else. I show that this maximally constrained, narrow-syntactic formulation of Agree is the only technology necessary to derive complex agreement patterns including optionality and PCC effects.

The formulation of Agree that I pursue here takes place in the narrow syntax, operating on nothing more than a basic set of syntactic features. In other words, I account for agreement facts such as the complex alternations of agreement with multiple subjects in inversion constructions solely using a minimal syntactic formulation of Agree. Postsyntactic operations such as feature impoverishment (Bonet 1991; Keine & Müller 2025) will not be used.

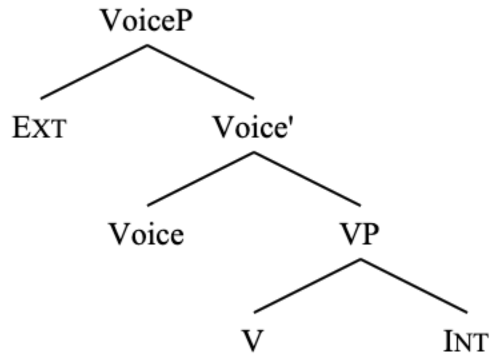
These background assumptions on Agree inform the theory of Agree described in detail of Chapter 2 of this dissertation. Furthermore, these assumptions on Agree, along with my theoretical assumptions on Argument Structure, inform my understanding of inversion and other voice alternations.

1.2.2 Theory of voice

In order to lay forth my theoretical assumptions on voice, it is first necessary to explain how the current dominant theory of voice is not adequate for the purposes of this thesis. This is necessary to identify exactly what we mean when we say that a given construction involves a syntactic voice alternation, which is what I argue for inversion constructions.

The popular (and near consensus-view) conception of syntactic voice can be broadly understood as a functional projection whose featural makeup is responsible for the projection (or not) of the external argument (Kratzer 1996; Embick 1998; Pylkkänen 2008; Legate 2014, Alexiadou et. al 2015; a.o). In other words, VoiceP is understood as the functional projection in the verbal domain which is responsible for introducing (or not introducing) the external argument. VoiceP under this view is tied to the projection of the external argument.

(34) Voice as the projection of the external argument



Much of this body of work treats VoiceP as if it were equivalent to little vP (in the Chomsky (1995) sense), and work that separates VoiceP and little vP still maintains that VoiceP is the projection responsible for introducing external arguments (Harley 2007; 2013; Sundaresan & McFadden 2017; a.o)[2].

The idea that the head responsible for the projection of the external argument is the same head that encodes voice alternations (i.e., active and passive) originates with Krazter (1996), who first names the head that introduces the external argument Voice, arguing that it "is truly at the heart of a theory of voice" (Kratzer 1996:120), though no empirical evidence is presented in this paper to suggest that the presence or not of the external argument is the same alternation as a voice alternation such as active vs. passive.

Empirical evidence for the theory of VoiceP is given later by other authors, such as Pylkkänen (2008), who suggests that there is no external argument projected in passives, giving evidence of this in the form of arguing that implicit agent external arguments of English passives cannot be modified by secondary depictive predicates. Of this, she gives the following example.

(35) *This letter was written drunk (Pylkkänen's judgment) (Pylkkänen 2008:22)

Despite the claim here that short passives *cannot* take agent-oriented secondary depictive predicates, there is a significant literature documenting the fact that such examples are in fact grammatical in English (Roeper 1987; 2020; Baker 1988; Stroik 1992; Collins 2005; 2024; Simpson 2005; Bhatt & Pancheva 2006; Müller 2008; Meltzer-Asscher 2012; a.o). I reproduce some of these counterexamples below, originally presented in Collins (2024).

(36) The game was played nude/drunk/sober/angry (Roeper 1987:297)

(37) a. This song must not be sung drunk (Baker 1988:318)
b. Such petitions should not be presented kneeling

[2] In these theories, v^0 is a verbalizing head (Harley 2007), or a causitivizing head (Sundaresan & McFadden 2017). Under the theory that I build up in this thesis, vP is responsible for introducing the external argument, and VoiceP has no role in the projection or not of the external argument, and in fact is not any kind of applicative head at all.

(38) At the commune, breakfast is usually eaten nude (Collins 2005:101)

(39) a. Traditionally, the koto was played seated on the floor (Meltzer-Asscher 2012:281)
b. The hula is danced barefoot

(40) a. The painting was painted blindfolded (Stroik 1992:132)
b. I can tell that this letter was written in a good mood
c. The bank job wasn't done alone

(41) a. Originally the game was played naked in the burning sun (Collins 2024:41)
b. Musical chairs should be played fully clothed
c. At the refugee camp, soccer was played barefoot
d. Our album is meant to be listened to stoned

Similarly, external arguments of passives can control adjunct clauses, which is characteristically not a possibility in constructions such as unaccusatives which do not involve an external argument.

(42) a. *The ship sank to collect the insurance (Mauner & Tanenhaus & Carlson 1995)
b. The ship was sunk to collect the insurance

This is also true of other voice constructions such as middles.

(43) a. Most physics books read poorly even after reading them several times (Stroik 1995:168)
b. Bureaucrats bribe easily after doing them a favor or two.

Additionally, the implicit external argument of passives can bind elements such as reflexive pronouns (Collins 2024; Sulemana 2024).

(44) a. Some things are better kept to yourself (Collins 2024a:17)
b. Emails sent to myself go to my junk folder (Collins 2024a:21)

As such, there is significant evidence to support the idea that external arguments are syntactically projected in voice alternations such as passives and middles in English. There is also work supporting this same conclusion for Spanish (Demonte 1986; Goodall 1999), Ewe (Gotah 2024), Bùlì (Sulemana 2024), Zulu (Halpert & Zeller 2016; Halpert 2022), Greek (Angelopoulos & Collins & Terzi 2020; Angelopoulos 2024), Czech (Karlík 2020), Italian (D'Hulst 1992), Hindi (Mahajan 1994), Albanian (Manzini et al. 2016), and Oshiwambo (Ndapo 2024). Furthermore, there is a significant body of work claiming more generally that voice alternations do not necessarily correspond to the projection or not of the external argument (Roberts 1987; 2019; Merchant 2013; Ramchand 2017; Newman 2020; Collins 2024a; a.o).

It is true that the data on secondary predication in short passives presented thus far in this section are controversial, and that there are explanations of them that do not rely on the projection of the external argument in passives (e.g., Williams 2015). Furthermore, it is certainly possible to have control of a PRO within a purpose/reason clause or secondary depictive predicate by something other than the external argument agent of a matrix clause (Williams 1974; 1985). Such observations, however, have no bearing on whether or not an external argument is present in passive derivations.

(45) Grass is green to promote photosynthesis (Williams 1985:310)

The fact that the PRO of the purpose clause here does not involve control by the agent of the matrix clause does not change the fact that (35) unambiguously carries the clear interpretation that the writer of the letter was drunk. Similarly, (42) carries the unambiguously clear interpretation that the implicit agent of the infinitival clause (the collector(s) of the insurance) are understood to be the sinker(s) of the ship. There is a relationship between the implicit agent of these sentences and the adjunct predicates that simply does not exist in (45).

That is, control of the PRO of an adjunct clause or secondary predicate by the syntactically realized agent of a passive does not mean that all PROs of adjunct clauses or secondary predicates must be controlled by an agent. In other words, the "Williams sentences" are simply not relevant to the question of whether or not an external argument must be syntactically present in the passive. It is not the case that, if one class of sentences does not involve control by an implicit agent, then no class of sentence can involve control by an implicit agent. Of the Williams sentences, Bhatt & Pancheva (2017:11) further argue that examples such as (45) do not definitively show that control into a purpose clause by an implicit argument is or is not possible.

The point of this discussion is to make clear that voice is not inherently associated with the projection (or lack thereof) of the external argument. The data, references, and discussion presented here strongly suggest that voice constructions such as passives actually do involve an external argument projected in Spec,vP ("VoiceP"). The data definitively demonstrate that significantly more evidence is needed to convincingly show that voice alternations are synonymous with alternations in the projection of an external argument (what has been called a *valence* alternation). As such, voice alternations are conceptually separable from the projection of the external argument. This fact and these data show the need for a definition of voice that does not center around the projection (or lack thereof) of the external argument.

I now turn to Oxford (2023), which analyzes the inverse voice construction in Algonquian languages. Despite using the VoiceP as the projection into which the external argument is externally merged (Oxford 2019:961-2; 2023:330), Oxford has the following to say about syntactic voice (46) and the inverse voice construction specifically (47). He cites the following works as background for these definitions: Klaiman (1991); Nichols (1992); Gildea (1994); Givón (1994); Zuñiga & Kittilä (2019).

(46) *Voice*: a contrastive grammatically marked pattern for mapping the arguments of a predicate to syntactic roles (e.g., active voice, passive voice) (Oxford 2023:313)

(47) *Inverse voice*: a transitive voice construction in which the patient is mapped to the subject position but the agent remains a core argument (Oxford 2023:314)

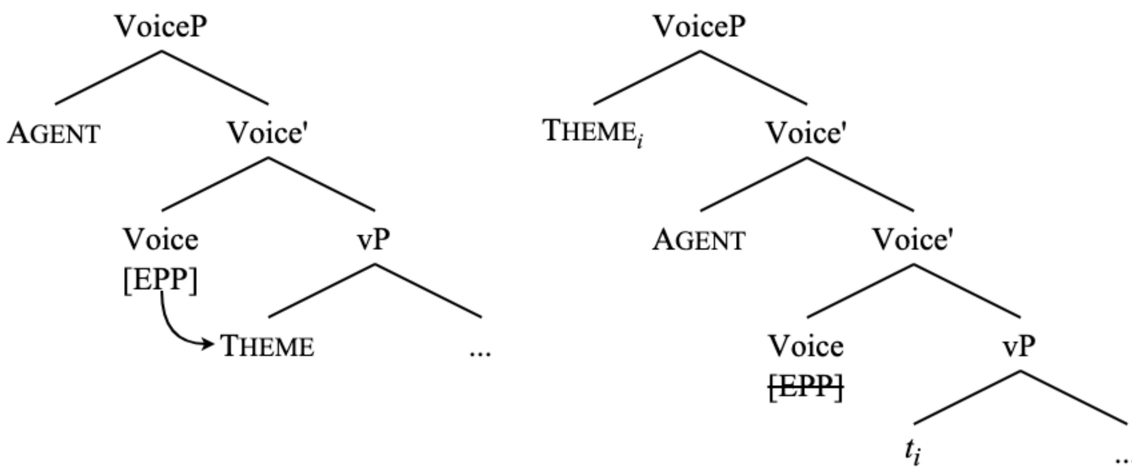
Here Oxford presents a definition of voice alternations – using a specific construction (inverse voice) as a case study – that is completely independent of whether or not an external argument is projected. Oxford has presented a definition of voice that does not rely on the projection of an external argument, despite using VoiceP as the projection that introduces the external argument. For Oxford, the derivation of the non-inverse and inverse voice constructions requires different VoicePs: one that does not induce movement of the object (internal argument) over the subject (external argument), and another that does[3]. The former derives the non-inverse construction (48), and the latter derives the inverse voice construction (49). See the following example from Oxford (2023) (ultimately from Nichols 1980:276-292).

(48) ni-wa:pam-a: (Ojibwe noninverse, Oxford 2023:312)
1- see- 3.OBJ
'I see her'

(49) ni-wa:pam-ikw (Ojibwe inverse, Oxford 2023:312)
1- see- INV
'She sees me'

The derivation of (48) involves the projection of a VoiceP that introduces the external argument AGENT and does not have any features that necessitate movement of the internal argument THEME to a position above the AGENT. In (49), however, a VoiceP is merged which bears an abstract [EPP] feature that triggers the movement of the THEME to Spec,VoiceP (which is already filled by the AGENT, and, as such, this derivation necessarily involves multiple specifiers).

(50) Movement of THEME to Spec,VoiceP in Oxford (2023)



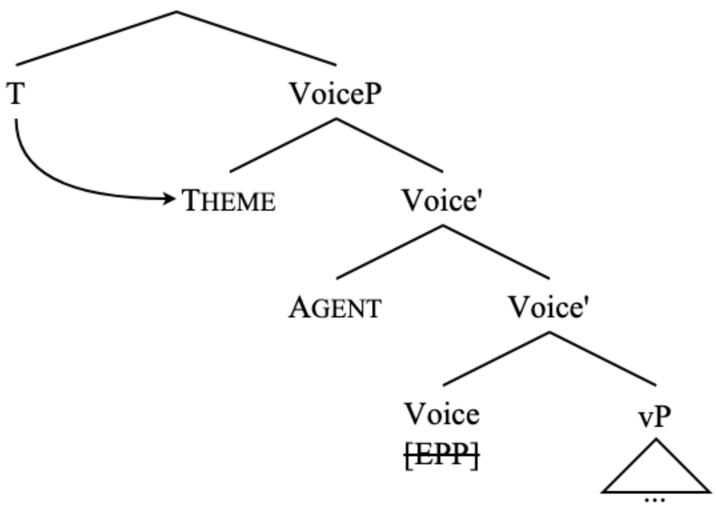
This movement of the THEME to a Spec,VoiceP above the Spec,VoiceP in which the AGENT is

[3] The idea that there are (at least) two different VoicePs with distinct featural makeups – one to derive canonical active/non-inverted, and (an)other(s) to derive marked voice constructions such as passives, middles, and inversion – is widespread in the literature that ties voice alternations to the projection of the external argument (i.e., Pylkkänen 2008). This kind of stipulation of multiple functional heads of the same flavor but of different featural makeups is an additional caveat for the number of heads that make up the verbal clausal spine that a language user must acquire. The VoiceP presented in this thesis, however, is uniform across constructions not only in its function but also in its featural makeup. This is another advantage that this formulation of voice bears over the theory that VoiceP is responsible for the projection of the external argument. Under that theory, there is a VoiceP in every single derivation, so it is not immediately clear how some constructions are more marked as involving "voice" over others.

located, triggered by the [EPP] feature on Voice^0, is essentially responsible for the inverse voice derivation. Crucial to these derivations of inverse voice is the allowance of multiple specifiers to create a landing site for the thematic object, which I show is an unnecessary complication that furthermore does nothing to account for verb movement and VP phrasal movement in inversion constructions, as this operation does not target any verbal projection or head of any kind, which I discuss in the following subsection.

From its moved position in Spec,VoiceP, the THEME then is the closest goal for undergoing phi-agreement with T^0.

(51) T^0 probes for Agree, finds THEME in Spec,VoiceP (Oxford 2023)



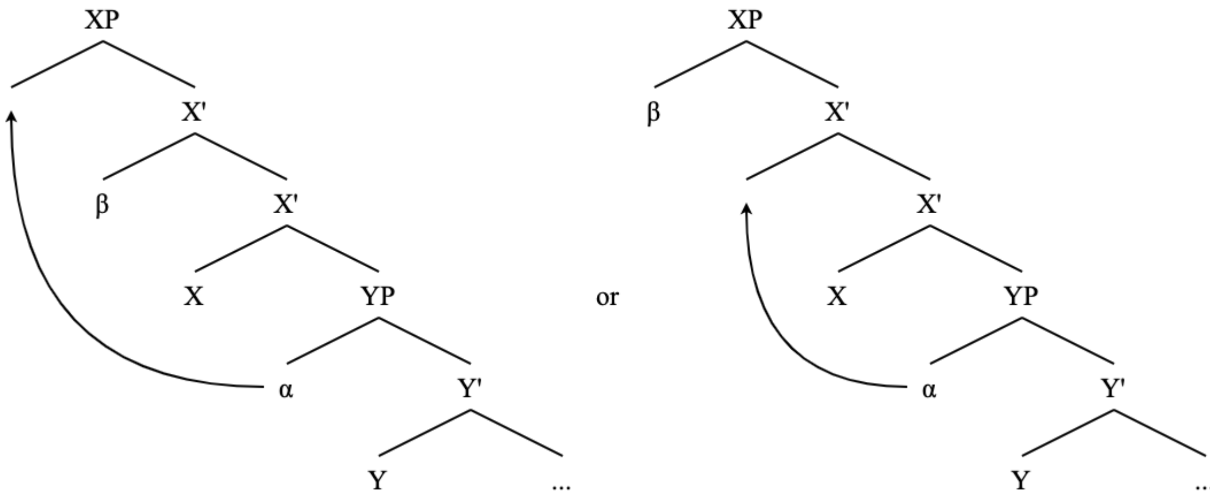
Additionally, under Oxford's analysis, the two specifiers and the head of a projection are considered equidistant from a probe located above that projection, so the THEME, AGENT, and Voice head are all accessible for the purposes of Agree[4].

Oxford's presentation of voice alternations effectively reduces to whether or not the VoiceP bears an [EPP] feature to allow raising of the THEME, but has nothing to with whether or not there is an external argument projected. In fact, Oxford claims that the "agent remains a core argument" (Oxford 2023:314) in these inverse constructions and shows it invariably projected in all VoiceP specifiers. In other words, Oxford shows that the projection of the external argument AGENT does not vary with syntactic voice alternations; the underlying theoretical assumption that syntactic voice alternations must be tied to the projection that introduces the external argument leads to an analysis that is empirically lacking (e.g., it does not work for English) and posits unnecessary devices such as equidistance.

[4] The incorporation of equidistance allowing for simultaneous Agree with more than one goal is a further complication which I show is unnecessary to derive the agreement interaction effects with multiple goals seen in these constructions, which I handle instead with my formulation of Agree as Minimal Search that allows for defective circumvention (which is also able to handle complicated facts concerning PCC effects in these Multiple Agree constructions in Algonquian (Coon & Keine 2023))

The entire formulation that Oxford uses crucially relies on equidistance. Further, any configuration using multiple specifiers necessarily assumes equidistance. See the following example.

(52) Possible multiple-spec configurations (with internal merge)



In the left tree, which is consistent with Oxford's derivation, equidistance of α to the head X^0 and β is necessary to derive α's landing site above β (in addition to the fact that, for Oxford, equidistance of these positions is necessary to derive the multiple agreement facts of Algonquian). In the right tree, which involves multiple specifiers and "tuckingin " – that is, α moves to a position between X^0 and β – equidistance of β and α would still be necessary to derive any further movement of α over β (or rather any operation that targets α at all). As such, multiple specifiers are an unnecessary complication for the same reason that equidistance is an unnecessary complication, which is that such a caveat to minimality is actually not needed to derive these complex agreement and inversion data.

This definition of voice that Oxford presents and derivations that involve voice alternations show that VoiceP as the external argument introducer is a misnomer, because Oxford nonetheless refers to the projection that introduces the external argument as "VoiceP". This misnomer and competing definitions of voice seem to work Oxford into a bind in which he must make use of technology such as multiple specifiers and equidistance in order to keep VoiceP central to this inverse voice alternation. This is technology that I ultimately show is unnecessary and can be done without, so long as the head that encodes voice alternations is separate and higher from the head that introduces the external argument. This kind of theoretical bind is also evident in accounts which must postulate yet another functional projection such as PassP to derive voice alternations separate of the VoiceP (i.e., Bruening 2013).

On this use of "VoiceP" being a misnomer, Myler & Mali (2021:3) have the following to say.

(53) "Since the head that introduces external arguments is in some languages clearly separate from and lower than the head that encodes passive-active alternations, the label Voice turns out to be something of a misnomer; see especially Merchant 2008, 2013. Nevertheless we retain this usage here for consistency with the works cited in the first

paragraph of this section and with other recent literature on causatives that uses the same framework." (Myler & Mali 2021:3)

Note that Myler & Mali continue to use VoiceP despite the misnomer, as does Oxford (2023), though they at least offer an explanation of it. Relatedly, of accounts such as Merchant (2013) and Johnson (2004) which show that passives can serve as grammatical antecedents to elided active VPs (and vice versa) (54), Merchant says the following (55).

(54) a The janitor must remove the trash whenever it is apparent that it should be
b. The system can be used by anyone who wants to (Merchant 2013:78-79)

(55) "The crucial element involved in these accounts is the separation of the head that determines voice from the head that determines the external valency of the predicate. There is in fact no conceptual reason these two should go together, and the ellipsis facts argue directly against this assumption." (Merchant 2013:98)

It thus follows that, conceptually and empirically, syntactic voice can be severed from the responsibility of some functional projection to introduce the external argument of a clause (see also Arregi & Nevins (2013; 2014) for motivation to separate VoiceP from the external argument introducing head (which they call *v**)). This, of course, leaves us with a very important question: what is a voice alternation? What is voice at all? Here we need a new working definition of voice, one that does not make reference to the projection of the external argument.

The idea that voice should be separated from valence is not new (see all the authors cited above), and this will guide the definition of voice presented here as a core assumption of this thesis. Returning to Oxford's (2023) definition of voice in (46), this is a good starting point, though it is not exactly clear from this definition (or the entire paper) what he means by "syntactic roles". As such, I instead make reference to the fact that voice alternations affect and create new possibilities for A-dependencies (by which I mean Case assignment, agreement, theta-role assignment, and binding (Fong & Halpert 2023)). A-movement and A-dependencies are also associated with linear precedence, as well as LF and information-structural phenomena scope (Bruening 2001; 2016) and even discourse prominence and topicality (Reinhart 1981; Thompson 1989; 1994; Nissim et al. 2004; Grimm 2009; Chierchia 2017), which voice alternations have been described as affecting (Bresnan & Dingare & Manning 2001; Doron 2015; Oxford 2023).

Voice alternations involve the creation of new A-dependencies – that is, A-dependencies which are not normally possible in an active, non-inverted derivation – which is reflected in patterns of Case assignment, agreement, binding, linear precedence, and scope, as well as information-structural properties that are influenced by A-dependencies such as topicality. Given this, I present the definition of a voice alternation that will be adopted and expanded upon in this dissertation.

(56) *Voice*
An information-structurally marked configuration in which a given thematic argument is mapped onto an A-position that would not be accessible to the same argument in a standard unmarked configuration. This alternation is dictated by the presence of a

functional projection VoiceP that mediates between the positions in which arguments of a predicate are externally merged (i.e., theta-positions) and A-positions that arguments move to via internal Merge (via Agree).

I expound on the exact nature of this functional projection VoiceP in the following section. This discussion on the nature of a voice alternation bears on another principle of SMT, which is the Duality of Semantics (Chomsky 2021).

(57) *Duality of Semantics* (Chomsky 2021:18)
EM is associated with θ-roles and IM with discourse/information-related functions.

This principle mean that the theta-role of a given argument is determined uniformly by external Merge: an AGENT is externally Merged in the position of an AGENT, a THEME in the position of THEMES, and so forth. Any internal Merge of an argument – that is, movement – determines not the thematic status of an argument, but its information-structural and discourse status, so long as an argument did in fact receive a theta-role in its external-Merge position. The thematic status of an argument remains consistent across voice alternations (including inversion), but the A-movement in voice alternations does affect the focus, topicality, and discourse-prominence of certain arguments over others. The typological definitions of certain voice alternations from Givón (1994) categorize voice alternation in this way, without making reference to valence at all.

(58) *Relative topicality of the agent and patient in the four main voices* (Givon 1994:8)

Voice	Relative topicality
Active/direct	AGT > PAT
Inverse	AGT < PAT
Passive	AGT << PAT
Antipassive	AGT >> PAT

These typological facts directly follow from the definition of voice alternations presented in this dissertation along with the Duality of Semantics principle. Thus, voice alternations are there to enable further relations between thematic arguments and information structure via A-movement that would otherwise be blocked by an intervening argument. This is technically implemented via smuggling, which I expound upon in the following section.

This definition of grammatical voice has the benefit of not making reference to the valency of a predicate at all, and thus does not rely on assuming that certain voice constructions lack an external argument[5]. This is beneficial given the empirical issues that come up when claiming that passives and middles lack an external argument. It is additionally beneficial as it allows for an expansion of what are considered voice constructions to include alternations that do not involve the linear displacement of the external argument (i.e., dative shift, antipassives), as well as constructions in which the external argument is uncontroversially present, such as Algonquian

[5] There is another advantage of this definition, which is that it does not make reference to overt morphology necessarily being part of voice alternations. Some work takes for granted that voice alternations must necessarily be encoded by dedicated verbal morphology (i.e. Dixon 1994:146; Embick 1998). This definition leaves room for constructions in which voice alternations are not marked on the verb, which can be anything from passives in languages such as Bùlì (Sulemana 2024) to inversion constructions in English to dative shift and antipassives.

inverse voice constructions and inversion constructions such as quotative inversion, locative inversion, and predicate inversion, which are the focus of this dissertation.

1.2.3 Smuggling (VP phrasal movement)

At this point, a potential conundrum emerges. The theory of argument structure taken up in this dissertation, which essentially reduces to the Argument Criterion, is radically simple. Thematic verbal arguments are uniformly introduced by argument-introducing heads in the hierarchy of verbal projections. The theory of Agree is also radically simple, as it is maximally constrained and effectively reduces to Minimal Search.

Given these assumptions, it is not immediately clear how to derive voice alternations (i.e., create new A-dependencies for verbal thematic arguments) with the simple formulation of argument structure and the maximally constrained formulation of Agree without violating locality constraints such as Relativized Minimality (Rizzi 1990) and without resorting to technology that carries a hefty weight of extra theoretical complications such as equidistance and multiple specifiers. If we want our theories of argument structure, Agree, and voice alternations to conform to SMT, then we need smuggling to get around locality constraints. I motivate this formulation of smuggling here.

For example, how does a THEME become the subject of a passive when it is not a viable subject in its base position? The solution here is that voice alternations necessarily involve phrasal movement. Let us first see some empirical support for this claim[6].

It is well-known that verb movement strands particles (Koopman 1991). In standard English active sentences, particles alternate between being able to immediately precede or follow the direct object of the verb.

(59) a. The coach summed (up) the argument (up) (Collins 2005:88)
b. John wrote (up) the paper (up)

In addition to involving the displacement of some arguments, many voice constructions also show overt displacement of the verb: i.e., in passives, the verb appears before the external argument. In these cases, particles may not appear on either side of the postverbal thematic argument.

(60) a. The argument was summed (up) by the coach (*up) (Collins 2005:88)
b. The paper was written (up) by John (*up)

This is additionally true of middles, assuming that the external argument can be realized with a *for*-phrase, which is similar to the *by*-phrase in passives (Stroik 1992).

(61) a. This paper cuts (up) easily for me (*up)
b. These lab reports write (up) super quicky for her (*up)

[6] It should be noted that a theory of voice alternations such as Oxford (2023) involving Multiple Agree and multiple specifiers does not account for these facts concerning VP phrasal movement.

This inability of the particle to follow the postverbal thematic argument shows that phrasal movement of the entire VP to a position over the postverbal thematic external argument.

A similar constraint is shown for the English double-object construction and dative shift (Collins 2021). In the double-object construction, a particle can follow the GOAL, but in the prepositional dative, a particle cannot immediately follow the GOAL.

(62) We sent the boys up some pizza (Collins 2021:103)

(63) a. We sent some pizza up to the boys (Collins 2021:103)
b. We sent up some pizza to the boys
c. *We sent some pizza to the boys up

According to Collins (2021), this contrast with the distribution of particles between the double-object construction and the prepositional dative construction shows that dative shift involves movement of a constituent containing the THEME and the particle over the GOAL argument. In other words, there is movement of some verbal XP.

Allowing the mixing of distinct registers for a moment, the semi-productive English antipassive construction also shows this contrast with the distribution of particles. Take a standard English transitive clause, which shows optionality in the realization of the particle.

(64) a. Whosoever drinks (up) my blood (up) shall live forever
b. Whosoever eats (up) all this bread (up) shall gain entrance to the kingdom of heaven

There is no such optionality in the antipassive.

(65) a. Whosoever drinks (up) of my blood (*up) shall live forever
b. Whosever eats (up) of all this bread (*up) shall gain entrance to the kingdom of heaven

These data demonstrate that four voice constructions in English involve VP phrasal movement, as the particle tracks the movement of the verb. This phrasal movement of a verbal XP over a thematic argument is exactly how new A-dependencies are created in voice constructions without violating locality constraints, and it is empirically supported (I give further evidence of this phrasal movement in the body of this thesis as it relates specifically to inversion constructions). The point of this phrasal movement, then, is to make a thematic argument visible to processes such as Agree that would otherwise be blocked by an intervening argument, via movement of a larger constituent that contains the moved argument, which is smuggling (see example (19)).

Smuggling, I argue, is at the core of all voice constructions, which is empirically supported by the distribution of particles in English voice constructions, including inversion. In fact, English inversion constructions show the same facts regarding the distribution of particles. See the following example with quotative inversion (see Collins 2003).

(66) a. Juan shouted (out) “Hey!” (out) to the crowd
b. “Hey!” shouted (out) Juan (*out) to the crowd

Smuggling as voice, then, is the movement of some verbal projection over a thematic argument, so that another thematic argument within the moved XP can be the target of an A-movement operation that it would not otherwise be a viable goal for. More simply, smuggling as voice involves phrasal movement of a verbal projection (Step 1), followed by A-movement of a thematic argument (Step 2), with the possibility of Step 2 being entirely dependent on the successful completion of Step 1. Smuggling has been used to analyze other voice constructions such as passives (Collins 2005; 2024; Sulemana 2024), dative shift (Collins 2024, Newman 2024; Thoms 2024), predicate inversion (Shlonsky 2024; Storment 2025), subject-object inversion (Shlonsky 2024), and quotative inversion (Collins 2003).

Smuggling is necessary given assumptions of this thesis that are in line with SMT. The one-to-one correspondence between the projection of thematic arguments and functional heads that introduce those arguments (Argument Criterion) and the formulation of Agree as Minimal Search necessitate a device to create new potential A-dependencies for thematic arguments without violating minimality. This device is smuggling, which is also empirically supported by the data on English VP movement (presented partially above, and more throughout the dissertation. According to SMT and these considerations, any OVS construction should be derivable by smuggling as voice.

VoiceP, then, is the projection that allows for smuggling (for a formal implementation of this with features, and more evidence of VP movement, see Chapter 3). Voice^0 as a non-applicative, non-thematic head, which is the formulation of it that I have here, given the discussion above on how voice alternations are separate from the projection of arguments, and also given the fact that this formulation of Voice^0 has a precedent in the literature (Collins 2003; 2005; 2024; Gotah 2024; Pietraszko 2024; Sulemana 2024; Storment 2025; a.o). See the following definition.

(67) *VoiceP*
A functional projection that targets the closest eligible verbal maximal projection for movement to its specifier, thereby causing the thematic argument contained within that maximal projection to become an accessible goal for A-movement.

Smuggling as voice shall be at the core of the analysis of inversion constructions in this thesis, as I show for English locative inversion, quotative inversion, predicate inversion, and also existential constructions. As I show in this thesis, smuggling as voice involves phrasal movement of a verbal projection to Spec,VoiceP, a projection which is necessarily present in voice constructions, and, in turn, not present in constructions which are not derived via smuggling. This fact touches on the markedness of voice constructions mentioned in Oxford (2023) (and ultimately in my own definition of syntactic voice).

A potential objection to any formulation of smuggling is that it poses a problem for the existence of freezing effects (Culicover & Wexler 1977; Wexler & Culicover 1980). Smuggling characteristically involves A-movement out of a moved constituent, which seems to be a violation of any generalization on freezing.

(68) *Freezing* (Bošković 2018:248)
Movement is not possible about of moved elements

The issue with freezing as it pertains to smuggling, though, is that the definition in (68) is an empirical generalization, not a condition that bans a certain class of movement wholesale. A minimalist account of the generalization in (68) should not ban outright any apparent exceptions. Counterexamples to the freezing generalization, then, should not only be expected, but should be able to be accounted for under the same system that derives freezing effects where they crop up. There is considerable work on freezing effects that allow for multiple kinds of freezing "violations", including smuggling (Corver 2017; Bošković 2018).

Using labeling and phasehood, Bošković (2018) replaces the generalization in (68) with the following generalizations to explain freezing effects when they appear and also to allow for counterexamples to be explained.

(69) a. Phases with nonagreeing Specs cannot undergo movement (Bošković 2018:272)
b. Unlabeled elements cannot undergo movement

Under these conditions for freezing, smuggling is not banned, as the no point of the smuggling operation involves the movement of a phase with nonagreeing Spec or an unlabeled element. Bošković says as much that this formulation of freezing does not rule out smuggling (Bošković 2018:272 *et seq*).

Furthermore, given that (68) is an empirical generalization, it is first necessary to show that every situation in which freezing holds does so solely because of some general ban on movement out of a moved element, and is not ruled out through some other means. Only then would it be possible to definitively claim that freezing poses a significant issue for any smuggling analysis. For now, however, that does not seem to be the case, nor would it be expected to be the case under any minimalist formulation of freezing effects that do not require positing freezing as an independently stated condition in the grammar.

Returning to smuggling, Voice concerns the mapping of thematic arguments from the positions in which they are externally merged onto A-positions to which these arguments move. As to why such an alternation should exist at all, I believe that it is tied to topicality and information structure more generally; that is, voice is responsible for the promotion (and subsequent demotion) of certain arguments in the clause, and ultimately in the discourse. This purpose of voice as a mediator to A-positions is certainly tied to the observation that A-movement can enable and facilitate further relations to the left periphery, which is the information-structural clausal domain. I make use of this fact in my analyses of locative and quotative inversion (indeed, information structure and its relevant projections such as FocP are relevant for all inversion constructions analyzed in this dissertation). As such, we have the following theoretical framework.

(70) *A theory relating agreement alternations to syntactic voice constructions*

a. Agree in these inversion constructions is characterized by *defective circumvention*
   i. Minimal Search – in accordance with SMT
   ii. Copying of syntactic features
   iii. No equidistance, no simultaneous Multiple Agree
   iv. Derives agreement optionality, PCC effects

b. Voice alterations involve *inversion via smuggling*
   i. Dependent on the Argument Criterion – in accordance with SMT
   ii. Not tied to projection of the external argument
   iii. Not tied to any overt morphological alternation
   iv. Expanded definition of what counts as a voice alternation
   v. Deriving complex data from core underlying principles of Agree and argument structure

This theory of voice as smuggling is advantageous in many ways. First, the separation of syntactic voice and the projection of the external argument allows for a more accurate account of voice constructions which clearly involve an external argument, and for an expanded definition of what kind of constructions count as voice constructions in the first place. This theory of voice also does not need to make use of technology such as multiple specifiers or equidistance to work around locality constraints. Using inversion constructions as the primary object of study, I convincingly show in this dissertation that inversion constructions such as locative, quotative, and predicate inversions are canonical voice constructions which are derived via smuggling and have a different set of A-dependencies than their non-inverted derivational counterparts.

1.3 A note on the data

The vast majority of the data in this section are from the Internet. They were either obtained via corpus search (e.g., Twitter) or via Google search. Written linguistic data collected from the Internet in informal settings (social media, blog posts, etc.) are quite valuable due to their qualities of often being spontaneous, low-register, colloquial, and minimally edited (Kilgarrif & Grefenstette 2003; Schütze 2011; Wikström 2017; Collins 2023). This type of data is ideal for data that would not otherwise appear in writing, as published work usually undergoes editing and prescriptive influence, and historically excludes more minoritized language varieties. These data are good for natural written utterances that avoid self-editing due to the informal setting in which they occur, and as such they are more likely to circumvent the Observer's Paradox (Labov 1970) than other written material.

Spoken linguistic data online also corroborates the written utterances, as there are informal spoken materials widely available on the Internet that are often transcribed and searchable. These materials have the same advantages as the written examples described above, with the added benefit of not having to control for issues such as typos in the same way. These data include YouTube videos, Facebook videos, podcasts, and video interviews. This corroboration also serves to highlight the ways in which written language online comes to closely resemble spoken language (Storment 2024: section 6).

Such data have been systematically collected and effectively utilized for linguistic research (Collins & Postal 2012; 2014; Hoff 2020; Fernández-Serrano 2022; Storment 2024; Collins

2024a; a.m.o.), though researchers working with these data must take certain steps to verify the validity of the data that may not be necessary when gathering data through other means. The considerations I outline below have been described by other researchers (i.e. Fernández-Serrano 2022, Collins 2023), but I repeat them here to indicate that these steps were also taken for the data presented in this thesis.

On obtaining the relevant data via Internet searches (or spontaneous encounters), it must be filtered for clarity, as it stands to reason that examples that are incoherent and massively full of typos are more likely to contain ungrammatical strings or be from bot accounts. It is also necessary to filter out examples that have clearly been copypasted over and over, as those are more difficult to confirm the native-speakerhood of the author. These issues generally do not apply to spoken utterances.

The status of the author as a native English speaker must also be verified. If the post containing the relevant example contains information about the author or a link to the author's profile, information such as their name and place of origin can be used to verify this, as well as reading other posts of theirs to verify that their written English does not contain transfer errors. For most of the data presented here, it was possible to obtain biographical information about the author, but for those which contained no biographical information about the author, it was still possible to comb through their other posts to see if their English approximated that of a native speaker.

After going through these steps, the data which have made the cut must be verified by a native speaker (or several) via a grammaticality judgment task. This is perhaps the most important step when verifying the validity of these data. The confirmation of sentences online as grammatical by native speakers of the language in which the utterances are written gives overt positive evidence that the written strings are grammatical in (at least some) speakers' I-languages. These positive confirmations also obviate the potential issues of typos or non-native-speakerhood of the original authors.

This discussion is to say that the data presented throughout this thesis of nonstandard, colloquial agreement patterns (such as the optionality of agreement with a third person plural postverbal subject) are valid for the purposes of scientific inquiry into the syntax of natural languages.

It must be noted that the existence of these examples and the fact that some speakers find them grammatical does not mean that all speakers find them grammatical, nor does it predict that. The presence of this intraspeaker variation is a point of *inter*speaker variation. The exact dimensions of this interspeaker variation are not the focus of this thesis; rather, the focus is the underlying syntax that creates the intraspeaker variation.

1.4 Outline

This thesis is structured as follows. In Chapter 2, I present data on agreement with postverbal subjects in English and a variety of other languages, as well as the theory of Agree that accounts for these facts. This conceptualization of Agree takes place in the narrow syntax, allows for sequential Agree with multiple goals, and gives rise to agreement optionality and weak PCC effects in various multiple subject constructions.

In Chapter 3, I discuss in detail the mechanism for deriving locative inversion as smuggling as voice, as well as the theoretical implications of such a technology. This analysis of locative inversion preserves the thematic status of verbs that participate in locative inversion constructions and involves A-movement of the fronted PP. The derivation of locative inversion is the general framework in which I analyze the other inversion constructions presented in this thesis. I also discuss the quotative inversion, as well as properties of inversion constructions such as the Transitivity Constraint (Collins & Branigan 1997) and heavy NP shift.

In Chapter 4, I argue that English existential sentences are also inversion constructions of the same variety as locative inversion, and I also show how predicate inversion is derivable from the same mechanism. Here I discuss the syntax and semantics of so-called expletive ‘there’, and discuss in more detail the information-structural properties of these inversion constructions.

Chapter 5 concludes. I discuss my main findings and theoretical contributions, as well as some open questions. In Chapter 5, I also discuss some potential extensions of this work, such as the derivation of more traditional PCC effects (e.g., those seen with multiple object clitics in Spanish (Bonet 1991), seeing if the mechanism of inversion as smuggling via VoiceP can sufficiently derive other inversion constructions crosslinguistically. I further speculate on the application of defective circumvention to derive other cases of optionality in language, as well as the application of smuggling to derive cases of apparent superiority violations.

2 Deriving optionality with Agree

In the colloquial English of many speakers there is an agreement alternation that occurs with postverbal thematic subjects and preverbal material such as existential 'there', a locative, or a quote. The alternation is between phi-feature agreement with the thematic subject and so-called default (that is, third person singular) agreement.

(71) a. There {was/were} problems
b. Into the room {walks/walk} four cats
c. "You pass!" {says/say} the committee members
d. What you need {is/are} some new shoes

Crucially, these alternations are not possible when the thematic subject is the preverbal argument structural subject.

(72) a. Problems {are/*is} rampant in his work
b. Four cats {walk/*walks} into the room
c. The committee members {say/*says} "You pass!"
d. Some new shoes {are/*is} what you need

Such alternations are attested elsewhere crosslinguistically, for example with dative subjects and postverbal nominative DPs in Spanish (73) (Lope Blanch 1971; Villa-García 2010; Fernández-Serrano 2022; a.o.), Icelandic (74) (Sigurðsson 1992; 1996; 2004; Holmberg & Hróarsdódttir 2003; Hoover 2021; a.o.), and Faroese (75) (Jónsson 2009), existential constructions in Romance varieties such as Catalan (76) (Rigau 1999a; b; 2005) and Spanish (77) (Rodríguez Mondoñedo 2006), and unaccusative verb constructions with postverbal subjects in Portuguese (78) (Costa 2000; Kato 2002) and Spanish (79)-(80) (Olarrea 1996; Mejías-Bikandi 1995; Mensching & Remberger 2006; Huidobro & Franco 2012).

(73) Nos encanta{-Ø/ -n} las películas de terror (Fernández-Serrano 2022:9)
us.DAT love {-3sg/-3pl} the films of terror
'We love horror movies'

(74) Henni {líkaði/ líkuðu} þeir (Icelandic, Sigurðsson & Holmberg 2008)
her.DAT {like.3sg/like.3pl}they.NOM
'She likes them'

(75) Nógvum kvinnum {dámar/ dáma} mannfólk (Faroese, Jónsson 2009)
many.DAT women.DAT {like.3sg/like.3pl} men.ACC
við eitt sindur av búki
with a bit of belly
"Many women like slightly fat men'

(76) Hi ha {-Ø/ -n} tres cadires (Catalan, Rigau 2005)
there have{-3sg/-3pl} three chairs
'There is/are three chairs}

(77) Hub{-o/ -ieron} dos hombres en la fiesta (Rodríguez-Mondoñedo 2006:326-7)
had {-3sg/-3pl} two men in the party
'There {was/were} two men at the party

(78) Fech {-ou/ -aram} muitas fábricas (Portuguese, Costa 2000)
closed{-3.sg/-3.pl} many factories
'Many factories closed

(79) Lleg {-ó/ -aron} los colectivos (Spanish, Ordóñez p.c.)
arrived{-3sg/-3pl} the buses
'The buses arrived'

(80) Lo que quiere {es/son}los privilegios para sus empresas (Spanish, Ordóñez p.c.)
it that wants {is/are} the privileges for his businesses
'What he wants is/are privileges for his businesses'

Some research has pointed out that postverbal subjects have a stronger tendency to not be agreed with than preverbal subjects (Samek-Lodovici 2002; Ortega-Santos 2008). Similar patterns have been noted for Italian varieties (Brandi & Cordin 1989; Saccon 1993) and Arabic varieties (Fassi Fehri 1993). This pattern has even been noted as a linguistic universal (Greenberg 1963; 1990; Corbett 1979; Barlow 1992; Manzini & Savoia 1998; Ortega-Santos 2008).

(81) *Greenberg's universal 33* (Greenberg 1963:94)
When number agreement between the noun and verb is suspended and the rule is based on order, the case is always one in which the verb precedes and the verb is in the singular.

The agreement alternation with plural associates in the English existential 'there' construction shown in (71a) has been described elsewhere in the literature (Rochemont 1978; Sobin 1997; Munn 1999; Schütze 1999; López 2007; Deal 2009; a.o), although with the caveats that the singular agreement is only acceptable with the contracted form of the singular copula *-'s* (Sobin 1997, Deal 2009) or that singular agreement is only possible via first conjunct agreement with a DP coordinate structure associate (Munn 1999). It also has been pointed out that singular verbal agreement with a plural associate is only possible in existential 'there' constructions with a copular verb (Rochemont 1978; Schütze 1999; Deal 2009). I demonstrate that all of these claims do not hold for many English speakers, and that the distribution of singular verbal agreement morphology with a plural associate in the existential 'there' construction – and other postverbal subject constructions in English – is significantly more widespread than previously reported.

Using colloquial speech data collected online in both written and spoken form, I show in this chapter that the above generalizations previously reported in the literature are not true for all English speakers. I also crucially show that this agreement pattern of singular verbal agreement with a plural postverbal subject is optional for those speakers who have it: that is, they may alternate between an agreeing form and a non-agreeing form. It is not the case, though, that every postverbal subject may freely show this agreement alternation. I show these constraints as well. These generalizations inform the theory of Agree presented in this chapter. These agreement optionalities and their distributions are accounted for under a formalization of Agree that effectively reduces to Minimal Search and that allows for a probe to undergo Agree with more than one goal so long as the features of each goal are compatible with each other and with the probe. These interactions create the agreement optionalities as well as Person-Case-Constraint (PCC) effects that vary crosslinguistically depending on the featural specification of the phi-probe, as well as the ways in which certain phi-feature values are parametrically represented.

Traditionally, minimalist grammars do not predict any sort of alternations in form which do not correspond to alternations in semantic interpretation (Chomsky 1995). In other words, there should be no pure syntactic optionality: that is, an alternation in agreement or word order that is semantically vacuous (Biberauer & Richards 2006). The optionality between two or more morphosyntactic structures in the exact same context, then, is potentially a problem for minimalist syntax. Obviously, such semantically vacuous alternations must be allowed in the grammar, not only to explain the agreement patterns in (1) but also to handle more well-described phenomena which are (erroneously) said to have no effect on truth conditions such as passives (Collins 2005; 2024a; Sulemana 2024) and dative shift (Collins 2024a; Newman 2024; Thoms 2024). How exactly to derive this optionality within the speech of a single individual requires careful consideration (and some may question whether the optionality is possible at all[7]). At the end of this chapter, I detail several possible alternative accounts of this agreement optionality in agreement, showing that they are not sufficient to derive the data.

2.1 On phi-feature deficiency

All of the relevant constructions mentioned in this thesis involve a configuration in which a non-canonical subject appears in the preverbal structural subject position (for our purposes it is sufficient to say that this position is Spec,TP) and there is also a postverbal DP which the verb can optionally track agreement with. In sentences such as the Spanish and Portuguese examples where nothing appears overtly in the preverbal position, I assume a null element *pro* preverbal subject position[8] (Suñer 1982; 2000; 2003; Goodall 2001; Kučerová 2014; a.m.o). The non-canonical subject that ends up in Spec,TP in these constructions is crucially featurally deficient, which is what allows for the agreement optionality. Here I clarify this notion of deficiency.

Deficiency is a broad notion in syntax, and in linguistics in general. Deficiency is a concept implemented in various areas of linguistic scholarship. A given linguistic element can be phonologically deficient, having a reduced or null phonological exponence (Cardinaletti & Starke 1996; 1999). An element can also be semantically deficient if it is underspecified for an index or lacks one altogether (Kratzer 1998; Paul & Travis 2022).

A syntactically deficient element is one that lacks a given set of features that elements of the same category are typically specified for. For example, a clause lacking a tense feature – and the structure associated with projecting a tense feature – is a deficient clause (Satik 2024). Additionally, a DP which lacks certain phi-features that DPs in that language normally are

[7] This is perhaps a valid concern. To these people, my initial reaction is to say, "Let the data speak for itself!", though I do not rule out the possibility that this point may be worthy of more significant discussion.

[8] It is worth noting that, in both of the languages mentioned here (Spanish and Portuguese), there are dialects in which these proposed null subjects are actually realized overtly, which is further corroboration of their existence in the null forms. See the *ello* of Dominican Spanish (Muñoz-Pérez 2014) and the *ele* of European Portuguese (Carrilho 2008). These overt DPs are phi-deficient in the same way as the null ones.

specified for is a phi-deficient DP; this is also called “defectivity” (Chomsky 2000:128).

The notion of deficiency/defectivity is a relative one. Elements are only deficient in relation to the specification of other elements of the same kind in the context of a given dialect/language. A DP that lacks a gender feature is only deficient in a language that systematically distinguishes gender on DPs. An arbitrary pronoun is only semantically deficient in relation to the referring power of other pronominal expressions in the language. A phonologically reduced pronoun – for example, ‘ya’ /jə/ (‘ya know’) or ‘y-’ /j-/ (‘y’know’) in English – is only deficient in comparison to the non-reduced form of the pronoun ‘you’ /ˈju/.

Deficiency is also gradient. In the phonologically reduced pronoun example above, ‘ya’ is less reduced than ‘y-’, which is in turn less reduced than a null pronoun.

Let us conceive of a syntactic element ε which is specified for features [X], [Y], [Z], and [W]. Now, imagine a second element ε′ of the same category as ε which is specified for features [X], [Y], and [Z]. Imagine once again a third element ε″ which is specified for features [X] and [Y]. We can also posit the existence of two more elements: ε‴ which bears only the feature [X], and ε⁗ which bears no features at all. This is a spectrum of syntactic deficiency starting with (what we can assume is) a minimally deficient element going up to a maximally deficient element which bears no features at all.

If a probe in this language is specified to only search for and be valued by elements of the same category as ε bearing only a feature [X] (in the popular literature we would say that this probe bears a feature [*u*X]), then only the element ε⁗ is deficient according to the specifications of this probe. Every element besides ε⁗ will be able to fully value (or “satisfy” (Deal 2015 *et seq*)) the needs of the probe.

Conversely, if a probe in this language bears the features [*u*X], [*u*Y], [*u*Z], [*u*W], and [*u*Q], then all the listed elements are deficient according to the needs of the probe. In other words, none of the elements listed would be able to fully value this probe.

Given that deficiency is such a relative notion, it makes sense to only talk about deficiency as it relates to a given scenario. Speaking specifically of phi-feature deficiency, it makes sense to only conceptualize of elements being phi-deficient with respect to the specifications of a relevant phi-probe. As such, I define phi-deficiency as the following:

(82) *Phi-deficiency*
A DP is phi-deficient iff the phi-features for which it is specified are a nonzero proper subset of the phi-features on the relevant phi-probe.

I take so-called “default” agreement to be the result of a phi-probe entering into an Agree relation with a deficient DP. The alternation between default agreement and agreement with the features of a postverbal subject DP in the data presented in this thesis should be then understood as the alternation between the phi-probe on T being valued only by the features of a deficient subject or by the features of a non-deficient postverbal subject.

Assuming arguments (including so-called "expletives") are externally merged in functional projections below the phi-probe on T responsible for subject-verb agreement, we end up with the following configuration (see also example (10) in the introduction).

(83) $[P_{[uF\ uG]} \ldots [XP_{1[F]} \ldots [XP_{2[F\ G]}]]]$

The agreement alternation in (71a), then, is taken to be the result of the phi-probe on T optionally Agreeing with just existential 'there' (XP_1), or agreeing past this first goal and also entering into an Agree relation with the postverbal DP 'problems' (XP_2). The first of these possibilities is spelled out as 'was', and the second is spelled out as 'were'. Agreeing with XP_2 is only possible after the probe Agrees with XP_1 because of minimality. When subject-verb agreement is spelled out according to the phi-features of XP_2, it is indicative of a configuration in which the probe has simultaneously undergone Agree with both the deficient XP_1 and the fully specified XP_2.

According to Chomsky (2000:128), English existential 'there' bears only a person feature. The idea that existential 'there' is specified only as third person and nothing else is corroborated elsewhere (Chomsky 1981; 2001; Béjar 2003:62; Deal 2009; Tsiakmakis & Espinal 2022; a.o). This fulfills the definition of featurally deficient because it does not lack any specification for phi-features at all, but it does lack a number feature, which English T^0 is also specified to find. This is how we will derive the fact that 'there' can optionally affect phi-agreement.

(84) a. There was problems
b. There were problems

It would be problematic to claim that deficient subjects such as existential 'there' are totally underspecified for all phi-features as then they would be completely invisible to the phi-probe (and thus we would never expect a sentence such as 'There was problems'), unless of course we posit that 'there' bears some abstract underspecified phi feature [ϕ] that the probe is specified to find (see such a probe-goal configuration in Deal (2021)). This is also problematic, though, because a probe specified to only Agree with any element bearing [ϕ] would have no reason to probe past a deficient DP bearing only [ϕ], and thus we would never expect any sentence in this configuration in which agreement happens with a postverbal subject.

The fact that these deficient subjects are specified only for person – third person – and nothing else is crucial to derive the constraints on what is and is not possible in these constructions, which I lay out in the remainder of this chapter.

Again, deficiency is a relative notion. If a given element is not deficient with respect to the features of the probe, then there is no possibility of probing past it. This is what we see with existentials in African American English and French, for example.

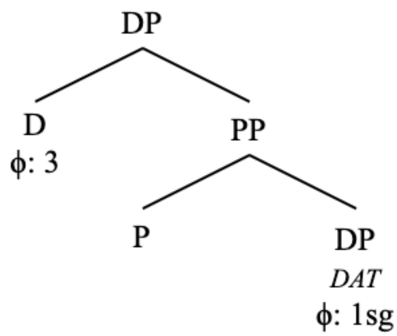
(85) It {was/***were**} a lot of things going on in this lesson (AAE; Green 2002:81)

(86) Il y {a/ ***ont**} deux hommes (French; Schütze 1999:478)
it LOC {has/***have**} two men
'There is two men'

The same kind of deficiency (i.e. only being specified as third person, and lacking any other phi-feature specification) holds not only of existential 'there' but also of wh-cleft subjects (Krapp-López & Gravely 2024), locative PPs, and, as I show, quotes. This deficiency also extends to dative subjects in Spanish, Icelandic, and Faroese.

Dative subjects are situated in a PP shell headed by a deficient DP (Rezac 2004; 2008; Bjorkman & Zeijlstra 2019; Fernández-Serrano 2022, a.o). As such, dative subjects are visible to the phi-probe on T^0 as bearing only a person feature, regardless of the features of the DP inside the shell which is actually marked as dative. See the following (simplified) structure of a first person singular dative DP.

(87) Dative shell

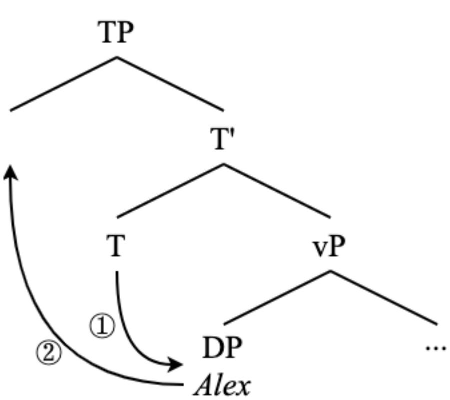


I make use of the idea of a deficient shell of sorts to also explain the deficiency of locatives and quotes (Chapter 3) and ultimately I connect this to the deficiency of existential so-called expletives as well (Chapter 4). Before that, though, it is necessary to explain in detail the mechanism of Agree by which these agreement alternations between a deficient and non-deficient subject are derived.

2.2 Deriving agreement with Agree

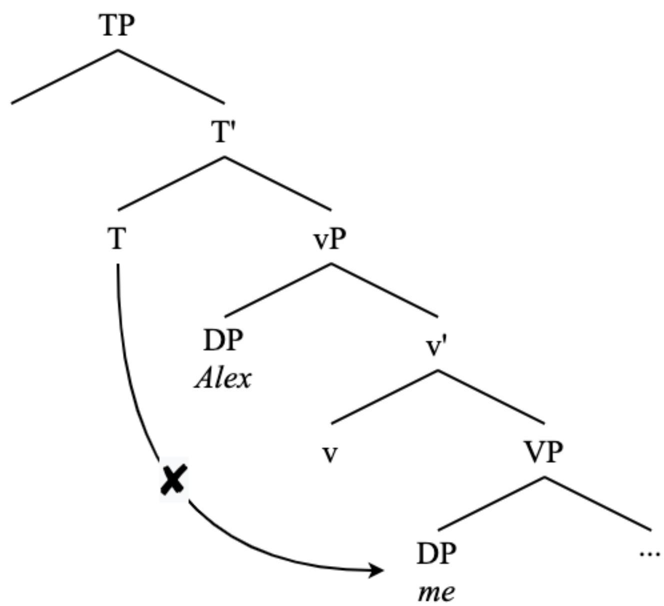
Let us start by detailing how agreement works in English. Typically, the relevant phi-probe for subject-verb agreement – which I assume to be T^0 – agrees with the closest phi-feature bearing goal in its search domain. This DP is usually fully specified for the phi-features that the probe is looking for. Take the following example, which includes agreement and EPP movement.

(88) Alex loves me



Here T probes down to find the subject DP 'Alex', which is specified as third person singular, T undergoes agreement with this DP (①), and the DP is moved to Spec,TP, where it is pronounced (②). The agreement morphology on the verb is spelled out as third person singular as a result of this Agree relation being established (Chomsky 2000; 2001). In this example, T is not able to agree with any DP besides the one closest to it. In other words, an Agree relation could never be established and spelled out between, say, T and the object DP, which bears first person singular phi-features, such as in the following example.

(89) *Alex love me



For this Agree relation to obtain would be a violation of minimality: it would involve probing past a closer DP with which an Agree relation could be established (Rizzi 1990; McGinnis 1998; Chomsky 2000; 2001). The phi-probe on T would never encounter the object DP in this example because there is a closer, equally suitable goal. This structure also brings up PIC-related issues.

It may seem, then, that the phi-probe on T in English is specified to find and Agree with the first phi-bearing element of any kind that it encounters on its search, and then stop. Such has been claimed in Deal (2021), and elsewhere; however, what happens when the closest potential goal to the probe under minimality is phi-deficient shows that the probe can search past its closest potential goal. Upon further examination, such as the data presented thus far in this thesis, it seems more likely that T is specified to find person features and then number features in that order (Béjar 2003, Preminger 2014, Coon & Keine 2021, Krapp-López & Gravely 2021, a.m.o.), with some sources claiming that this probing order on T is universal across languages (Béjar & Rezac 2009; Preminger 2014; Deal 2015; Kalin & van Urk 2015; Půskar 2019) (see Section 2.1.3.1). This apparent ordering can be explained by the nature of deficiency and the hierarchical organization of phi-features both on DPs and phi-probes.

As such, consider the following example of a phi-probe on T that only searches for a phi-bearing goal (90a), and a phi-probe on T that is relativized to search for an element bearing person and number features, such as what we have in English (90b). As I show in the following section, the probe in (90b) actually entails the probe in (90a) due to the hierarchical nature of phi-features.

(90) a. T[$u\phi$]
b. T[[uPers][uNum]]

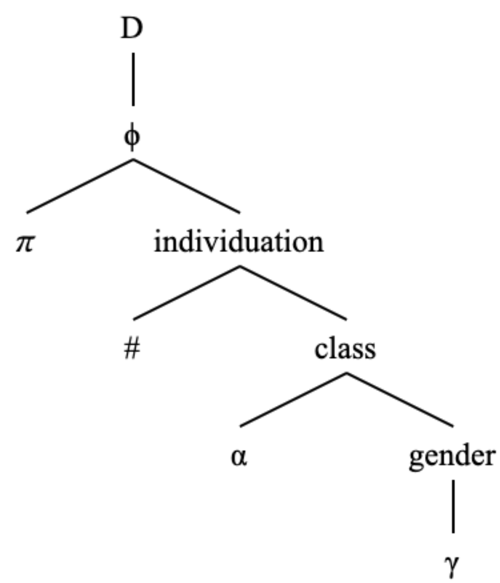
2.2.1 On the representation of features

Phi-features do not exist in a vacuum. They exist in the context of one another. Therefore, it is necessary to develop a systematic representation of phi-features that captures the dependencies and interactions that exist among phi-features. Here I discuss how I represent phi-features, keeping these desiderata in mind.

2.2.1.1 Feature geometries

Phi-features are hierarchically arranged in geometries (Harley & Ritter 2002) of privative features (Béjar 2003; Béjar & Rezac 2009; Preminger 2014; Coon & Keine 2021; a.o) whose presence (or absence) causes DPs to be interpreted a specific way. These feature geometries are syntactic objects (Preminger 2014). I assume a (simplified from Harley & Ritter (2002)) hierarchy of phi-features (ϕ), shown in the following example, with π representing Person, # representing Number, α representing Animacy, and γ representing Gender. Harley & Ritter present this hierarchy as universal, a claim that I am sympathetic to. This means that π will always be ordered above # in any language that encodes both person and number in its syntax.

(91) Hierarchy of phi-features (phierarchy)



In this hierarchy, bearing a person feature [π] is the minimum value that a phi-set may be specified for while still carrying any kind of internal structure. This is why the phi-deficient DPs discussed thus far are only valued for person and nothing else. This is corroborated by the fact that there exists languages which do not make morphosyntactic gender distinctions, animacy distinctions, and even number distinctions (Harley & Ritter 2002), but there are no languages in this world which do not encode any kind of person distinction.

Languages vary with respect to which phi-features they project. For example, it is well-known that not every language shows grammatical gender or animacy distinctions, and, although it is rarer, there are languages which do not show number distinctions. As such, any claim about the purported feature geometries of one language cannot be expected to hold of another language without evidence, aside from some universal constraints on ordering. I assume entailment relations between phi-features: features on lower nodes entail the presence of features above them (Coon & Keine 2021:664, a.o).

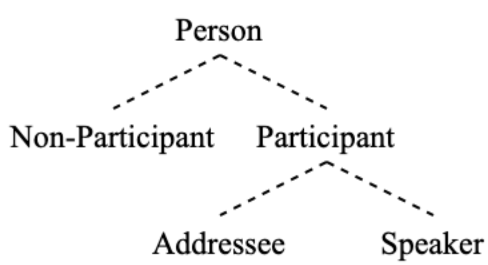
In this representation, phi-features are privative, as opposed to being specified for some value. As opposed to values on a specific feature [F:*val*] (Pesetsky & Torrego 2007), the interpretation for a given subset of features are computed from whether or not certain features are present in the geometry. About this, Béjar (2003) says the following:

(92) "Features in the system are privative categories with no 'value' per se. In a sense, entailment dependencies introduce something like values. For instance, we could think of a [Part[Spkr]] as a PARTICIPANT category with the value SPEAKER. We will see later in this chapter that in certain respects entailment relations do function as though dependents were 'values' for their dominating features. However it is nevertheless unmotivated to claim that there is a substantive distinction between categories and values; we could just as easily say of PARTICIPANT that it is a value of the category π. Thus, SPEAKER looks to be a value from one perspective, and a category from another" (Béjar 2003:51)

As such, the question of phi-features as privative versus valued features is rendered unimportant in this representation. The advantage of feature geometries with privative phi-features, in my opinion, is that they allow for a more intuitive representation of phi-features as syntactic objects, and they make it easier to encode dependencies such as hierarchical asymmetries or entailment relations between different kinds of features. As I show in this chapter, they also allow for a clear representation of what kinds of feature combinations are not permitted.

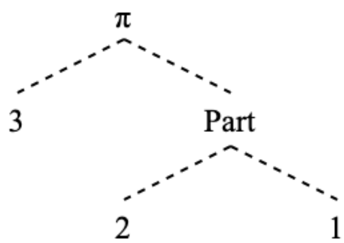
I adopt a representation of a possible mother-daughter relations for person that looks like the following, with only one terminal node permitted per DP (in English, at least; the story there may be more complicated for languages that make inclusive/exclusive person distinctions).

(93) Range of possibilities for person feature geometry (English)



If we think of the Non-Participant – Participant distinction as a kind of binary, then it is even clearer why these two features at the same time could not be permitted (for an implementation of this system using binary features, see Nevins (2007)). These nodes can be abbreviated as follows.

(94) Range of possibilities for person feature geometry, abbreviated (English)



Instead of a first person English DP being represented as [Pers: 1], it is represented here as [Pers[Part[Spkr]]], or [π[Part[1]]]. A third person DP, instead of bearing a valued feature [Pers:3] contains a segment of features [Pers[NonPart]] (or [π[3]]). Note that, in this structure, third person is still the least specified possible iteration of Person, as both first and second person involve additional structure that third person lacks (namely the presence of [Part]).

This representation of person features in English bears on the "3yesP" vs "3noP" question (Nevins 2007 *et seq*; Bondarenko 2020; Grishin 2023; a.o). Namely: does third person correspond to an actual feature – such as NonPart – under the feature Person, or is third person computed from the absence of a further feature specification under Person? Under the 3yesP view (Nevins 2007), there is a feature corresponding to third person specifically, which I have represented here as [NonPart] or [3]. Under the 3noP view, there is no specific feature for third person: instead, third person is account for in terms of the absence of a more specified person feature (e.g. [Part]). 3noP– the idea that third person is not a person (Benveniste 1966; Kayne 2000) – is the consensus view currently, due to "strong evidence that 3rd persons are totally underspecified" (Béjar 2003:48) such as the alleged invisibility of third person, defined as follows.

(95) *Invisibility* (from Grishin 2023)
In no language can third person explicitly be targeted to the exclusion of first and second person in various kinds of morphosyntactic processes.

There is, however, various exceptions to Invisibility with third person, such as the "spurious *se*" PCC effects of Spanish (Nevins 2007) and third-person marking in some Algonquian languages (Trommer 2008; Grishin 2023; a.o). Another piece of evidence that third person is distinctly represented from other Persons in English is the insertion of the present tense agreement morpheme for singular DPs, which only happens for those which are third person.

(96) a. {He/she/it/Gollum/The Ring} want*(-s) to be found
b. {I//you} want(*-s) to be found

Another example of a violation of Invisibility of third person in English comes to us from LGBT English, with the binding of anaphora and singular 'they'.

(97) a. They$_i$ saw themself$_i$
b. They$_i$ left their$_i$ phone at the bar

For individuals with ‘he/they’ or ‘she/they’ pronouns (that is, individuals who can be felicitously referred to with a gendered set of third person pronouns (‘he’ or ‘she’) as well as singular ‘they’), the use of singular ‘they’ as an anaphor is possible with a ‘he’ or ‘she’ as an antecedent.

(98) a. He saw$_i$ {him/them}self$_i$
b. She$_i$ left {her$_i$/their$_i$} phone at the bar
c. I$_i$ saw {my/*them}self$_i$
d. You$_i$ left {your$_i$/*their$_i$} phone at the bar

The story goes that singular ‘they’ is deficient in phi-features (Conrod & Schultz & Ahn 2022), at least number. If third person is the lack of Person, and this kind of pronominal coindexation is derived via Agree (Sag & Wasow & Bender 2003:208; Collins & Postal 2012:89), there is no clear reason why singular ‘they’ should not be able to serve as an anaphor in (98c-d).

For these reasons, it is advantageous to hold that third person is a distinctly represented instance of Person, at least in the languages which show these anti-invisibility third person effects. Perhaps the generalization that there are many processes which target first and second person to the exclusion of third person (stated in the Invisibility definition in (95)) is sufficiently explained by the fact that, in the feature geometry given in (93)/(94), first and second person form a constituent to the exclusion of third person, as they are both dominated by the [Part] feature.

Béjar (2003:48) states that “the feature specification of 3rd persons must be a point of variation”, the variation being between third person being totally underspecified or third person being specified with some distinct feature.

Looking at second person, it is represented in some languages as [Pers[Part[Addr]]], other languages as [Pers[Part]], with first person being in both cases represented as [Pers[Part[Spkr]]] (Harley & Ritter 2002; McGinnis 2005; Preminger 2014).

So, in a given language, second person is underlain by [Part] plus an additional addressee feature, or just by [Part]. In other words, second person is either a specified “value” of Participant, or it is computed from a bare [Part] feature with no “value”. See the following possibilities, keeping in mind that, in both possibilities, [Part] is still embedded under [Pers].

(99) Two possible representations of second person
a. [Part] b. [Part]
|
[Addr]

This is mirrored, then, with third person and [Pers]. Languages have the possibility of representing third person in one of two ways: either as the presence of a feature that entails [Pers] (3yesP), or as a bare [Pers] feature (3noP).

(100) Two representations of third person
a. [Pers] b. [Pers]
|
[NonPart]

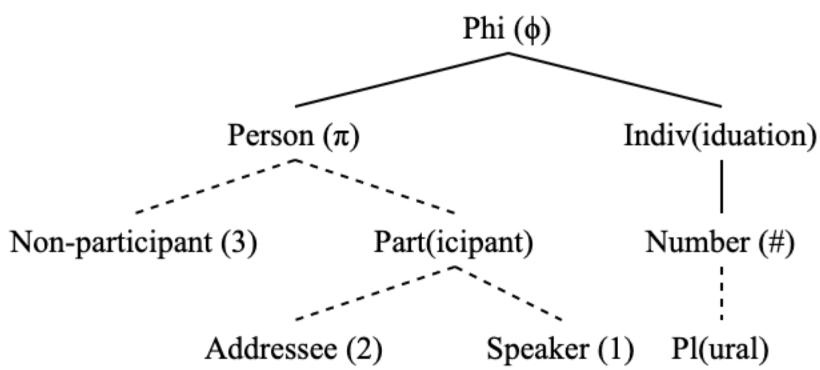
This is advantageous as there should be more than one way to represent how third person is represented across – and perhaps even within – languages (Furuya 2017). Turning now to number, I assume that a bare Number feature by itself is interpreted as singular, and a Plural feature below the Number feature is interpreted as plural[9].

(101) Feature geometry for Number
[Num(#)]
|
[Plural(Pl)]

Thus, a feature geometry of English showing all of the possibilities for featural exponence looks as follows. I assume that features for person and for number are represented in the same feature geometric structure and that entailment relations can hold between person and number features just as they can between various features that are subsets of either person or number.

For example, only third person DPs may lack a number feature. First and second DPs must be specified for number in languages that project number features in the syntax. In a way, then, number specifications are sensitive to person specifications. As such, I assume that person and number are indivisibly part of a larger feature geometric structure ϕ.

(102) Possible ranges for feature geometry of English

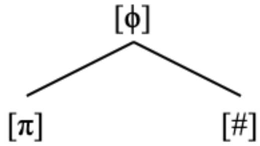


From here out, I simplify the representation of the features π and # with respect to ϕ as showing them as both daughters of ϕ, as in (103), with π on the left. While this shows π and # as sisters for the sake of not making the representations too complicated, it must be kept in mind that π is

[9] There is a question of whether or not singular should be treated as the absence of a feature, just as this has been asked of third person (and, to some degree, second person as well). While it certainly is a logical possibility that singular could be represented as [#[Sg]] just as plural is represented as [#[Pl]], this question of representation of singular number actually does not make a difference for the analysis presented in this thesis. In every language analyzed, singular appears unambiguously to be the least specified instantiation of number. Three seminal works on the representation of phi-features (Harley & Ritter 2002; Béjar 2003; Preminger 2014) do not comment on this issue.

still hierarchically situated above #; π is closer to ϕ than # is, as in (102).

(103) Simplified feature geometry of English (just showing π and #)



I assume that Agree involves the copying of entire phi-feature geometries, not just “snippets” or “segments” of it; in other words, the copy of phi-feature geometries is coarse (Preminger 2014; Coon & Keine 2021). For example, a search operation Find(ϕ) results in the entire feature geometry containing [ϕ] being copied back to the probe, and Find(#) results in the same for the entire geometry containing [#]. For a formal implementation of this using the idea of feature valuation as set union over two feature geometries, see Preminger (2017).

The coarseness of feature copying in this way is related to the assertion that Match is evaluated at the root node of a phi-feature geometry (Béjar 2003:53). If [π] and [#] are connected in such a way that the root of all feature geometries is [ϕ], then it makes sense that a Match for [#] and [_#] triggered by Find(#) would result in the copying of an entire feature geometry dominated by the root node [ϕ]. With these considerations in mind, I move on to discuss how these features are represented on probes and goals, and how these two elements communicate with one another and undergo Agree.

2.2.1.2 Probes and goals

These feature geometries are present both on the probe and the goal. What differentiates a phi-probe from a phi-goal – since they both contain feature geometries – is that the specification on a phi-probe to search and copy information (that is, features) from other, more specified phi-feature geometries (the goals).

Phi-probes contain the same hierarchical feature geometries that are present in phi-goals (Béjar & Rezac 2009; Preminger 2014; 2017). The feature geometries on phi-probes contain specifications to find and copy back specific features from other phi-bearing elements. These phi-feature search specifications have been represented in different ways in the literature, namely with unvalued/uninterpretable feature designations (Chomsky 2000; 2001; Pesetsky & Torrego 2007; Bjorkman & Zeijlstra 2019; a.o), with interaction and satisfaction feature designations (Deal 2015; 2019; 2021 a.o), or with some kind of “placeholder” feature on the probe that triggers searching for a corresponding feature (López 2007; Preminger 2014).

In the traditional probe-goal approach (Chomsky 2000; 2001; *et seq*), the uninterpretable/ unvalued features on phi-probes must search for and find interpretable/valued phi-features on goals. The reason in this view that Agree must take place is because unvalued features cause problems for interpretation and spellout at interfaces (LF and PF).

Preminger (2014) introduces numerous arguments as to why this is not the correct way to think about Agree: the failure of a probe to value all of its unvalued features in many cases does not cause ungrammaticality at all. Keeping in mind the discussion in Section 3.1, see the following.

(104) Strákunum {leiddist/ *leiddust} (Icelandic, Sigurðsson 1996:1)
boys.DAT {bored.3sg/*bored.3pl}
'The boys were bored'

Here the phi-probe on T only agrees with the deficient dative DP *strákunum* 'the boys', and the sentence is completely grammatical, despite the fact that a specification to find a number feature is also present on the phi-probe in Icelandic (Taraldsen 1995; Hoover 2021), which we know because agreement is possible with a postverbal subject that contains a number feature, when one is present.

(105) Henni leiddust strákarnir (Icelandic, Sigurðsson 1996:1)
her.DAT bored.3pl boys.NOM
'She found the boys boring

This and similar situations are extremely well-attested crosslinguistically, which is (in addition to all the evidence from Preminger (2014)) damning of any framework of Agree that is grounded in the need of unvalued/uninterpretable features to be removed from a probe in order for a derivation to be grammatical.

A theory of Agree which makes use of uninterpretable features as "derivational time-bombs" must make use of additional machinery such as Partial Default Agreement (PDA) (Béjar 2003) to explain how a sentence like (104) could be grammatical. PDA and other proposals like it serve to mark unvalued features to receive a default value at the interfaces in order to not cause ungrammaticality. In Béjar's original formulation of PDA, it is only available once a probe has exhausted its search domain for the relevant feature it needs; however, the data presented in this thesis suggests that this does not account for all of the facts, as 'default' agreement forms are often optionally available, and not some kind of last resort to save a derivation that is otherwise doomed to fail.

In a theory where unvalued features do not cause a derivation to crash, the introduction of a mechanism such as PDA (and subsequent constraining of it) is not needed, the lack of which is preferable because it reduces the amount of machinery needed to compute these utterances. Theories of Agree such as reactive Agree (López 2007), Interaction & Satisfaction (Deal 2015 *et seq*), and the obligatory operations model (Preminger 2014) move away from the idea of uninterpretable features as "derivational time-bombs".

These theories treat probing (with the desired outcome of Agree) as an inherent consequence of the featural specifications of the probe. López (2007) refers to this process as the probe's "reaction" to the unvalued features of the probe, and Preminger (2014) refers to this as an obligatory operation. In these theories, ungrammaticality does not necessarily result if the probe's obligatory reaction to search for phi-features does not actually result in any Agree relation being established.

In the traditional probe-goal framework, the process of *valuation* makes reference to the copying (or sharing (Frampton & Gutmann 2000; 2006; Pesetsky & Torrego 2007)) of specific feature values from the goal to the probe; however, in a privative phi-feature system there are no feature values to be copied in the same way. As such, any discussion of valuation in a privative representation of phi-features must be understood as the actual copying of feature-geometric structures from a goal onto a probe, instead of the copying of a feature value *val* from a valued goal [F:*val*] to an unvalued probe [F:__] (Preminger 2014:47,96).

Thus, an unvalued feature should be understood as a featural specification on the probe to find and copy back a given piece of a phi-feature geometry: an instruction to probe, as it were. A valued feature, then, should be understood as any instance of a phi-feature embedded in a feature geometry that could be visible to a phi-probe. In other words, an unvalued feature is a specification to search for a given feature, and a valued feature is an actual occurrence of said feature. This conceptualization of valuation does not make reference to interpretability.

Given this, perhaps it does not make much sense to talk about the valuation of features in this way, but terminology that makes reference to feature *values* and *valuation* is so widespread in the literature – even in the literature that explicitly rejects the traditional probe-goal unvalued-valued approach to Agree – that there is no point in fighting against it. Let us then define very carefully what a probe is, what a goal is, and what it means for "valuation" (and ultimately, Agree) to take place.

Preminger (2014:96) defines the operation that causes probing as follows, Find(F), with the same considerations on the terminology around valuation applying to this definition as have been discussed here, namely that valuation involves the copying back of phi-feature geometries from a goal to a probe.

(106) Find(F) (from Preminger 2014:96)
Given an unvalued feature F of a head H°, look for an XP bearing a valued instance F and assign that value to H°.

So, an unvalued feature [F] triggers the operation Find(F), and, crucially, failure to find [F] does not cause a derivation to crash. Unvalued features, according to Preminger (2014:96), are "placeholder(s) for snippet(s) of feature geometry rooted in [F]". In other words, unvalued features are specifications to Find(F). From here on out I represented unvalued features as [_F]. Other researchers working in the same framework represent the same unvalued features as [*u*F] (e.g. Coon & Keine 2021), but I avoid this given the discussion of uninterpretability above.

As such, we can define a phi-probe as the following.

(107) *Phi-probe*
A phi-probe P is any head H° that contains an instance of an unvalued feature [_F].

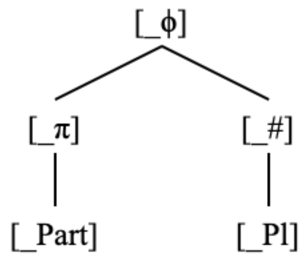
Then, we can define a phi-goal in relation to a phi-probe.

(108) *Phi-goal*
A phi-goal is any XP bearing a feature [F] in the search domain of a probe P which initiates the operation Find(F) due to the presence of an unvalued feature [_F] on P.

As goals are defined in relation to a relevant phi-probe, it is possible that an XP may be a goal with respect to one probe, but not another (or in a given derivation, but not another). The definition of a probe is *not* relative in this way, as any H° containing a [_F] will trigger the operation Find(F), even if there is no suitable goal in the search domain of P.

Just as valued phi-features have entailment relations between them, so do unvalued phi-features. That is, [3] entails [π] entails [ϕ], [Pl] entails [#] entails [ϕ], and so on. Probes that are relativized to features that are lower down the phi-hierarchy also upwardly entail higher features. So, a probe that is relativized to have [_Part] and [_Pl] can actually be understood as the following due to the entailment relations that exist between phi-features.

(109) Probe that is relativized for [_Part] and [_Pl]



A probe initiates searches hierarchically. In other words, the highest unvalued node [_F] initiates the operation Find(F). So, the first search operation triggered by the probe in (46) would be Find(ϕ). Only if Find(ϕ) is unsuccessful in valuing all of the other unvalued features on the probe will other features trigger subsequent searches. This is formalized in (110).

(110) *Hierarchical Search*
Phi-probes search first for the features located highest in their feature geometries, then work their way downward if the first search is unsuccessful in valuing all features.

This hierarchy condition on Search explains the observation that person probing before number on T is a linguistic universal (Béjar & Rezac 2009; Preminger 2014; Deal 2015; Kalin & van Urk 2015; Pŭskar 2019), as person is situated above number in the hierarchy of phi-features in (102).

At this point it is necessary to ask why probing even happens in the first place, especially if it is not the case that it happens only to delete uninterpretable features. Preminger (2014) and López (2007) treat probing as an obligatory consequence of bearing unvalued features, yet, the failure to remove the unvalued features does not lead to ungrammaticality. This begs the question: why probe at all? The Interaction and Satisfaction model of Agree (Deal 2015; *et seq*) "grounds Agree in its ability to create redundancy, rather than in the need to remove…'derivational time-bombs'" (Deal 2021:47). I believe the answer lies here: probing – and, ultimately, Agree – takes place to create redundancy in the grammar. If probing never happens at all, then there is no possibility of redundancy ever being established.

2.2.2 Search, Match, and Copy

Now, let us go through the mechanism of Agree that happens when a probe finds a suitable goal. Following a phi-probe initiating a search, formalized as the operation Find(F) in (40), the probe seeks to Match with a goal and copy back its features.

Agree is typically decomposed into three sequential steps: Search, Match, and Value (Chomsky 2000; 2001; Béjar 2003; Bhatt & Walkow 2013; Kalin 2020; a.m.o). Even conceptualizations of Agree that do not use this terminology still make use of some machinery in which the probe looks for values to be copied, identifies values to be copied, and copies over actual values (see Checking and Valuation in Bjorkman & Zeijlstra (2019), for example).

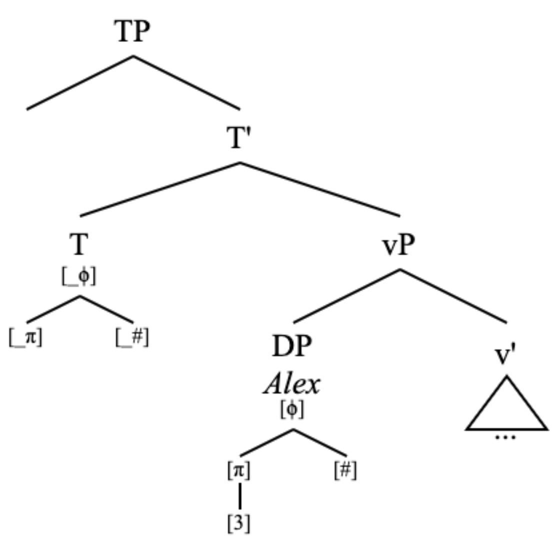
As such, I define the steps of Agree as the following (Chomsky 2000;2001).

(111) *Agree*
a. *Search*
Probe P on head H° initiates the operation Find(F) for a given unvalued feature [_F] present on H°.
b. *Match*
Probe P bearing an unvalued feature [_F] identifies the closest (under c-command) instance of [F] in its search domain
c. *Copy*
The feature geometry containing [F] is copied onto the probe, replacing (and thereby deleting) any instances of [_F]

The most important steps here are Search and Match, and the formulation of Agree that I present in this chapter crucially relies on unvalued features of a probe initiating an operation Search that finds and Matches with the closest instance of that relevant feature. This technology is sufficient to derive the complex agreement patterns presented here.

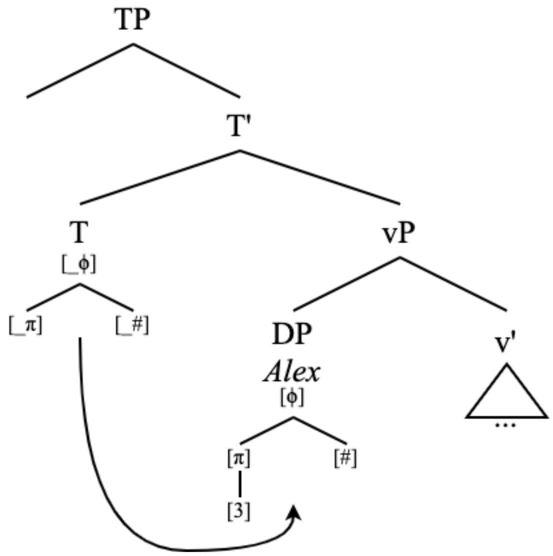
Now let us see an implementation of this, with the sentence from (23), repeated below, now indicating the syntax of the sentence before any agreement or movement has taken place.

(112) Alex loves me



Here, T has an unvalued person [_π] and number [_#] feature, as well as an unvalued feature [_ϕ] above them both. Because of the Probe Search Hierarchy, Find(ϕ) is initiated. This search finds the subject DP 'Alex', which has third person and singular features.

(113) Alex loves me: Search



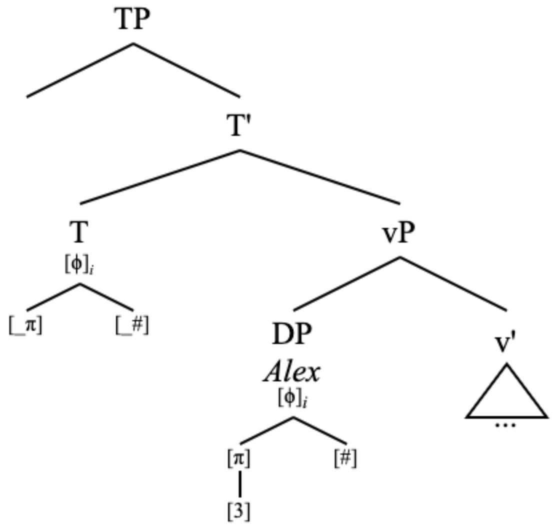
At this point, Match is evaluated. I adopt the following definition of Match.

(114) Match (Chomsky 2000:122)
a. Matching is feature identity
b. D(P) is the sister of P
c. Locality reduces to "closest c-command"

In this definition, Match obtains between a goal and a probe that bear a matching feature iff the goal is in the probe P's search domain D(P) and iff the goal is the closest element to the probe that bears the relevant feature. In other words, Match holds between a probe and a goal if they both bear an instance of the same feature, the probe c-commands the goal, and there is not another element between the probe and the goal that bears the same feature.

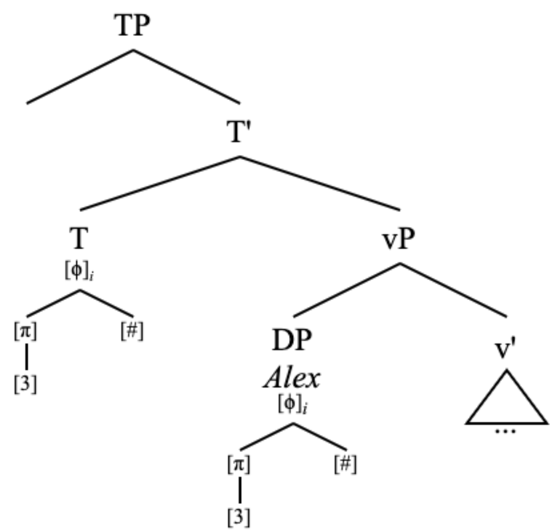
Here, Match only applies between [ϕ] and [_ϕ] since Find(ϕ) is the search taking place. I represent Match in the following structure with subscripts on the phi-feature geometries (this is purely notational; I do not assume any actual coindexation in the syntax).

(115) Alex loves me: Match



Once the probe has identified its suitable goal via Match, the feature geometry of the goal is Copied to the probe. This is what is also known as valuation. The relevant unvalued features are removed ("checked") as a result of this.

(116) Alex loves me: Copy

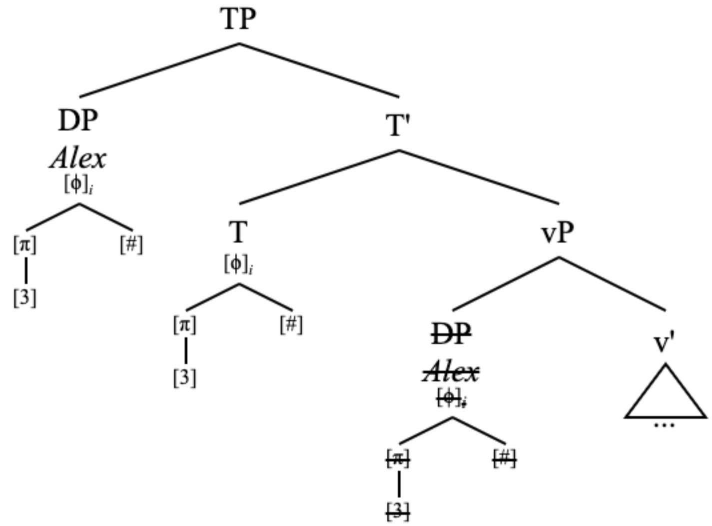


This derivation misses something, though. It misses the fact that the DP 'Alex' which undergoes Agree with T appears before T at spellout.

2.2.3 Move and Case

Now, the feature geometry of the goal DP 'Alex' is present on the phi-probe T as well, which leads to the verbal agreement being spelled out in correspondence with the features of the subject DP. This representation, however, does not account for the fact that the DP 'Alex' moves to Spec,TP, which is typically referred to as EPP (Chomsky 1982) movement.

(117) Alex loves me: Move



In the original formulation of EPP movement (Chomsky 1995), there is an abstract EPP feature which is unvalued (or "strong") on T that forces the target of agreement to move to Spec,TP. Agreement and checking of this EPP feature is said to be independent of phi-agreement and Case assignment, which are also derived with valued-unvalued feature correspondences on the T probe and the relevant DP goal. Under Chomsky (1995), an EPP feature, Case feature, and phi-features which all need to be checked in a given probe-goal configuration form a trifecta in which one type of feature is accessed for checking and valuation and the others are checked along with it as "free riders", but they are still necessarily separate features.

A separate EPP feature that must enter into a valuation relation solely to derive subject movement to Spec,TP is highly stipulative, and is not necessary to derive movement, especially when the relevant probe and goal are already in dialogue with one another for the checking and valuation of other features: phi and Case.

Several accounts derive the requirement for EPP movement via the need of T to assign Case to the DP with which it undergoes phi-agreement (Boeckx 2000, Bošković 2002; Epstein & Pires & Seely 2004; Epstein & Seely 2006; a.o) or, alternatively, the need of the DP that undergoes phi-agreement with T^0 to receive Case from it (Bošković 2007; Bjorkman & Zeijlstra 2019). These analyses make use of the Inverse Case Filter and the Case Filter, respectively.

If, however, Case assignment itself is reduced to a consequence of phi-Agree (Chomsky 2000; 2001; Platzack 2006; Legate 2008; Georgi 2017; Poole 2024, a.o), then perhaps it no longer makes sense to conceive of EPP movement as necessary in order to derive Case assignment since Case is explainable via phi-agreement. Indeed, Case is not a motivation for movement at all. This would allow for a uniform analysis of instances in which an element receives Case without moving (e.g. postverbal nominatives in Icelandic) and those in which an element moves without receiving Case (e.g. wh-movement). This allows for a more inclusive theory of movement in which elements that cannot and do not receive Case can still be the target of Agree and movement operations.

Case, then, tracks an Agree dependency between a goal DP and some functional head which was valued by it.

Case is not only related to Agree, but to movement as well. For example, the association between nominative Case and the Spec,TP position in English is well-established. Even if the verb tracks agreement with a postverbal subject, it is not the case that that subject will appear nominative. See the following data.

(118) a. What their families really need are them (Blog)
b. Honestly the only thing about muncie that i'll miss are them (Instagram)
c. The demographic who does it the most are them (Blog)

(119) a. There are them that cannot sing, and them that cannot weep (The Crucible)
b. There are them and then there's #ME ! (Instagram)
c. There were them and there was us; and he never ever saw it any differently (Interview)

At least in English, it must be the case that nominative Case assignment tracks only the element that moves to Spec,TP, either because T has only one nominative Case feature to give out (and it gives it to the constituent which moves to Spec,TP first, since that is the first element to undergo Agree with T), or because Case assignment is only possible in a Spec-Head configuration, brought about by Agree attracting a viable Case recipient to the specifier position of a Case-assigning functional head (Ura 1996; 2000a). I adopt the former position.

There still remains a question: Why does movement happen at all? For this I turn to Miyagawa's (2010) work on the underlying motivations for both movement and agreement.

(120) *Why Move?* (Miyagawa 2010:33)
Movement triggered by agreement takes place in order to keep a record of functional relations for semantic and information-structure interpretation. Given the architecture of human language, movement is the only way to preserve functional relations beyond the interface to interpretation of semantics and information structure.

This question of why movement happens has major consequences for our formulation of the EPP. Instead of representing EPP movement as being motivated by an abstract feature (Chomsky 2000; 2001), it can now be subsumed under phi-Agree just as Case assignment as been.

The underlying motivation for movement is to indicate functional relations (which he ultimately argues is the function of Agree as well). Crucially, movement is "triggered by agreement" in this system. That is, we can think of Movement as a necessary consequence of Agree, which helps to carry out the ultimate goal of both operations, which is to indicate and preserve functional relations between elements in the syntax (Miyagawa 2010:9,33). Movement of a goal to the specifier of its probe is a reflex of the Agree operation itself. This is defined as the following.

(121) Probe-Goal Union (PGU) (Miyagawa 2010:35)
A goal moves in order to unite with its probe.

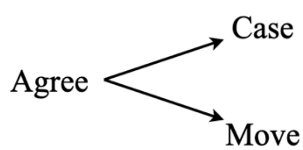
Assuming the movement that takes place for PGU is the movement of the goal to the specifier of the probe head (which Miyagawa indeed assumes), this is consistent with the following definition of a specifier.

(122) α is located in a Spec of X° iff (*i*) and (*ii*): (Ura 1996:20)
(*i*) α is excluded by X° (i.e., α is not dominated by $X^{\circ Max}$).
(*ii*) a. α enters into a feature-checking relation with X°, or
b. α is assigned an external θ-role by X°.

In other words, the Spec,XP position is reserved for thematic arguments of X that are externally merged in that position and for goals that have undergone Agree with a probe on X.

This system orients movement and Case assignment both as consequences of Agree, not as operations that are dependent on one another. They are both united as consequences of Agree. The "free riders" of phi-agreement under Chomsky (1995) are now achieved not by separate sets of features that must be stipulated, but as reflexes of a single Agree operation.

(123) Agree and her "free riders" (either of which can not come to fruition for some reason)



I assume that Case assignment or Movement of a goal may fail for various reasons, just as Match and Value are also fallible operations (Preminger 2014). For example, Case may fail to be assigned by a given functional head to a DP which has already received Case from another functional head. More crucially for this analysis, PGU may not be able to hold of a given goal with respect to a probe if another goal has already moved to the specifier of that probe at an earlier point in the derivation.

I am not so much interested in the relative timing of valuation, Move, and Case assignment with respect to one another and with respect to transfer beyond the fact that Case and Move take place after – crucially, because of – Agree. Miyagawa (2010) locates Move (PGU) after Agree and before transfer/spellout. This places the entirety of the machinery of Agree in the syntax (*cf.* Kalin 2020; a.m.o). PGU ultimately happens to preserve the Agree relation before spellout, since the probe is ultimately deleted at spellout (i.e., T is not interpreted as bearing phi-features).

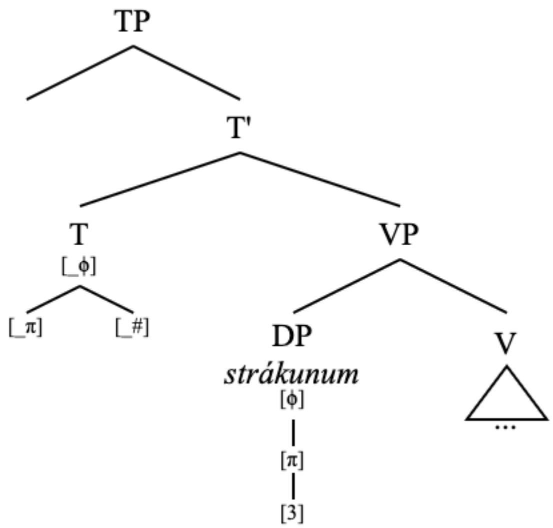
Agree is an operation which consists of three steps: Search, Match, and Copy (111). If these steps are successful, PGU and Case assignment from the probe to the goal may obtain. Given that Match, Copy, Case assignment, and Move may all fail or be blocked for various reasons, the only obligatory operation for a derivation to be grammatical remains Search, which is consistent with Preminger's (2014) obligatory operations model of Agree.

2.3 Agreement optionality in action

Now, let us go through a derivation in which the goal of the phi-probe is deficient, given the discussions on the mechanisms and results of Agree. I start with the Icelandic example from

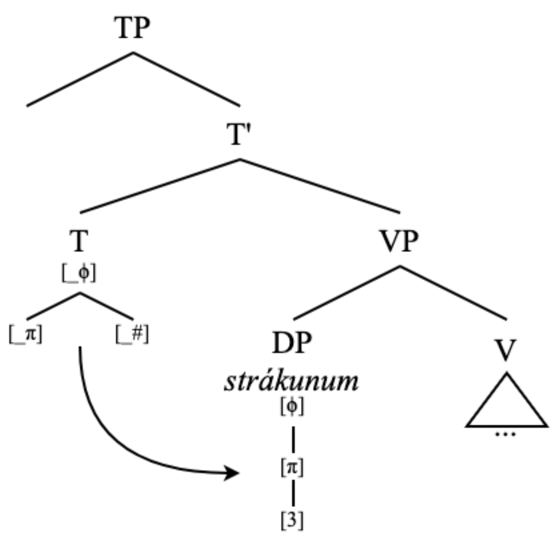
(104), repeated here. I indicate its underlying structure. The dative subject is located in a shell that is visible to the phi-probe on T only as third person.

(124) a. Strákunum {leiddist/ *leiddust} (Icelandic, Sigurðsson 1996:1)
boys.DAT {bored.3sg/*bored.3pl}
'The boys were bored'
b. Basic underlying structure



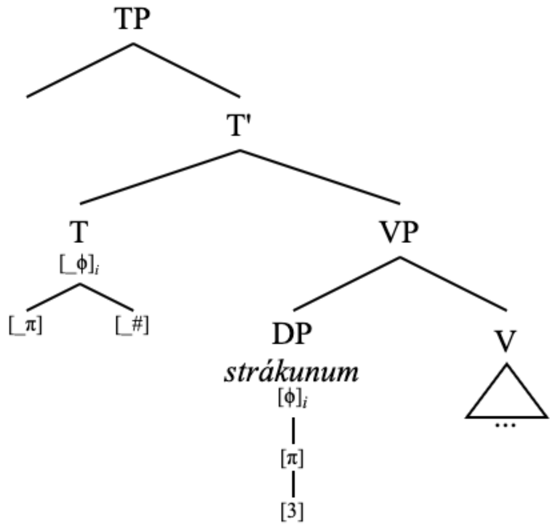
The first step for Agree here is that the phi-probe on T^0 initiates a search based on its feature [_ϕ], which triggers the operation Find(ϕ).

(125) *Strákunum leiddist* 'The boys were bored': Search



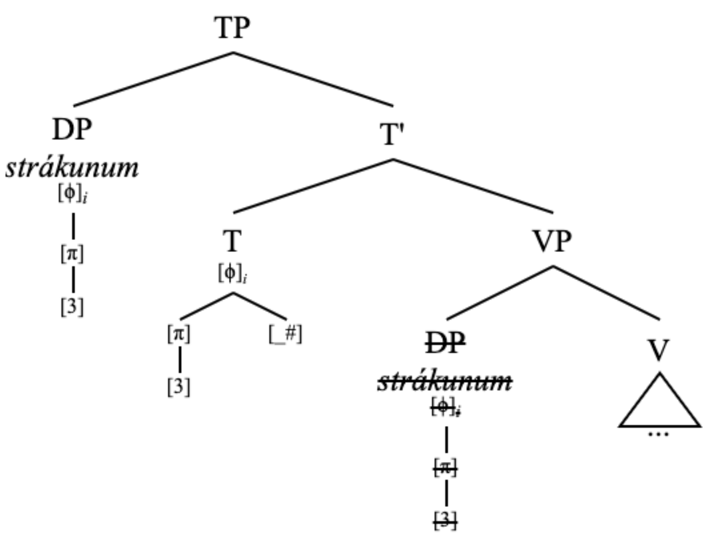
The presence of a [ϕ] feature on the DP *strákunum* 'boys.DAT' and the presence of an unvalued [_ϕ] feature on T lead to Match.

(126) *Strákunum leiddist* ‘The boys were bored’: Match



Since the goal has been identified via Search + Match, valuation (Copy) takes place. This involves copying of the valued features on the goal over to the probe and the subsequent removal of any unvalued features on the probe that correspond to the valued features being copied over. As a result of all three steps (Search, Match, Copy) of the tripartite operation Agree having successfully taken place, the goal DP *strákunum* ‘boys.DAT’ moves to the specifier of the head containing the probe: Spec,TP.

(127) *Strákunum leiddist*: Copy + Move



Here probe never ends up finding a [#] feature and getting rid of its unvalued [_#] feature. There is at no point a [#] feature (or [Pl] feature, which would entail the presence of a [#] feature) visible to the probe in this derivation. This is spelled out as singular verbal agreement ‘*leiddist*’.

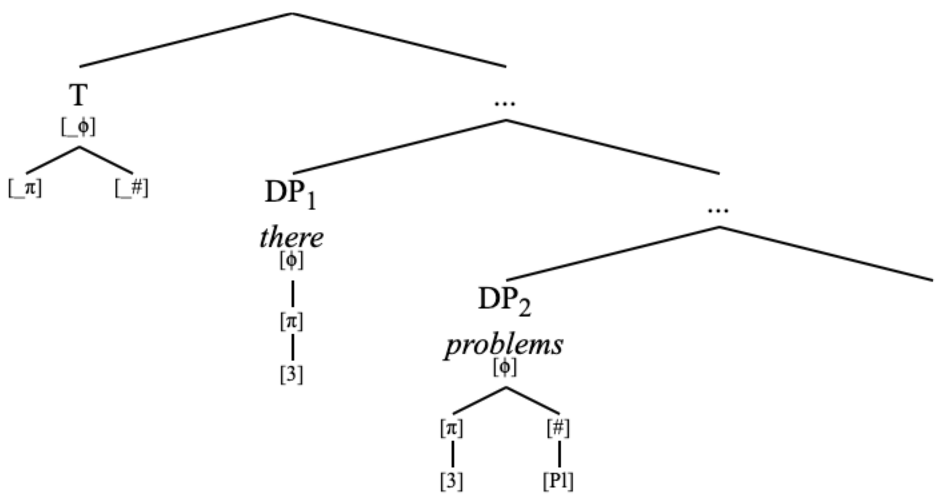
At this point, the remaining unvalued feature [_#] could trigger yet another search, or it could not, as the probe has already searched for its root node [_ϕ] and so the obligatoriness has been met, but nothing would happen if it did, so it does not make a difference in this derivation. The probe still has no number feature accessible to it, so valuation of [_#] is not possible.

When the valuation of [_#] is possible, despite the lack of any number feature on the closest accessible goal to the probe, there is the possibility of agreement with a lower goal that does bear the relevant feature the probe is searching for, thus leading to defective circumvention. With this, agreement may track the higher deficient goal or the lower specified goal. What agreement options are possible in these configurations are constrained by the featural requirements of the probe and the features that the two goals bear.

2.3.1 Defective circumvention

Let us now return to the English existential 'there' construction, which exhibits optional agreement with a third person plural postverbal subject. This is the same agreement pattern that surfaces with Icelandic postverbal nominative 3pl subjects with dative preverbal subjects. See the following simplified structure, showing only that, at the time of T^0's probing, the deficient goal existential 'there' is located closer to the probe than the postverbal argument DP.

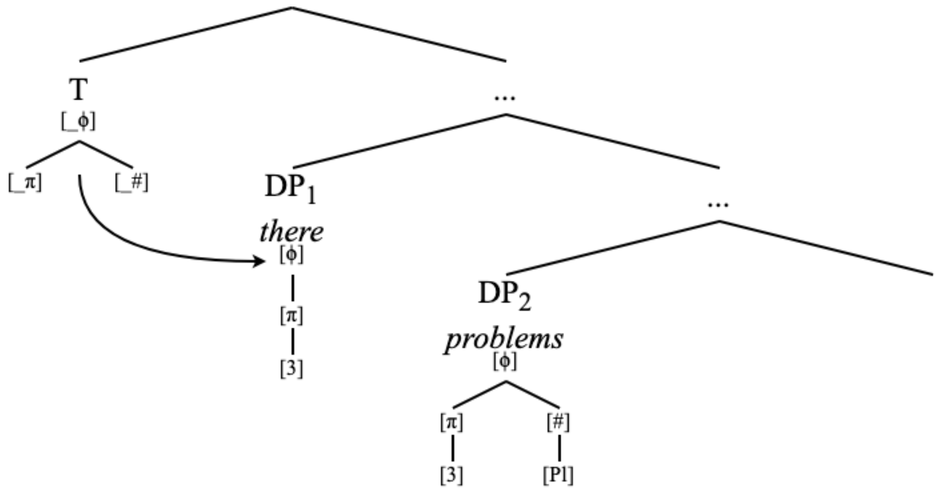
(128) a. There {was/were} problems
b. Basic underlying structure[10]



Just as before, the probe on T searches for a [ϕ]-bearing goal.

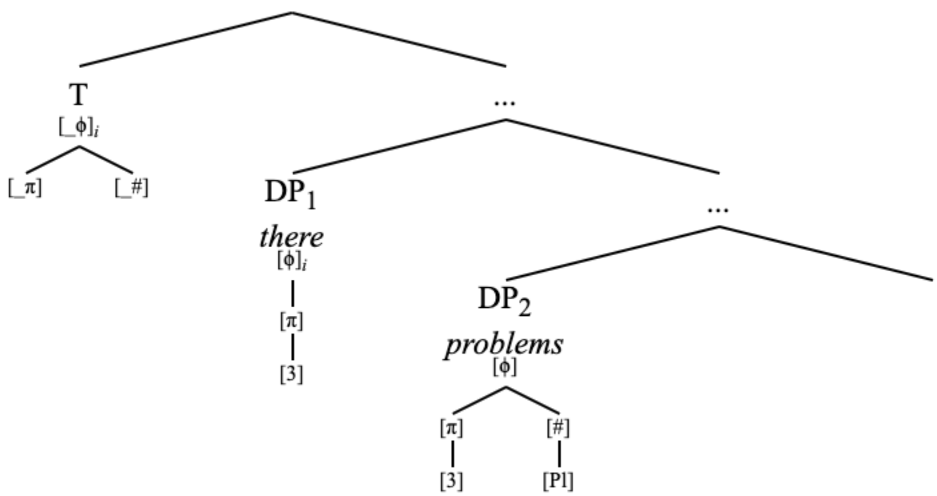
[10] The structures presented here for English existential sentences are intentionally ambiguous, as I will end up showing that these derivations actually involve smuggling of the higher goal over the lower one, which I do not show here just yet so not to distract from the analysis of Agree. The major point for now is that deficient existential 'there' is in a position where it is closer to the phi-probe on T^0 than the non-deficient thematic subject. This structure shall be expounded upon and disambiguated in Chapter 4.

(129) There {was/were} problems: Search



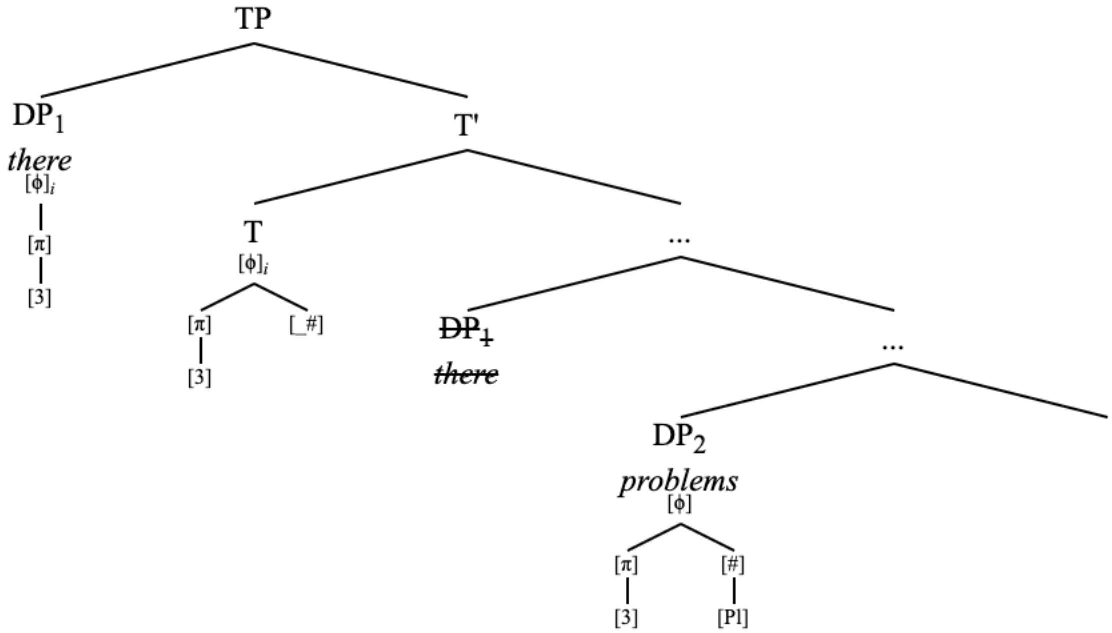
Again, the deficient DP is first encountered by the probe because it contains [ϕ].

(130) There {was/were} problems: Match



The probe and the deficient DP match for [ϕ]. As a result, the probe and 'there' enter into an Agree relation, and 'there' moves to Spec,TP. Up to this point, this Agree operation plays out as the exact same process as when there is no agreement with the postverbal DP.

(131) There {was/were} problems: Copy + Move



The probe has searched and undergone Agree with one DP. The probe bears phi-features for third person, and an unvalued number feature, and nothing else. This is where the optionality arises. At this point, the phi-probe on T can either stop probing or it can probe again due to the presence of the unvalued number feature on T, thus undergoing defective circumvention. If the former, then verbal agreement is spelled out as third singular ('was'). If the latter, then, upon agreeing with the third person plural postverbal subject, verbal agreement is spelled out as third plural ('were').

Under the obligatory operations model of Agree (Preminger 2014), it is not clear why the probe should be able to optionally stop here, since in this theory the unvalued number feature (or any unvalued feature) on the probe is only able to remain after the probe has exhausted its entire search space looking for a match and failing to find one. The glaring issue here is that there is a valued number feature in the derivation (on 'problems') that is optionally accessible to the probe; so, to say that there is a licit derivation in which the number feature on the probe never gets valued is perhaps confusing.

My answer to this is that the probe agreeing with 'there' actually does exhaust its search space, in a way, since at this point it has already found the closest goal phi-goal under minimality. Assuming that all locality can be reduced to minimality (Halpert & Zeijlstra 2024), the closest phi-goal to the probe marks the extent of the probe's search domain if the probe must be minimally valued for [ϕ]. The postverbal subject DP may be the closest goal for number, but it is still not the closest phi-goal. The syntax then has the option to choose to probe beyond the locality domain in order to value another feature (this is, in some sense, phase unlocking (Branan & Davis 2019; Halpert 2019; Tsz-Ming Lee & Yip 2024) without the phases).

Initiating another search is a costly operation for the syntax, as it involves a separate Search cycle. Conversely, *not* initiating another search and leaving an unvalued [_#] feature is also a kind of violation, especially when there is a valued number feature in the derivation that is in some way accessible to the probe. There is also undoubtedly some additional work needed to be done at transfer in order to spell out any unvalued features left in the syntax[11].

In this way, both 'There was problems' and 'There were problems' involve the violation of some kind of economy condition in the syntax, which contributes to the situation of having two optional spellout choices: both options are less economical. Those speakers who claim to not accept default agreement with a third person plural postverbal subject have a grammar in which leaving the unvalued number feature in the syntax is more costly of a violation than probing a second time for number[12].

As for why probing for [_ϕ] is obligatory while probing for [_#] is not, this is because [_ϕ] is the root node of the geometry here. That is, the obligatoriness condition of searching applies to the entire geometry. Any non-root nodes, such as [_#], do not carry the same obligatoriness condition (unless they are specially marked to do so, such as [_Part], which I discuss further on).

As such, the probe has the option to stop after Agreeing with the deficient DP 'there'. This is the minimality respecting option, and the option which does *not* involve defective circumvention. This option further involves some kind of mechanism in which a "default" singular number feature is available on the phi-probe at transfer, meaning that there is an extra mechanism involved beyond Agree, which is less economical than simply finding a number feature in the syntax.

The other option is that the phi-probe on T probes again as a result of it still bearing an unvalued feature [_#], which triggers the operation Find(#). The successful execution of the operation Find(#) in these cases is known as defective circumvention. Again, this is costly for the syntax since it involves a whole other cycle of probing. See the following formal definition, given the definition of deficiency presented in (82).

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[11] I believe this intuition is the motivation for positing mechanisms such as Béjar's (2003) Partial Default Agreement (PDA).

[12] It must be noted that there are speakers for whom '*There were us'* is not as ungrammatical as '*There am me'* or '*There were you*(sg.)', though it is certainly not a well-formed utterance and is considerably worse than '*There were them'*, for example. I agree with these judgments.

i. a. **There am me
b. **There were you(sg.)
c. */???There were us
d. There were them

These judgments could indeed be represented by a "higher ranking" of the preference to value number for speakers for whom there is a contrast between (*i*a-b*)* and (*i*c). Perhaps this points to the emergence of some kind of Split Repair last resort strategy (Fernández-Serrano 2022) in which the probe copies back *only* the number feature of the postverbal DP in order to assure that the probe may receive a value for number. This potential loss of coarseness in Agree still rules out (*i*a-b) as those would involve the copying of first or second person features to the phi-probe.

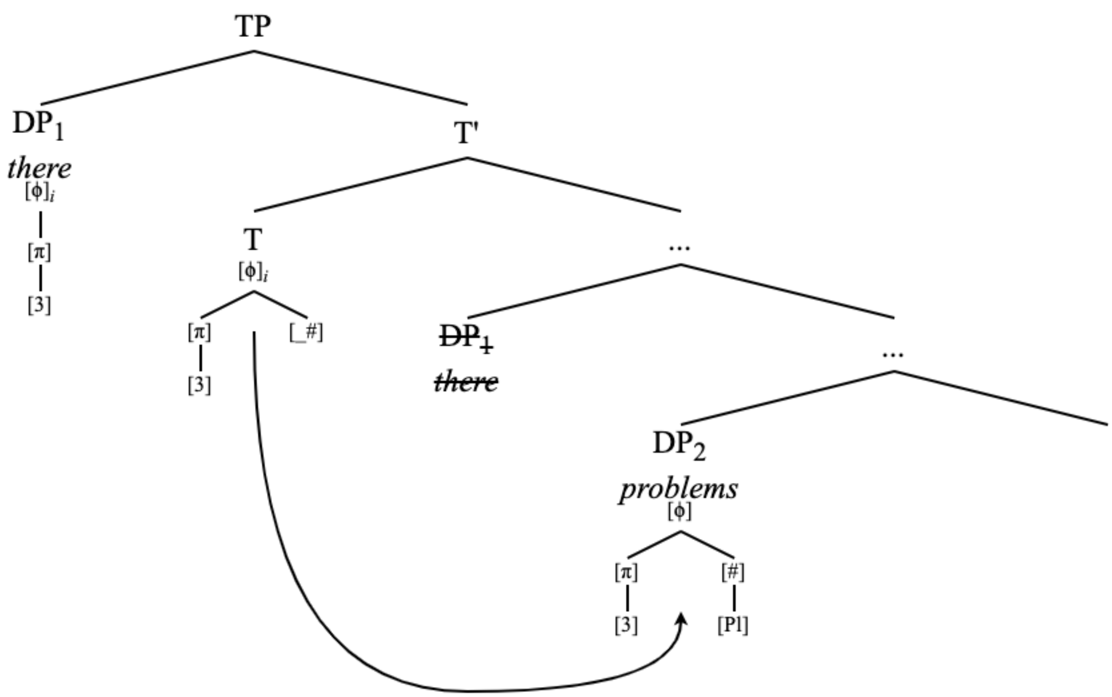
(132) *Defective Circumvention*
A probe P optionally undergoes Agree with a goal XP_2 that has the full set of features that P is specified to search for after first entering into an Agree relation with a deficient goal XP_1 that intervenes goal between P and XP_2.

Defective circumvention necessarily involves agreeing past a deficient goal that a probe has already undergone Agree with. The successful application of defective circumvention leads to verbal agreement tracking the features of the second, non-defective goal. The optional second search Find(#) of the phi-probe in these cases is what kickstarts this operation.

The optionality itself is explained in terms of the fact that it is typically only the root node of a feature geometry that carries the obligatoriness condition of probing, as well as the fact that general economy principles in syntactic derivations create preferences for or against certain derivations.

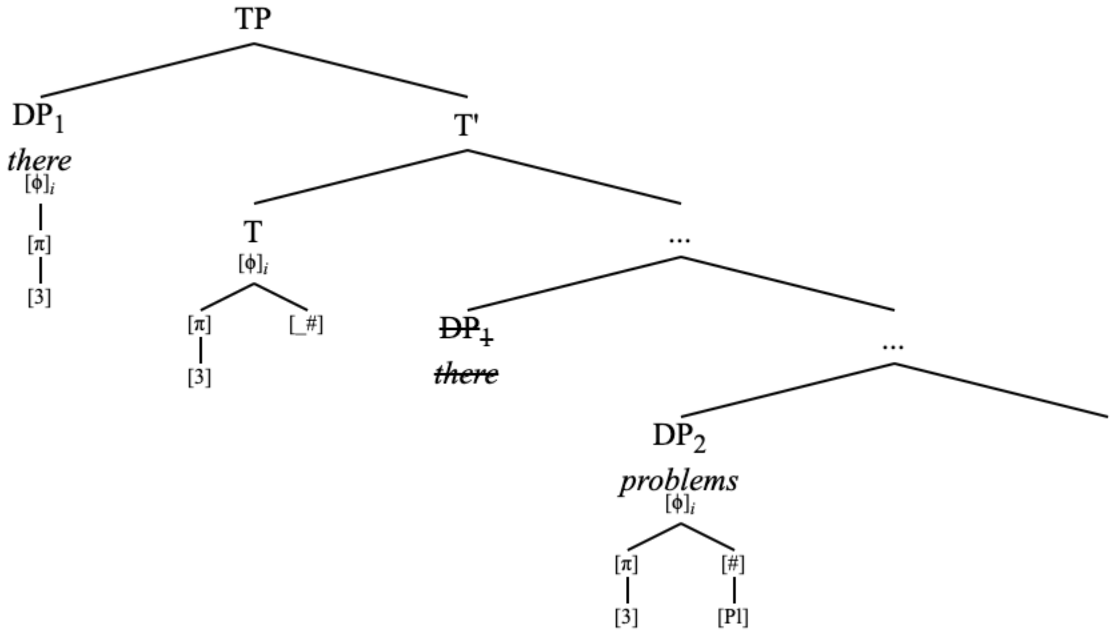
One must take care when discussing these kinds of preferences, as it can quickly sound as if one is attributing some explanatory power in the syntax to some kind of Optimality-Theory (OT) constraint rankings. Crucially, that is not what I am doing here; rather, these preferences are reflections of underlying principles of economy.

(133) There {was/were} problems: Second Search



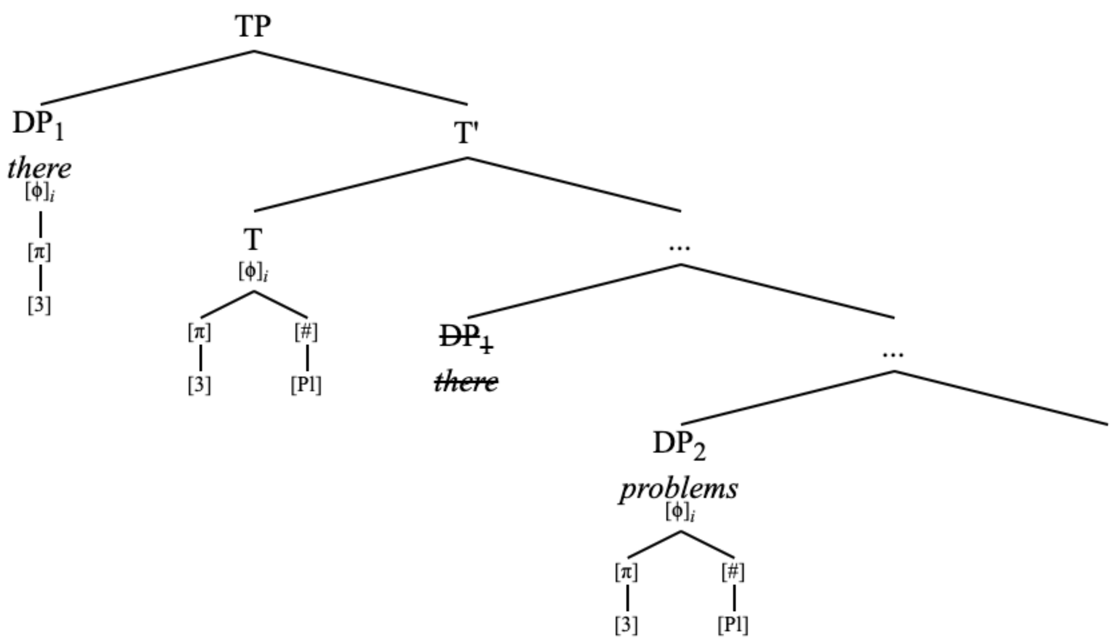
The feature geometry of the lower DP 'problems' is Matched with the probe due to the number feature on 'problems'.

(134) There {was/were} problems: Second Match



After Match, the feature geometry of the lower DP is copied to the probe, which already contains the features that were copied from the higher deficient DP. In this case, the features from the higher DP were third person and nothing else, represented geometrically as [ϕ[π[3]]]. Note that these features are a proper subset of the features on the lower DP. In other words, there is complete overlap. There are no "conflicting" features which may cause a problem for interpretation, which I detail further on.

(135) There {was/were} problems: Copy features of 'problems' (no Move)



Despite the probe entering into an Agree relation with the lower DP ‘problems’, this DP is not moved to Spec,TP. This is due to the simple fact that something else (the higher DP) already occupies that position. T^0 ends up with third person plural phi-features, causing the verbal agreement morphology to be spelled out as ‘were’. Defective circumvention derives the prescriptively correct agreement pattern in English existential ‘there’ sentences. Thus, without defective circumvention, the verb only agrees with the deficient element, resulting in singular agreement morphology. When there is defective circumvention, agreement tracks the non-deficient goal. This is summarized in the following table.

(136) Agreement outcomes

Defective Circumvention?	*Agreement morphology*
No	Third singular
Yes	Tracks postverbal subject

Defective circumvention is the mechanism by which agreement with postverbal subjects is derived any time there is a deficient DP occurring between the probe and the postverbal subject DP including in English existential ‘there’ sentences as well as locative inversion, quotative inversion, and predicate inversion. Defective circumvention also extends to dative subject constructions in Icelandic and Faroese, Spanish psych verbs and unaccusatives, Portuguese unaccusatives, and existentials in Spanish and Catalan.

Not all postverbal subjects, however, can be agreed with. That is, there are instances in which defective circumvention may be attempted but fail depending on the featural specification of the two goals with respect to the probe and to one another. I now show how this is possible under the formulation of Agree laid out thus far.

2.3.2 Failing defective circumvention – weak PCC effects

Recall the following data on agreement in English existential ‘there’ sentences with first and second person pronouns in postverbal positions.

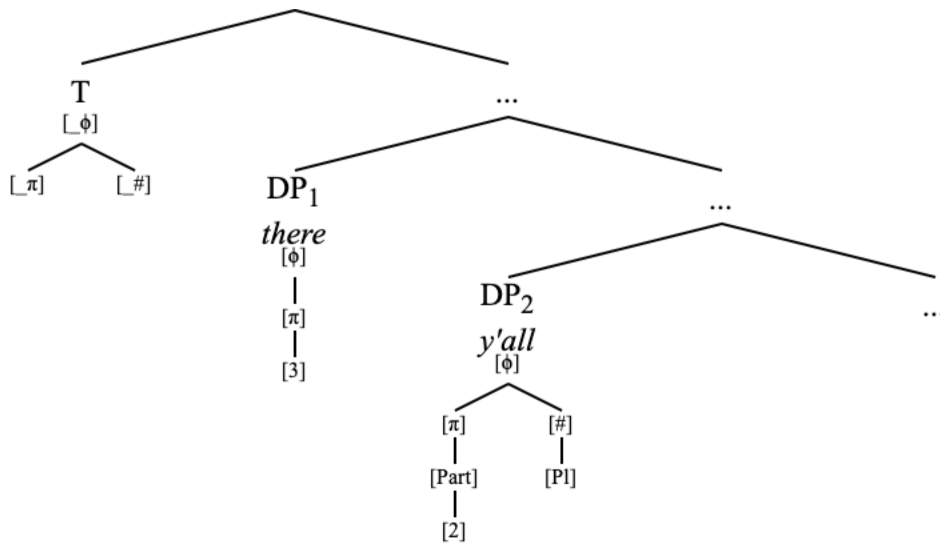
(137) a. There {is/*am} me
b. There {was/*were} you
c. There {was/*were} us
d. There {was/*were} y’all

This is not a ban on pronominal agreement as a whole in existential ‘there’ sentences. Recall that the following is still acceptable (Francez 2006; Deal 2009).

(138) There {was/were} them

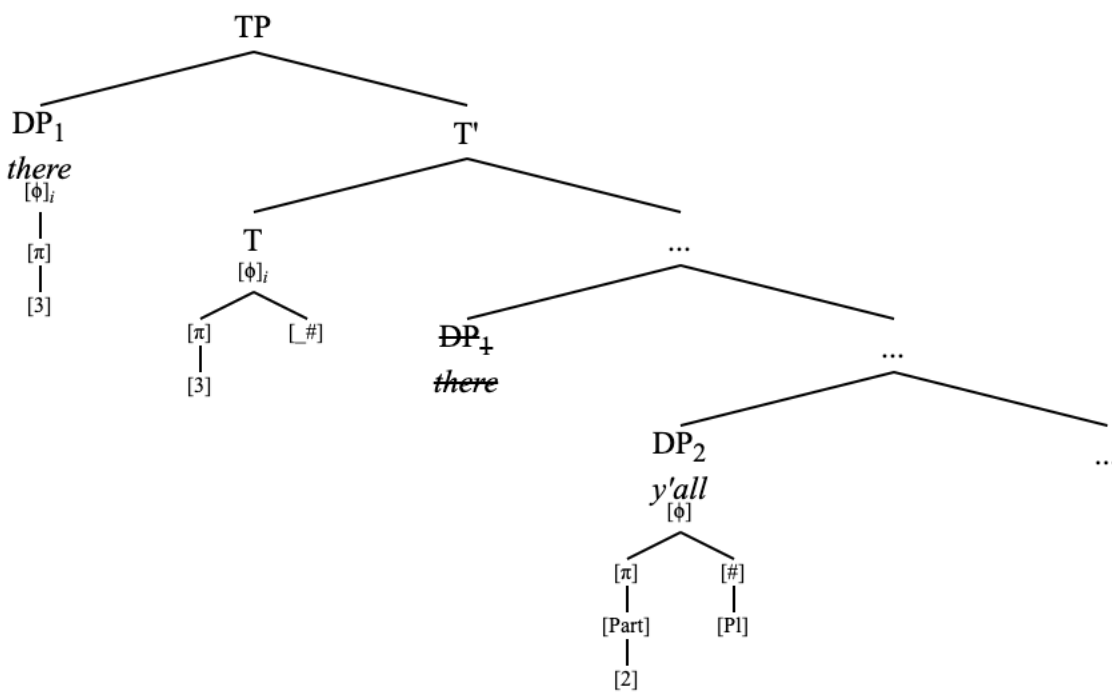
Rather, agreement with first and second person (i.e., [Part]-bearing) DPs in English is incompatible with a probe that has already entered into an Agree relation with a third person DP. Defective circumvention may be attempted in these cases, but it will create a featural conflict on the probe that leads to ineffability. I show this now with a second person plural DP ‘y’all’ in the second DP position.

(139) There {was/*were} y'all



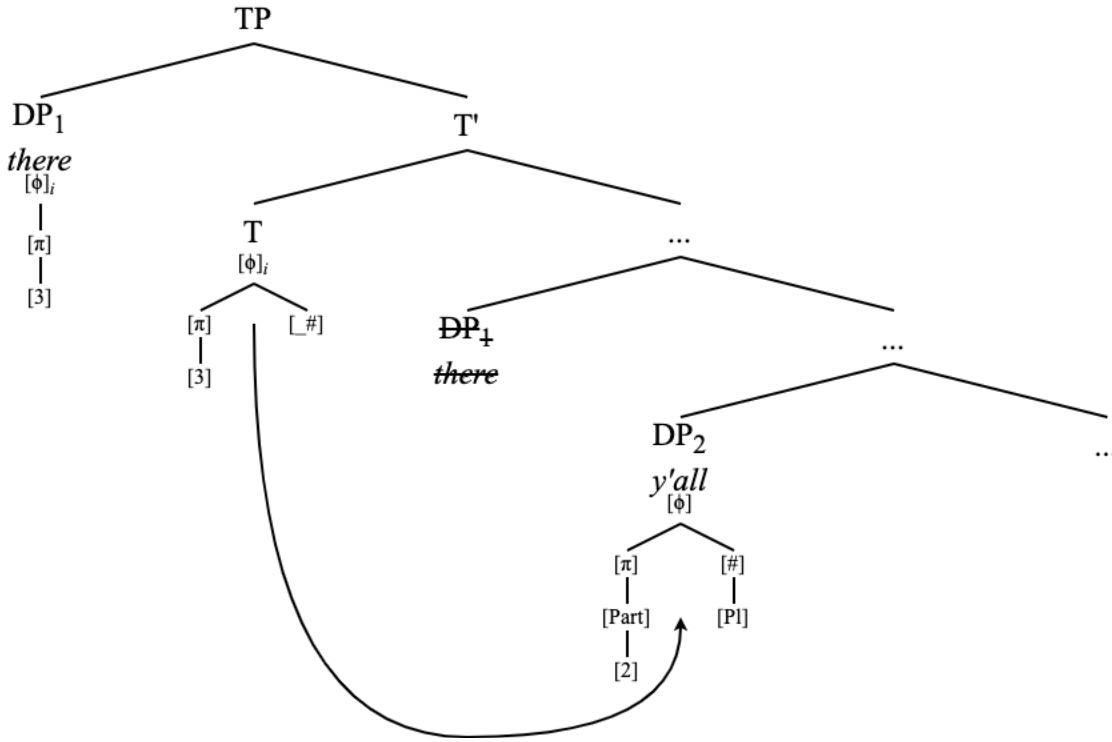
Search, Match, Copy, and Move happen with 'there' just as it did in the previous derivation.

(140) There {was/*were} y'all: Search, Match, Copy + Move



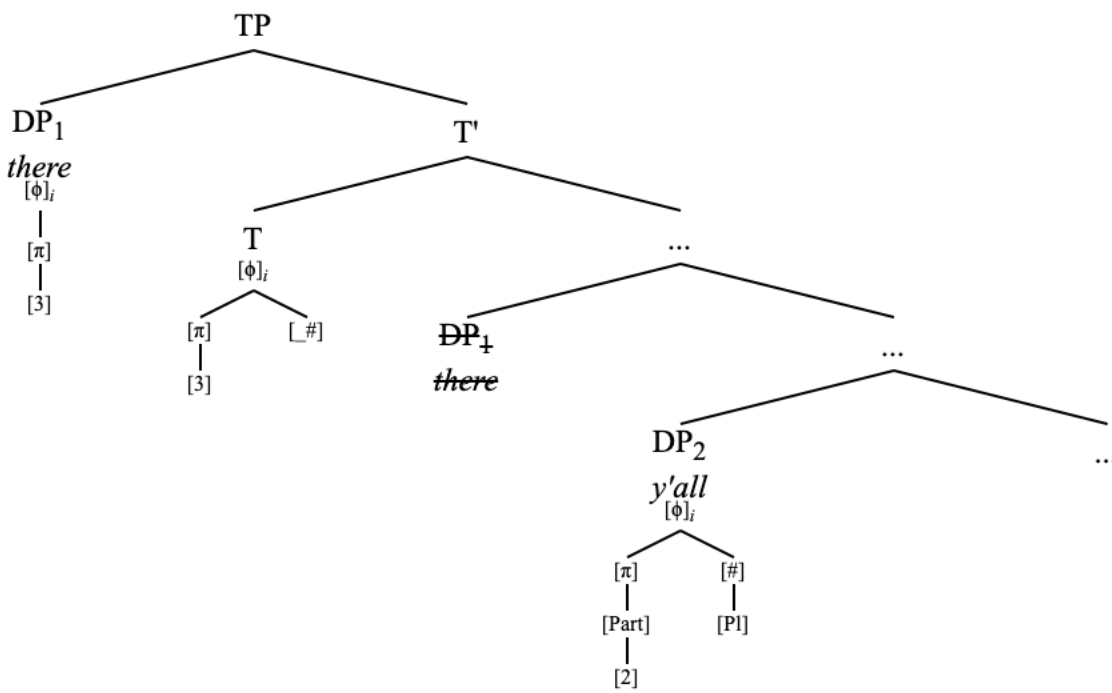
The phi-probe now bears a third person feature and an unvalued number feature. The probe has the option to not probe again, or to probe again for the valuation of [_#]. If the probe does not initiate a second search, then the verbal agreement is spelled out as 'was'. If it does attempt defective circumvention, though, it leads to ungrammaticality. For the sake of possible grammatical derivations, T undergoing Agree with just 'there' is the only option. I now show what goes wrong when T attempts to value its number feature with a first or second person DP

(141) There {was/*were} y'all: Second Search



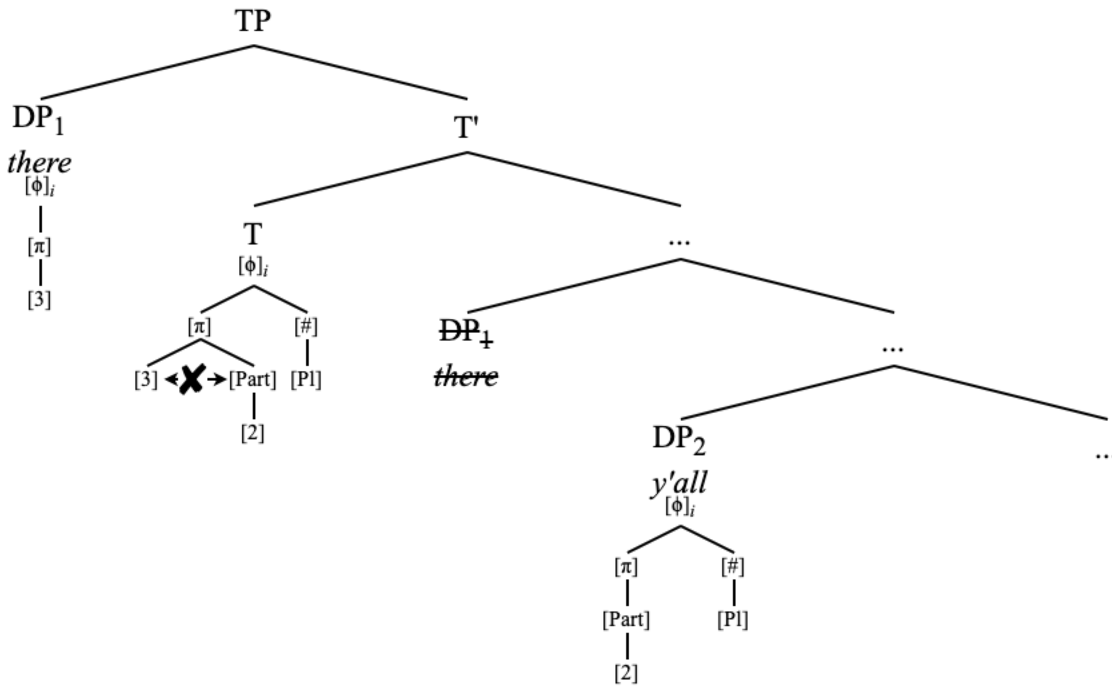
Search identifies the DP 'y'all' which contains a number feature.

(142) There {was/*were} y'all: Second Match



Match occurs between 'y'all' and the probe based on the presence of [#] on the goal and [_#] on the probe. The valued features on the probe (inherited from higher DP) are not a subset of the features on the lower DP. The higher DP contains [3], while the lower does not. The lower DP contains [Part[2]], which the higher does not. While this does not prevent the probe and the goal from undergoing Match at this point, it will cause problems at the Copy stage. See the following.

(143) There {was/*were} y'all: Fatal Copy



After having been identified as a matching goal, the feature geometry of the goal 'y'all' is copied to the probe. Remember that the copying of phi-features is coarse, so the entire geometry is copied over, even though it was only identified under a search for a number feature. Once the features of 'y'all' are copied to the probe that already contains a valued third person feature, the probe bears plural number [ϕ[#[Pl]]], second person [ϕ[π[Part[2]]]], and third person [ϕ[π[3]]].

This feature geometry on the probe contains a conflict as [π] has two daughters: [3] and [Part[2]]. Recall that [3] is shorthand for [Non-participant], so this second Agree cycle has caused the probe to contain both a [Non-participant] and [Participant] feature[13]. These noncompatible features create issues at the level of interpretation, especially considering that English does not mark clusivity in this way.

The third person feature [3] is not compatible with the presence of another person value such as [2] or [1]. Recall that English only allows one terminal node under [π]. English does not distinguish between first person inclusive and exclusive forms, and no language distinguishes between second person inclusive and exclusive forms (Harbour 2016). In any case, first person

[13] Perhaps this can be conceived of as a binary feature to better capture how [3] and [Part] are incompatible, something like Nevins (2007), so [±Part].

inclusive is typically analyzed as a first person feature [1] plus a second person feature [2][14], and first person exclusive is analyzed as a first person feature [1] by itself (Harley & Ritter 2002; Béjar 2003; Nevins 2007; Harbour 2016; Bobaljik & Sauerland 2023; a.o). There exist languages for which first person inclusive forms contain unambiguously second person morphology (e.g. Guaraní (Zubizarreta & Pancheva 2017; Johnson 2024)), but there are no languages for which first person exclusive forms contain overtly third person morphology. That is, [2]+[3] and [1]+[3] (generalized as [Part]+[3]) are not licit person feature combinations[15].

In languages that do not mark third person with a distinct feature and for which third person is the underspecification of person (represented as [Pers]), third person is not compatible with first or second person in a different way: for the simple reason that any specification of [Part] will overwrite the third person interpretation that comes from just having a [Pers] feature by itself. Note that this does not cause a conflict of two features trying to occupy the same position; rather, it simply renders one of the feature values inaccessible at the level of interpretation.

There is a question here of why [1] and [2] can be allowed together in languages that mark clusivity, but [3] and [1]/[2] is not permitted. This is because [1] and [2] form a constituent together under [Part] whereas [3]+[1] or [3]+[2] do not form a constituent in this way. So, features under [Part] are able to coexist in a way that [3] and [Part] cannot. This is ultimately a claim that [Part] and [Nonpart] are not compatible in the same feature geometry as the presence of both features cannot be interpreted or spelled out. In other words, [Part] and [Nonpart] compete for the same slot. There is also the issue of the probe no longer matching either of the DPs with which it was supposed to be agreeing in the first place.

As such, the copying of the feature geometry of 'y'all' to the phi-probe which already contains the features of 'there' in (143) creates a fatal conflict for Person as the feature geometry on the probe now contains [3] and [Part]. In a sense, this probe has become gluttonous (Coon & Keine 2021), though I do not claim that the only issue created here is one of morphological insertion.

The probe here is not opaque (in the sense of Kalin (2020)) to any step of Agree with the lower DP 'y'all' as I do not claim that any steps of Agree (Search, Match, or Copy) are prevented from happening. Rather, it is the "successful" application of all three steps of Agree attempting to perform defective circumvention that lead to ungrammaticality.

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[14] It should be noted that this configuration of multiple sequential Agree via defective circumvention does not leave open the possibility that a probe could acquire a [1] and [2] feature from two separate goals, thus leading to a first person inclusive-marked probe when neither goal is marked as such. Aside from creating a general issue of matching, such a situation is ruled out in the first place due to the nature of phi-deficiency and what kind of DPs we expect to serve as deficient interveners/circumventees in this schema. There are no number-deficient elements which are marked as first or second person, which suggests a situation in which [Part] is actually situated below [Num] in a phi-entailment hierarchy (Shlonsky 2023).

[15] It must be noted that, in languages for which third person is deficiently represented (i.e., "3noP" languages), the inclusion of both [Part] features and a third person feature would not lead to a conflict on the probe, since third person is represented only as [π]. In this way, in 3noP languages, there is no conflict in the simultaneous representation of participant and non-participant representations. 3noP languages, then, could allow for the simultaneous presence of first person and third person features in a single geometry to mark first person exclusive (Harbour 2016), if that is indeed how first person exclusive is marked. It would be interesting in future work to investigate the connection between 3yesP/3noP languages and languages that mark clusivity.

The only possible verbal agreement morphology in an existential ‘there’ construction with a first or second person postverbal subject is third person singular ‘is’/‘was’. This is because T has the option to Agree with ‘there’, and trying to Agree past ‘there’ with the [Part]-bearing postverbal subject results in ungrammaticality. Effectively, T can only Agree with ‘there’ in these cases.

These data lead to the following generalization on agreement in these English constructions involving postverbal subjects.

(144) Agreement and Case assignment of English postverbal subjects in inversion constructions

Phi-features	*Verbal agreement*	*Case*
1^{st} person	Impossible	Accusative
2^{nd} person		
3^{rd} person	Optional	

What is and is not possible in these configurations in a given language are constrained by how person is represented and what the featural specifications of the phi-probes are. What I have shown here with the impossibility of defective circumvention with a [Part]-bearing goal in English is the existence of PCC (Person-Case Constraint) effects in multiple-subject (i.e., multiple viable potential goals in the search domain of T^0) constructions in English. PCC effects are “a family of restrictions on the person features of different arguments in relation to one another” (Preminger 2016:1).

See again the (b) and (c) examples from (119), in which agreement via defective circumvention with a postverbal (accusative) third person DP is possible, but not with a [Part]-bearing one.

(145) a. There are them and then there's ME !
b. There were them and there was us; and he never ever saw it any differently

This is a weak PCC pattern, in which agreement with a lower 1^{st} or 2^{nd} person DP is blocked by the presence of a higher 3^{rd} person DP (Coon & Keine 2021). Here I have described a weak-PCC configuration involving English multiple subject constructions, adding to the literature on PCC effects with multiple subjects (Coon & Keine & Wagner 2017). I demonstrate in subsequent chapters that the same effects hold of locative inversion, quotative inversion, and predicate inversion constructions. I assume that other classes of PCC effects (e.g., strong, ultra-strong, me-first) can be derived from this same framework depending on the way that features are relativized on the probe (Nevins 2007) and the parametric representation of different feature values crosslinguistically (see examples (99)-(102)).

This formulation of Agree involving multiple sequential Agree relations by a single probe driven by Minimal Search and defective circumvention is capable of deriving PCC effects by collecting multiple feature sets on a single probe. In this way the analysis is partially similar to Coon & Keine’s (2021) *feature gluttony*, though my analysis here does not make reference to spellout and vocabulary insertion to derive these effects; it is done squarely within Agree.

A further issue with the feature gluttony approach is that their system predicts all forms in which a probe first acquires features from a featurally-underspecified goal and then features from a more specified goal to lead to ungrammaticality unless the derivation can be saved with morphological syncretism; however, such Agree configurations are exactly what we get with examples such as ‘There were problems’, despite the lack of syncretism between the third singular and third plural verbal spellout forms. This is an even bigger issue for feature gluttony when we see the Spanish data, in which defective circumvention is not only possible but obligatory with [Part]-bearing goals which are accessible to T^0 past a deficient goal, despite the fact that this necessarily involves a probe acquiring two different sets of features from a deficient goal and then a fully specified goal, and the fact that Spanish verbal morphology is notably poor in syncretic forms. I detail the Spanish data now.

2.4 Obligatory defective circumvention in Spanish

Refocusing now on the syntax of optional agreement with deficient DPs and fully specified DPs, let us look at the facts in Spanish. In a variety of constructions in Spanish (the exact derivational details of which can be left aside for now), the same optionality is realized with third person plural postverbal subjects. Some examples are repeated below.

(146) *pro* llega {-Ø/ -n} los gatos (Ordóñez p.c.)
pro arrived{-3sg/-3pl} the cats
‘There arrive(s) the cats’

(147) Lo que quiere {es/son} los privilegios para sus empresas (Ordóñez p.c.)
it that wants {is/are.3pl} the privileges for his businesses
‘What he wants is/are privileges for his businesses’

(148) Nos encanta{-Ø/ -n} las películas de terror (Fernández-Serrano 2022:9)
us.DAT love {-3sg/-3pl} the films of terror
‘We love horror movies’

(149) Hub{-o/ -ieron} dos hombres en la fiesta (Rodríguez-Mondoñedo 2006:326-7)
had {-3sg/-3pl} two men in the party
‘There {was/were} two men at the party

Just as in English, the optionality of agreement disappears when the postverbal DP is first or second person; however, the option to realize third person singular verbal agreement is what is impossible. Verbal agreement must be with the [Part]-bearing DP. In English, defective circumvention is impossible with a [Part]-bearing postverbal DP, while it is obligatory in Spanish.

(150) *pro* lleg {-amos/*-ó} nosotros
pro arrived{-1pl/ *-3sg} we
‘There arrived us’

(151) Lo que quiere {soy/*es} yo
it that wants {am/ *is} I
‘What he wants is me’

(152) Me gusta{-s/ *-Ø} tú
me.DAT like {-2sg/*-3sg} you
'I like you'

I do not address the issue of agreement in Spanish existential sentences with the verb *haber* in much detail as Spanish has a strong definiteness effect – much stronger than English or even Catalan, for example – as well as significant dialectal variation with agreement in existential *haber* sentences, both of which significantly complicate the analysis of agreement optionality with first and second person postverbal DPs there. One interesting fact about Spanish existential sentences is that the prescriptively correct pattern of agreement is to have no agreement with a third person plural postverbal subject (153). Having verbal agreement track the postverbal subject in these cases is degraded (154).

(153) No habrá testigos
no have.FUT.3sg witnesses
'There will be no witnesses'

(154) No habrán testigos
no have.FUT.3pl witnesses
'There will be no witnesses'

The phi-probe on T in Spanish is relativized to find [_Part], not just person (Gravely & López & Greeson 2024). This looks like [_ϕ[_π][[_Part]]] (Coon & Keine 2021). Evidence of this comes from the fact that, in situations in which a probe has two accessible goals in its search domain and only one of them bears a [Part] feature, the probe ultimately agrees with the [Part] bearing DP, even if the potential goal without a [Part] feature is not deficient for number and is closer to the probe. See Gravely & López & Greeson (2024) for an in-depth analysis of such examples.

(155) Mi esperanza {eres/ *es} tú
my hope {are.2sg/*is} you
'My hope is you'

(156) Ese {soy/*es} yo!
that {am/ *is} I
'That's me!'

(157) Las víctimas {fuimos/ *fueron} nosotros
the victims {were.1pl/*were.3pl} we
'The victims were us'

Furthermore, existentials in Spanish with the verb *haber* show the same restriction whenever there is a first or second person postverbal subject. Again, Spanish existential sentences have a strong definiteness effect, so it is not possible to have pronouns in this position with *haber*, but some indefinite DPs may bear a [Part] feature. When these DPs are in the postverbal position, agreement with them is obligatory.

(158) Habe-mos dos estudiantes en la clase (Rodríguez-Mondoñedo 2006:343)
have -1pl two students in the class
'There are two students(+Spkr) in the class'

(159) Habé-is dos estudiantes en la clase (Rodríguez-Mondoñedo 2006:343)
have -2pl two students in the class
'There are two students(+Addr) in the class'

In these sentences, there is necessarily an interpretation that the postverbal subject DP describes a group that contains a speech-act participant (a speaker and/or addressee). While the variant of these two examples with third person singular agreement on the verb is grammatical, in that sentence there is no possibility of having an interpretation in which the DP contains a participant.

(160) Ha -y dos estudiantes en la clase
have-LOC two students in the class
'There is two students(–Part) in the class'

When an existential postverbal subject DP in Spanish has a participant feature, the verb must obligatorily show agreement with it. Spanish existential *haber* sentences thus show the same agreement restrictions as these other constructions in Spanish.

Additionally, third person in Spanish is not distinctly represented with a [Non-participant] feature as it is in English. Spanish third person is represented as a [Pers] feature with no further specifications below it. While English is 3yesP, Spanish is 3noP (Nevins 2007). As evidence of this, I present the verbal agreement paradigm for Spanish verbs in the present tense.

(161) Present tense verbal agreement paradigm for *llegar* 'come'

<table>
<tr><th>ϕ</th><th>Verb root</th><th>Theme vowel</th><th>Agreement suffix</th></tr>
<tr><td>1sg</td><td rowspan="7">lleg-</td><td>-Ø[16]</td><td>-o</td></tr>
<tr><td>2sg</td><td rowspan="6">-a</td><td>-s</td></tr>
<tr><td>3sg</td><td>-Ø</td></tr>
<tr><td>1pl</td><td>-mos</td></tr>
<tr><td>2pl</td><td>-is</td></tr>
<tr><td>3pl</td><td>-n</td></tr>
<tr><td>Infinitival</td><td>-r</td></tr>
</table>

The representation of third person singular as the lack of any agreement suffix is consistent across tenses, and it is the only feature specification for which agreement morphology is zero. This is distinctly the opposite of English present tense, where third person singular is the only feature specification for which there is any phi-agreement suffix.

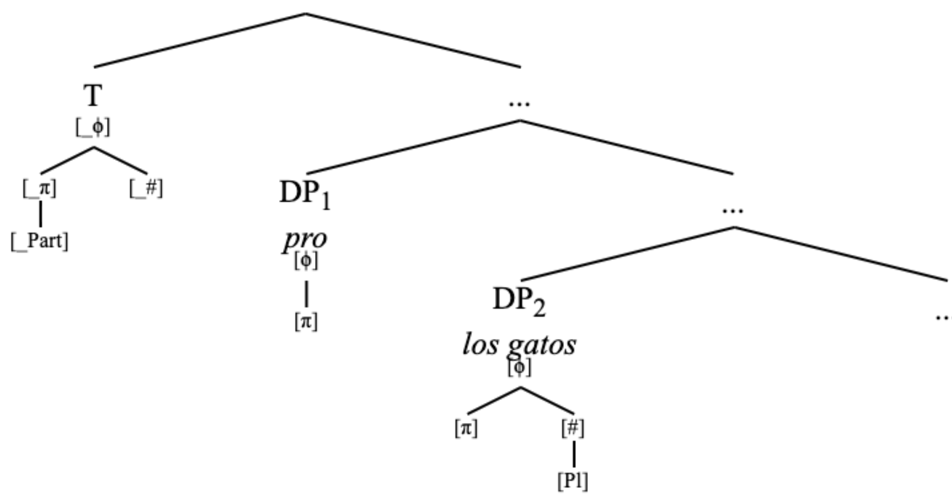
[16] The "absorption" of the theme vowel by the first person singular present agreement suffix, while notable on this table, is irrelevant and uninteresting for the discussion at hand.

2.4.1 Optionality with third person plural

Keeping in mind that Spanish is a 3noP language, and that the phi-probe on T in Spanish is relativized for [_Part], it is possible to analyze the data presented above in the same framework as the English existential 'there' construction. I use the unaccusative verb sentence as the primary example for now.

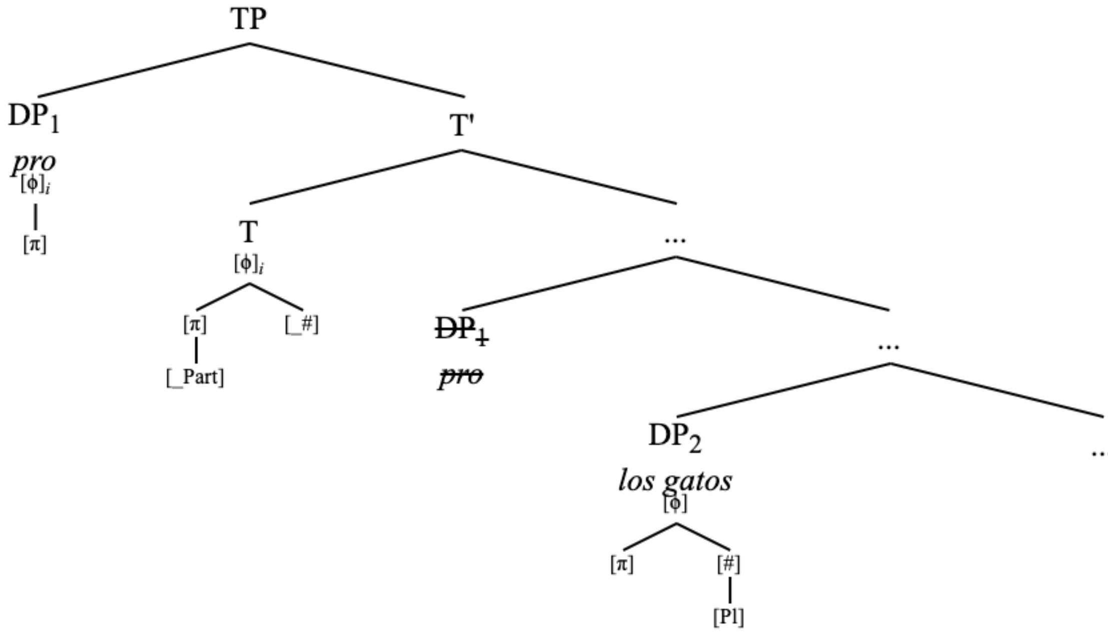
First I detail how the optionality is derived in agreement with third person plural postverbal subjects. This derivation is essentially the same as how it worked for English. I assume a null deficient argument *pro* that stands in a position as the intervener/circumventee between the phi-probe and the thematic overt subject argument (Pinto 1994; Adger; 1996; Zubizaretta 1998; Goodall 2001; Kučerová 2014; a.o.)

(162) a. *pro* llega {-Ø/ -n} los gatos
pro arrive{-3sg/-3pl} the cats
'There arrive(s) the cats'
b. Basic underlying structure



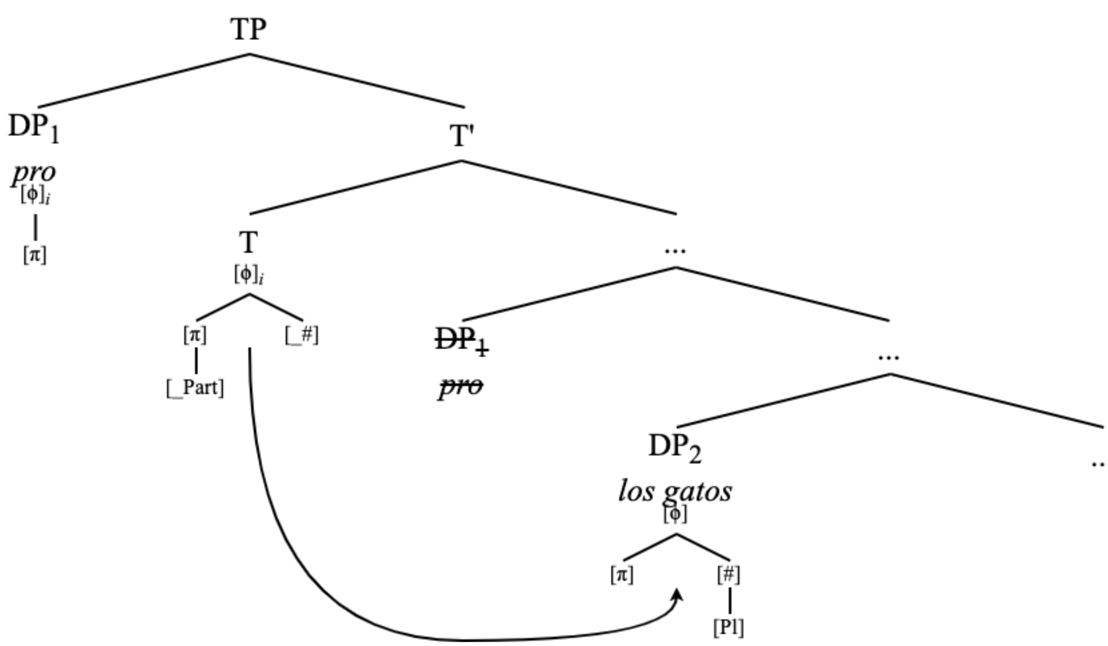
The phi-probe on T searches based on its specification to initiate the operation Find(ϕ). This causes the probe to find the deficient DP *pro*, which it Matches with, and then copies back the features of *pro*. As a result of successfully Agreeing with the probe on T, *pro* moves to Spec,TP.

(163) *Llega(n) los gatos* ‘There arrive(s) the cats’: Search, Match, Copy + Move



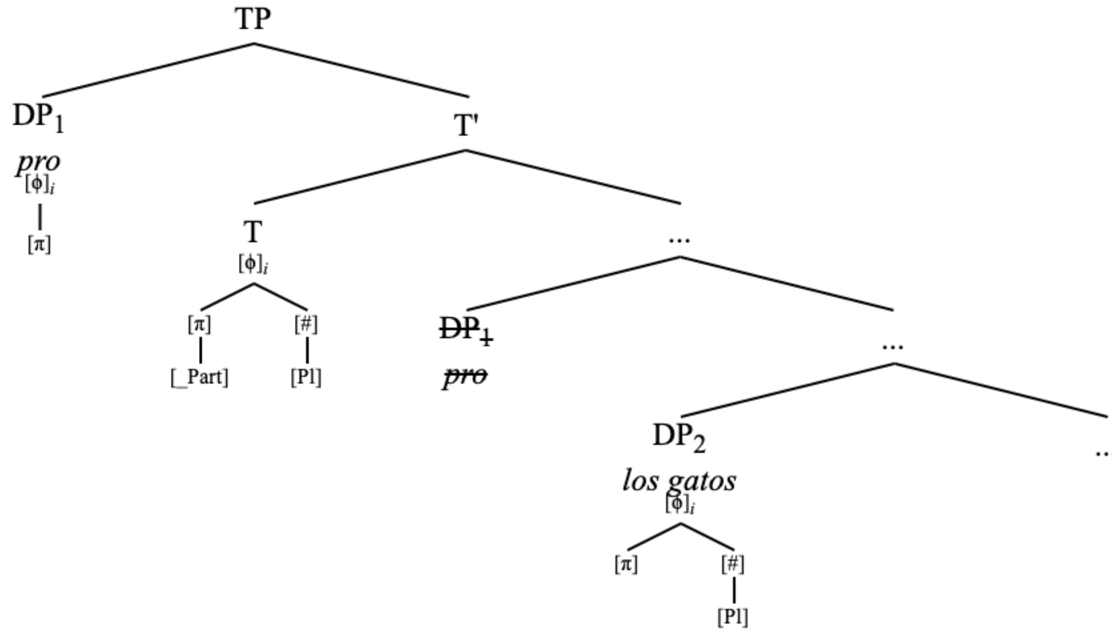
Now the probe bears the features [ϕ] and [π]. It still has unvalued features [_Part] and [_#]. A second search Find(Part) triggered by a [_Part] feature will find nothing. At this point the probe can either stop probing and spell out third person singular default agreement, or the probe can search again for number, and undergo defective circumvention.

(164) *Llega(n) los gatos* ‘There arrive(s) the cats’: Second Search



If the probe searches again, it finds, Matches, and successfully Copies back the features of the second DP *los gatos* ‘the cats’. This defective circumvention creates no conflict on the phi-probe.

(165) *Llega(n) los gatos* 'There arrive(s) the cats': Second Match + Copy (no Move)

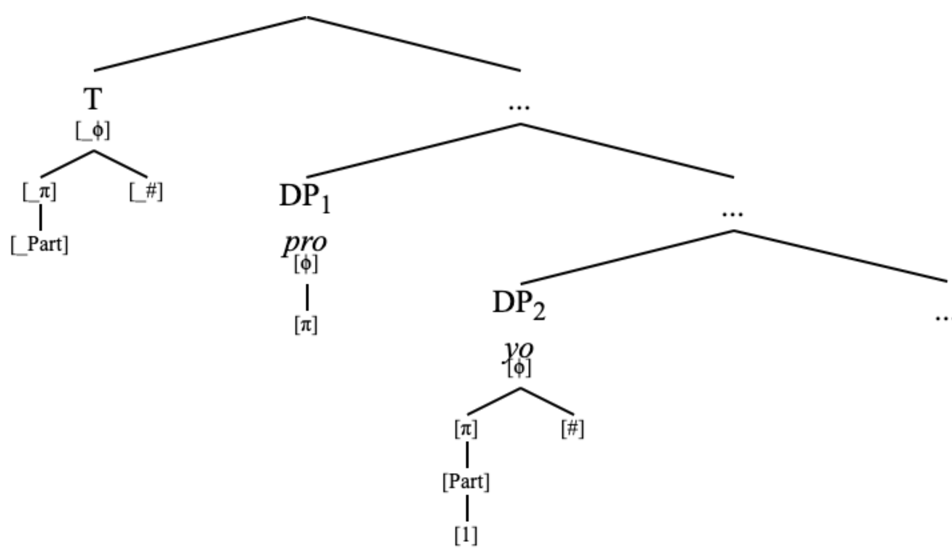


The valued features on the probe are identical to the valued features on the second DP. This causes verbal agreement to be spelled out as third person plural. Just as in English, no defective circumvention leads to third person singular verbl agreement, and defective circumvention leads to verbal agreement that tracks the features of the postverbal subject DP.

2.4.2 Agreeing with [Part]

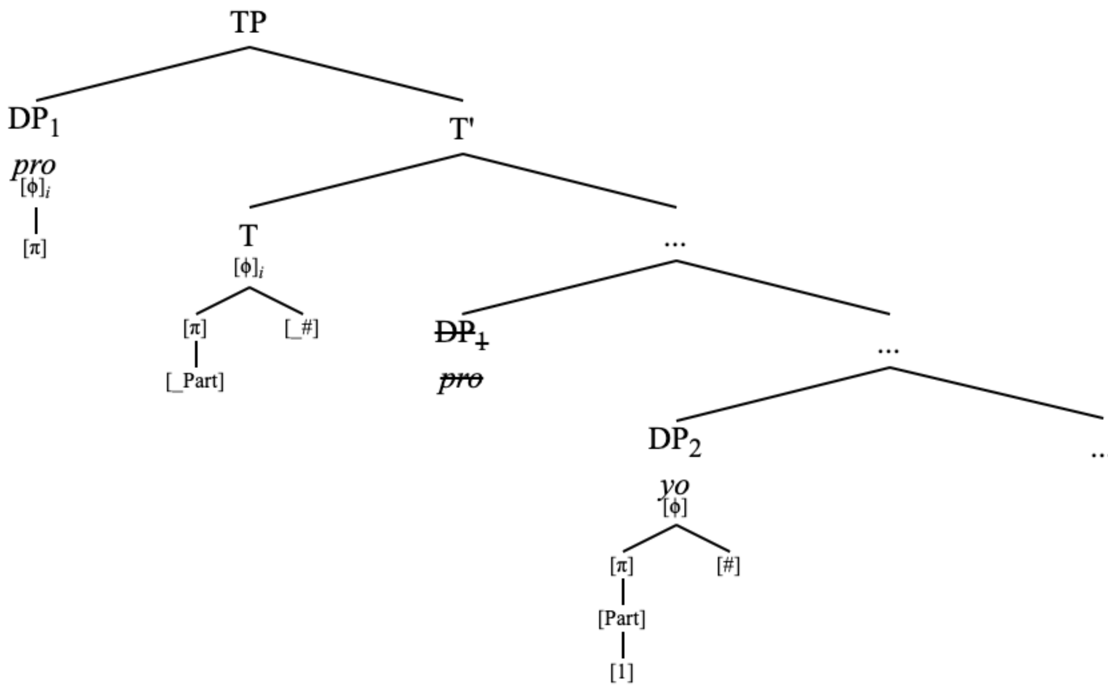
First or second person DP obligatory must be agreed with via defective circumvention.

(166) a. *pro* lleg {-ué/ *-ó} yo
pro arrived{-1sg/*-3sg} I
'There arrived I'
b. Basic underlying structure



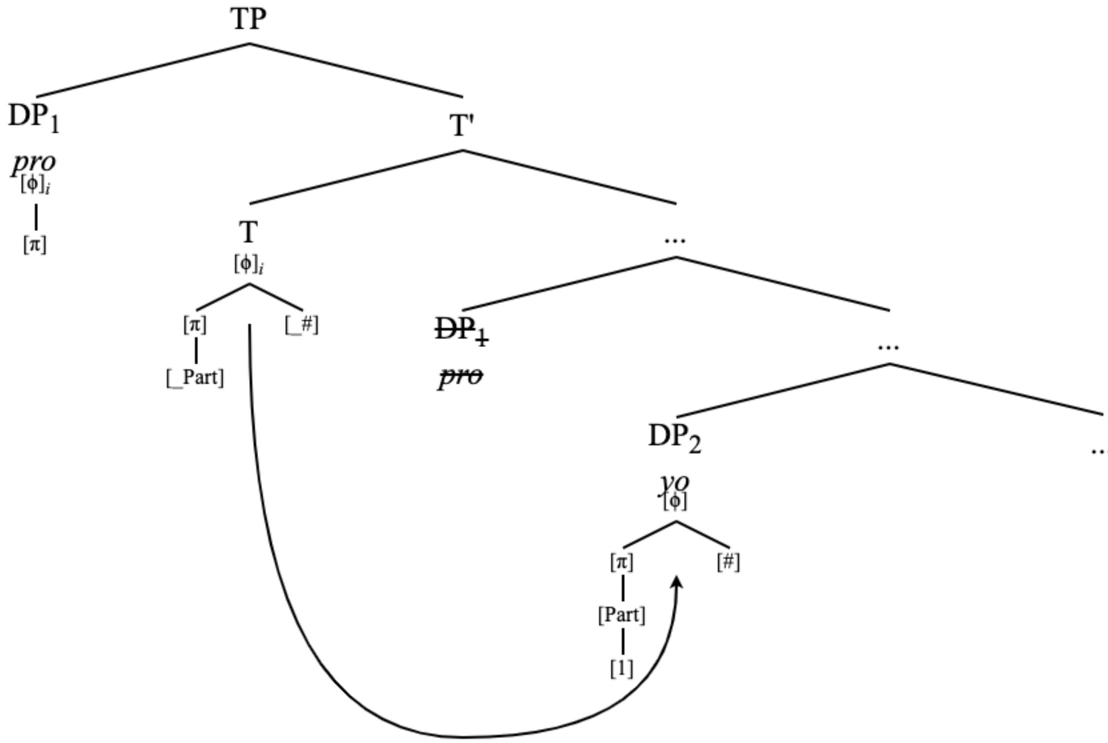
As in the previous cases, the phi-probe on T initiates the operation Find(ϕ). The higher, deficient DP *pro* is identified via Match, then its feature geometry is copied to the probe. As a result, *pro* is moved to Spec,TP.

(167) *Llegué yo* 'There arrived I': Search, Match, Copy, Move



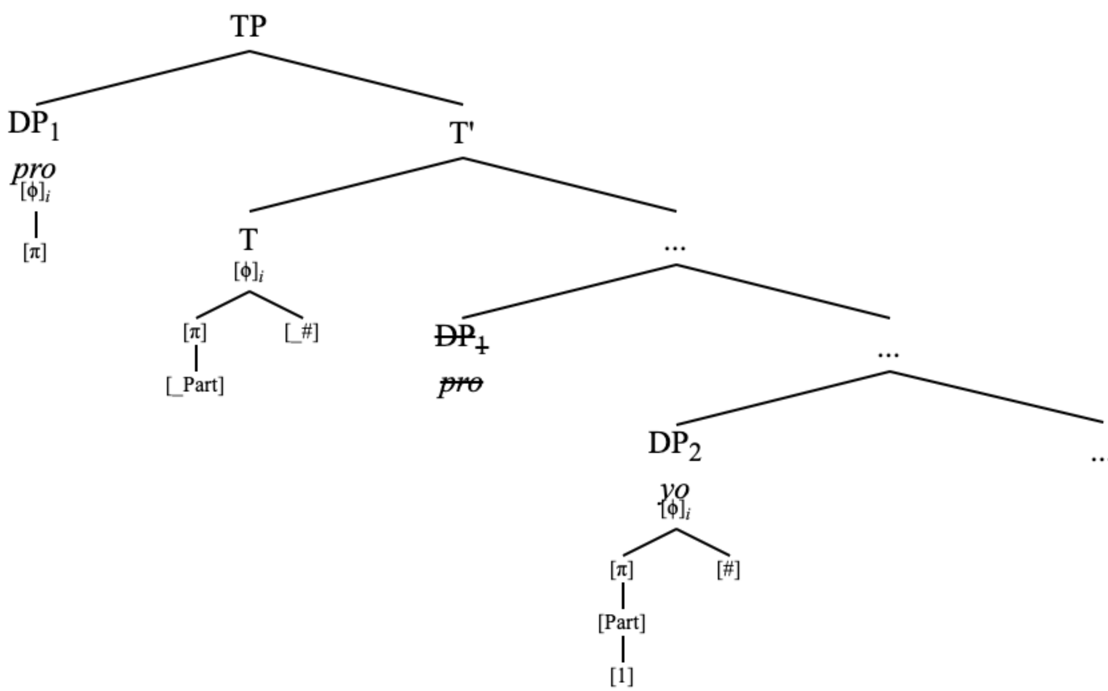
At this point the probe contains all the features that were on *pro*: [ϕ[π]]. The probe also still has the unvalued features [_Part] and [_#]. As such, Find(Part) is initiated, and a second Agree cycle is initiated, starting with Search.

(168) *Llegué yo* 'There arrived I': Second search



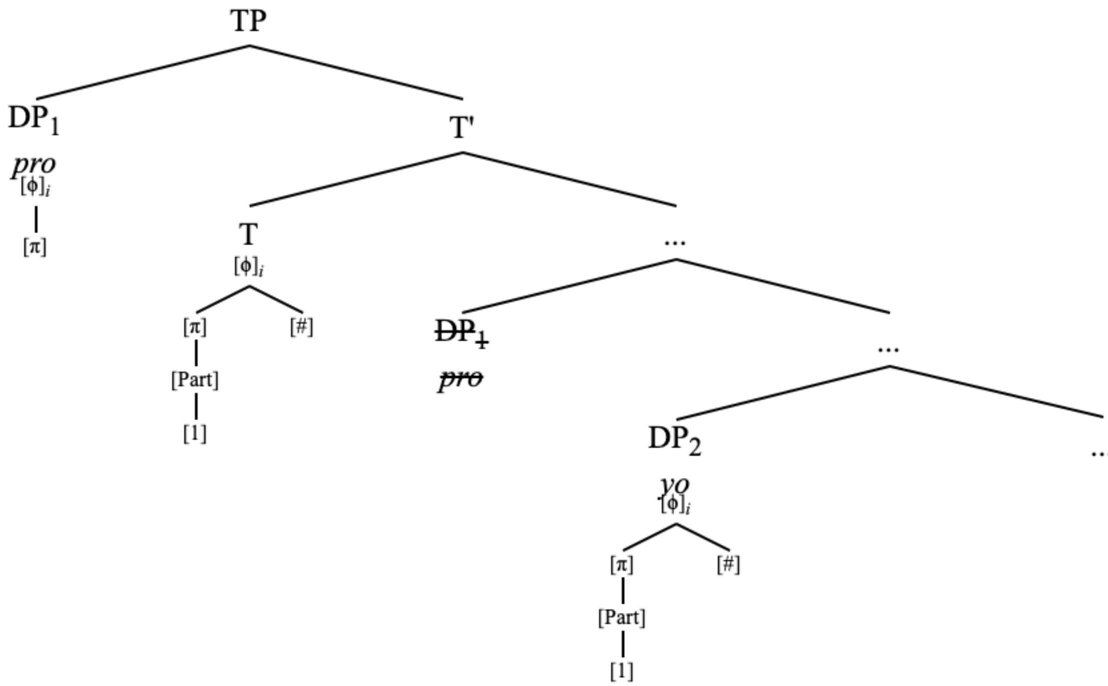
Here the participant features on the probe and on the second DP *yo* 'I' cause the two feature geometries to undergo Match.

(169) *Llegué yo* 'There arrived I': Second Match



Now the feature geometry of the lower DP is copied over to the probe, which already contains the features of the higher DP, and defective circumvention has obtained. Note that there is no conflict at this point: the valued features on the probe that were copied over from the higher DP are a subset of the features on the lower DP. This is due to the deficient representation of third person in Spanish.

(170) *Llegué yo* 'There arrived I': Second Copy (no Move)



The probe now bears an identical feature geometry to the geometry of the lower DP, even though it has undergone Agree with two distinct DPs. The features of the lower DP bearing the [Part] feature totally eclipse the features of the deficient DP *pro*, so to speak. As such, the verbal agreement in this configuration is spelled out as first person singular.

See the following generalizations on agreement with postverbal subjects in Spanish inversion configurations.

(171) Agreement and Case assignment of English postverbal subjects in inversion constructions

Phi-features	*Verbal agreement*	*Case*
1st person	Obligatory	Nominative[17]
2nd person		
3rd person	Optional	

The crucial difference between this configuration – a deficient third person DP over a non-deficient participant DP – in English and in Spanish is that the representation of third person with a distinct non-participant feature in English causes a fatal conflict when the probe attempts Agree

[17] These postverbal subjects in Spanish are all nominative, whereas in English they always surface as accusative. This difference can simply be explained by the fact that nominative Case assignment works differently in these two languages. It is well-known that nominative Case is the default Case in Spanish, but not in English.

with the participant DP, whereas no such conflict arises in Spanish due to the representation of third person as a lack of further specification on [π].

Furthermore, the option to stop probing and spellout default agreement after the probe has only undergone Agree with the higher deficient third person DP is not available in Spanish, whereas it is in English.

(172) a. What I love most in the world {**is**(=*default*)/*are} you
b. Lo que más quiero en el mundo {eres/ ***es**(=*default*)} tú
it that most want.1sg in the world {are.2sg/*is} you
'What I love most in the world is you'

It is not immediately clear why third person singular default agreement is not available in Spanish when the postverbal DP bears a [Part] feature, even though default agreement is an option when the postverbal subject is third person (and it is also the only option in the same configuration in English). To answer this, I turn to the necessity of [Part] to value an unvalued feature, as well as the novel interconnected necessity of [_Part] to (attempt to) undergo valuation.

The Person Licensing Condition (henceforth PLC) is a requirement that any [Part] feature must participate in an Agree relation (Béjar & Rezac 2003; Baker 2008; Preminger 2014).

(173) *Person Licensing Condition* (Béjar & Rezac 2003:53)
Interpretable 1st/2nd-person features must be licensed by entering into an Agree relation with an appropriate functional category.

The category [Part] must be licensed via Agree. This explains the ungrammaticality of default agreement in a sentence such as (172b), as default agreement would mean that the [Part] on the postverbal DP never enters into an Agree relation. The PLC does, however, present a problem for the English data, in which postverbal subjects bearing a [Part] feature may not be agreed with.

(174) a. There {is/*am} me
b. What I love most in this world {is/*are} you
c. Into the room {walks/*walk} us

These data are puzzling if the PLC is a universal condition on [Part] features on DPs (Nichols 2001; Ormazabal & Romero 2001), since the [Part] features in the sentences in (97) are not participating in an Agree relation with T. Indeed, Béjar & Rezac (2003:53) present the PLC as an axiom. While it is possible to work around this by postulating that the [Part]-bearing DPs in (97) participate in some valuation relation with another functional head (i.e., PredP or vP), such a claim would present problems for the activity of these DPs as goals (Chomsky 2000; 2001; Carstens & Diercks 2013) since third person DPs in the same position can be Agreed with by T.

While the PLC in (96) helps to derive the patterns of agreement in Spanish, it complicates the story for languages such as English in which agreement past a defective goal to a first or second person DP is not possible.

There is another version of the PLC represented in work by Preminger (2011; 2019) and expanded upon elsewhere (Coon & Keine 2021) that makes explicit reference to phi-probes; however, this formulation of the PLC still does not explain the English data.

(175) *Person Licensing Condition (Revised)* (Preminger 2011:931; Coon & Keine 2021:661)
A [Part] feature on a DP in the same clause as a person phi-probe must be agreed with by that phi-probe.

If anything, this more explicitly formulates why the English cases are an apparent violation of the PLC. It makes it even more puzzling. The Premingerian response here would perhaps be to say that the Case of the postverbal DPs in English make them ineligible for phi-agreement with T (Preminger 2014:144-170; 2019:14b), but this is not the view of Case assignment that I adopt (see Section 2.2.3). Furthermore, such an explanation is inconsistent with the fact that third person non-nominative DPs may control phi-agreement when they appear as the second goal of a phi-probe that has already undergone agreement with a defective DP (176), as well as the fact that overtly nominative DPs in the same positions may not be agreed with in certain constructions such as quotative inversion (177).

(176) a. There {was/were} them
b. What she loves most in this world {is/are} them
c. Into the room {walks/walk} them

(177) "Beans!" says {I/we/you}

As such, any of the existing formulations of the PLC which place the burden of undergoing Agree on the [Part] features of the phi-goal do not accurately account for the agreement possibilities of postverbal subjects in English. As opposed to getting rid of the PLC for these reasons, I instead propose that this special status of participant features is better conceived of as a condition on phi-probes, not DP goals. I thus propose an Inverse Person Licensing Condition meant to replace any prior formulations of the PLC.

(178) *Inverse Person Licensing Condition*
An unvalued participant feature [_Part] on a phi-probe must trigger the operation Find(Part) if it has not been valued by some prior search (i.e. Find(ϕ)).

This explains how agreement with a 1st or 2nd person postverbal subject DP in Spanish is obligatory – because [_Part] always triggers a search if it has not been valued by the preverbal deficient DP. Then, if there is a [Part]-bearing goal in the search domain of the probe, Agree will happen no matter what. The Inverse PLC does not affect the optionality of agreement with a postverbal third person plural DP because that DP is not visible to an operation Find(Part) since it does not bear a participant feature.

Replacing the PLC with the Inverse PLC does not cause problems for prior analyses which make use of the PLC since the languages described under these analyses are languages which the authors claim also have unvalued participant features on the phi-probes, namely Georgian,

Basque, and Kichean (Béjar 2003; Béjar & Rezac 2003; 2009; Preminger 2011; 2014; 2019; Coon & Keine 2021, a.o). That is, the PLC effects in these languages can still be adequately explained under the Inverse PLC[18].
As English phi-probes do not contain a [_Part] feature, the Inverse PLC is not a relevant condition on the distribution of first and second person DPs in English. Any [Part] feature on a goal may be copied back to a phi-probe during any other Agree operation (e.g., Find(ϕ) or Find(#)), but it is not the case that there is a [_Part] feature on English phi-probes with the special status of requiring valuation.

This is puzzling still for a language like Icelandic in which not only is agreement with a [Part] bearing DP ungrammatical in these configurations, but the presence of this feature on the postverbal subject causes the whole sentence to be ungrammatical.

(179) *Henni {líkaði/ líkaðir} þu (Sigurðsson & Holmberg 2008)
her.DAT {like.3sg/like.2g} you.NOM
int: 'She likes you'

Icelandic is unlike both English and Spanish in this way. It has been claimed independently that phi-probe on T in Icelandic is relativized to [_π] and [_#] and does not contain a [_Part] feature (Taraldsen 1995; Hoover 2021).

But, what if Icelandic has a [_Part] feature like Spanish and also represents third person with a distinct [3] feature like English? Icelandic, then, is partially like Spanish and partially like English. This explains why first and second person DPs are generally forbidden as the nominative postverbal subject in this construction: they always create a feature conflict since third person is distinctly represented and since [Part] probing is obligatory due to the Inverse PLC. This analysis also has the potential to explain the fact that syncretic verbal agreement forms may be able to save a sentence like (179) from ungrammaticality (Coon & Keine 2021) as some kind of repair strategy to license the otherwise incompatible person features collected on the probe.

As a fourth case, let us look at Brazilian Portuguese. Brazilian Portuguese shows the same agreement optionality with postverbal third person plural subjects.

(180) Cheg {-ou/ -aram} as cadeiras (Costa 2000)
arrived{-3sg/-3pl} the chairs
'There arrived the chairs'

For some speakers, agreement with a [Part]-bearing postverbal subject is optional in the same way.

[18] There is still a question of why a condition like this should exist in the first place. This question is not addressed by Béjar & Rezac (2003), Preminger (2011; 2014), or Coon & Keine (2021), and the Inverse PLC as it is formulated here does not address it either. Baker (2008:ch 4) relates the PLC to the special status of first and second pronouns needing to be licensed via binding by an operator, which is ultimately related to the discourse licensing properties of [Part]-bearing DPs. The Inverse PLC better captures this licensing-via-binding requirement as it places the responsibility on the licensor (agreement) rather than the licensee (the [Part] features of a goal).

(181) Cheg {-ou/ -uei} eu (Silveira 2021)
Arrived{-3sg/-1sg} I.NOM
'There arrived me'

These patterns are different from English, Spanish, and Icelandic. In Brazilian Portuguese, agreement with all postverbal subjects is optional, regardless of their person-feature makeup. Like in Spanish, third person is deficiently represented in Brazilian Portuguese, making it a 3noP language. Like English, though, T^0 in Brazilian Portuguese lacks a [_Part] feature. This means that any probing for a postverbal subject is optionally driven by [_#] and no person conflict will arise with [Part]-bearing DPs.

In sum, agreement with postverbal first and second person DPs by a phi-probe that has already undergone agreement with a defective third person DP is obligatory in Spanish because of the requirement of the unvalued participant feature on the phi-probe to be valued. Agreement with first and second person DPs in the same configurations in English is not possible because (a) English lacks a [_Part] feature on T^0, and (b) the representation of third person in English is such that simultaneous agreement with a third person DP and a [Part]-bearing DP by the same phi-probe is not possible. In Icelandic, agreement with a [Part]-bearing postverbal subject is always obligatory because of [_Part] on T^0 , but it always leads to a feature conflict as third person is distinctly represented in Icelandic. As for Brazilian Portuguese, all agreement with postverbal subjects is driven by [_#] (like in English), and no conflicts arise on the probe with [Part]-bearing DPs as third person is deficiently represented (like in Spanish).

The differences here between English and Spanish are perhaps related to the fact that the distribution of participant features on goals is larger in Spanish than it is in English, as English only really has first and second person pronouns, whereas Spanish has first and second person pronouns as well as participant features on non-pronominal plural DPs (Torrego & Laka 2015). Spanish is more liberal than English regarding the distribution of its [Part] features, on goals and on probes as well.

It must be asked at this point why Find(#) is necessarily optional, but Find(Part) is obligatory. Looking at this contrast, there seems to be variation of which unvalued features must initiate a search, aside from the feature denoting the whole geometry [_ϕ], which always initiates Find(ϕ). The definition of the Inverse PLC in (178) sets apart Find(Part) as unique in its requirement to find a corresponding feature (and [Part] has been described as unique in this way as well (Béjar & Rezac 2003; Preminger 2011; 2014; 2019; Coon & Keine 2021)). It could very well be the case that doing any subsequent search after Find(Phi) is optional for every feature dominated by [_ϕ] except for [_Part]. So, any [_F] which entails [_ϕ] may be optional in whether or not it initiates Find(F) given that Find(ϕ) has already taken place[19]. Any probes which do not initiate an operation Find(Part) only do so because they lack a [_Part] feature altogether, the presence of which is parameterized in different languages, such as English and Spanish.

[19] Although, Preminger (2014) presents derivations which convincingly rely on the operation Find(#) to derive the relevant agreement patterns. Perhaps it is the case that probes are parameterized as to which features MUST initiate a Find(F) beyond the requirement of the whole feature geometry Find(ϕ) to do so, except for [_Part] whose search is always obligatory when present.

This point of variation in the distribution of [_Part] is not random; rather, it is connected to the fact that Spanish is a null subject language, and English is not. Rizzi (1982) claims that T bears a [+Pron(ominal)] feature in null subject languages which licenses ability to not pronounce subjects overtly. This [+Pron] feature is absent from T in languages in which subjects must be overt. This assertion reflects the accurate intuition that phi-agreement probes in null subject languages are more pronoun-like than they are in non-pro-drop languages, but I argue that a [±Pron] feature on T is not necessary to encode this into the syntax. It can instead be done with phi-feature specifications that already exist on phi-probes. This has the advantage of reducing the number of abstract features that must be posited in the grammar.

While T itself is certainly not a pronoun – that is, a DP – in null subject languages, it is able to absorb some of the responsibilities of an overt subject DP, so to speak. There exists formulations of the EPP in which the requirement to have an overt DP in Spec,TP is satisfied by phi-agreement on T (Alexiadou & Anagnostopoulou 1998), and it has been noted more generally that the availability of null subjects correlates with the richness of phi-agreement on T^0 (Chomsky 1981; Rizzi 1982; Holmberg 2010; Roberts 2019). In these languages in which T^0 has more of a pronominal function, it makes sense that T^0 would be encoded with more pronominal phi-feature values (e.g., Participant), as opposed to T^0 in a language such as English where it has no such function. In other words, Rizzi's formulation of a [+Pron] feature on T^0 is reduced to the presence of a more robust feature geometry on T^0.

Given this connection between having a [_Part] feature on T^0 and the null subject parameter, as well as the connections here with the "3yesP' vs "3noP" question, we can categorize languages according to their features in this way.

(182) Constraining factors on defective circumvention by language

	"3yesP"	*"3noP"*
[_Part] probe	Icelandic	Spanish
No [_Part] probe	English	Br. Port.

On two poles we have English and Spanish, English being not pro-drop at all and Spanish being fully pro-drop. If pro-drop is something associated with T^0 having a [_Part] feature and with third person being deficiently represented (3noP), then it makes sense that Spanish is fully pro-drop as it meets both of these criteria, while English is completely not pro-drop as it lacks them both.

As for Icelandic and Brazilian Portuguese, which are both in a way like English and like Spanish with respect to the presence/absence of [_Part] and the 3yesP/3noP question, these languages have both been described as partial pro-drop languages (Alexiadou & Carvalho 2017; Wurmbrand 2017). Given the association between these features and being a null subject languages, it naturally follows that languages which contain a mixture of these features should be partially null-subject languages.

To quickly test these predictions, let's quickly look at copular sentences in French and Italian that contain a first-person postverbal subject. French, a non pro-drop language, is like English in that agreement with the postverbal subject is impossible. Italian, a pro-drop language, is like Spanish in that agreement with the postverbal subject is obligatory.

(183) a. C' est moi (French)
that' is me
b. *Ce suis je
that am I
'It's me'

(184) {Sono/*é} io (Italian)
{am/ *is} I
'It's me'

Compare this with English 'It's me' vs Spanish *Soy yo*, a parametric difference which this system of agreement explains perfectly.

2.5 Interim Summary

I have thus far presented a theory of syntactic Agree as Minimal Search which leads to multiple simultaneous Agree via defective circumvention and its explanatory power to derive a variety of complex agreement phenomena including optionality and PCC effects in English and Spanish. This analysis uniformly explains the same agreement optionality seen with postverbal third person subjects in both English, Spanish, and Icelandic via the process *defective circumvention*. Parametric variation in the realization of defective circumvention – such as the fact that it is not possible with [Part]-bearing postverbal subjects in English, but it is obligatory in the same configurations in Spanish – boils down to the featural representations of the probes and goals involved in the configuration in each language, and these features' compatibility with each other.

This analysis also connects the parametric differences in agreement with English and Spanish here to the null subject parameter, using other languages such as Icelandic, Brazilian Portuguese, French, and Italian as evidence.

This analysis has the advantage of deriving these complex phenomena from a radically simple formulation of Agree and the representation of phi-features. As such, there is no need to appeal to additional complicative technology such as equidistance, simultaneous multiple Agree, postsyntactic feature evaluation and vocabulary insertion, or feature impoverishment. This theory thus achieves explanatory adequacy in compliance with SMT.

Before moving on to an account of other derivations in which defective circumvention occurs, it is first necessary to discuss some possible alternatives to the account of agreement optionality presented here in which the two possible outputs are the result of a single set of grammatical operations.

2.6 (Im)possible alternatives

Here I present some possible alternative analyses of the phenomenon of agreement optionality. Namely, I refute the idea that these alternations in English should be explained by an account that explains such optionalities via multiple grammars – each with their own set of rules – that coexist within the same speaker. I show that my analysis presented here is theoretically and empirically optimal.

2.6.1 Multiple grammars

Much work on intraspeaker morphosyntactic variation relies on the idea of having two separate, competing grammars in the same speaker (Kroch 1994; Tortora 2014a). It is important to clarify why this is not applicable here, as others have done for similar situations of intraspeaker agreement alternations (Adger 2006; Fernández-Serrano 2022).

The fact that a single speaker may use both variants of this agreement alternation within the same register and even the same utterance is not problematic for the idea of multiple grammar variation (Kroch 1989; 1994). Indeed, this is quite well-attested. See the following naturally-occurring examples.

(185) a. If there was problems at the turnstiles how is that Newcastle's problem!!!, there were problems at their last game with away fans having the same problems was that Newcastle's problem as well (Twitter)
b. If there are negative marks, then why isn't there grace marks? (Forum)
c. If there are capital letters, why isn't there capital numbers? (Twitter)
d. There is people who borrow money and avoid you so they dont need to pay it back. Then there are pricks like this who commit high level fraud which. (Twitter)
e. In today's political environment, there are 2 types of people. People that think the FDA could ever protect us, people that think the EPA could ever protect the environment, people that think govt can be the answer and then there is people who do not. (Twitter)
f. There is kids in townships who haven't even seen a white person in their life. But there are other issues in those areas that need to be tackled (Twitter)
g. Well it's not like there was many options for him to play against there were 8 teams in the league (Twitter)
h. I saw that there were marks on the floor but there was several in a line and i couldn't see what they were. (Twitter)
i. If there was two shooters, which it now seems there were, then they both would not have missed Trump. Staged (Twitter)
j. A reminder that there is two sides…There are two opposing vicious sides (Twitter)

More troubling, though, is the idea that a single speaker may be said to be multidialectal in two grammars that converge on everything except this. Certainly there are situations in which a language user may alternate between one grammar that has single, underspecified form for all verbal agreement patterns and another that has more robust verbal inflection, for example African-American English (AAE) speakers (Green 2002).

For a speaker that accepts both variants in (186), the example in (187b) is never grammatical. It is strikingly much worse than (186), and even more strikingly worse than (187a), which would be surprising if the speaker is said to have access to one grammar in which third person plural verbal agreement always surfaces as the default form for whatever reason.

(186) a. There have been a lot of people here
b. There has been a lot of people here

(187) a. A lot of people have been here
b. *A lot of people has been here

An analysis which attributes the alternation in (186) to a result of multiple grammars interacting in the same language user fails to account for the cases in which the optionality of agreement between (third person) singular agreement and (third person) plural agreement fails to hold, not just in existential expletive constructions, but elsewhere as well.

It is also not clear how an account that posits multiple competing grammars should deal with the fact that there are no speakers who *only* accept singular agreement in these constructions (although there are speakers who claim to only accept plural agreement).

A lexical variationist account may predict that, at times, the (third person) singular agreement form may appear instead of a (third person) plural agreement form, but this is not the case outside of a limited number of situations in English.

(188) a. *My four pet cats is very cute
b. *A lot of you is getting on my nerves
c. *The flowers blooms in spring
d. *The queens reads to us

It must be remarked that (188) do not improve with contraction. This suggests that, while the uncontracted copula *is* and the contracted copula *-'s* may be derivationally distinct on some level (MacKenzie 2013), there is not sufficient reason to believe that they are the result of different agreement relations.

(189) a. *My four pet cats's very cute
b. *A lot of you's getting on my nerves

Alternative analyses do nothing to account for the PCC situations in which agreement with the associate is not possible at all.

(190) a. Somewhere in the crowd there's you (Song lyrics)
b. *Somewhere in the crowd there are you

It does not make sense to say that the ability to switch between two grammars to get a given agreement pattern is somehow conditioned by what the associate is. Multiple extra stipulations on the properties of these grammars and when using one over the other(s) is permitted must be posited.

If there really were two grammars interacting to produce this optionality, we should expect to find a monolingual group of people for whom agreement with the associate is always bad, but no such dialect exists. It is not the case that there is a dialect which agrees with postverbal DPs in every single sentence, which I show further on.

As such, deriving this alternation via multiple grammars (or multiple lexical items (or both)) and conditions on when to switch between them is not sufficient to account for the data. Additionally, it raises plenty of theoretical concerns. It is not ideal to attribute such a simple morphosyntactic agreement alternation to the idea that a language user must acquire multiple grammatical systems in order to do this. Also, positing multiple homophonous, semantically equivalent lexical items that show up in the exact same contexts and vary only in their featural makeup is not ideal for acquisition. Either way, these analyses rely heavily on the language user doing some complicated feats of acquisition.

2.6.2 Change in progress? Not likely

Another reason to not attribute this specific agreement alternation to multiple grammars converging in one speaker is because of the nature of the alternation itself. There is no evidence to suggest that this agreement alternation is dialectal. In other words, lack of agreement with the associate is not particular to one specific dialect of English. The data presented in this chapter come from speakers from various regions within the United States, the United Kingdom, Canada, Ireland, Australia, South Africa, and potentially other areas as well (for those whose origin could not be determined). As such, were there two grammars converging to create this intraspeaker variation, it is not likely that the two grammars belong to two different geographic dialects.

Another dimension of variation aside from geography is changes over time, or diachronic variation (Lightfoot 1979). The convergence of multiple grammars within the same speakers is said to drive syntactic changes in progress (Kroch 1989; 1994; 2001; Niyogi & Berwick 1997). As such, the convergence of multiple grammars creating this agreement alternation can be taken to mean that the alternation itself is a change in progress – moving, say, from one grammar in which the verb must agree with the postverbal subject to one in which it does not[20]. It is not clear, though, if there has ever been a point in Early Modern or Modern English in which singular agreement with a plural postverbal subject in the existential 'there' construction was not possible.

Corpora such as University of Michigan's *Corpus of Middle English Prose and Verse* reveal numerous existential 'there' constructions with singular verbal agreement and a plural associate, as do various other sources on the Internet. I repeat some of these examples here.

(191) a. Also ther was men off gretare alyaunce (c. 1450, Decameron)
b. For there ys many wordes in Latyn that we haue no propre englyssh accordynge therto (1530, Early English Books Online)
c. Concludinge then that as there ys two deathes the fyrst and second / in yerth and in hell: so there vs a fyrst and second resurrection (1597, Early English Books Online)
d. The Duke had lodged him in the City. Soon after there arrives three Princes of the House of Savoy, Philip Lord of Bresse, the Count de Romont, and the Bishop of Geneva, then the Mareschal of Burgundy, the Lords du Lau, and d'Urfe, and some others, all Enemies to the King (1683, Early English Books Online)

[20] I assume that, in this hypothetical situation, the innovative grammar is the one in which associate agreement does not occur, as obligatory associate agreement is what is considered prescriptively correct (Sabin 2011).

e. As from time to time there Arrives Several Strangers here & not knowing where to get their Bread they are Generally Sent to me to be Employ'd in your Garden, till they can Elsewhere find Employment (1735, Colonial Records of the State of Georgia)
f. There is three separate strings of Wampum, which the Half-King has desired me to send (1754, The Writings of George Washington)
g. There was 3 French Deserters (1754, The Writings of George Washington)
h. There has arrived two Indians from Moskingam (1754, The Writings of George Washington)
i. if a man said he was union his life was threatened and nearly all the men was union but ther was men sent from the army to quell this union sentiments (1863, Bryan Tyson Papers)
j. Ther ys two chapelles of her more (1866, Corpus of Middle English Prose and Verse)

It is not the case that singular agreement was historically the only agreement pattern available. The above sources still show numerous instances of the existential 'there' construction with verbal agreement that matches a plural associate. In fact, the associate agreement examples far outnumber the non-agreeing ones, just as is the case in contemporary English. See the following examples, also from The Writings of George Washington (examples (191) above).

(192) a. They inform'd me, that there were four small Forts between New Orleans and the Black-Islands (1753, The Writings of George Washington)
b. That at New Orleans, which is near the Mouth of the Mississippi, there are 35 Companies, of 40 men each (1753, The Writings of George Washington)
c. There are also several other Houses, such as Stables, Smiths, Shop,&c. (1753, The Writings of George Washington)

Given the facts from historical attestation and current grammatical possibilities, an attempt to account for this agreement alternation using an analysis of multiple competing grammars converging within the language user does not sufficiently constitute a diachronic or synchronic analysis of the data.

2.6.3 Against derivational timing and ordering of operations

A popular analysis nowadays is to say that intraspeaker morphosyntactic variation is a result of variation in derivational timing: specifically, that a head probing for one feature before another feature creates one result, and the same head probing in a different order creates a different result (Obata & Epstein & Baptista 2015; Obata & Epstein 2016; Puškar 2017; Halpert 2019; Hoover 2021; Fernández-Serrano 2022; Driemel 2024; Streffer 2024; a.m.o.). For example, T^0 probing for person (or EPP, or something) over number produces the agreement pattern in (193a), but T^0 probing for number over the other feature produces the agreement pattern in (193b).

(193) a. There was things that I could've done differently
b. There were things that I could've done differently

This derivational timing account of intraspeaker variation is presented as a more advantageous alternative to the "multiple competing grammars" intraspeaker variation; namely, it is said to not involve placing the locus of this type of variation in the lexicon or attributing it to speakers switching between multiple grammars. It is not clear to me why that is the case, both because I am not sure how positing two different probe timing derivations actually evades the problem of positing multiple lexicons or multiple competing grammars that the language user must oscillate between, and also because there is compelling evidence that processes such as agreement and movement take place in a specific, invariable order (Preminger 2014) and that phi-probes must be hierarchically ordered (Béjar 2003; 2011; Béjar & Rezac 2003; 2009; Preminger 2012; 2019; Coon & Keine & Wagner 2017; Coon & Keine 2021). I outline these two issues here.

2.6.3.1 Back to square one

How does one derive optionality of what feature a head probes for first, if it is understood that a head probes for more than one feature? How does one formalize the variation in order of probing for given features?

Say there is a probe which searches for a goal bearing features X and Y. The claim is that the probe can either search for X then for Y (call this P(X>Y)), or search for Y then for X (call this P(Y>X)). If the order in which a probe searches for features must be specified in some way, then these are essentially two different probes. We are back to the acquisition of two minimally distinct lexical items problem. It is not clear how a language user would acquire two different probes P(X>Y) and P(Y>X) that show up in the exact same places and vary so minimally.

If we want to say that the difference between P(X>Y) and P(Y>X) does not come down to the difference between specific probes but is actually a difference in the grammar itself (i.e. *all* probes can be either P(X>Y) or P(Y>X)), then we are back to the competing grammars problem. Do we really want to say that every language user with this kind of variation has access to one grammar in which probes have one specification and another grammar in which probes have the opposite specification, and they are in completely free variation with one another and converge on everything else?

The other option is to say that the features on a given probe are inherently unordered, and the optionality arises not because of two different specifications in the order but from a lack of specification altogether. This is problematic for the reasons to be discussed in the following subsection, namely that there is a clear hierarchy of the ordering of phi-features that shows up in many places in the syntax, including probes.

If the probes are inherently unordered, it is also not clear why the following contrast between (194) and (195) should hold.

(194) a. *Then there were we in the median
b. *There are you in the way

(195) a. Then there was us in the median
b. There's you in the way

If 'there' has only a person feature and 'you' has a person and a number feature, then it is not clear why P(π>#) can derive the (195) pattern but P(#>π) is not available to derive the (194) pattern. Similarly, if P(#>π) is always available (and coarse), why should the (193) examples be ungrammatical while the following are acceptable?

(196) a. There were orcs in Mordor
b. There were hobbits in the Shire

These derivational timing analyses may neatly account for some data, but I have yet to see one that effectively explains how or when a language user ends up going with one order over the other (let alone comment on this issue at all). Claiming only that there is one order to derive one agreement pattern and another order to derive another agreement pattern is completely circular. Why would this kind of variation be present in the grammar of a language at all? How is it acquired? Georgi (2017:fn 27) posits four different probes with different orderings with respect to Move and Agree features that must all be acquired by a child in order to derive certain phenomena. Essentially, this means that a child must acquire four distinct abstract heads of the exact same category and distribution, which is a lot of work for alternations that could be handled by a single head which induces a uniform set of operations.

Furthermore, It has been effectively demonstrated that probes can search for multiple features, that probes initiate multiple searches in a given order, and that probes do different things with different features (Deal 2015; 2024). Aside from the considerations relevant to the data presented here, there is also compelling evidence elsewhere that phi-probes (at least on T^0) initiate searches in a fixed hierarchical order with person preceding number (Béjar & Rezac 2009; Preminger 2014; Deal 2015; Kalin & van Urk 2015; Půskar 2019). To claim that these specifications can freely vary on a given probe is not a minor stipulation. This is either a claim that the language has two or more sets of probe specifications, or that the language user has access to two or more different grammars – each with its own probe specifications – and can freely alternate between the two. Once again, this brings up the problems of positing lexical variation or positing multiple competing grammars to explain intraspeaker variation. The other option is that these probe specifications are fundamentally unordered, which leads to a different set of issues.

2.6.3.2 Hierarchy of operations

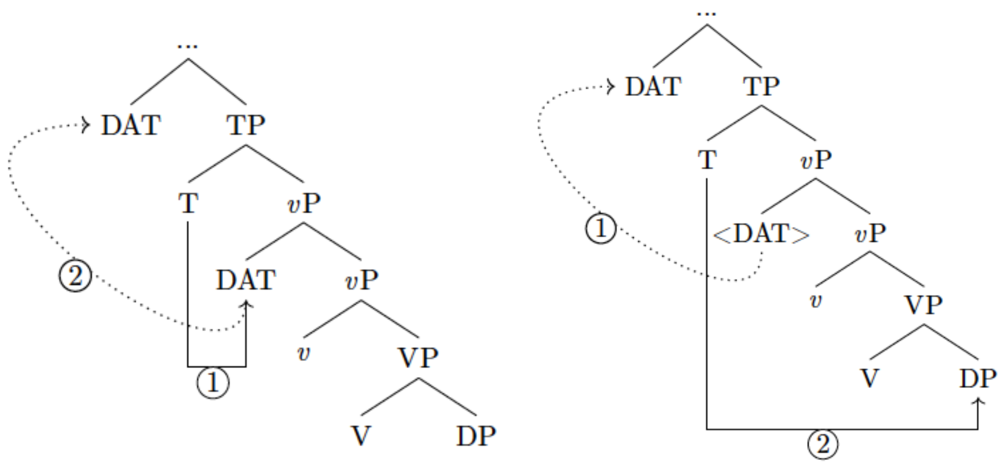
In addition to claiming difference in specification of the way the phi-probe initiates searches is the cause of intraspeaker variation of this kind, there is another claim that the timing of Move with respect to Agree can derive these different agreement patterns (Fernández-Serrano 2022; among others).

Take the following Spanish data showing a similar point of intraspeaker morphosyntactic variation involving alternation between default third singular agreement and specific agreement with a postverbal DP (Fernández-Serrano 2022).

(197) a. Nos encanta las películas de terror (Fernández-Serrano 2022:43)
us.DAT love.3sg the films of terror
b. Nos encantan las películas de terror
us.DAT love.3pl the films of terror
‘We love horror movies’

Fernández-Serrano (2022) presents a possible analysis in which the dative subject triggers default agreement, and the (197b) agreement pattern is a result of T^0 probing for phi-features after the dative subject has moved to a higher position, so it only finds the plural DP *películas de terror*. For the (197b) agreement, T^0 probes for phi-features before the dative subject moves above T^0, so it gets default agreement.

(198) Derivation of (197a) and (b), respectively (Fernández-Serrano 2022:85)



While an analysis such as this accounts for the data, it raises several problems. First of all, T^0 agreeing with a clitic pronoun is never possible in Spanish. Furthermore, the presence of a dative clitic (or any clitic) actually is not needed to get this agreement alternation. See the following.

(199) a. Llegaron las niñas (Francisco Ordóñez, p.c.)
arrived-3.pl the girls
b. Llegó las niñas
arrived-3.sg the girls
'The girls arrived'

The major issue with this analysis of these Spanish examples that attributes the agreement alternation to a variation in probe search order that finds two different goals for agreement, though, lies in the fact that the derivation of the optionality relies on the variation of ordering Move and Agree. There is sufficient evidence to say that movement must be fed by phi-agreement (Miyagawa 2010; Preminger 2014). Derivationally, agreement precedes movement. What would the motivation be to move before Agree takes place? The claim that agreement takes place before movement (A>M) has far more support than the claim that movement happens first (M>A), and there seems to be no independent motivation that the ordering of these operations can vary (*cf.* Georgi 2017).

Theoretically, the M>A order is not necessary or even advantageous at all. How can one conceive of movement taking place without any kind of agreement having happened before it? That is, movement of any given element must be the result of a probe agreeing with it and "attracting" it (Chomsky 1995). Without Agree, what is the underlying force that drives movement? Again, this is not clear. A separate operation that precedes both Move and Agree would have to stipulated.

Given the discussion above, the best way to account for this kind of intraspeaker syntactic variation is to consider it the output of a single set of grammatical operations, which is what I have done in this chapter. This has the advantage not only of positing less strain on the acquisition of this alternation by speakers who have it, but also of positing less strain on the syntactic theory of the grammar that is responsible for generating the alternation.

2.7 Summary

In this chapter I have presented a theory of Agree which allows for optional agreement in configurations in which a phi-deficient DP goal is located between a phi-probe and a second potential goal that is fully specified for all relevant phi-feature values: defective circumvention. This minimal technology handles a variety of complex agreement phenomena in a variety of languages, and does so as a single set of grammatical operations and computational principles in accordance with SMT.

The differences in which third person is represented across languages and the differences in what features are present on phi-probes lead to variation in agreement possibilities in these configurations crosslinguistically.

This model of Agree is highly local – phi-probes have no ability to search beyond the closest DP(s) that bear the relevant phi-features that the probe needs to find. This model of Agree is purely syntactic; no part of it takes place at LF or PF, nor does it predict that that should be possible at all. As I will demonstrate in the following chapters, this configuration of Agree is local in such a way that it forces a smuggling analysis of inversion and a greater articulation of thematic structure.

In the following chapters I analyze inversion constructions across a variety of languages, centering the discussion on English, and propose a theory of inversion which, when combined with the theory of Agree proposed in this chapter, can derive the facts about agreement English inversion constructions presented here. These accounts crucially involve A-movement of the fronted constituent from a position in which it is a viable goal for searches by T^0 and an intervener between T^0 and the thematic subject.

3 Locative Inversion

Given the analysis of Agree presented in Chapter 2, it is possible to discuss the syntax of specific inversion constructions, starting with locative inversion (henceforth LI), which shows the agreement alternation and restrictions shown in the previous chapter.

(200) a. Off the plank {walk/walks} several bad pirates
b. Off the plank {walks/*walk} you

These agreement realizations are only possible in the LI configuration.

(201) a. Several bad pirates {walk/*walks} off the plank
b. You {walk/*walks} off the plank

In this chapter, I develop a syntax of LI constructions in which the agreement realizations shown in (200) can be derived. As part of this analysis, I consider: (a) the thematic status of verbs that participate in LI, (b) the word-order status of elements in the clause, and (c) agreement variation with postverbal subjects in LI sentences. Points (a) and (b) have been inaccurately characterized in the extant literature on LI, and point (c) has not considered at all in the literature on the syntax of LI in English. As such, the analysis of LI presented here accounts for a wider array of facts than its predecessors, and also presents a generalized machinery that can account for a variety of inversion constructions, which I demonstrate in subsequent chapters of this thesis. I extend this analysis to account for so-called Heavy NP Shift LI constructions, which have distinct syntactic properties from typical LI.

In section 3.1 I deal with (a), showing that LI verbs can do not have to be unaccusative. In 3.2, I address point (b), showing that unergative and unaccusative verbal structures have the same basic properties in LI constructions, and from there I develop the analysis of LI as smuggling as a Voice phenomenon. The agreement facts (c) are brought up in 3.3, where I connect the analysis of LI back to the theory of Agree introduced in Chapter 2. In section 3.4, I discuss the differences between standard LI constructions and LI constructions which involve Heavy NP Shift. In section 3.5, I discuss quotative inversion and the transitivity constraint.

3.1 The unaccusativity question

A central question in the literature on the syntax of LI constructions is the thematic status of the verbs that participate in LI. The view in much of this literature is that LI is only possible with unaccusative verbs, and thus all verbs in LI constructions are unaccusative (Bresnan & Kanerva 1989; Bresnan 1994; Culicover & Levine 2001; Doggett 2004; Rizzi & Shlonksy 2006; Diercks 2017; a.o), though there are those who argue that the postverbal subject in LI constructions can be agentive as well (Coopmans 1989; Hoeckstra & Mulder 1990; Levin & Rappaport Hovav 1995). I defend the latter position.

The analysis of LI presented in this chapter is compatible with both unergative and agentive verbs, and does not require any unusual stipulations that the thematic status of a given verb may vary depending on the surface word order of the sentence in which it appears.

I first discuss the previous analyses of LI as a purely unaccusative construction, and address where these fall short (3.1.1). I then show that agentive verbs such as 'walk', 'run' and 'fly' are indeed agentive, even when they appear in LI constructions (3.1.2). These facts set up the need for an analysis that works both for agentive and unaccusative verbs, which I develop further on as smuggling (3.2).

3.1.1 Prior analyses

There are existing accounts of LI constructions which rely on the unavailability of the projection of an external argument by LI verbs (Culicover & Levine 2001; Doggett 2004; Mendikoetxea 2006; a.o). This is, of course, helpful in deriving the fact that the subject appears postverbally. In Culicover & Levine (2001) (henceforth C&L), every verb that appears in an LI construction must be unaccusative, though they distinguish true LI constructions from Heavy NP Shift, which they do analyze as allowing an external argument. Thus, C&L analyze the canonical LI sentences in (202) as unaccusative, while analyzing the Heavy NP Shift examples in (203) as unergative.

(202) a. Into the room walked John
b. Down from the wall leapt Gimli
c. In the room slept Robin

(203) a. Into the room walked the very famous and handsome actor from my favorite movie
b. Down from the wall leapt the very last dwarf in all of Moria
c. In the room slept the students in the class who had heard about the social psych experiment that we were about to perpetrate

Each set of examples has different structure, which C&L show with the placement of adverbs.

(204) Into the room walked (*carefully) John (carefully)

(205) Into the room walked (carefully) the very famous and handsome actor from my favorite movie (carefully)

There is more reason to believe that 'light' LI and Heavy NP Shift LI have a different syntax, which I leave until Section 3.4. These differences, however, are not sufficient to claim that the two constructions involve different verbs. For this more evidence is needed.

C&L give the following example, which they assign a judgment of ungrammatical, to show that agentive verbs cannot participate in light LI. This example is repeated throughout the literature (Rizzi & Shlonsky 2006; Diercks 2017; a.o) as ungrammatical.

(206) *In the room slept Robin (asterisk reflects judgments of C&L)

It is the intuition of myself and several other native English speakers who I consulted with that this sentence is not ungrammatical. Indeed, I find it no worse than any of the examples in (202). Additionally, many examples of this kind (without heavy NPs) are attested online.

(207) a. In the crib slept a tiny babe (Fanfiction)
b. In the manger slept a lovely babe (Catholic News Archive)
c. In a manger slept a stranger (Song)
d. There on a bed slept a boy (Fanfiction)
e. On the bed slept the boy (Fanfiction)
f. There on the bed slept the old man (Reddit)
g. Separate from them and also on the floor slept the appellant (Court Database)
h. On the floor slept a dozen children (Blog)
i. On the floor slept a bum (Harper's Magazine)
j. On my bed slept a rose (Fanfiction)

Furthermore, if it were the case that all LI verbs must be unaccusative, it is unclear to me why 'sleep' should be forbidden from ever being unaccusative, while verbs such as 'walk' are able to do so. The verb 'sleep' has no more of an agentive flavor to it than 'walk'; if anything, it has less of one (not that agentivity should necessarily have a one-to-one correspondence with unaccusativity). It strikes me as an ad hoc conjecture to claim that some unergative verbs may actually be unaccusative in the presence of a locative PP, but not others (Coopmans 1989: 741; Mendikoetxea 2006). This claim additionally gets us nowhere if the examples reported as ungrammatical are actually acceptable for many speakers and if there is a way to derive LI without making unnecessary stipulations on the thematic status of LI verbs. That is, there is no need to make the claim that LI verbs must be unaccusative since it is supported by neither the data nor the theory.

There are two necessary steps to take from this point: first, to demonstrate definitively that LI verbs do not have to be unaccusative in all cases; second, to set up a theoretical analysis to which an external argument postverbal subject poses no challenge. I take on the first step now.

3.1.2 Tests for agentivity

There are broadly two classes of intransitive verbs: unaccusatives and unergatives (Perlmutter 1978). Unaccusatives are those which do not project an external argument, and unergatives are those which only project an external argument (Burzio 1986; Hale & Keyser 1993). Following a set of theoretical assumptions on uniformity within argument structure that gets us to a descriptive generalization such as UTAH (Baker 1988; 1997), unaccusatives only project a THEME argument, and unergatives only project an AGENT argument.

The first series of tests are rather straightforward: to show that verbs which may participate in LI constructions are not underlyingly unaccusative. It is known that there are various syntactic and semantic differences in unergative and unaccusative verbs (Dowty 1979; Baker 1997; Yang 1997, a.o). I outline some of them here, showing that manner-of-motion verbs with locative PPs may indeed be unergative.

For example, it is known that unaccusative verbs do not license pseudopassives, while unergative verbs do (Perlmutter & Postal 1984).

(208) *Pseudopassive test*: unaccusative verbs of motion do not license pseudopassives
a. The room was {walked/ran/swam/flown/etc.} into (by Chris)
b. *The station was arrived at (by the bus)

This test shows that manner-of-motion verbs such as 'run', 'swim', 'walk', and 'fly' are indeed unergative. Thus, the claim that these verbs are underlyingly unaccusative no longer holds. A test such as this one, however, does not provide any evidence against the claim that these verbs may selectively become unaccusative when they appear in LI constructions (Coopmans 1989; C&L 2001; Mendikoetxea 2006; Diercks 2017:fn. 9). It is therefore necessary to show these tests of agentivity with inverted sentences as well.

The pseudopassive test does not apply to LI sentences because of independent constraints on inversion constructions that have nothing to do with the thematic status of their arguments. That is, pseudopassivization and LI are separare derivations that are not compatible with one another. Because of this, I introduce new tests here that apply to LI.

Unaccusatives may be modified by resultative phrases, while unergatives may not be (Levin & Rappaport Hovav 1995).

(209) a. The river froze solid
b. The vase broke into pieces
c. Her body burned to a crisp

(210) a. *Gabriella walked tired (*int*: Gabriella walked herself tired)
b. *Rebecca ran high (*int*: Drew ran to the point of getting runner's high)
c. *The old man shouted hoarse (*int*: The man shouted himself hoarse)

In the same way, unaccusative verbs in LI sentences can be modified by secondary resultative predicates.

(211) a. Out in the cold there froze (solid) an entire bottle of vodka (?solid)
b. Deep in the foundation there cracked (wide open) a fissure (??wide open)
c. In the fire burned (to a crisp) all of our belongings (??to a crisp)
d. On the sidewalk broke (into pieces) my mother's favorite vase (???into pieces)

Unergative verbs still cannot be modified by resultatives in LI sentences[21].

(212) a. *Through the woods walked tired Grandma
b. *Across the field ran sweaty the marching band
c. *Out of the fire shouted hoarse a voice
d. *In the cold, murky water swam sick the young boy

If all LI verbs were unaccusative, then resultatives should be predicted to be acceptable with far

[21] There is a version of these sentences which is grammatical, in which the adjectival predicates are understood as depictive, not resultative. Crucially, these are ungrammatical here as *resultatives*.

more verbs than they are.

Furthermore, passivization is possible with LI sentences.

(213) a. Up into the sky was thrown a ball
b. Into the mailbox was hastily shoved a letter
c. Deep in the dungeons were revealed many dark secrets
d. On the table was set out all the dinner plates and napkins

Given that it is impossible to passivize unaccusatives in English (Perlmutter 1978), and that passives involve the projection of an AGENT argument in Spec,vP[22] (Collins 2024), it cannot be the case that only unaccusative verbs are licit in LI derivations. See Section 4.5 for a detailed analysis of the structure of passivized LI sentences.

In this section I have presented various diagnostics which clearly demonstrate that LI constructions do not need to involve an unaccusative verb. I have shown in this section that any analysis of inversion (specifically LI) that bans a given thematic structure is not adequate at explaining the data. As such, an account of LI must be developed in which multiple verb types and their arguments can participate in LI.

In the following section I present a smuggling analysis of LI in which the locative is smuggled over the thematic subject – AGENT or THEME – inside of a larger constituent which includes the verb as well. This derives the subject-final word order that is hallmark of LI constructions, and also explains numerous other facts which I present in the following section.

3.2 Smuggling

Smuggling is an operation by which syntactic elements are moved within larger constituents in which they are embedded, rather than directly being the target of a movement operation (Belletti & Collins 2021; Collins 2005; 2024; Roberts 2010; 2019; 2024; Gotah 2024; Shlonsky 2024; Stegovec 2024; Storment 2025; Sulemana 2024; Thoms 2024; a.o). This happens so that an element can be targeted for some syntactic operation that it would not be a viable target for in its base position without violating minimality.

The definition of smuggling given in the introductory chapter is repeated here.

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[22] I only show short passives here (that is, without a by-phrase), but it is possible to have passive LI sentences that also have the agent appear overtly.

ii. Into the mailbox was hastily shoved a letter by our angry mailman.

There are several pieces of strong evidence supporting the claim that passive sentences always involve the projection of a null AGENT argument. Indeed, there are too many to list here. For justifications of this point I refer the reader to Collins (2024), as well as the discussion from the introduction of this dissertation.

(214) *Smuggling* (Collins 2005:97)

Suppose a constituent YP contains XP. Furthermore, suppose that XP is inaccessible to Z because of the presence of W (a barrier, phase boundary, or an intervener for the Minimal Link Condition and/or Relativized Minimality), which blocks a syntactic relation between Z and XP (e.g., movement, Case checking, agreement, binding). If YP moves to a position c-commanding W, we say that YP smuggles XP past W.

I represent a basic smuggling derivation below in (215), with Z representing a probe searching for an element bearing the feature [F] with two potential goals: W and XP. With W being closer to the probe than XP, XP cannot be a viable target for X without violating minimality unless it is moved to a position above W. Since there is no operation that can target XP to the exclusion of W while their features are indistinguishable to a probe, XP must be smuggled to a position above W within a larger constituent. I detail this movement operation in this section, and how it successfully derives the data from LI constructions.

(215) a. $[\mathrm{Z}_{[u\mathrm{F}]} \dots [\ \mathrm{W}_{[\mathrm{F}]} \dots [_{\mathrm{YP}}\ \mathrm{XP}_{[\mathrm{F}]}]]]$

b. $[\mathrm{Z}_{[u\mathrm{F}]} \dots [[_{\mathrm{YP}}\ \mathrm{XP}_{[\mathrm{F}]}]_i \dots [\ \mathrm{W}_{[\mathrm{F}]} \dots [t_i]]]]$

Smuggling is – among other things – largely responsible for deriving surface word order in inversion constructions: that is, constructions where two elements which canonically appear in the order (A B) surface with the order (B A). Thus, this definition of inversion includes passives (Collins 2005; 2024; Sulemana 2024), dative shift (Collins 2024, Newman 2024; Thoms 2024), predicate inversion (Shlonsky 2024; Storment 2025), subject-object inversion (Shlonsky 2024), and quotative inversion (Collins 2003), to name some constructions. As I have demonstrated throughout this thesis (with specific examples in the introduction), the notion of all of these alternations in the realization of arguments being derived by smuggling of the VP is empirically supported by the data concerning the distribution of particles – which normally alternate in their ability to appear immediately before or after a postverbal thematic argument – in these constructions. See the following examples from the introduction on passives, middles, dative shift, antipassives, and inversion, repeated here.

(216) a. The argument was summed (up) by the coach (*up) (Collins 2005:88)

b. The paper was written (up) by John (*up)

(217) a. This paper cuts (up) easily for me (*up)

b. These lab reports write (up) super quicky for her (*up)

(218) a. We sent the boys up some pizza (Collins 2021:103)

b. We sent some pizza up to the boys

c. We sent up some pizza to the boys

d. *We sent some pizza to the boys up

(219) a. Whosoever drinks (up) my blood (up) shall live forever

b. Whosoever drinks (up) of my blood (*up) shall live forever

(220) a. In the field toiled (away) the laborers (*away)
b. "Hey!" chimed (in) Jason (*in)

These facts directly follow from the VP-movement analysis of voice alternations and inversion via smuggling that I present here in this section. The view of inversion via smuggling that I present here is motivated by the ordering of arguments and their mappings onto A positions from the positions in which they are externally merged. All the aforementioned constructions, then, involve smuggling.

Under the definition of smuggling presented in (19), smuggling involves two applications of internal merge, the second application of which is facilitated by the first application. As such, smuggling is *not* an additional technological stipulation on the grammar in the same way that allowing for equidistance or multiple specifiers would be. It is simply internal merge followed by internal merge.

Why such a device would be available specifically to derive these voice alternations (although there are other applications of smuggling (Belletti & Collins 2021)) follows from constraints on argument structure, which is the positions into which thematic arguments are externally merged. Given the maximally local and minimal formulation of Agree presented in Chapter 2, some additional operation is needed to move and manipulate from their base positions in order to feed new possibilities for A-relations without violating minimality and locality conditions. It is smuggling via VoiceP, then, that is responsible for changing the potential A-dependencies of arguments from the positions into which they are externally merged. These changes allow for different mappings of arguments to LF, which can be seen in the way voice and inversion constructions behave with respect to phenomena such as prominence and topicality, or scope and binding relations. In other words, voice alternations derived via inversion as smuggling are there to maximize possibilities for mapping argument structure to LF. Voice is a way to mediate between argument structure and LF, and this is formally implemented via smuggling. I now discuss the featural motivations and formal implementation of voice as inversion as smuggling.

Inversion via smuggling is triggered by the presence of a functional projection (called InvP in Collins 2003; Storment 2025). As this functional projection mediates the ordering of arguments as they are mapped onto A-positions, it is a type of Voice phrase (Chomsky 1995; Collins & Thráinsson 1996; Collins 2005; 2024; Pietraszko 2024); as such, I call this functional projection VoiceP throughout. The definition of VoiceP from the introductory chapter is also repeated here.

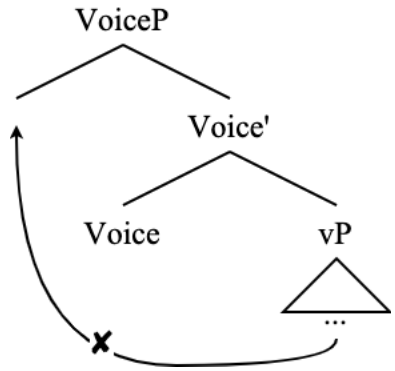
(221) *VoiceP*
A functional projection that targets the closest eligible verbal maximal projection for movement to its specifier, thereby causing the thematic argument contained within that maximal projection to become an accessible goal for A-movement.

Movement to VoiceP is a maximally local operation[23] that targets the closest eligible verbal projection to its Spec. If VoiceP is merged directly above vP, and vP is its complement, it does not attract vP to its Spec as that would violate anti-locality; rather, it would attract the verbal

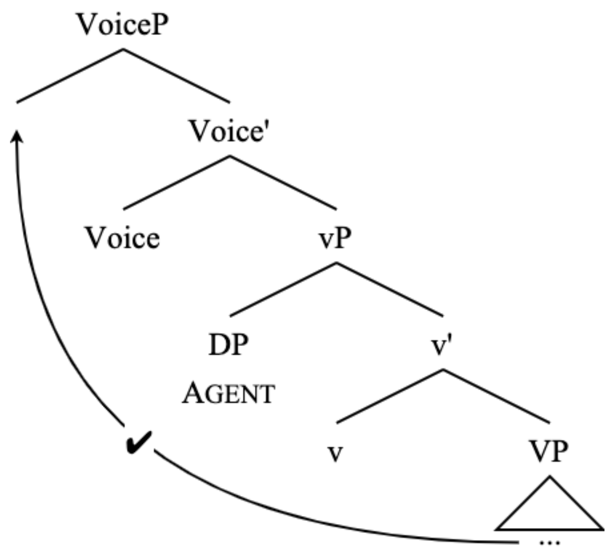
[23] The locality of this smuggling operation suggests that this VP smuggling is not an A-bar operation, as A-bar movement is able to facilitate long-distance movement, which is notably absent here.

projection that vP takes as its complement.

(222) Movement to VoiceP: anti-locality



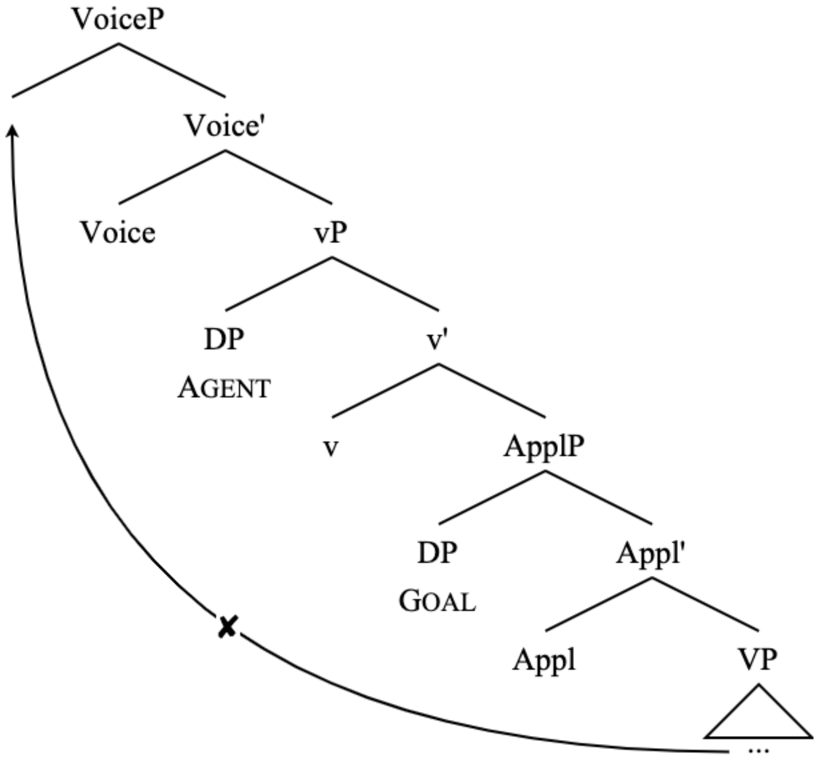
(223) Movement to VoiceP: no anti-locality



The goal for movement to Spec,VoiceP must also not violate minimality: a verbal projection cannot move to Spec,VoiceP if there is another eligible target preceding it.

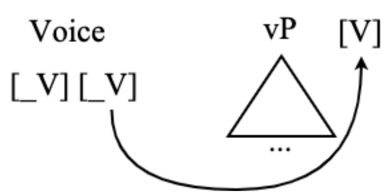
The featural motivations for smuggling have not been systematically explained before, so I do so in this section.

(224) Movement to VoiceP: anti-minimality



I explain this fact by positing two unvalued features on Voice^0: one that selects its complement, and another that selects a target for movement to its specifier. These are both [_V] features. One of these [_V] features selects the complement of VoiceP. I assume that anything bearing a relevant [V] feature is a maximal projection in the verbal clausal domain (e.g. vP, VP, ApplP)[24]. First, one of these [_V] features selects the complement of Voice^0 [25].

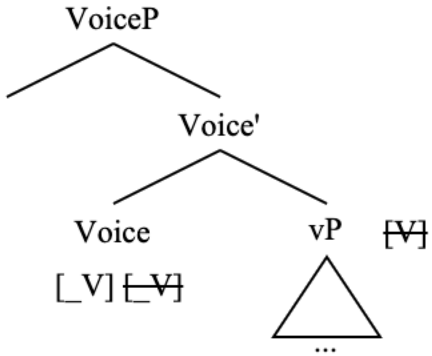
(225) [_V] selects complement of VoiceP



[24] I further assume that Voice^0 bears its own [V] feature, since it is part of the hierarchy of verbal projections. I do not show this here as it is not necessarily relevant for this question of selection and probing.

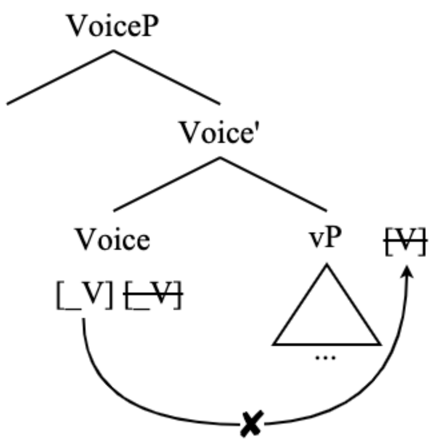
[25] This feature does not have to be the category feature [V] *per se*; rather, it must be some feature that is common to all verbal projections in the clausal spine. We know that there is such a feature common to all these projections as they can all be selected by T^0, for example. T^0 can take as its complement VoiceP, vP, VP, ModP, AuxP, etc. Unless we are in the business of positing multiple different T^0 heads for all these different derivations, it must be the case that some feature is in common to all these different projections that can be selected by T^0. This featural uniformity is consistent with the idea that all these different verbal functional projections are ultimately "shells" of the same VP, and thus share a kind of verbal identity (Larson 2014).

(226) Voice^0 and vP project



At this point, the remaining [_V] feature is still seeking valuation. For this, Voice^0 must probe to find an element with a [V] feature to move to its specifier position[26]. The complement of VoiceP is not a viable target for this search as it is too local, and would trigger anti-locality (Grohmann 2003). The reason it is too local in this framework is because the complement of VoiceP has been rendered inactive to further searches by Voice^0 as a result of already having been selected by it as its complement. The probe cannot see it.

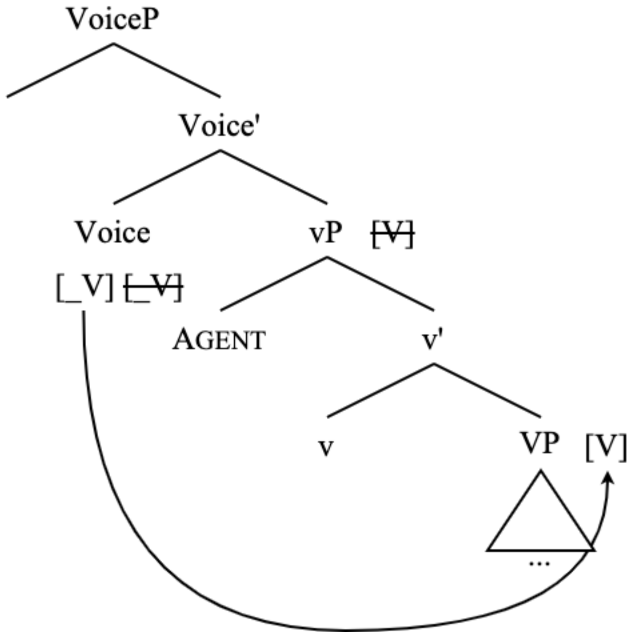
(227) vP is not visible to Voice^0 for this operation



As such, the probe must search within the c-command domain of its complement for a projection containing a [V] feature which is not inactive to probing operations by Voice^0.

[26] There is, of course, the possibility that a new verbal projection could be externally merged in Spec,VoiceP. This must be ruled out. There seems to exist a limit of just one verb (and its associated functional projections) per clause outside of coordinate structures (e.g., 'John ate dinner and drank wine') and serial verb constructions (which English notably lacks), so that is one way to rule out external merge into Spec,VoiceP: it would create a sentence with too many verbs, the upward limit of which can be computed from general parameters on the basic architecture of a clause. Furthermore, external merge into Spec,VoiceP would surely introduce a verbal projection that introduces an argument. This would introduce an issue of having (a) too many arguments that are related in some way to the event of the verb, and (b) too many arguments to be licensed. For example, an argument of a VP that has been externally merged in Spec,VoiceP would be the first target for A-movement to Spec,TP and would thus introduce a complication for the licensing of the arguments contained within the complement of VoiceP. This would also cause issue for licensing of the *verb* of the clausal "snippet" stuck in the Comp,VoiceP position. In short, the impossibility of externally merging a verbal projection in Spec,VoiceP is ruled out independently of the properties of VoiceP.

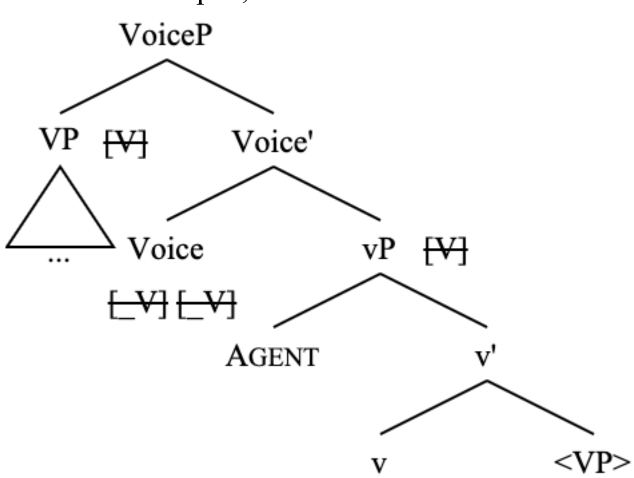
(228) VoiceP probes within the c-command domain of its complement



Here it finds a viable goal for movement to Spec,VoiceP and the checking of its [_V] feature, which is the complement of its complement (which was selected by the complement of VoiceP at some earlier point in the derivation, but crucially has not already been selected by VoiceP).

As a result of this, the complement of the complement of VoiceP moves to Spec,VoiceP.

(229) VP moves to Spec,VoiceP



This movement has checked all of VoiceP's unvalued features. This derives exactly what VoiceP does in terms of selection and triggering VP phrasal movement. Any projection further away than the complement of VoiceP's complement would be ruled out by Minimal Search: the complement of the complement of VoiceP (VP, in the example) is the closest viable goal for this operation.

There is, however, a potential issue that comes up with the fact that this projection VoiceP contains only unvalued [_V] features (and its own category feature). Such projections are typically conceived of as unnecessary given that they have no role in case licensing, information structure, or introducing arguments (Chomsky 1995) (though others simply conceive of these heads as "functional heads" (den Dikken 2006:22)). This issue has been commented on by Collins (2005), who says the following.

(230) "An immediate objection to this analysis is that it postulates a functional head consisting entirely of uninterpretable features (see Chomsky 1995:349 for arguments against Agr). However, Collins 2003a, 2005b, and Baker and Collins 2003 discuss the syntax of linkers in great deal, showing that they are precisely functional heads composted purely of uninterpretable features. Similarly, even in the standard theory of the passive, it must be admitted that the preposition *by* in the structure [$_{PP}$ *by* DP] does not have interpretable features" (Collins 2005:95).

This issue is not obviated by theories of voice alternations which do not utilize this conception of VoiceP. There is a similar issue with the PassP of Bruening (2013) and indeed any PassP-like projection whose sole purpose is to derive movement of an internal argument in passive configurations (Wood & Tyler 2023).

There are similarly other functional projections with non-zero morphological exponence whose purpose relates to the surface order realization of arguments. Such projections also bear only unvalued features responsible for deriving movement (as well as their own category features). An example of this is the projection associated with verbal linkers in Khoisan and Bantu (Baker & Collins (B&C) 2006; Collins 2019).

(231) a. Mo- n- a-hir- ire okugulu k'- omo- kihuna (Kinande, B&C 2006:311)
AFF-1sS-T-put-EXT leg.15 LK.15-LOC.18-hole.7
b. Mo- n- a-hir- ire omo- kihuna m'- okugulu
AFF-1sS-T-put-EXT LOC.18-hole.7 LK.18-leg.15
'I put the leg in the hole'

(232) a. Koloi g‖on-a ǂ'amkoe ki gyeo na (ǂHoan, B&C 2006:331)
car hit- PERF person LK road in
b. Koloi g‖on-a gyeo na ki ǂ'amkoe
car hit- PERF road in LK person
'A car hit a person in the road'

These linkers have no effect on the semantics and only relate to the ordering of arguments. Evidence that they contain unvalued features can be seen by the fact that the linker in Kinande agrees with the argument that moves to its specifier. B&C (2006) and Collins (2019) present a cohesive analysis of the Bantu and Khoisan linkers which convincingly empirically demonstrates that these linkers are the realization of functional heads composed only of unvalued features whose role relates purely to the movement of arguments.

Another example of a kind of head of a functional projection that has no features other than its category feature and unvalued feature(s) comes from linker-adjacent elements in English, such as the 'of' in the following example.

(233) a. A physics teacher
b. A teacher of physics

This 'of' in the (b) example has no clear semantic contribution, nor is it involved in the licensing of either of the two nominals. If it were, then we should expect an 'of' in the (a) example as well. In fact, this alternation in (233) looks like a kind of nominal inversion. That is, the 'of' and its associated projection only plays a role in the surface realization of these nominals with respect to one another. This is another case of a functional preposition which do not have a semantic contribution and lack any "interpretable" features. See also certain formulations of this 'of' as a kind of nominal copula, which, indeed, raises an issue that all copulae may be fall under this definition of a head which does not assign a thematic role and is defined only by its "uninterpretable" features (den Dikken 2006).

Furthermore, there are numerous examples in the literature of functional projections composed only of unvalued features whose only function is to create Agree dependencies (and subsequently potentially induce movement). Various examples of these agreement heads (and their associated projections) include π-AGR and #-AGR (Béjar 2003), Pn and Nr (Sigurðsson & Holmberg 2008), and π^0 and $\#^0$ (Preminger 2014), all of which essentially serve the same purpose. These are examples of heads composed only of unvalued features – [_π] and [_#], respectively – that are clearly separate from heads which bear features such as [Tns] or [C] (Sigurðsson & Holmberg 2008:258). The motivation for these heads is to house probes; i.e., unvalued features (Preminger 2014:34), and they serve no other purpose in the clausal domain. There are still more cartographic approaches which expand significantly upon this idea, such as Shlonsky (2023) who introduces a distinct head for every distinct privative feature that is relevant for verbal agreement. As such, there is an undeniable precedent for this variety of functional head.

It cannot be the case, then, that there is no such thing as a head of a projection that has only unvalued features. There exist a class of these projections which seem to all play a role in the agreement and movement of arguments. These include VoiceP, the linker phrase, and certain uses of functional prepositions such as 'of' in in (233)[27].

For LI, smuggling involves the movement of a constituent – the VP (3.2.1) – over the thematic

[27] If one still does not want to concede for theoretical purposes that there should exist a class of heads which have only unvalued features, it is indeed possible that VoiceP could be endowed with some kind of information-structural feature, as voice alternations seem to be highly associated with alternations in information structure such as topicality (Gildea 1994; Givón 1994; Oxford 2023), and there is precedent for associating VP movement with topicality (Alshamari & Jarrah 2016). As such, it could easily be the case that VoiceP is endowed with some kind of pragmatic information-structural feature; however, this feature would have to be very pragmatically "weak" and not directly associated with any given information-structural property such as topic or focus given that, across all voice alternations and inversion constructions, it is not the case that every single one of these alternations is "unambiguously associated with a particular pragmatic configuration" (Quesada & Skopeteas 2010:2594). Alternatively, there could be multiple different VoiceP each with its own distinct pragmatic feature, but this loses out on the explanatory power of having a single projection responsible for deriving all these different alternations.

subject (AGENT for unergatives and THEME for unaccusatives) that contains a new target for phi-agreement and movement to Spec,TP: the locative (3.2.2). This is the exact same role that VoiceP has in a smuggling analysis of the passive (Collins 2005; 2024). Let us see some empirical motivation for this claim now.

3.2.1 Movement of the VP

I now identify the constituent that is targeted for smuggling movement in LI constructions as the VP. I show this for several tests with both unergatives (3.2.1.1) and unaccusatives (3.2.1.2), ultimately demonstrating that the same mechanism derives LI with both verb types.

These tests involve showing that VP-internal elements raise to a position above the thematic subject, while elements that are higher than VP, such as vP-level adjuncts, do not raise to a position above the thematic subject.

3.2.1.1 Unergatives

First, I show the tests with unergative verbs. Then, once the pattern is established, I move on to unaccusatives, showing that they work much the same. These tests define the smuggled structure for unergative verbs with LI as follows:

(234) $[_{\text{VoiceP}}\ [_{\text{VP}i}$ V $[\text{PP}_{\text{LOC}}]]\ [_{\text{Voice'}}$ Voice $[_{\text{vP}}\ \text{DP}_{\text{AGENT}}\ [_{\text{v'}}$ v $[_{\text{VP}}\ t_i]]]]]]$

The first test is particle verb constructions. Particles such as 'up', 'in', 'out', and 'away' in (235) originate as internal arguments of the VP (Collins 2003; 2005; Collins & Thráinsson 1996), and may appear directly following or preceding a THEME argument, if there is one.

(235) a. I wrote (up) a paper (up) during my short time in Botswana
b. The hospice nurse wheeled (in) her patient (in) for the visit
c. She works (out) her legs (out) every day in the gym
d. The aristocratic parents sent (away) their troublesome son (away) to boarding school

Collins & Thráinsson (1996) give the following underlying structure in (236) for particle verb constructions with a THEME argument, which I amend further on, though the observation that Prt is generated as a VP-internal argument remains the same.

(236) $[_{\text{vP}}$ Subj $[_{\text{v'}}$ v $[_{\text{VP}}$ Obj $[_{\text{V'}}$ V Prt$]]]]$

As such, when the VP is smuggled to Spec,VoiceP, the particle moves with it.

(237) a. To the top of the mountain climbed (up) Stephen (*up)
b. Through the doors walked (in) the president (*in)
c. From the side of the cliff face jutted (out) a long branch (*out)
d. In the field toiled (away) the laborers (*away)

If only the verb – and not the entire VP – were moving above the thematic subject, this would be quite unexpected, as verb movement is well-known to strand particles (Koopman 1984; 1991; 1995). The exact same fact holds of the passive, as well as the other voice alternations mentioned in the introduction of this thesis, which is evidence of the same kind of VP-smuggling derivation deriving the voice alternation in all these constructions.

(238) a. The argument was summed (up) by the coach (*up) (Collins 2005:88)
b. The paper was written (up) by John (*up)

These data on particle verbs clearly show that a constituent larger than just the verb raises above the thematic subject: the VP. Now, the inability of vP-level adjuncts to raise above the thematic subject in unergative LI constructions shows that nothing larger than the VP is moving to Spec,VoiceP. Subject-oriented secondary depictive predicates (239) and adjunct purpose clauses controlled by the external argument of the matrix clause (240) are vP-level adjuncts (Roberts 1988; Willams 1980; Mallén 1991; Pylkkänen 2008; Collins 2024) and must appear below the thematic subject[28].

(239) a. Into the room stumbled (*drunk off his ass) the man (drunk off his ass)
b. Onto the field ran (*naked) the streaker (naked)
c. Across the lawn marched (*barefoot) the band (barefoot)

(240) a. Into the operating room sprinted (*to get the doctor) the nurse (to get the doctor)
b. Through the crowd pushed (*to get a better view) the spectator (to get a better view)

Another piece of evidence that it is the VP that raises to Spec,VoiceP is the distribution of control clauses, which are complements of VP (Williams 1980; Chomsky 1981).

(241) a. ?Right into the palace attempted to walk the commoners before they were stopped by the guards
b. *Right into the palace attempted the commoners to walk before they were stopped by the guards

The sentence in (241a) is perhaps not perfect, though (241b) is significantly worse (similar judgments are presented for quotative inversion in Collins (2003)). These data on particles, secondary predicates, adjunct clauses, and control clauses show that the entire VP moves above the thematic subject in LI constructions (at which point part of the locative PP is extracted for movement to Spec,TP; more on that in subsection 3.2.2). This movement is facilitated by the functional projection VoiceP. In this section I have given empirical evidence to support the claim that the VP is what moves to Spec,VoiceP in unergative LI derivations.

There are clear A-movement properties of LI, such as the obviation of weak crossover (242) (C&L 2001) and the failure to license parasitic gaps (243), which is only a property of A-bar movement (Murphy 2022).

(242) a. Right into Natalie's$_i$ room walked her$_i$ long-lost twin brother
b. Into every dog's$_i$ cage peered its$_i$ owner (C&L 2001:289)

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28 These elements can appear after the thematic subject when that subject undergoes Heavy NP Shift in LI. I cover this in section 3.4.

(243) a. *Ali peeked into that very room [without actually entering __]
b. It was into that very room that Ali peeked [without actually entering __]
c. Into which room did Ali peek [without actually entering __]?
d. Into that very room Ali peeked [without actually entering __]
e. *Into that very room peeked Ali without actually entering __]

These data in (242) and (243) do not directly show that the smuggling itself is A-movement. These more clearly diagnose that the movement of the locative PP to the subject position is A-movement. It is sufficient to say that the smuggling operation facilitates A-movement of a constituent inside of the moved VP (Belletti & Collins 2021). I return to these questions of the status and movement of the locative PP after showing the same movement tests with unaccusative verbs in LI sentences.

3.2.1.2 Unaccusatives

In this section, I show the same tests to diagnose VP movement above the thematic subject in LI in the case of unaccusative verbs. First, the same generalizations can be made about particles with unaccusatives as they can with unergatives.

(244) a. On my screen popped (up) an annoying ad (*up)
b. In the fire burned (up) all the evidence (*up)
c. Inside Mount Doom melted (away) the One Ring (*away)

Adjunct purpose clauses are not grammatical with unaccusatives (which further cements their status as vP adjuncts), even outside of inversion. The following famous sentence shows this.

(245) *The ship sank to collect the insurance (Mauner & Tanenhaus & Carlson 1995)

As such, this test is not applicable to unaccusatives as it was to unergatives.

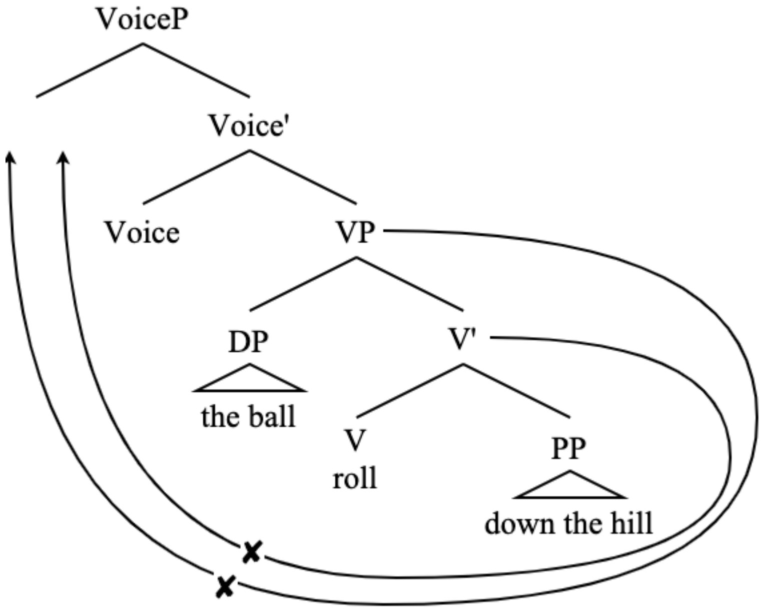
The data on control clauses with unaccusatives are also identical to those with unergatives, although the reading of control clauses and unaccusatives is admittedly marked.

(246) a. ?Down the hill managed to roll the ball, despite the low incline
b. **Down the hill managed the ball to roll, despite the low incline

These data clearly show that, just like with unergatives, unaccusative VPs are moved over the thematic THEME subject in LI as well. This creates some problems, though. If the THEME argument is base-generated inside VP (as it is quite commonly assumed to do (Baker 1988; 1977; Larson 1988; 1989; Chomsky 1995; 2000; 2001; a.o)), and if vP is not projected in unaccusative structures (Hale & Keyser 1993 *et seq.*), then there is no way to any thematic verbal projection to Spec,VoiceP without either violating anti-locality or moving something that is not a maximal projection or a head (i.e. a bar-level). See the trees for the following unaccusative example.

(247) Down the hill rolled the ball

(248) VP-internal theme: anti-locality

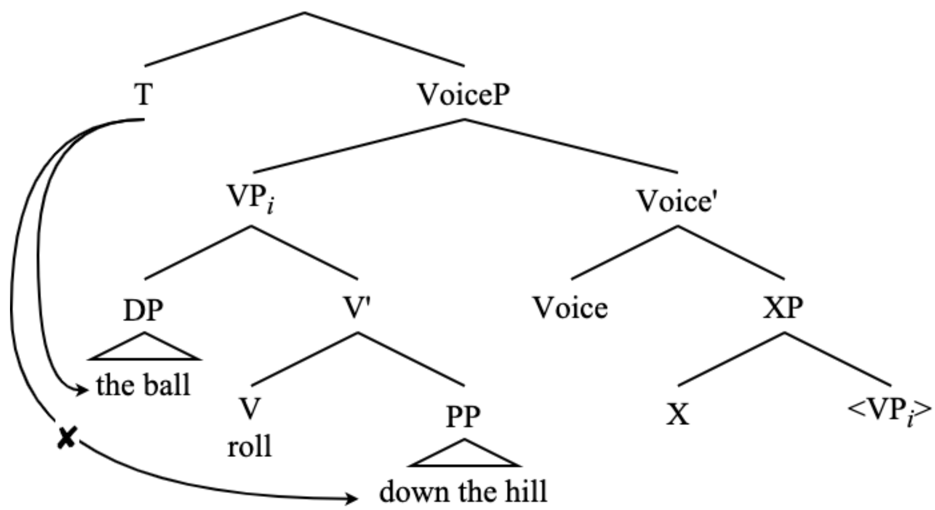


Additionally, if it was possible for there to be some other projection (XP) that could appear between VP and VoiceP (though there is sufficient evidence that this is not the case, such as the ungrammaticality of (245)[29]), then raising VP to Spec,VoiceP does nothing to smuggle the locative PP over the THEME, assuming that the THEME and the locative PP have the same features relevant for the phi-probe on T (which I develop in the following subsection).

[29] There is work suggesting that vP is projected in all derivations, regardless of whether or not there is an AGENT argument present in the derivation (Harley 1995; Kratzer 1996, Marantz 1997; 2001, Arad 2003, a.o). While data like (245) suggest that there is really no implicit AGENT anywhere in the unaccusative structure (and by extension, no space for one), they do not necessarily rule out the possibility that an argumentless vP could be projected by unaccusative verbs. I reject this notion on the theoretical grounds that our representations of these structures should, ultimately, be as radically simple as possible (see SMT) and – most importantly – be driven by the data. That is, if the data does not mandate the stipulation of a functional projection primarily responsible for introducing an external argument, then it is not advantageous to claim that it is there, especially if there is another way to derive the data. There is also an incongruity with the work I cite above regarding the purpose of the head of vP.

This line of argumentation is informed by the notions of Poverty of the Stimulus and Plato's Problem (Chomsky 1986), as well as Occam's razor. These guiding principles are crucial to the Strong Minimalist Thesis and the notion of Economy in grammar (Chomsky 1995; Collins 1997), which are fundamental to answering the question of why human language exists in the form that it does and how children acquire it so easily (Berwick & Chomsky 2016). I believe that these questions should be the driving force behind every line of syntactic inquiry. That is, we have no need to posit vP in this structure, so there is no vP in the structure.

(249) VP-internal theme: anti-minimality

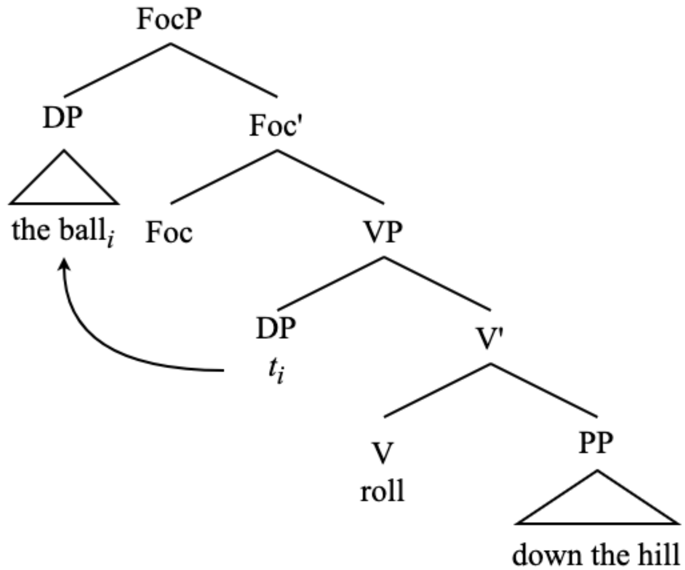


Given these issues of movement and selection of a VP-internal THEME argument, and given the abundant evidence that there is movement of the VP in LI constructions, there are two options forward. First, we say that the THEME is not externally merged in Spec,VP, and is instead merged in a higher functional projection. Second, we say that the THEME evacuates the VP before there is smuggling of VP to Spec,VoiceP. Keeping with tradition, I assume that the THEME is indeed an argument of VP and that it must evacuate the VP before smuggling[30]. This movement handles the minimality and locality problems introduced above.

Given the association between LI and a pragmatic configuration in which the postverbal thematic subject receives a focus interpretation, the THEME must first move to the specifier of a FocP that takes VP as its complement before smuggling takes place.

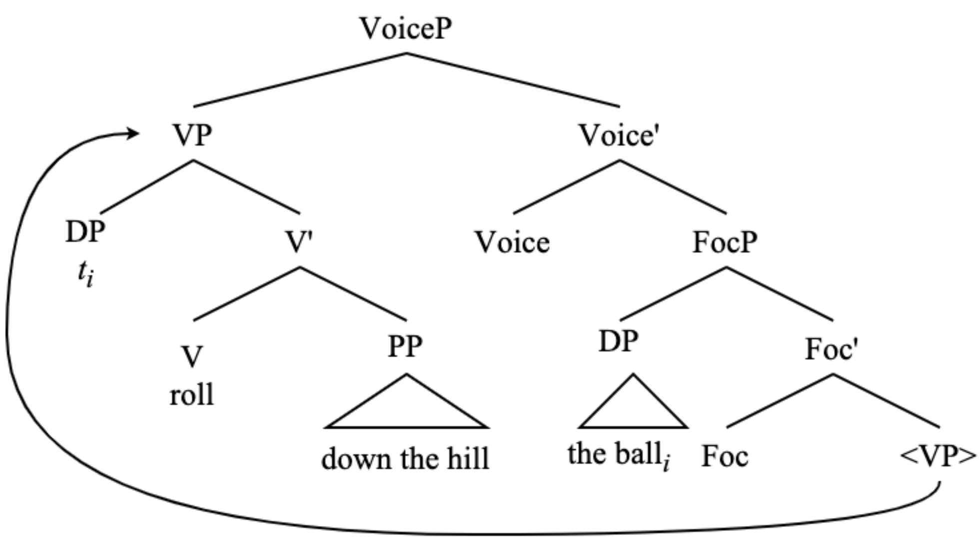
[30] Claiming that the VP introduces both the theme and a locative argument creates a complication for the Argument Criterion (Collins 2024:5), which states that each argument-introducing head is only responsible for introducing one argument. Furthermore, given the structure of particle-verb configurations assumed in (236), there is a further complication of how VP should have room for a particle, a locative argument, and a THEME in examples such as the ones in (244). Therefore, I leave open the possibility (indeed, the likelihood) that what I refer to throughout this work as VP actually contains more structure (i.e., more functional heads) than what I show here.

(250) Movement of the THEME to Spec,FocP



From here, a VoiceP which takes this FocP as its complement is able to select the VP which does not contain the THEME for movement to Spec,VoiceP. In this analysis, the FocP also carries a [V] feature as it is part of the verbal "spine" and is able to be taken as complement of VoiceP.

(251) Movement of unaccusative VP to Spec,VoiceP



The presence of this FocP also explains an interesting contrast regarding the distribution of secondary depictive predicates. Unaccusative verbs are more permissive with where they allow depictives to appear (contrast the following with (239)).

(252) a. At the station appeared (?drunk) the suspect (drunk)
b. Deep in the cave sat (?almost completely naked) a creature (almost completely naked)
c. Down the hill rolled (?barefoot) the children (barefoot)

Certainly, there is a contrast with the acceptability of secondary depictive predicates that precede the thematic subject in LI when comparing unaccusatives and unergatives.

(253) a. ?/^{OK}At the station arrived drunk a suspect
b. */??Across the field ran naked the streaker

This follows from the structural derivation of each variant of LI. With unaccusatives, the depictive adjunct has two possible attachment sites: the FocP, or somewhere within what we are calling the VP. When it adjoins to FocP, the depictive appears following the thematic subject, but when it adjoins to the VP, it appears preceding the thematic subject in LI (because the VP moves above the THEME).

With unergatives, however, there is only one attachment site (given that there is no Heavy-NP Shift (see section 3.4)), which is the vP. Since a vP-level adjunct does not undergo smuggling as it is not part of the VP, this always results in the depictive following the thematic subject of unergatives.

Note that, in all three of these licit derivations that I have just described (depictive>THEME, THEME>depictive, and AGENT>depictive), the secondary depictive predicate adjoins to a maximal projection whose specifier the depictive modifies.

As further evidence for the presence of this FocP with unaccusatives in LI is the fact that unaccusative LI can be used as the answer to a wh-question (Dik 1997; van der Wal 2016).

(254) Q: What arrived last night?
A: Last night arrived several presents from Santa

It is my intuition that this structure is much more readily available as an answer to a wh-question than unergative LI sentences (outside of a specific narrative register), which supports the view that unaccusative LI obligatorily contains this FocP, while unergative LI does not.

(255) Q: Who walked into the room?
A: #Into the room walked John

Given these structures, it is now possible to present a derivation of LI for both unergatives and unaccusatives, which both rely on VP smuggling via VoiceP.

3.2.2 The status of the locative argument

I have thus far presented a conception of argument structure in whic the locative PP – which becomes the structural subject of LI constructions – is a VP-internal argument with the thematic status LOC. This theta-role is distinguishable from GOAL/RECIPIENT (introduced by ApplP) and from SOURCE (whose argument structure I do not discuss here) (Bresnan 1994; Tortora 2014a). The following example shows that LOC is structurally distinct from GOAL and also in a lower position. This is evidenced by the fact that LOC arguments must always follow the THEME

(256) a. I sent the package to New York / *I sent New York the package (Bowers 2010)
b. I sent the package there / *I sent there the package
c. I sent the package to Mary / I sent Mary the package
d. I put the book on the table / *I put on the table the book

Structurally and semantically, LOC is the closest argument to the event (Larson 1988; 1989), which is further evidence of its status as the VP-internal argument. We can see that these PPs are arguments – and not adjuncts – by their obligatoriness to be selected by certain verbs (to the degree that something like obligatoriness can diagnose an argument (*cf* Koenig & Mauner & Bienvenue 2003).

(257) a. I sent the package *(to New York)
b. Erin put the book *(on the table)
c. The ball rolled *(down the hill)

The apparent optionality of the locative argument to appear with some verbs may cast some doubt on this observation,

(258) a. Robert walked (into the room)
b. Paul ran (across the field)

As there is no thematic alternation of the subject in these examples depending on the presence of the object (as there is with unaccusative/transitive alternating verbs), so these sentences are more reminiscent of a verb such as 'eat' which can take an arbitrary null object (Rizzi 1986; Authier 1989; Cardinaletti 1990).

(259) Ed ate (bacon-wrapped filet mignon)

More evidence of the argument status of these locatives is that they cannot be preposed over subject-oriented *wh*-questions, which locatives adjuncts can do (Reinhart 1983; Brensan 1994).

(260) a. On the corner, who drank? (Bresnan 1994:82)
b. *On the corner, who stood?
c. On the platform among the guests of honor, who knitted?
d. *On the platform among the guests of honor, who sat?

Thus, a revised theta-role hierarchy (Baker 1988) considering the status of the locative argument looks as follows (*cf* Ura 1996).

(261) AGENT > GOAL > THEME > LOC

Moving now to the internal structure of these locative PP arguments, it is clear from the tests in (242) and (243) that the LOC arguments undergo A-movement to the subject position in LI constructions. A-movement is predicated on the mover bearing phi-features (van Urk 2015; Lohninger & Kovač & Wurmbrand 2023; Fong & Halpert 2023; Keine & Zeijlstra 2024). Additionally, LOC subjects can control other A-dependenices such as phi-agreement.

(262) a. Off the plank {walks/walk} several bad pirates
b. Into my life {walks/*walk} you

As an argument, LOC should be able to receive Case as well. Given the connections between the ability to receive Case and bearing phi-features (Chomsky 2000; 2001; Platzack 2006; Legate 2008; Georgi 2017; Poole 2024, a.o), as well as the other A-dependencies of LOC arguments mentioned above, it stands to reason that the PP LOC arguments are specified for phi-features.

It is fully feasible that a non-nominal could be specified for phi-features (Halpert 2019; Keine & Zeijlstra 2024). Furthermore, the presence of a PP blocks phi-probing of a nominal situated inside of a PP, with or without movement.

(263) a. *Through the doors enter John
b. *The doors through enter John

There is further reason to believe that the PP complement nominal would not be accessible to the phi-probe on T^0 in its base position (Van Riemsdijk 1978; Citko 2014), or that a PP forms a phase edge. Given the idea that phasehood is actually connected to the presence of phi-features, it makes further sense that a PP constitutes a phase.

There is a precedent for analyzing locative argument PPs as either containing a light noun in their structure (Terzi 2005 *et seq*.; Collins 2007; Noonan 2010; Fraga 2020), or themselves being nominal (Marácz 1984; Collins 2004b; Aboh 2010). Following the latter idea, the proposal that locative deictic PPs themselves contain phi-features explains the nominal-like properties of locative PPs.

This PP is phi-deficient (bearing only a third person feature), and only appears with locative prepositions. There is a class of prepositions which are non-locative and are more functional in nature, which Collins (2024:84) calls argumental KPs. These PPs cannot so easily participate in LI.

(264) a. *For charity ran Paula
b. *With her husband walked Becky
c. *At Natalie looked Melissa
d. *By the house fell Lisa

Inversion and nominal status (and therefore locative argumenthood) favors locative 'lexical' prepositions over argument functional preposition, which are non-spatial/temporal PPs. These functional prepositions perhaps, then, do not contain phi-features.

The locative PPs need to be phi-deficient (which other people have claimed of light nouns contained within PPs (Kishimoto 2000; Collins 2007)), namely because of fact that LI sentences with PP subjects have the characteristic agreement optionality that is central to this dissertation. As phi-deficient elements, (Section 2.1), these PPs lack a number feature and must be third person. This can be seen in the way that locatives cannot be pluralized.

(265) a. *I want to go someplaces fun for my birthday
b. *Somewheres over the rainbow include the Yellow Brick Road and the Emerald City

(266) a. *This house and this city bring me so much joy; I never want to leave heres
b. *This house and this city brought me so much joy; I want to go back theres

There are several locative elements that cannot undergo pluralization or have any sort of [PARTICIPANT] feature, and this is consistent with the larger class of locatives[31]. Indeed, I take the existential 'there' to be a kind of deficient locative that lacks number in the same way as these locative PPs (more on this in Chapter 4). Dative subjects as well, which I have mentioned in section 2.1, also involve a PP "shell" structure that is featurally deficient, as evidenced by the fact that dative subjects cannot ever trigger plural verbal agreement.

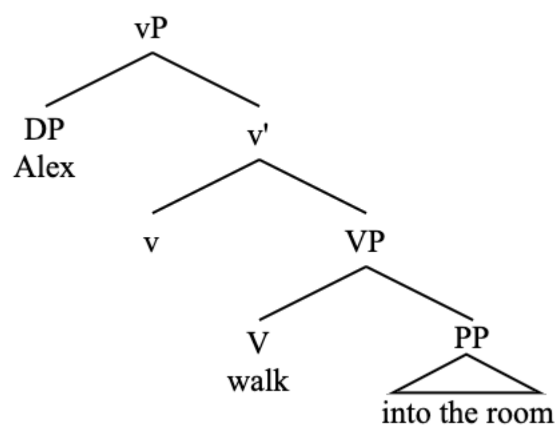
3.2.3 The smuggling derivation of Locative Inversion

Now that the nature of smuggling via VoiceP, thematic structure of verbs that participate in LI constructions, and the internal structure of the LOC argument have been established, it is possible to run through an entire derivation of LI as smuggling, using the following sentence.

(267) Into the room walked Alex

First, see the base position of the arguments and the verb.

(268) Into the room walked Alex: underlying structure



VoiceP is then merged above vP, where is selects the closest eligible verbal projection to move

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[31] Though these light nouns cannot be pluralized, plural verbal agreement with light nouns is acceptable in coordinate structures.

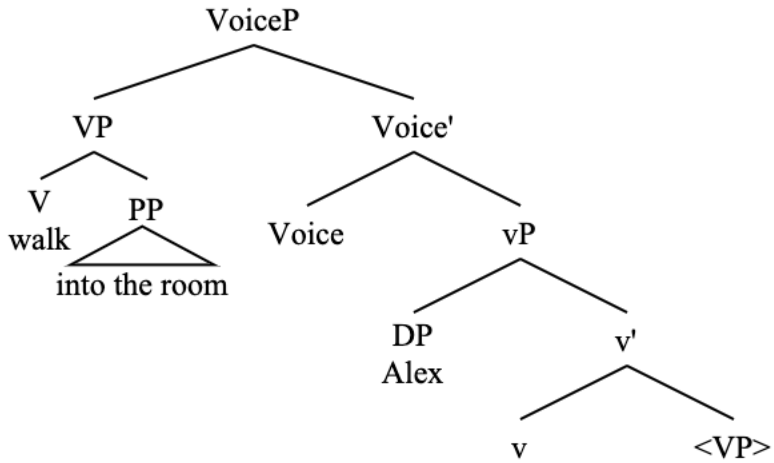
iii. Someplace warm and someone I love {are/??is} all I want in this life

This can be forced with PPs and nominal locative elements such as 'here' and 'there' as well, even though all of these elements resist pluralization.

iv. Under the bed and behind the dresser {are/?is} very dirty
v. Over here and over there {are/?is} always too cold

to its Spec, which in this case is VP.

(269) Into the room walked Alex: movement of VP to Spec,VoiceP



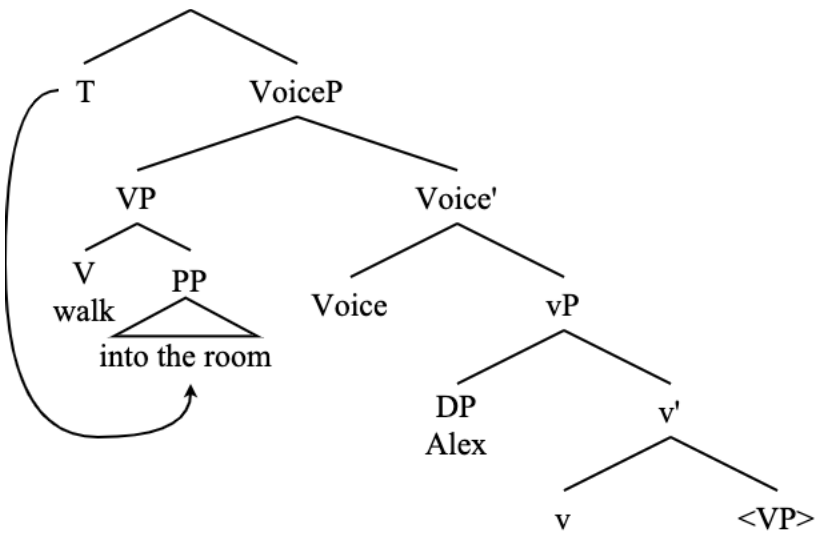
In this structure, there is no head movement of the verb from V to v, which is typically assumed to be obligatory in English (Adger 2003). There is no V-to-v movement in this structure because the VP has moved to Spec,VoiceP, thus breaking the head movement chain required for V-to-v movement (Dékány 2018). After the movement of VP to Spec,VoiceP, the v head no longer c-commands the V head, so head movement cannot take place (see the Head Movement Constraint (Travis 1984:131)).

Smuggling only functions as a prevention of head movement if the smuggling movement derivationally precedes head movement. In Chomsky's (2000; 2001) formulation, head movement takes place at PF. As A-movement happens in the narrow syntax, and smuggling derivationally precedes A-movement (as it creates the possibility for new A-dependencies). It therefore follows from this account that head movement does indeed derivationally follow A-movement. There are many analyses of head movement which claim that it follows the narrow syntax (Boeckx & Stjepanović 2001; Zwart 2001; Harley 2004; Matushansky 2006; Vicente 2007; Gallego 2010; Adger 2013; Ramchand 2014; a.o).

Even accounts of head movement which place it in the narrow syntax also place it after other narrow syntactic operations such as Agree (which is itself an A-dependency) (Roberts 2010). Therefore, head movement does not pose a significant issue to inversion as smuggling. Additionally, the head containing the verb (V) in the structure outlined here is still in a sufficiently local c-command relation with the T head in order for tense and phi-agreement to appear on the verb (Adger 2003).

Moving on to the following step, T^0 is merged above VoiceP. T^0 selects (via phi-Agree, to be expounded upon in the section 3.3) the closest eligible phi-bearing goal for movement to its specifier. This happens to be the locative PP.

(270) Into the room walked Alex: PP is selected by T^0



Crucially, the thematic subject DP ‘Alex’ is not selected here. If it were, the result would be ungrammatical.

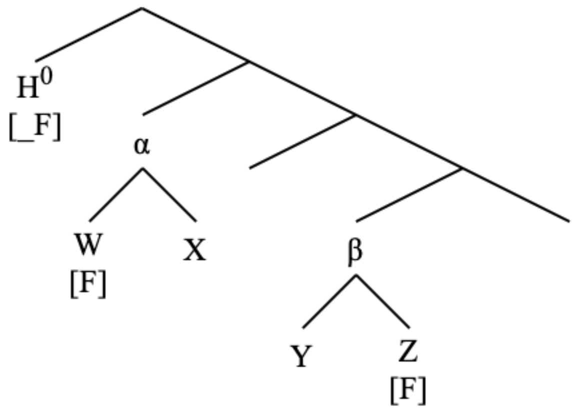
(271) *Alex into the room walked

The thematic subject DP is not selected here because the Loc argument is structurally closer to the phi-probe on T than the thematic subject, even though there is no direct c-command relation between the two DPs. This forces a reevaluation of a definition of closeness that only makes reference to the direct c-command relation of two elements in the syntax (Chomsky 1995; Ura 2000b). As such, I propose the following amendment to the definition of closeness when it comes to a probe evaluating its search space (see Storment 2025 for an implementation of this).

(272) Closeness Without (Direct) C-Command
For a configuration in which a probe on a head H^0 asymmetrically C-commands two nonterminal nodes α and β, and α asymmetrically C-commands β, the probe will search the entirety of the structure dominated by α for a relevant goal before it searches the structure dominated by β.

As an example of this, in the following structure, the probe on the head H finds W before Z, and the probe targeting Z would violate minimality.

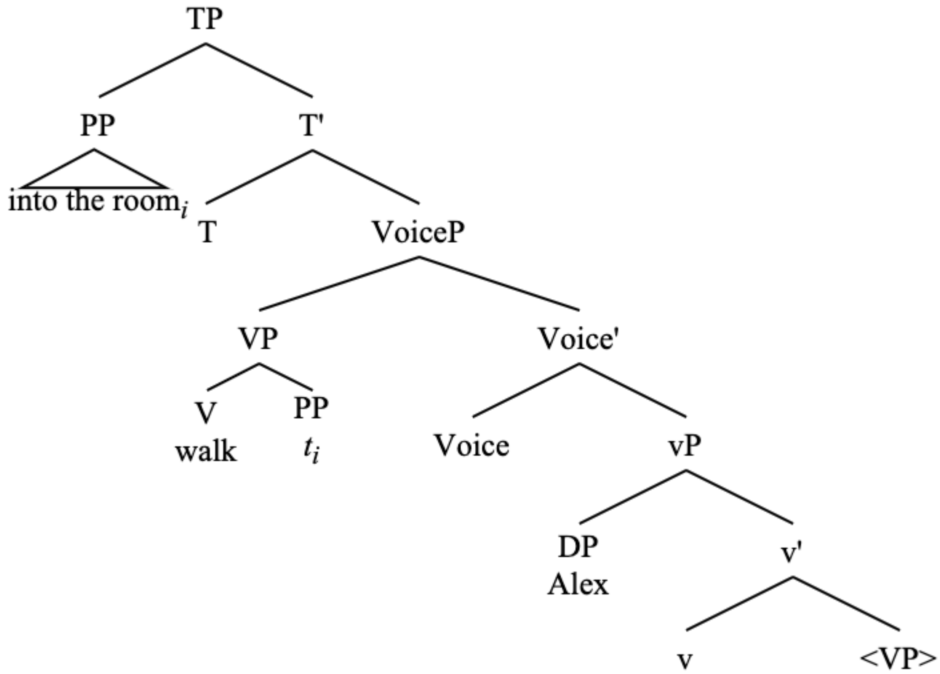
(273) Closeness without c-command



This conceptualization of closeness without c-command in Agree presented here bears on the issue of parsing algorithms in syntax: namely, whether or not probes perform Depth-First Searches (DFS) or Breadth-First Searches (BFS) (Milway 2023). This data certainly support a DFS algorithm for searches for Agree, which "starts at the root of an object and searches to a terminal node before backtracking" (Milway 2023:13). Without a a DFS algorithm, it is not clear how closeness could be determined between two elements that do not stand in an asymmetrical c-command relation to one another, especially concerning the accessibility of these elements to the operation Agree. DFS in Agree has been suggested elsewhere in the literature, with stipulations (Preminger 2019; Ke 2019; Chow 2022; Branan & Erlewine 2022). Branan & Erlewine (2022:6-8) specifically mention the necessity of a DFS algorithm in smuggling derivations, which is what I adopt here.

As a result of T^0 probing for a suitable goal in the LI derivation with this consideration on closeness in mind, the LOC argument moves to Spec,TP.

(274) Into the room walked Alex: LOC moves to Spec,TP.



At this point. T^0 probes again as a result of the phi-feature deficiency of LOC and may optionally undergo phi-agreement with the thematic DP 'Alex' (2.3.1), the details of which I explored in the previous chapter and leave aside for now until the following section in which I specifically address agreement in LI constructions.

It has been noted in the LI literature that the final resting place of the PP must be higher than Spec,TP (Bresnan 1994; Rizzi & Shlonsky 2006; Diercks 2017). As such, the derivation is not yet complete. Evidence of this is that an auxiliary cannot appear above the locative PP in cases such as question formation.

(275) a. *Did into the room walk Alex?
b. *Who did into the room walk?

It is possible for the LOC PP to raise above the subject position; see the following raising examples (Postal 1977; Bresnan 1994), which show that the fronted PP may undergo further A-movement.

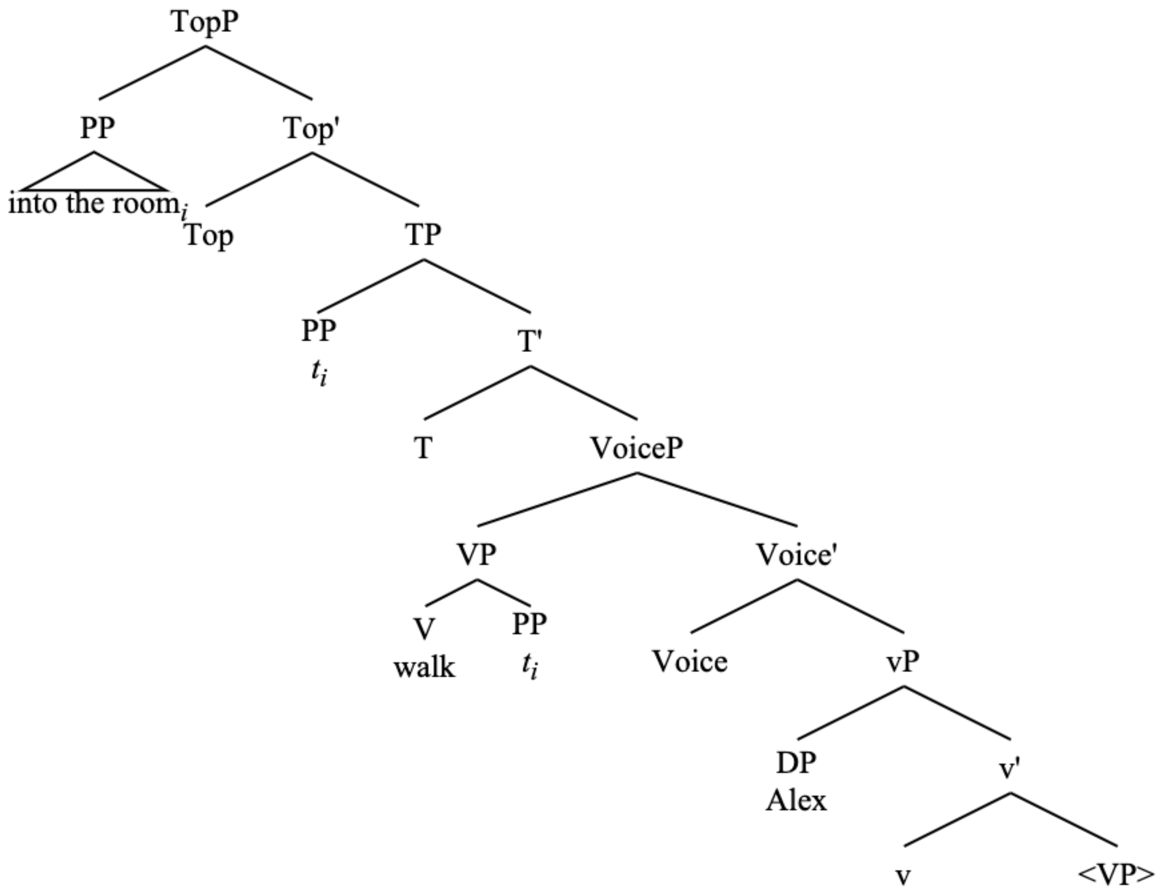
(276) a. Over my window seems to have crawled an entire army of ants (Bresnan 1994:96)
b. On that hill appears to be located a cathedral
c. In these villages are likely to be found the best examples of this cuisine

Movement into the left periphery is certainly possible as well. See the following.

(277) a. It was a perfect storm of media exposure and money, into which walked the 17-year-old Becker from a small town in Germany (News)
b. Here they formed a large ring, into which walked the Rolling Thunder and Han-na-nos-ko-a, the Prophet (Blog)
c. We cut to the Doomsday Comics store, into which walked the expressionless duo of Arby and Lee (Blog)

Indeed, I argue that this movement into the left periphery for information-structural reasons is obligatory in the LI derivation. the PP complement of the LOC argument DP raises to a position above Spec,TP. I represented this here with TopP (Rizzi 1997).

(278) Into the room walked Alex: movement of PP to Spec,TopP



It is not always the case, however, that a PP must undergo movement position above TP. The following examples involving focused postverbal subjects.

(279) a. There walked into the room a man I have never met before
b. There arrived at the station none other than ME!

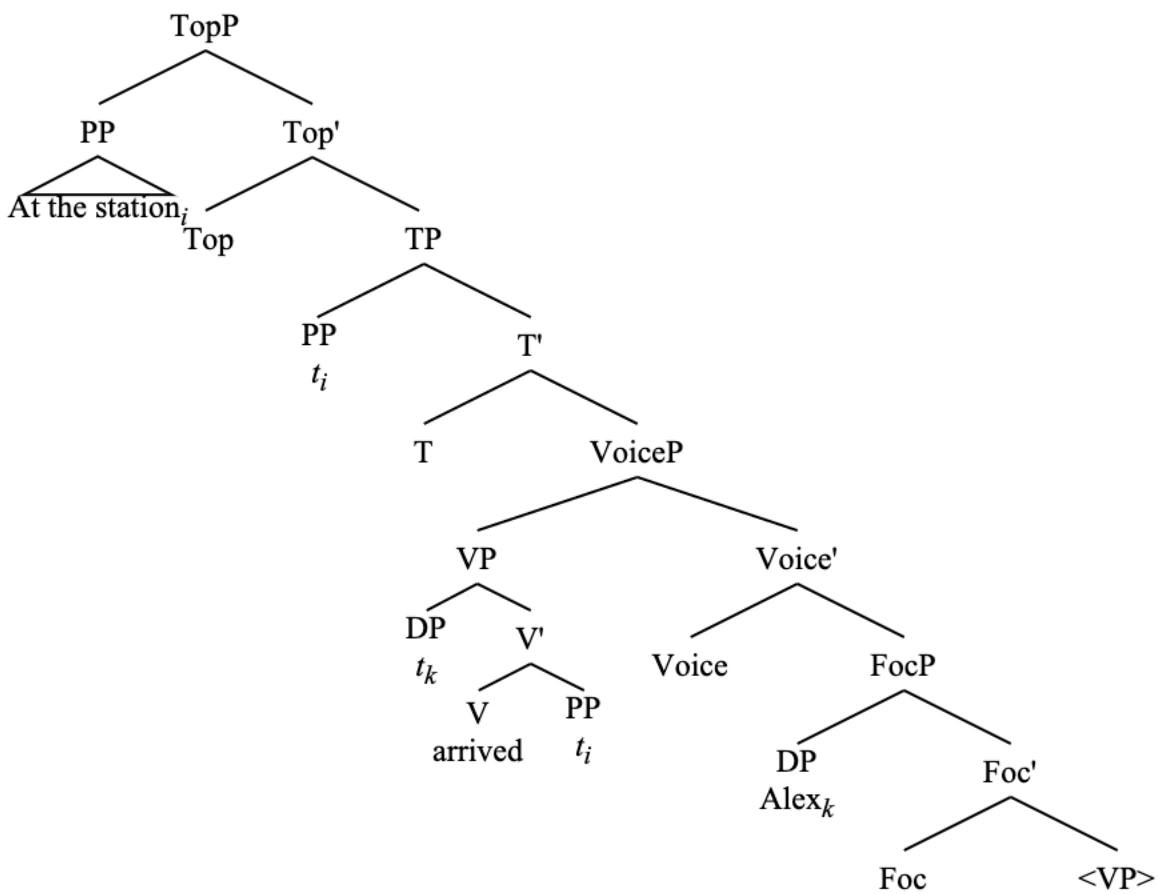
In these cases, there is no movement of a PP into Spec,TopP. Note that this structure is only licit with a focus reading on the subject.

(280) *There walked into the room John (no focus)

As such, the availability of the PP to not undergo movement to the left periphery is dependent on the focus properties of the postverbal subject, which further highlights the connection of inversion via VoiceP and the information-structural status of arguments at LF.

This derivation works much the same with unaccusatives, the major difference between the derivation of LI with unaccusatives and unergatives being that the unaccusative thematic subject (the THEME) must obligatorily move out of the VP into a FocP before smuggling can occur. I show this in the following example.

(281) At the station arrived a train: LI structure



In this structure, the thematic subject ‘a train’ is moved from its base position in Spec,VP to Spec,FocP. Then, the VP is smuggled via VoiceP over the FocP. From this point, the PP is selected by T^0 for movement to Spec,TP (at which point T^0 can optionally probe again, since ‘there’ is deficient, which would cause it to probe again and find the thematic subject). After this, the entire PP is moved to TopP.

Further evidence of the obligatoriness of the movement of the PP above Spec,TP is what happens when the locative subject is existential. When the locative subject is ‘there’ – especially with unergative verbs, which may not otherwise select an existential locative argument on its own (more on this in Chapter 4) – it must be accompanied by an overt locative deictic PP adjunct.

(282) a. *There walked a clown
b. Into the room there walked a clown

Furthermore, the locative PP must precede the ‘there’ subject.

(283) *There walked a clown into the room

We know that this ‘there’ is in Spec,TP because, when a structure like (282a) is licit, such as in the case of unaccusatives, an auxiliary in yes-no question formation comes to precede it.

(284) Did there arrive a clown?

Thus, even if the element in Spec,TP is not the locative that moves into the left periphery, there is still a requirement in LI derivations that holds in most cases for a locative to move into the left periphery. This requirement captures the connection between grammatical voice and information structure, which is that inversion driven by VoiceP is there to enable further relations to the left periphery and information structure via A-movement.

This analysis captures the fact that the fronting of the locative PP has A-properties such as the failure to license parasitic gaps and the obviation of weak crossover, as well as the fact that raising is possible for locative subjects and that they can control agreement.

The obviation of weak crossover specifically is still somewhat puzzling, though, as the two coreferent DPs do not actually stand in a c-command relation with one another from the position of the A-moved PP, since the quantified DP is buried within the locative PP.

(285) Into every boy$_i$’s room walked his$_i$ mother

This fact becomes even more puzzling when we compare this LI example with an example of a non-LI PP subject (the copula in these examples being interpreted as an identity copula, not an “is located”-type copula)[32].

(286) a. *Under every boy$_i$’s bed is his$_i$ favorite place to hide
b. Under his$_i$ bed is every boy$_i$’s favorite place to hide

This analysis also captures the apparent Ā-properties of the fronted PP in LI derivations (Diercks 2017), which include the information-structural status of this fronted element as well as the fact that it appears in a position above Spec,TP.

Additionally, when a PP moves directly into Spec,TopP without passing through Spec,TP, such as in the case of (282b) where Spec,TP is occupied by ‘there’, it notably lacks some of the A-properties that these locative subjects in LI normally have. For instance, the obviation of weak

[32] It is my belief that this issue can be solved by allowing more complex PP-internal structure, which I do not indicate here.

crossover seems significantly harder when Spec,TP is occupied by 'there'.

(287) a. Into every boy$_i$'s room walked his$_i$ mother
b. ??Into every boy$_i$'s room there walked his$_i$ mother

Additionally, parasitic gaps appear easier to license in the presence of a 'there' structural subject.

(288) a. *Into that very room peeked Ali [without actually entering _]
b. ?Into that very room there peeked Ali [without actually entering _]

These tests showing Ā-properties of a fronted PP in LI that does not pass through Spec,TP present a complication to the view that the landing site of these elements in the left periphery could be an A-position (Slioussar 2011; Van Urk 2015), at least in English.

Returning to the trees in (278) and (281), they show the complete derivation of LI as smuggling with unergatives and unaccusatives, respectively. There is a question to ask here of why the final movement of the PP to an information-structural position such as Spec,TopP should take place at all. Other analyses of LI treat this movement as obligatory, saying that Spec,TP is not a licit final landing site for the PP, so it must evacuate it (Bresnan 1994; Rizzi & Shlonsky 2006).

I find this explanation unsatisfactory, and it is difficult to make work in my analysis where the elements that end up occupying Spec,TP and Spec,TopP are distinct. This makes the question of a given element being licensed in one position but not the other irrelevant. I choose instead to explain the movement of the PP to the left periphery by relating this movement to the reason why inversion should occur in the first place.

Voice (of which inversion such as LI is a subset) concerns the ordering of arguments as they are to be mapped onto A-positions in the clause (Collins 2024:ch. 8; see also my discussion on voice in Chapter 1). In essence, it is the job of Voice^0 to promote certain arguments over others (and, by extension, demote the non-promoted arguments). This is transparently seen in passives, anti-passives, middles, and dative shift. In passives, for example, the THEME (or GOAL) is promoted over the AGENT. Similarly, in LI, the LOC argument is promoted over the AGENT/THEME.

Why such a mechanism of the promotion of certain arguments from their base positions should be available at all is innately tied to information structure and discourse prominence (Erteschik-Shir 2007). While active and passive derivations of the same sentence are typically taken to be truth-conditionally equivalent, there are pragmatic and discourse-functional differences between the two (Thompson 1987; Seoane 2013; Liu 2022).

I propose that the movement of the PP to the left periphery accomplishes the same goal as smuggling, which is to promote the internal argument of the verb. That is, smuggling, while it facilitates the creation of new A-dependencies, also enables further relations to the left periphery.

The movement of the locative PP to the left periphery also licenses the element in Spec,TP to have a null exponence, namely the trace of the PP, which is famously known to not normally be possible in English.

(289) *Walked John into the room

This is similar to the account of a null element in Spec,TP in LI put forth by Rizzi & Shlonsky (2006), although they posit that this movement to the left periphery must necessarily be A-movement. My analysis carries no such stipulation. Here, the movement of a constituent that contains a trace of the moved subject in Spec,TP to an adjacent specifier position that c-commands the Spec,TP position licenses the null trace in that position, whereas it would not normally be licensed there. This is because, in this configuration, the interpretation of the null structural subject can be easily recovered due to the structural proximity of the locative PP and its trace to one another, which is not the case in (289). Such a condition is needed in English to allow for a null element in Spec,TP, which is normally not allowed since English is not a null subject language. In a null subject language such as Spanish, something like this configuration has nothing to do with whether or not a null element can be in Spec,TP as that is always allowed.

As for why this specific technology of inversion as voice as smuggling is only available for a limited set of predicates in English (e.g., locatives, verbs of quotation, and some predicative DPs and adjectives), I tie this to the deficiency of the element that comes to occupy Spec,TP. In all of these constructions, this element is characteristically phi-deficient. Without the deficiency of this inverted structural subject, there is no possibility that the postverbal thematic subject could ever be visible to the phi-probe on T^0, as it would have no reason to probe past a non-deficient argument. It seems, then, that T^0 at least needs to retain the possibility of probing past the inverted subject in these inversion constructions, even though the probing operation does not always lead to grammaticality (see the PCC effects from Chapter 2).

Perhaps, diachronically, there was a point in English where it was necessary for the postverbal subject to be licensed by undergoing Agree with T^0. Although that is no longer the case, such a history would explain the current distribution of these inversion constructions (i.e., English lacks the subject-object inversion of Bantu (Shlonsky 2024) or Russian (Bailyn 2004)), and how the only option for inversion via VoiceP for non-deficient internal arguments is to have a passive.

(290) a. Beans were eaten by Drew
b. *Beans ate Drew (*int*: Drew ate beans + inversion)

Here, the passive verbal morphology Case-licenses the postverbal external argument (Collins 2005), since there is (nor was there ever) a situation in which T^0 could probe past the non-deficient internal argument to license another argument. These considerations bring up a question of how the postverbal subjects are licensed in these inversion constructions such as LI when it is not always the case that they undergo Agree with T^0. I explain this in the following section.

The smuggling structures presented in this section allow for an analysis of the agreement facts in LI under the theory of Agree set forth in Chapter 2, which I detail now.

3.3 Agreement in Locative Inversion

Agree in LI is subject to the same number agreement optionality and PCC effects seen in existential 'there' sentences in English, presented in Chapter 2. While these agreement facts have not been reported in the literature except to note their apparent ungrammaticality (Bresnan 1994:95), I present in this section a systematic overview of data collected online that show optionality in agreement with third person plural thematic subjects and impossibility of agreement with first and second person thematic subjects in LI constructions.

3.3.1 Agreement optionality

Extant literature on LI only acknowledges the possibility that verbal agreement must track the phi-features of the thematic subject (Diercks 2017), but the data I present in this section show that that is not the case. Take the following alternation.

(291) Off the plank {walks/walk} several bad pirates

Third person singular verbal inflection with a third person plural thematic subject is grammatical for many speakers and is quite attested online, with agentive (292) and unaccusative (293) verbs.

(292) a. Into the room walks three men, London gangsters, Gadder in charge with his musclemen Tosher and Freddy. (Blog)
b. Equally as panicked, into the room walks three orderlies. They've been assaulted, each of them carrying some kind of wound. (Blog)
c. Rita was getting ready, and into the room walks two girls, Stefanie and Nadinè from Berlin (Blog)
d. Just as Jamie and Claire are going to kiss, into the room bursts two little girls shouting "DADDY?" (Blog)
e. In through the door walks two enlisted Airman, one young and one old (Blog)
f. It was Valentine's Day, and in through the door walks these attractive looking girls, one who became my wife. (Blog)
g. One afternoon in 1967, into the store walks several long-haired hippy looking strangers, who announced that they were here to make a documentary film on Cajun Louisiana (Blog)
h. Earlier today as I ended my class & prepared to leave, in walks several students waiting for their professor to arrive (Twitter)
i. In walks the people you blame for it all (Reddit)
j. In through the door walks all the ideas, the dreams, the wishes you ever had, everything you wanted to do so you would have a wonderful life (Reddit)

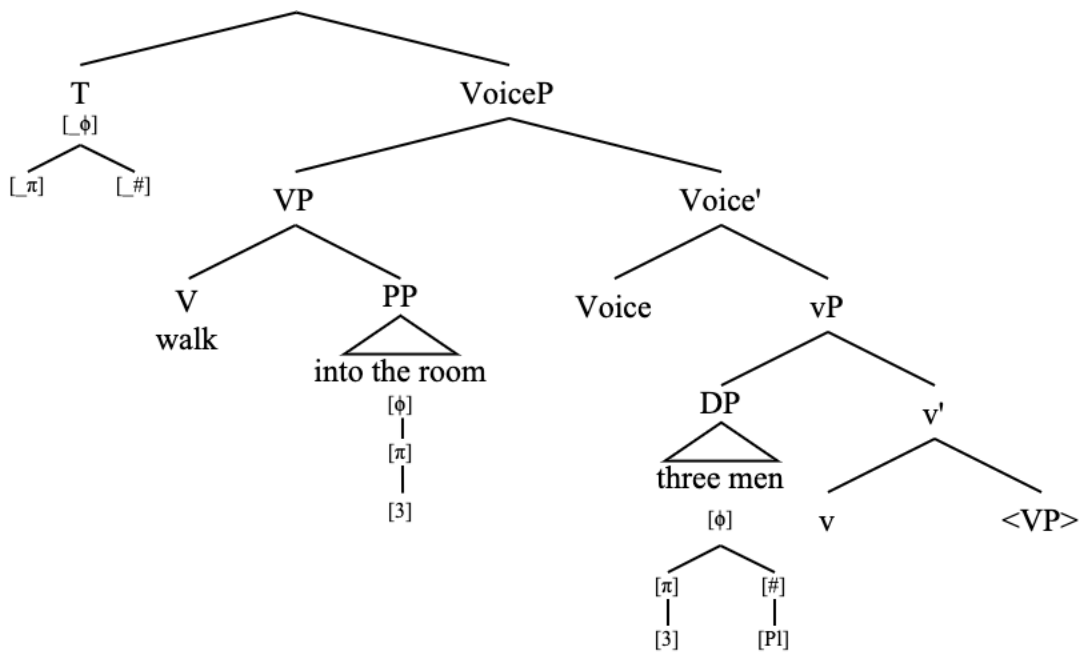
(293) a. Into the room comes THREE firemen in full gear (Blog)
b. As if we haven't had enough of red, white and blue…along comes these placemats! (Blog)
c. Then through the door comes these creatures, half organism, half machine (Game)
d. We were out at the pub drinking and having a good time, and through the door comes these three large Mexican dudes. (Blog)
e. So I was in the store getting some snacks for my lil one and through the door comes these two youngsters wearing their pants all the way down to their knees and cursing up a storm while these elderly people were standing right next to me (Facebook)
f. In the restaurant sits two guys... It seems that the restaurant was closed especially for

them (Blog)
g. On the bench sits three judge-type people (PDF)

For all of these sentences with third singular verbal agreement with a third plural thematic subject, third plural agreement is also grammatical. Given the condition on closeness without c-command (272), I now cast the LI agreement variation in the Agree framework from Chapter 2.

Using a truncated version the first sentence in (292), I present the derivation of Agree that gives rise to the following optionality, beginning with the structure after the smuggling movement of the VP containing the LOC argument to Spec,VoiceP

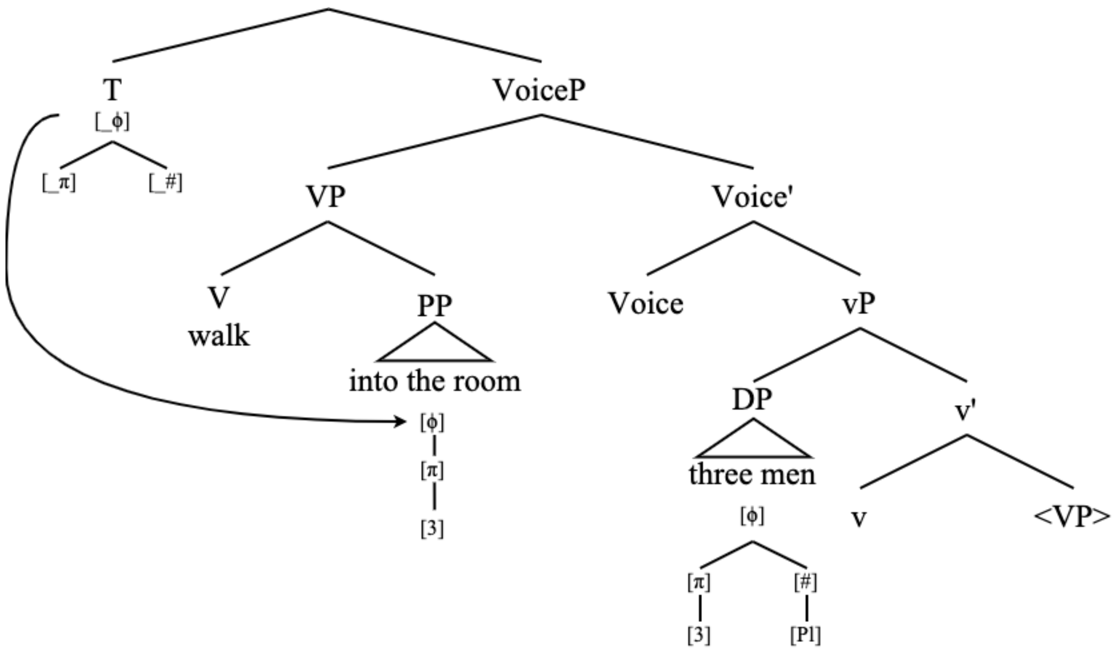
(294) Into the room {walks/walk} three men



Recall that phi-probes in English are specified to search for a goal bearing [ϕ], [π], and [#], with the feature [_ϕ] initiating its search first as it is the highest feature in the hierarchy of the probe. Also recall that the LOC argument is phi-deficient and lacks a number feature.

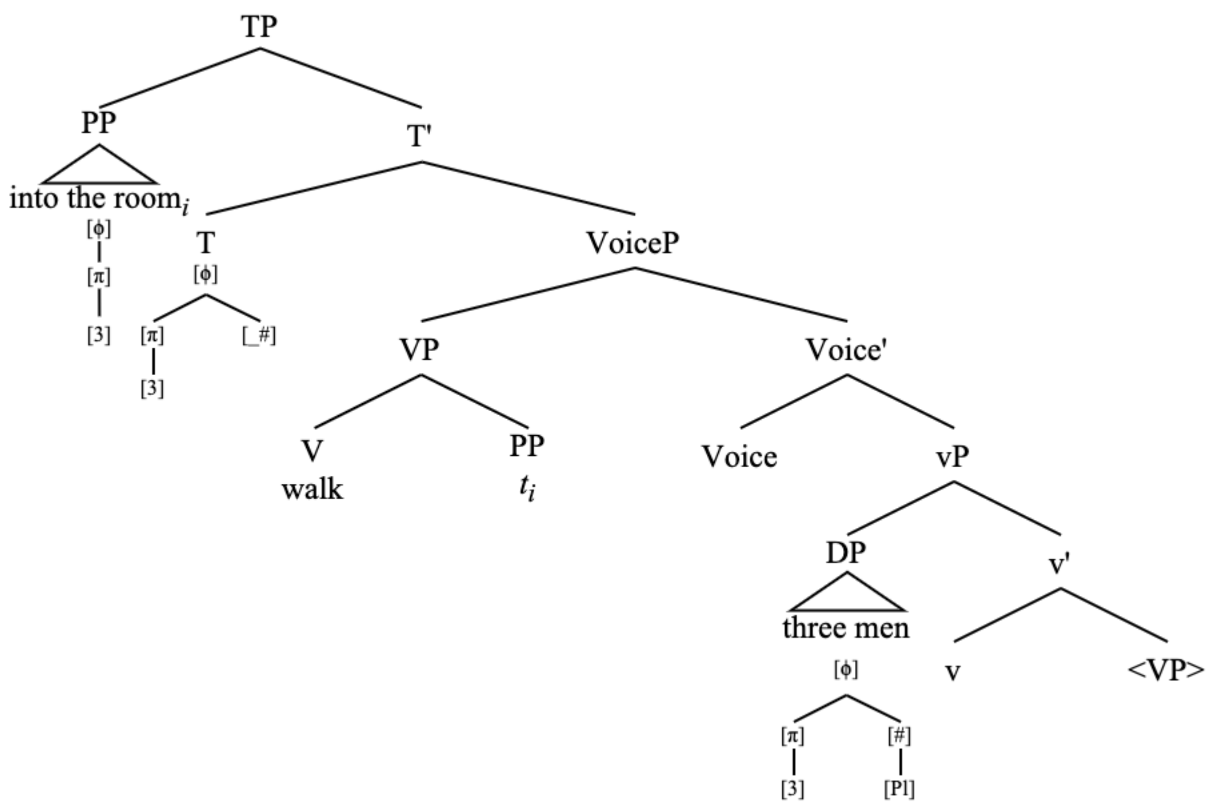
The phi-probe on T initiates a search for a goal bearing [ϕ]. Under Closeness without C-Command, the first element it finds is the LOC argument 'into the room'.

(295) Into the room {walks/walk} three men: probe encounters deficient goal



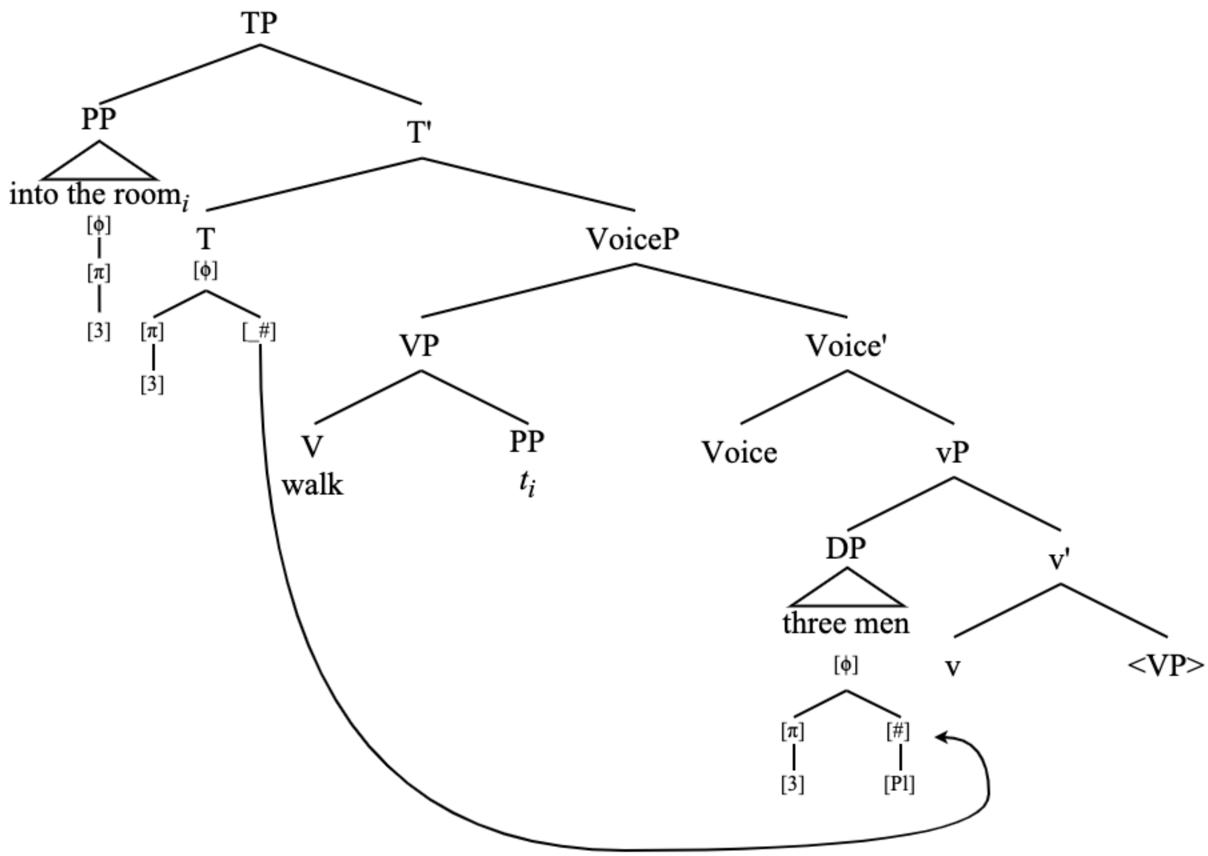
As a result of this search, the phi-probe undergoes Match and Copy of the features of the PP. As they now are in an Agree relation, the PP moves to Spec,TP, thus satisfying the EPP on T^0.

(296) Into the room {walks/walk} three men: Agree with PP, PP in Spec,TP



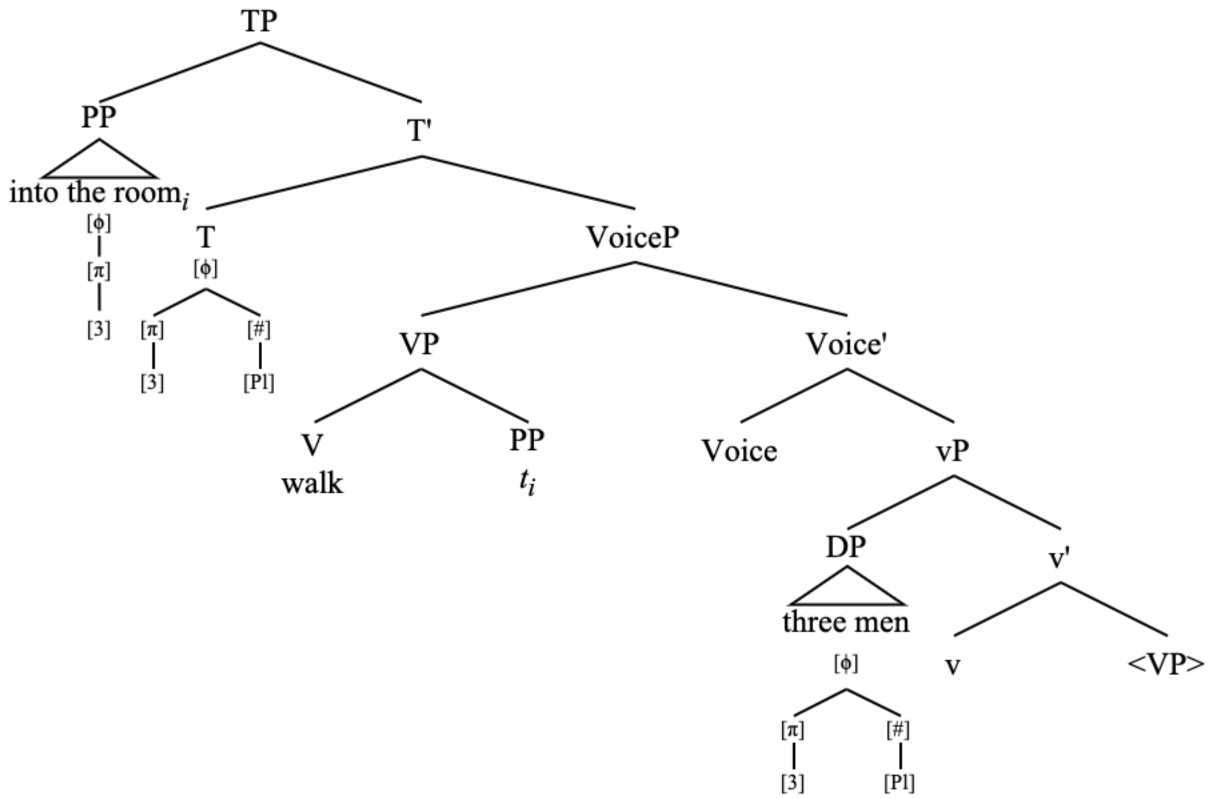
Now, there is only an unvalued number feature on the phi-probe. The phi-probe can either stop here and remain without a valued number feature, in which case singular agreement is spelled out on the verb, or it can probe again for a number feature (which causes a kind of minimality violation). Again, either option is slightly uneconomical. The PP-internal argument ‘the room’ is not visible to the probe of this operation as searches into the PP are opaque to probing operations; only the edge is accessible (van Riemsdijk 1978; Citko 2014).

(297) Into the room {walks/walk} three men: Second Search for [_#]



After the second probing, Agree is established with the postverbal thematic subject ‘three men’. This causes the verbal agreement to be spelled out as third person plural. Both the features of the LOC structural subject ([ϕ], [π]) and the AGENT thematic subject ([ϕ], [π], [#], [Pl]) have been copied over to the probe, and there is no structural conflict in the phi-probe that would cause problems for the syntax or interpretation. Defective circumvention has thus obtained and the verbal agreement is spelled out as ‘walk’ (plural).

(298) Into the room {walks/walk} three men: Agree with Agent, third person plural inflection



This is how optionality of agreement with third person plural postverbal subjects in LI constructions is derived. No defective circumvention (that is, only Agreeing with the deficient PP) leads to singular default verbal agreement. Defective circumvention leads to the verbal agreement tracking the features of the postverbal external argument. It is the exact mechanism of Agree laid out in Chapter 3 of this thesis.

3.3.2 PCC effects and Case

First and second person postverbal subjects in LI cannot control verbal phi-agreement. See the following data obtained via Internet search (Collins 2020) with 'you' as the thematic subject.

(299) a. Then along comes you, who has voluntarily turned away from the Force (Reddit)
b. You do a great big long rude burp, and up comes the gas and down comes you! (Charlie and the Chocolate Factory, ch. 22)
c. In walks you, and you make me feel like everything's okay (YouTube)

Although 'you' does not show Case distinctions, first and second person pronouns in this position cannot be Nominative.

(300) a. *Into the room walk(s) I
b. Into the room walk*(s) me

Furthermore, verbal agreement with 'you' in these cases is unattested and judged to be ungrammatical.

(301) *In walk you

This is corroborated by online data, and the forms given as ungrammatical in (300)-(301) are not attested online.

(302) a. Through the door walks me, her date (Clarkson 2008)
b. Then, in through the door walks me (Blog)
c. My work wanting a diversity hire and in walks me // A nonbinary audhd bitch with a collection of Shrek shirts (Twitter)
d. He would just have to look at me and then bam down goes me (Twitter)
e. All of the very stuffy mums with very professionally massaged babies in a circle and in walks us two (Blog)
f. Through the door of that restaurant, in walked us (Blog)
g. So up goes me to the roof to remove the bucket and clean out the hole that lets water through (Forum)
h. So, in comes me, an 8 year old *mocosa*, already forgetting everything he told me earlier (Blog)
i. when in walks me and falls arse backwards into the easiest lay I've ever had (Forum)
j. In walks us, a large couple, from NY, with our boys, 19 and 14. (Yelp Review)

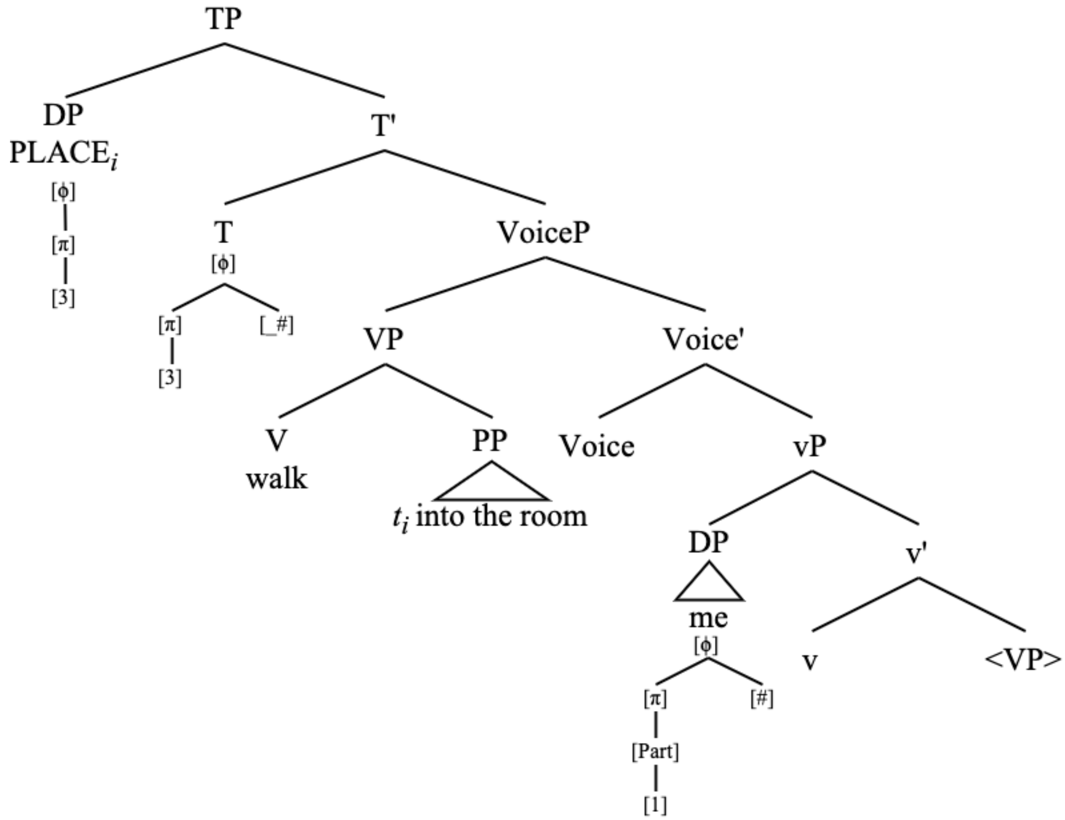
Once again, this is not a ban on agreement with a postverbal accusative DP. Agreement with third person postverbal accusative DPs is very possible.

(303) a. One raises fares, up go them all (Blog)
b. …and then in come them riding on a Harley (Forum)
c. Edwin(a) and Jan cast some spells and in come them (Forum)
d. Down go them all! (Play)
e. So, in go them! What's next? (Facebook video)

It is therefore the incompatibility of the probe to Agree with the [Part] feature of first and second person postverbal DPs that gives rise to the ungrammaticality of (300) and (301). Agreement with a first or second person postverbal thematic subject is not possible in LI given the PCC effects with multiple subject constructions outlined in chapter 3. I quickly go through a derivation of this now, with the grammatical sentence in (300b).

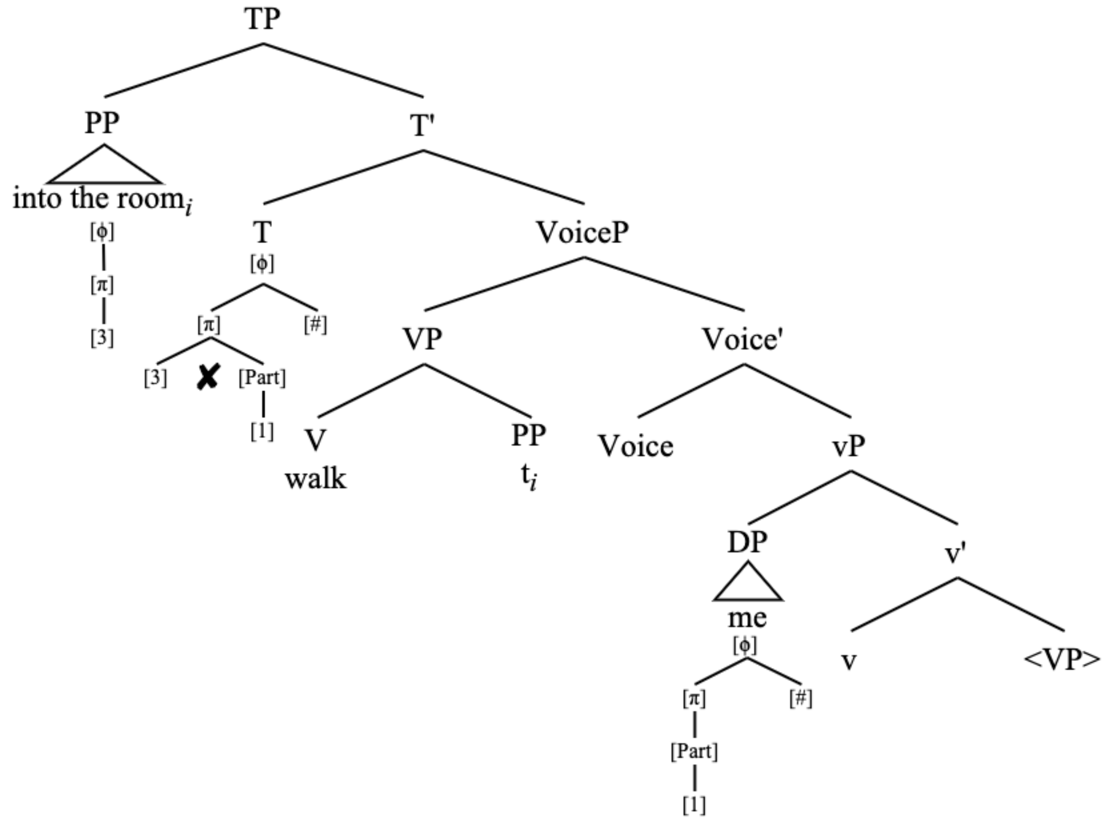
Agree is established between the phi-probe on T and the LOC argument in the exact same way as in (294) to (296), so I skip ahead to that point here.

(304) Into the room walks me: LOC in Spec,TP, in Agree relation with T



At this point the probe can stop and have an unvalued [_#] feature left over. This results in third person singular agreement on the verb. If the probe searches again for number, ungrammaticality results because incompatible person features will have accumulated on the phi-probe. I omit the visual representation of the step of the probe's second search and jump straight to the ungrammatical structure. Specifically, the [NonPart] or [3] feature and the [Part] feature are not compatible. This creates the impossibility of defective circumvention.

(305) *Into the room walk me: incompatible person features on T



As [Part]-bearing thematic subject DPs in LI cannot undergo Agree with T in a licit derivation, they cannot receive nominative Case. Furthermore, nominative third person pronouns do not seem so good with LI, even if the verb tracks agreement with them.

(306) a. Into the room {walk/walks} {them/*they} with all their luggage
b. Into the room walked {him/*he} with all his luggage
c. Into the room walked {her/*she} with all her luggage

The association between nominative Case and phi-Agree with T is well-established (Chomsky 2000; 2001; Platzack 2006; Legate 2008; Georgi 2017; Poole 2024), so it is not immediately clear why the data in (306) should be the way they are, aside from the idea that English nominative Case assignment is only possible in a Spec-Head configuration, brought about by Agree attracting a viable Case recipient to the specifier position of a Case-assigning functional head[33] (Ura 1996; 2000a). I described this in Chapter 2 (2.2.3), and, again, it is a kind of backwards conceptualization of Upward Agree and Case assignment as described in Bjorkman & Zeijlstra (2019). Nominative Case in LI, then, is assigned to the locative PP.

[33] I am aware that this discussion contradicts the concept of double nominative Case assignment in languages which allow that (Hiraiwa 2001; Pesetsky & Torrego 2004; 2007). It could easily be the case that nominative Case assignment is parameterized to work differently across different languages. Relatedly, the "default" Case in many languages seems to be nominative, which is not the case in English, so it is possible that English is relatively restrictive of the situations under which it licenses nominative Case.

The Spec-Head conceptualization of English nominative Case assignment also explains the Case of the postverbal thematic subject DPs in LI constructions, which receive accusative Case from the vP head with unergative verbs. Accusative Case assignment from v^0 explains the accusative Case morphology on the postverbal subject pronouns, though it must be the case that accusative Case is different from nominative Case in that accusative does not need a Spec-Head configuration in order to be assigned (undergoing Agree with v^0 is enough).

(307) a. Into the room walks {me/you/us/y'all/him/her/them}
b. *Into the room walks {I/we/he/she/they}

Unaccusative verbs are a different story, as there is no vP to assign accusative Case (hence the unaccusativity). I additionally note that the examples in (308) do not seem so wonderful unless they are marked with strong prosodic focus (which is not a restriction in LI with unergatives).

(308) a. ??Out of the shadows appeared you
b. ??At the station arrived us with all of our luggage
c. ??At your feet lays me
d. ??In a hole in the ground lives him

(309) a. Out of the shadows appeared YOU (not me)!
b. At the station arrived none other than US with all our luggage!
c. At your feet lays ME, and only me!
d. In a hole in the ground lives HIM (not her)!

This suggests that direct movement to FocP without any intervention from another Case-licensing head can license DPs that would not otherwise be able to receive Case. This also tracks with the analysis presented in this chapter of unaccusative verbs in LI constructions which necessarily involve movement of the THEME to Spec,FocP. The distribution of FocP is not just limited to unaccusatives, however.

As such, movement to Spec,FocP is able to license those arguments that appear as postverbal subjects in these inversion constructions which do otherwise receive accusative Case – for instance, when there is no vP. This is consistent with observations that a DP undergoing Agree with a Focus projection can value that DP's need for Case (Carstens 2015; 2025).

I now go on to discuss cases in which a FocP may appear with unergative LI verbs as well.

3.4 Locative Inversion vs Heavy NP Shift

The literature on LI constructions notes that there are two distinct varieties of LI: standard LI, or 'light' inversion, and Heavy NP Shift (henceforth HNPS) LI (Culicover & Levine 2001; Rizzi & Shlonsky 2006; Diercks 2017). In this section I show that there is indeed a distinct HNPS version of LI with unergatives with a distinct syntax, and that it can be accounted for with the same mechanism of smuggling, and the movement of the thematic subject to Spec,FocP, just as there is with unaccusative verbs in LI derivations. I show two diagnostics for HNPS LI: the same word order tests that were used in section 3.2.1, and the distribution of modals and auxiliaries in LI constructions.

3.4.1 Word order tests

There is a class of adjuncts which must follow the thematic subject in standard unergative LI constructions, such as secondary depictive predicates and adjunct clauses[34] (3.2.1). Keep in mind that both positions for depictives – immediately preceding or following the THEME – are available for unaccusative verbs, as discussed in section 3.2.1.2.

With a heavy NP thematic subject in unergative LI sentences, the depictive may precede the postverbal subject.

(310) a. Into the room strutted (drunk) the old man who was supposed to have been on the wagon for decades (?drunk)
b. Onto the field ran (naked) the unidentified and still-at-large serial streaker (?naked)
c. Across the lawn marched (barefoot) the marching band who were practicing a quite unusual technique (?barefoot)

This position for depictives is crucially unavailable without HNPS.

(311) a. *Into the room strutted drunk Mark
b. *Onto the field ran naked the streaker
c. *Across the lawn marched barefoot the band

Similarly, purpose clause adjuncts may precede HNP thematic subjects in unergative LI.

(312) a. Into the operating room sprinted (to get the doctor) the overworked nurse from Pennsylvania who didn't get paid enough (?to get the doctor)
b. Through the crowd pushed (to get a better view) the diminutive spectator of 5'4 who was tired of never being able to see a damn thing (?to get a better view)

Again, this position is unavailable without HNPS.

(313) a. Into the operating room sprinted (*to get the doctor) Jessica (to get the doctor)
b. Through the crowd pushed (*to get a better view) Xavier (to get a better view)

At this point it is necessary to ask: what even is a heavy NP[35]? It seems strange to posit that a syntactic operation should be sensitive to the phonological weight of a given constituent given the linearity of the Y-Model architecture of grammar. HNPS, then, is not about the heaviness of the constituent *per se*, but rather about the fact that heavy NPs are discourse-marked and have a special information-structural status. Contrastive focus DPs have the same distribution as heavy NPs with respect to the adjuncts mentioned above.

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[34] See similar tests with adverbials in Culicover & Levine (2001); Rizzi & Shlonsky (2006); Diercks (2017) to diagnose HNPS LI vs standard LI.

[35] I say heavy "NP" instead of "DP" purely because of convention and a precedence in the literature of this terminology. I am aware that most (if not all) heavy NPs are more accurately characterized as DPs.

(314) a. Into the room walked drunk ME, not you!
b. Onto the floor stepped barefoot JOHN, and only John!
c. Across the field streaked naked THAT man!

(315) a. Into the operating room sprinted to get the doctor JESSICA!
b. Through the crowd pushed to get a better view XAVIER!

To capture this fact, I analyze heavy NPs as moving to a FocP in the verbal projection spine (Collins 2024c). I show this with the following sentence. That is, the special status of heavy NPs is owed to their being focus-marked, syntactically computed as the movement to FocP.

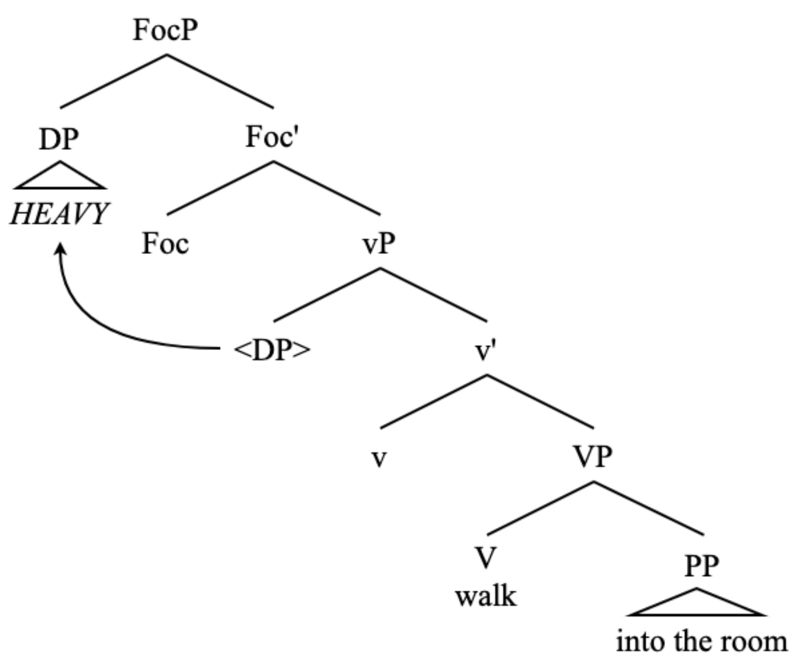
(316) Into the room walked the very esteemed and handsome gentleman known as Cecil

As evidence of this FocP, let's once again look at these data as the answer to a wh-question. HNPS in LI, even with unergatives, can function as an answer to a wh-question.

(317) Q: Who walked into the room?
A: Into the room walked the very esteemed and handsome gentleman known as Cecil

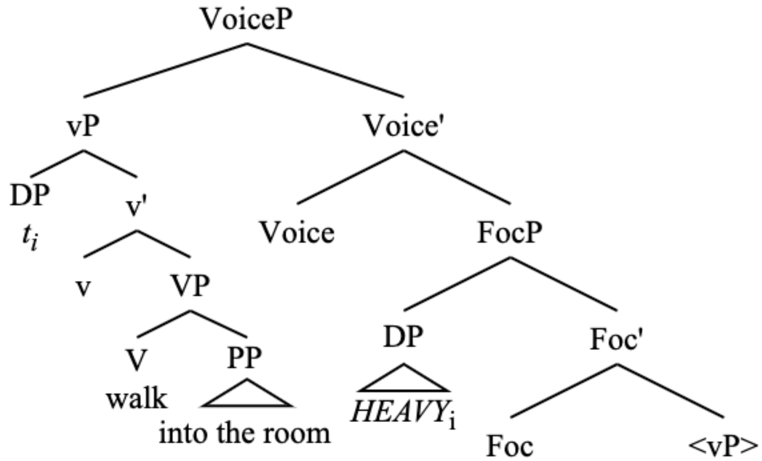
Rather than representing the entire heavy NP in the structure, I use the shorthand *HEAVY*.

(318) Into the room walked *HEAVY*: movement to FocP



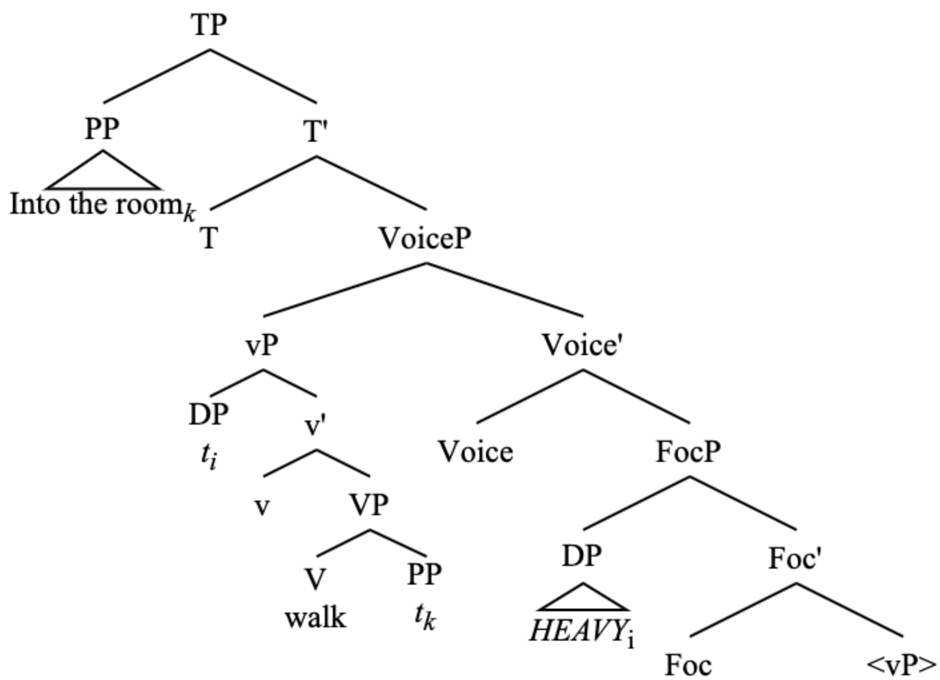
VoiceP is then merged above FocP, and the closest eligible verbal functional projection is selected to move to Spec,VoiceP. Unlike in standard LI, the closest eligible verbal projection here is vP, not VP, due to the presence of FocP.

(319) Into the room walked *HEAVY*: smuggling of vP over FocP



T^0 and its phi-probe are then merged above VoiceP. The probe on T^0 selects the closest phi-goal, which here is the locative PP, the internal argument of VP. The AGENT DP is not the closest goal because it moved out of vP into Spec,FocP.

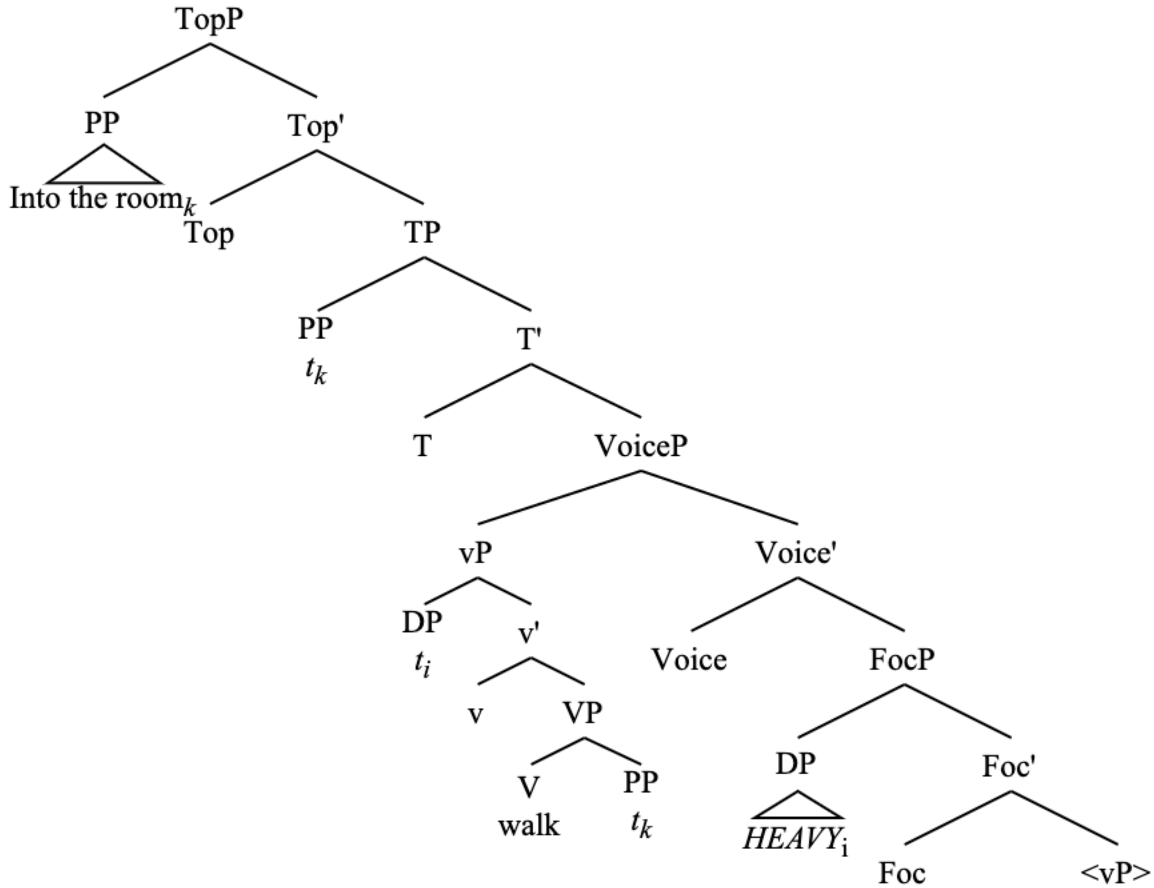
(320) Into the room walked *HEAVY*: PP moves to Spec,TP



Note that, in this structure, the entire vP – not just the VP – come to precede the thematic subject. This accounts for the word order of the data in (310) through (314) in which the vP-internal adjuncts come to precede the thematic subject.

Then, from here, the PP argument raises to the left periphery (again represented with TopP).

(321) Into the room walked *HEAVY*: PP moves to Spec,TopP.



The smuggling mechanism here is the exact same as it is in standard LI with unergatives. The only difference is the movement of the thematic subject to FocP, which causes the entire vP to precede the thematic subject. This movement of the subject to FocP is responsible for the differences between standard LI and HNPS. Also, though I do not overtly represent it in this structure, V-to-v movement is possible in this structure. The same technology of smuggling to VoiceP derives the inversion in both cases.

3.4.2 Modals and auxiliaries

Another striking difference between standard LI and HNPS LI is that HNPS LI allows for the presence of modals, which are not so acceptable without HNPS (Laparle 2020).

(322) a. ??/*Into this room could walk a man
b. Into this room could walk absolutely anybody at random from off the street
(323) a. ??/*Down the hill can roll the kids
b. Down the hill can roll every single kid in the whole class

(324) a. ??/*Through the doors will burst Deane
b. Through the doors will burst the most competent Air Force General in all 50 states

The same is true of auxiliaries (Richards 2016).

(325) a. ??/*Across the field is marching the band
b. Across the field is marching the incredibly well-rehearsed high school band

(326) a. ??/*Up Into the mountains have climbed the travelers
b. Up into the mountains have climbed countless people foolishly looking for adventure

These facts follow from the accounts of LI and HNPS LI given in this chapter.

Modals and auxiliaries are base-generated relatively low compared to where they usually end up (Homer 2011; Iatridou & Zeijlstra 2013; Jeretič & Thoms 2023). Although there is plenty of evidence for this in the referenced literature, one piece of evidence is that modals may appear below negation in multiple modal constructions in varieties of English such as Ozark English.

(327) We might not could leave

This is exactly parallel to what happens with multiple auxiliaries in standard English.

(328) Judith has not been eating

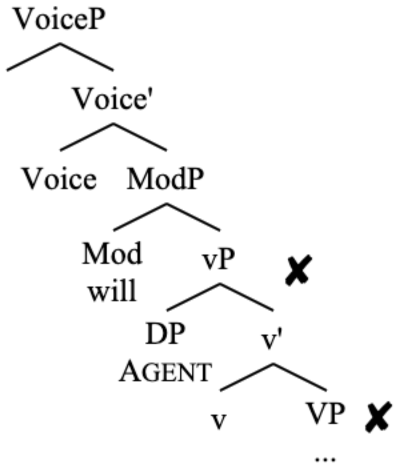
Given this low origin of modals and auxiliaries, it is possible to stipulate some constraints on the combinatorics of modals and auxiliaries.

Modals and auxiliaries respectively project the phrases ModP and AuxP. These phrases may take only argument-introducing verbal functional projections (as well as other ModPs and AuxPs) as their complements. As such, ModP cannot take VoiceP as a complement as VoiceP does not fit the criteria given here. It must therefore be the case that ModP appears below VoiceP[36]. This also speaks to the lowness of ModP/AuxP.

Trying to do inversion via smuggling with VoiceP in the presence of a ModP without HNPS is not possible. It would result at the modal appearing sentence-finally. There is no way that a constituent can be selected for movement that does not contain the thematic subject without violating minimality. Were a constituent containing the AGENT to be moved to Spec,VoiceP, the derivation would be functionally indistinguishable from one that does not involve smuggling via VoiceP.

[36] This means that ModP and AuxP must also bear a [V] feature, which is not an issue. They are both verbal projections and they both can appear as complements of T^0, which verbal projections such as vP and VP are also able to do. Therefore, whatever feature VP and vP have (e.g., [V]) must also be present on ModP and AuxP (lest we posit the existence of multiple different TPs that vary only in what complement they take).

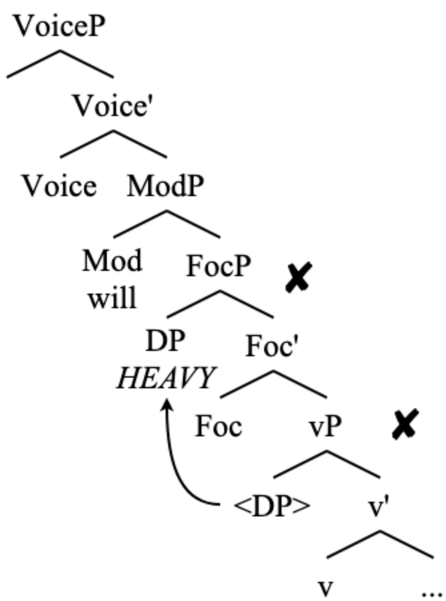
(329) Modals in LI without HNPS: minimality violation to move over AGENT



In this structure, the only targets for movement to Spec,VoiceP either contain the thematic subject or are not sufficiently local to Voice^0.

The addition of FocP allows a way out of this bind, but only if FocP is above ModP (again, FocP is not something that ModP can take as a complement). If ModP is above FocP, the same problem of either selecting a constituent that contains the Agent or creating a minimality violation needs to be dealt with.

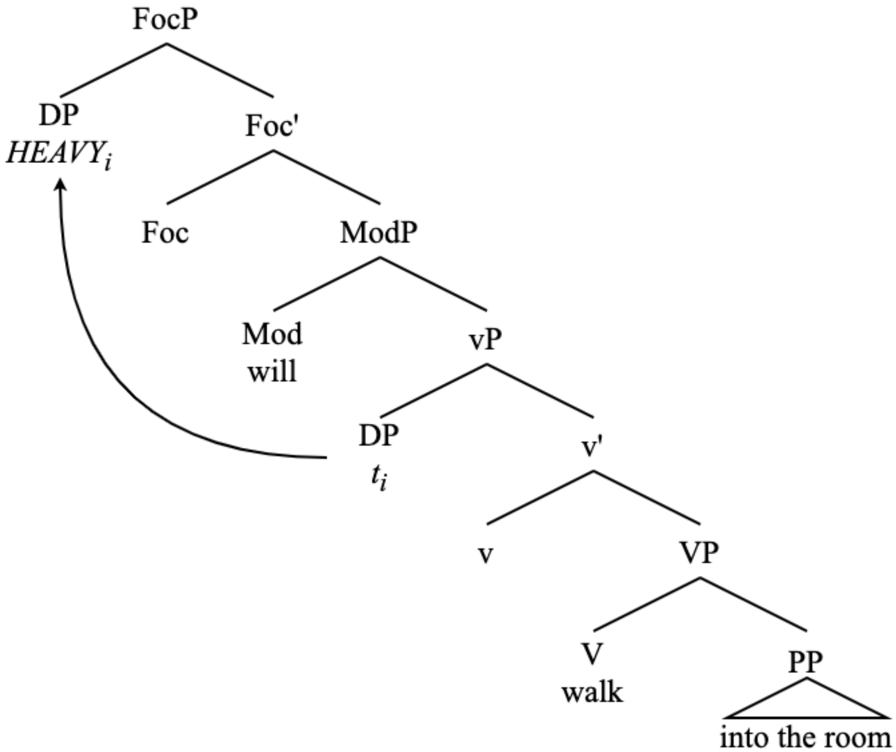
(330) ModP above FocP: minimality violation to move over AGENT



ModP is only licit in a smuggling derivation if it is below both FocP and VoiceP, and if it takes an argument-introducing thematic projection (or other ModP or AuxP) as its complement.

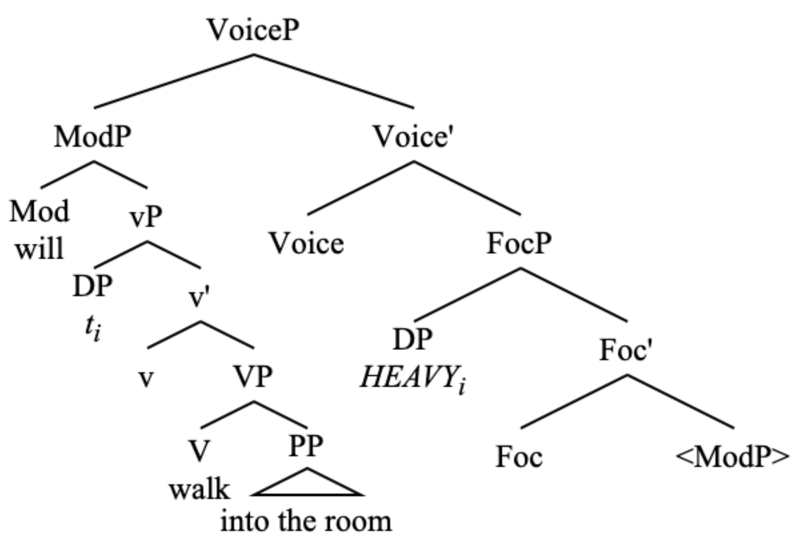
(331) Into the room will walk the amazing and illustrious superstar named Ted

(332) Into the room will walk *HEAVY*: heavy NP moves to Spec,FocP.



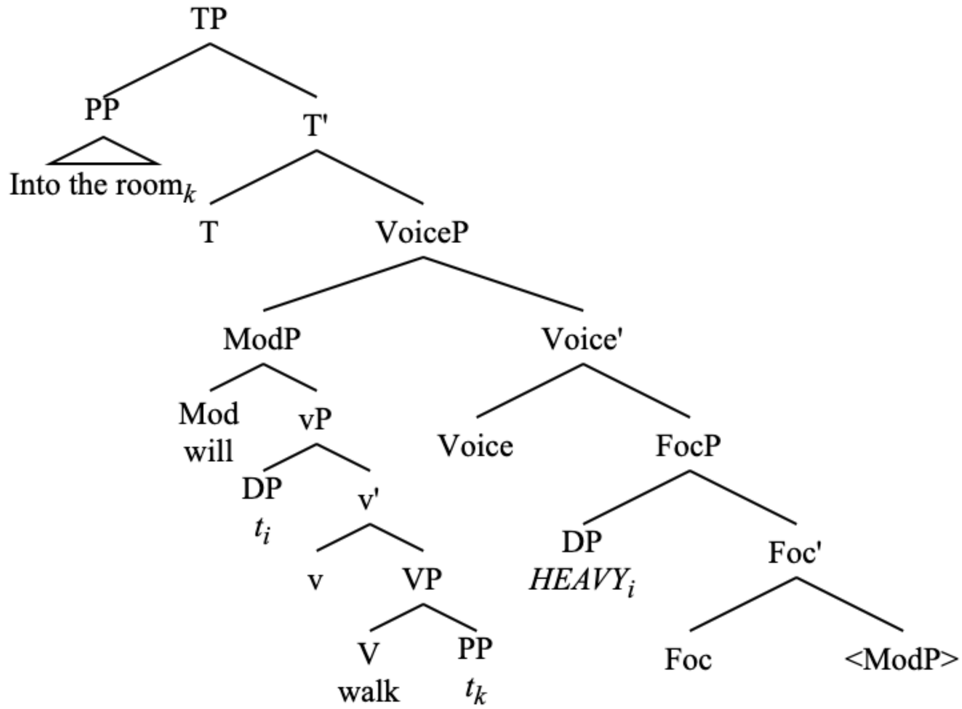
After this, VoiceP is merged above FocP and selects the closest eligible verbal projection for movement to its specifier, which in this case is ModP.

(333) Into the room will walk *HEAVY*: ModP moves to Spec,VoiceP



From here, T^0 is merged above VoiceP, and the PP argument is selected as the closest phi-goal for movement to Spec,TP, since the AGENT remains in Spec,FocP.

(334) Into the room will walk *HEAVY*: LOC moves to Spec,TP



After this, the PP moves to the left periphery. I do not show this here, but it works exactly as it does in all previous derivations.

Essentially, ModP and AuxP pose issues for LI constructions without HNPS as they interrupt the locality conditions necessary for LI to occur. This interruption is only remedied by the presence of a FocP to which the AGENT or THEME subject moves. That is why modals and auxiliaries do not appear without FocP.

This explanation, of course, carries the stipulation that all unaccusative LI constructions are essentially HNPS/focus constructions as well. This is certainly supported by the data showing that secondary depictive predicates may appear on either side of the THEME (252), which behaves identically to HNPS with unergatives, but not normal unergative LI derivations.

It is additionally true that modals and auxiliaries seem to be much more compatibles with unaccusatives without a heavy NP THEME than they are with non-HNPS unergative LI sentences.

(335) a. At the station has arrived a train
b. On your screen will appear an ad

These sentences can be contrasted with the unacceptable (a) examples in (322) through (326).

This analysis involving FocP will be especially relevant to the discussion in the following chapter on existential constructions and specificational predicate inversion. Before moving on to that, however, it is necessary to discuss an aspect of certain inversion constructions (aside from existentials and specificational predicate inversion), which is the transitivity constraint.

3.5 Quotative Inversion and the Transitivity Constraint

While it appears that many of the verbs that can participate in LI can also be transitive verbs, transitive objects are forbidden in LI constructions in English (Diercks 2017). This is known as Transitivity Constraint[37].

(336) a. Cristie walked the fluffy dog out into the hallway
b. The criminal ran the car into the lake

(337) a. *Out into the hallway walked the fluffy dog Cristie
b. *Into the lake ran the car the criminal

Switching the position of the subject and the direct object does nothing to improve these sentences.

(338) a. *Out into the hallway walked Cristie the fluffy dog
b. *Into the lake ran the criminal the car

There is a similar property in quotative inversion (henceforth QI) (Collins & Branigan 1997; Collins 2003).

(339) a. "What's on?" John asked Mary (Collins & Branigan 1997:20)
b. "The wind is too strong," the navigator told the skipper
c. "The plane will arrive on schedule," the stewardess assured the mob

(340) a. *"What's on?" asked John Mary (Collins & Branigan 1997:21)
b. *"The wind is too strong," told the navigator the skipper
c. *"The plane will arrive on schedule," assured the stewardess the mob

Again, switching the position of the subject and indirect object does not lead to grammaticality.

(341) a. *"What's on?" asked Mary John
b. *"The wind is too strong," told the skipper the navigator
c. *"The plane will arrive on schedule," assured the mob the stewardess

It is not immediately clear why the mechanism that derives LI should be unavailable for transitive verbs. The claim that all LI verbs must be unaccusative provides an easy explanation for this. If LI requires an unaccusative verb, and transitive verbs are by definition not unaccusative (Burzio 1986), then transitive verbs are automatically barred from participating in any LI derivation. This explanation, however, is no longer adequate since I have already shown that there is no requirement that LI verbs be unaccusative. Additionally, there is no reason at all to believe that verbs of quotation do not project an AGENT in Spec,vP.

However, there is a way to circumvent the Transitivity Constraint in quotative inversion, by using dative PPs.

[37] The terminology of 'Transitivity Constraint' comes from Collins & Branigan's (1997) analysis of quotative inversion.

(342) "Are you hungry?" asked Frodo to Sam

Thus, it is not the case that all tri-argumental structures are forbidden from undergoing inversion, nor is it the case that they are disallowed from doing so because of their thematic status. It is sufficient for now to point out that the Transitivity Constraint in inversion constructions cannot be accounted for under a general ban on certain thematic structures (be it non-unaccusatives or any class of transitives).

In this section, I first show that QI is derivable from the exact same mechanism of smuggling via VoiceP, and the empirical support for this claim. Then, I discuss a formulation of the Transitivity Constraint (henceforth TC) that accounts for its apparent exceptions.

3.5.1 Quotative inversion

QI is similar to LI in many ways; so much so that they warrant the same analysis. Here I show that QI is best accounted for as a voice construction involving smuggling. I motivate this word order tests and A-movement diagnostics, agreement alternations and restrictions, and a discussion of the syntactic category of the quote.

3.5.1.1 Word order tests and A-movement diagnostics

The word order facts concerning elements such as particles, secondary depictive predicates, control clauses, and purpose clauses are the same in QI as they are in unergative LI sentences. First, particles (which are VP-internal) cannot follow the postverbal thematic DP in QI (Collins 2003).

(343) a. "Where are you going?" called (out) Robert (*out)
b. "I want to go too!" chimed (in) Laurie (*in)

Secondary depictive predicates – which are vP-level adjuncts when they modify an AGENT – must follow the thematic subject (when there is no HNPS).

(344) a. "D'oh!" said (*drunk) the man (drunk)
b. "Whoa, man" said (*high) Madeline (high)

Control clauses, which are VP complements, must precede the thematic subject, just as in LI.

(345) a. "Help!" tried (to yell) Mary Kate (*to yell)
b. "I miss you," wanted (to say) Stephen (*to say)

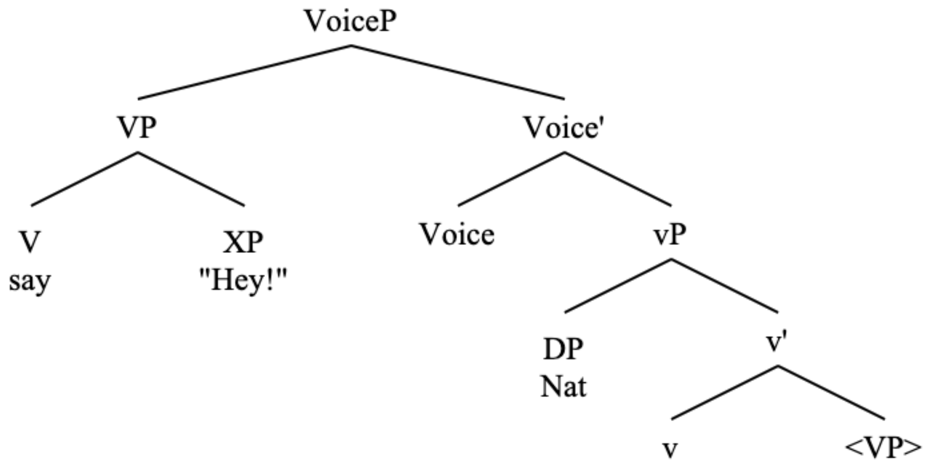
Adjunct purpose clauses, which are vP-level adjuncts when the controller of the embedded subject is the matrix AGENT, must follow the thematic subject when there is no HNPS.

(346) a. "Fire!" cried (*to get the crowd's attention) Taylor (to get the crowd's attention)
b. "Hey!" yelled (*to establish his dominance) Al (to establish his dominance)

These tests show that, exactly as in LI, there is phrasal movement of the VP to a position over the thematic subject. This is once again accomplished via smuggling to Spec,VoiceP. We do not need a FocP in QI because there is no issue of minimality that comes up with these agentive verbs and smuggling. Furthermore, it is definitely not the natural state of quotative inversion to have a focus reading on the postverbal subject. Let’s return to our wh-question test.

(347) Q: Who said “I love you”
A: #“I love you” said John

(348) a. “Hey!” said Nat
b. Smuggling to Spec,VoiceP (Collins 2003)



From here, the quote is able to undergo A-movement to a position above the verb, which I claim is Spec,TP. Evidence that the quote undergoes A-movement is that QI structures fail to license parasitic gaps (Murphy 2022).

(349) a. “We should leave,” Max thought without actually saying (Murphy 2022:4)
b. *“We should leave,” thought Max without actually saying

Another piece of evidence of the A-movement of the quote comes from raising predicates. See the following data.

(350) a. “Where did the wheat go?” seemed to say the heron (Flickr)
b. “He…hello…” seemed to say someone behind the door (Wattpad)
c. “We shall live and see”, seemed to say Lenin in the Great Initiative or in the State and Revolution (Revista Cogito)
d. “Who are you, what do you want,” seems to ask this little guy feeding under my bird feeder (Forum)
e. “Sorry”, seemed to say the elephant with the cry elephants usually do (Forum)

Additional evidence that QI involves A-movement of the quote comes from the fact that the movement of the quote to the preverbal position creates new A-dependencies for Agree and Case assignment. See the following section.

3.5.1.2 Agreement and PCC effects

Just as with LI, the data on QI conform to the generalizations on agreement laid out in Chapter 2, and are analyzable under the framework presented there. I support this again with data collected en-masse from Internet searches and confirmed as grammatical by native English speakers, which fall under the following paradigm.

(351) a. "Hey!" {say/says} the kids
b. The kids {say/*says) "Hey!"

Verbal agreement in QI with a third person plural AGENT may either track the features of the AGENT or appear as a default (third singular). I present the following examples of default agreement with a third person plural postverbal subject understanding that these sentences are all grammatical with third person plural verbal agreement as well. See the following examples with plural postverbal subjects and third person singular verbal agreement with full DPs (352) and bare NPs (353).

(352) a. "God bless you" says the men (Database)
b. "Go ahead! Spread that seed! Make more of me!" says the men's genes (Blog)
c. "Where does it come from" asks the children (Museum piece)
d. "What for?" asks the detectives (Newspaper)
e. Quack Quack goes the Ducks, Moo Moo goes the Cows // Hop Hop goes the Rabbits and Cluck Cluck goes the Chickens (Pet food store)
f. "Thank you very much" - says both organisms (Twitter)
g. "Mrs carter! these games were so fun thank you!" says the kids who were absolutely behaving during the halloween party in your class and not hyper at all (Etsy review)
h. It's my Grandfather's birthday!! "Happy birthday to you!" sings the singers. (Blog)
i. "Virginity is made up" says the people who think there are 200 genders. (Twitter)
j. "Time to get up" says the boys (Twitter)

(353) a. "It can send a big clot running up to your lung, where it can kill you," warns Doctors (Blog)
b. Do you squeeze your spots? Doing so could KILL you, warns experts (Blog)
c. "Many aspects of our psychology have evolved to be flexible rather than best suited to one specific way of life", warns experts (Blog)
d. Elderly motorists may be caught out by new DVSA eyesight proposals in a concern for those later in life, warns experts. (Blog)
e. Trump's nuclear weapons strategy could be dangerous, warns experts (Blog)
f. "Umm, can you really do that and not kill your career" asks parents everywhere (News)
g. "Let me help you create a killer resume to beat the ATS" - says people who sell resume writing services. (LinkedIn)
h. "Anyone who doesn't share my exact opinion can't be a member of the same broad political grouping as me" says people who continue to import terms like TINO directly from other Parties in other countries (Reddit)
i. "Hey, I bet they're out to get me!" says people who are out to get others. (Blog)
j. "My5tery Music will bring out the Musical My5tery in you" says three of our new step-team participants of Old West End Academy (Facebook)

The same weak PCC effects outlined in Chapter 2 are also present in QI derivations. See the following examples with the first person singular pronoun (354), other pronouns (355), and with verbs other than 'say' (356).

(354) a. I wash my car too much, says me (Facebook)
b. Hope I didn't lame it up too much, says me (Blog)
c. Island gorls we up! (Says me, a person who has never been on an island) (Reddit)
d. Fire that sucker up, says me! (Reddit)
e. "Sorry I can't do PE today my knee is playing up" said me with my tubular knee bandage on (Instagram)
f. "I love you" said me to me (TikTok)
g. The industry could learn a thing or two about purpose-driven initiatives like these, says me (LinkedIn)
h. "Well yes, but see if you can get a fee out of them", says me (Blog)
i. "I'm not gay" - says me, a gay guy (Reddit)
j. "I don't get it," says me. "How could Derrida die? He was a social construct." (Blog)

(355) a. "Probably too much" says you! (Facebook)
b. "What are we up to? Nothing much" said us never (Instagram)
c. "Giddy up" says us (Instagram)
d. mail slots on the street? Mail and packages go there, says them (Reddit)
e. "I'm up. I'm up" says him, very OBVIOUSLY NOT UP. (Blog)

(356) a. So what to do? What to do? wondered me. (Literary collection)
b. "Someone keeps moving the pot of gold, someone is not getting the memo, I'd rather be close to you." sings me to my India (Blog)
c. "How are you?" asks me (Blog)
d. "Pop!" goes me (Song lyrics)
e. "Hello!" answered me from beneath the blankets (Art website)

This weak-PCC configuration is the only possible configuration of agreement and Case morphology in retort-style sentences which do not overtly feature a preverbal quote, but make reference to a quotation in the discourse (Lee-Goldman 2011).

(357) a. Says us! (Video)
b. Says them! (Video)
c. Says me! (Video)
d. Says you! (Video)
e. Oh yeah? Says y'all (Arkansas Democrat Gazette)
f. Oh yeah, says him, and her. (Blog)

(358) a. *Say {we/they/I/you/y'all}!
b. *Says {he/she}!

In contrast with all of the previously mentioned constructions, however, third singular verbal agreement with a nominative postverbal subject pronoun is also possible, quite well attested[38].

(359) a. "I bet you can't spell my name," says I (Huckleberry Finn)
b. "I bet you what you dare I can," says he (Huckleberry Finn)
c. "All right," says I, "Go ahead" (Huckleberry Finn)
d. "To be sure we do" says we; for we never thought of playing such a dirty trick (The Catholic Layman)
e. "How far you going?" says she with a toss of her rainbow mane (Blog)
f. "You never give us none" says they. (Newspaper)
g. "There's a twenty pound reward" says you (Besant 1902)
h. "You're a fool" says you (Besant 1902)
i. "Let me only just find a door-mat," says you to yourself (Blog)
j. Oh, yes, says we to ourselves, Sydney's the place! (Blog)

(360) a. "Now what do we do?" asks I (Blog)
b. "Should I be afraid of you?" asks I over plastic bowls of salad later in the day (Blog)
c. Where are you from? Asks I (Twitter)
d. *Well*, asks I of myself, *what shall I find?*(The National Magazine)
e. "Why," thinks I, "what's the row? It's not a real leg, only a false leg." (Moby Dick)

It is not the case, however, that verbal agreement may track these postverbal nominatives.

(361) a. *"What do we do now?" ask we
b. *"I wonder what's over there" think I to myself
c. *"I feel great!" scream you after a few drinks

Also impossible is agreeing in phi-features with a non-nominative postverbal first or second person pronominal.

(362) a. *"What do we do now?" ask us
b. * "I wonder what's over there" think me to myself

To explain the data in (359) and (360), I propose that the postverbal subjects in these examples do not receive nominative Case at all, just as they don't in (354) to (357), despite their seemingly nominative morphology. Caha (2024) notes that pronominals may alternate between nominative

[38] Third person singular agreement on the verb with a nominative subject with differing phi-features is also possible in a non-inverted context in some dialects.

vi. a. "Waiter!" they says "We're in a 'urry!" (Newspaper)
b. "What?" I asks (Babes in the Jungle)
c. Like you says to me, if you get in any trouble they'll sen' you back to McAlester (Grapes of Wrath)
d. That's wot I whispers to meself at night. (The Songs of a Sentimental Bloke)

I take this to be a separate puzzle from the agreement phenomena seen in quotative inversion, as it appears in a far more limited number of (usually archaic as well as highly colloquial) contexts in a more limited number of dialects. Perhaps this possibility for agreement in the non-inverted sentences was retroactively influenced by the agreement possibilities that inversion creates. Most agentive verbs do not allow this.

and accusative case morphology in certain “default” environments where DPs are said to get unmarked Case (Schütze 2001). See the following environments in which pronouns may show either nominative or accusative case morphology.

(363) a. {Us/we} linguists are a crazy bunch (Schütze 2001; Caha 2024)
b. She likes rice, and {him/he} beans
c. {Her/she} and Sandy went to the store yesterday

Although, this is notably not possible in every “default” Case environment.

(364) a. Who wants to try this game? – {Me/*I}. (Schütze 2001; Caha 2024)
b. {Me/*I}, I like beans
c. Poor {me/*I}

The alternation in (363) has been described elsewhere (Quinn 2005). The alternation between nominative and accusative case morphology in “default” Case environments is analyzable as a difference across registers – a high-prestige register in which the pronouns surface with nominative Case, and a low-prestige (or “normal use”) register in which the pronouns do not appear with nominative Case (Emonds 1986; Sobin 1997; Grano 2006; Parrott 2009; 2021; Lemon 2017). This seems to be related to the fact that, in historical varieties of English, the postverbal subject of QI constructions could appear as nominative and Agree with the verb (which suggests a “true” nominative in these varieties).

(365) a. “I will do this or that,” say I to myself (Confessions of St. Augustine, trans. 1800s)
b. “We shall still drink water” say they to themselves (Freemason’s Magazine, 1795)
c. Say you to yourselves, then, “Our Lord will not always endure this idolatrous popery” (Dagon’s Ups and Downs, 1887)

The availability of this kind of Case morphology for postverbal subjects in QI – but not in LI or existentials – seems related to the fact that QI is highly restricted to a formal, literary/narrative register in a way that these other inversion constructions are not (though LI also has a strong contextual restriction). The nominative morphology of this postverbal subject is then a kind of diachronic fossil and does not actually reflect underlying syntactic Case.

These data show that Agree and Case assignment work exactly the same in QI as they do in the other constructions presented thus far in this dissertation. This is further evidence that these constructions warrant the same analysis. After briefly discussing the syntactic category of the quote, I present the analysis of QI as smuggling.

3.5.1.3 The quote
In QI, the quote typically appears preverbally, but may also appear postverbal or may be split up across a preverbal or postverbal position.

(366) a. Said the night wind to the little lamb, “Do you hear what I hear?”
b. “But,” said Frodo, “That is a long way off”

Other types of subjects – e.g., DPs and CPs – cannot do this. Thus, the question of what syntactic category a quote belongs to is a tricky on, to the point where many authors simply designate it as "XP" (den Dikken 2017; and has I have done earlier in this section) or do not comment on the category of the quote at all (Collins & Branigan 1997).
Quotes – or direct speech reports – can be thought of to be in complementary distribution with indirect speech reports. Indirect speech reports cannot participate in QI.

(367) a. Judith told me "I love you"
b. Judith told me that she loves me

(368) a. "I love you" said Judith
b. *That she loves me said Judith

Indirect speech reports are in complementary distribution with nominals.

(369) a. Judith said that she loves me
b. Judith said {something/it/that/etc.}.

These nominals cannot participate in QI.

(370) *{Something/it/that/etc.} said Judith (*int*: Judith said {something/it/that/etc.})

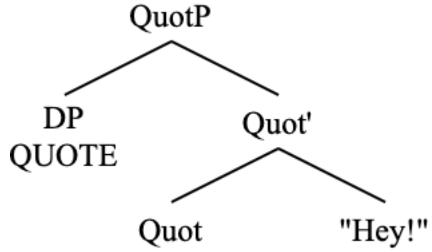
Some nominals, however, my cooccur with a quote (but not an indirect speech report) (Staps & Rooryck 2023).

(371) a. Ann said something shocking: "I love you"
b. Ann said this: "I love you"

(372) a. *Ann said something shocking: that she loves me
b. *Ann said this: that she loves me

Given the nominal status of quotes (Partee 1984; Rabern 2022), I propose a functional projection QuotP that pairs a referential light null DP QUOTE with the actual content of the quotation (which itself does not bear a syntactic category). This DP can be compared with the quotative operator of Collins & Branigan (1997). The DP may alternate with overt ones, like in (371).

(373) Internal structure of QuotP



The light noun QUOTE is phi-deficient (only specified as third person) in the same way as the locative PP and other light nouns. The fact that QuotP itself is not a DP – but rather contains a DP – explains how quotes can appear with predicates that do not select DP internal arguments, such as 'boast'.

(374) a. Merry boasted, "I'm the taller one"
b. "I'm the taller one," boasted Merry

(375) *Merry boasted {something/that/it/etc.}

I assume that the quote (and the DP contained within the quote) are phi-deficient in the same way that the locative PP of LI constructions is. Evidence of this is that quotes cannot be pluralized.

(376) * "Get out of here!"s yelled Pippin

Even in a coordinate structure, quotes cannot trigger plural agreement in QI constructions.

(377) *"I'm sorry" and "I love you" say Sam

Furthermore, the nouns which can appear alongside quotes in (371) cannot be pluralized, either.

(378) a. *Ann said somethings shocking: "I love you" and "I'm sorry"
b. *Ann said these: "I love you" and "I'm sorry"

As such, the DP 'QUOTE' is phi-deficient for number, and is third person.

This functional structure of QuotP is especially necessary to account for the facts in (366), especially the (b) example. In no other derivation – even in other inversion constructions – is it possible to have the subject material extraposed to the right like this.

(379) a. *Walked John into the room
b. *Is Trump the president (non-question)

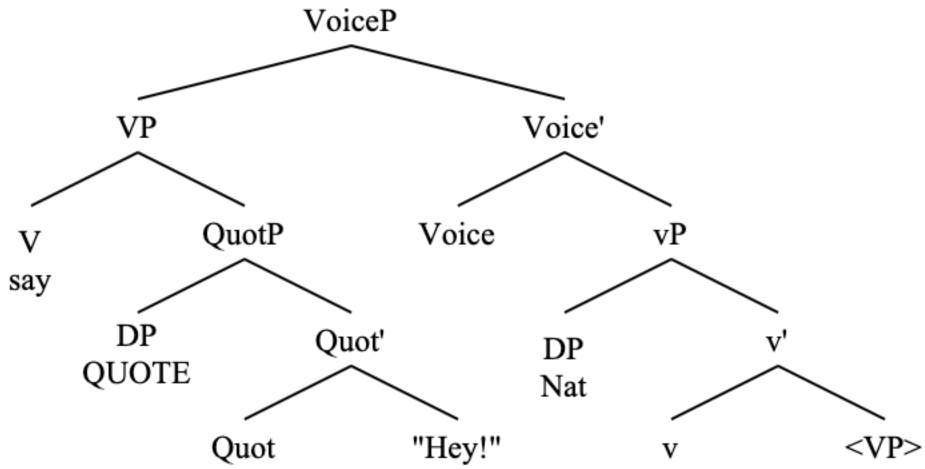
This is unique to quotative inversion. As such, QI must have a structure in which an element can occupy Spec,TP that is distinct from the quote itself. As for this element being null in (366) (and indeed in most QI derivations), it must be the case that the movement of a constituent that contains a trace of the moved subject in Spec,TP to an adjacent specifier position that c-commands the Spec,TP position licenses the null DP in that position, whereas it would not normally be licensed there. This is because, in this configuration, the interpretation of the null structural subject can be easily recovered due to the structural proximity of the light noun and QuotP to one another, which is not the case in (288). Such a condition is needed in English to allow for a null element in Spec,TP, which is normally not allowed since English is not a null subject language. In a null subject language such as Spanish, something like this configuration has nothing to do with whether or not a null element can be in Spec,TP as that is always allowed.

This structure is what I assume for the internal structure of quotes that appear as VP-internal arguments. As such, we can now revisit the QI derivation.

3.5.1.4 QI derivation

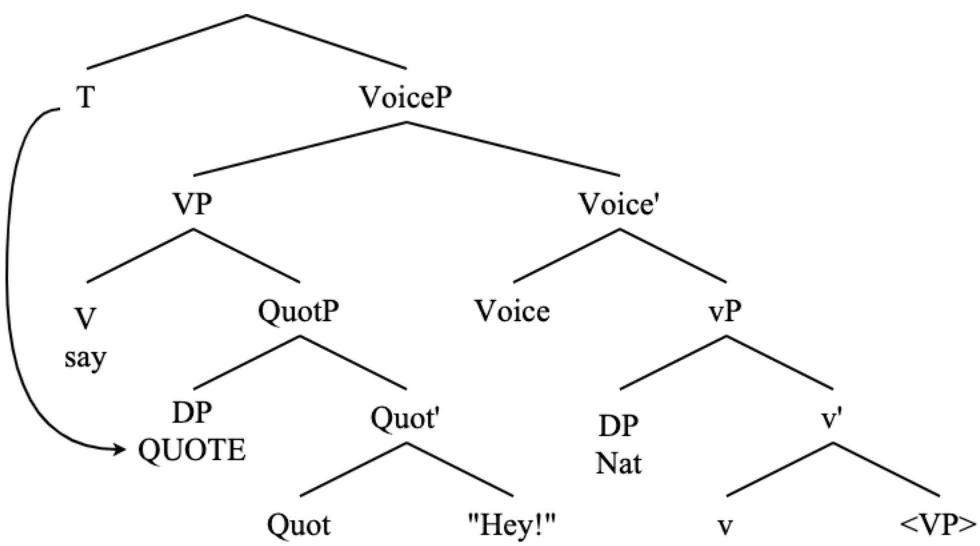
I now return to the structure presented in (348), this time with the internal structure of the quote.

(380) '"Hey!" said Nat' with internal structure of the quote



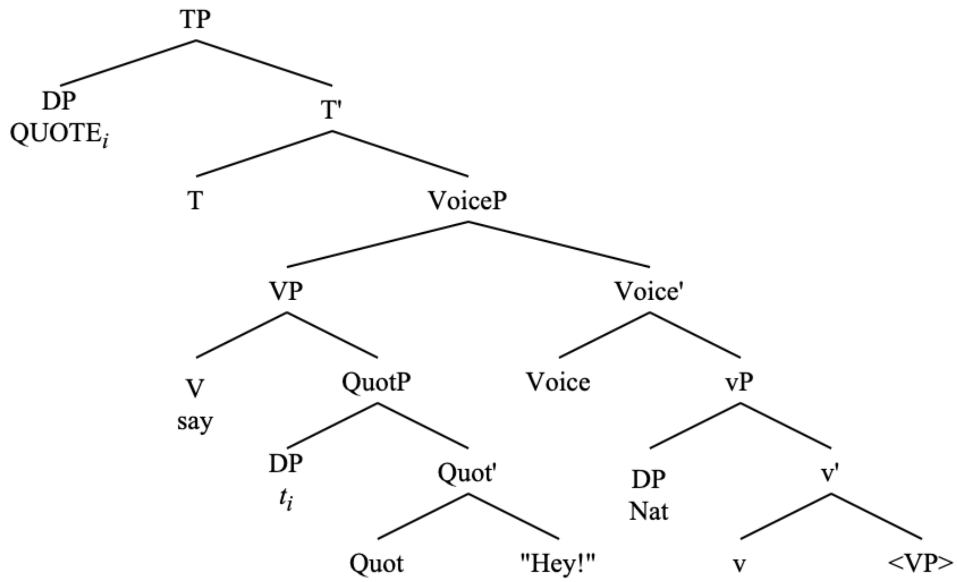
At this point, T^0 is externally merged above VoiceP and probes for a phi-bearing goal, at which point it finds the quotative DP.

(381) T^0 probes, finds QUOTE



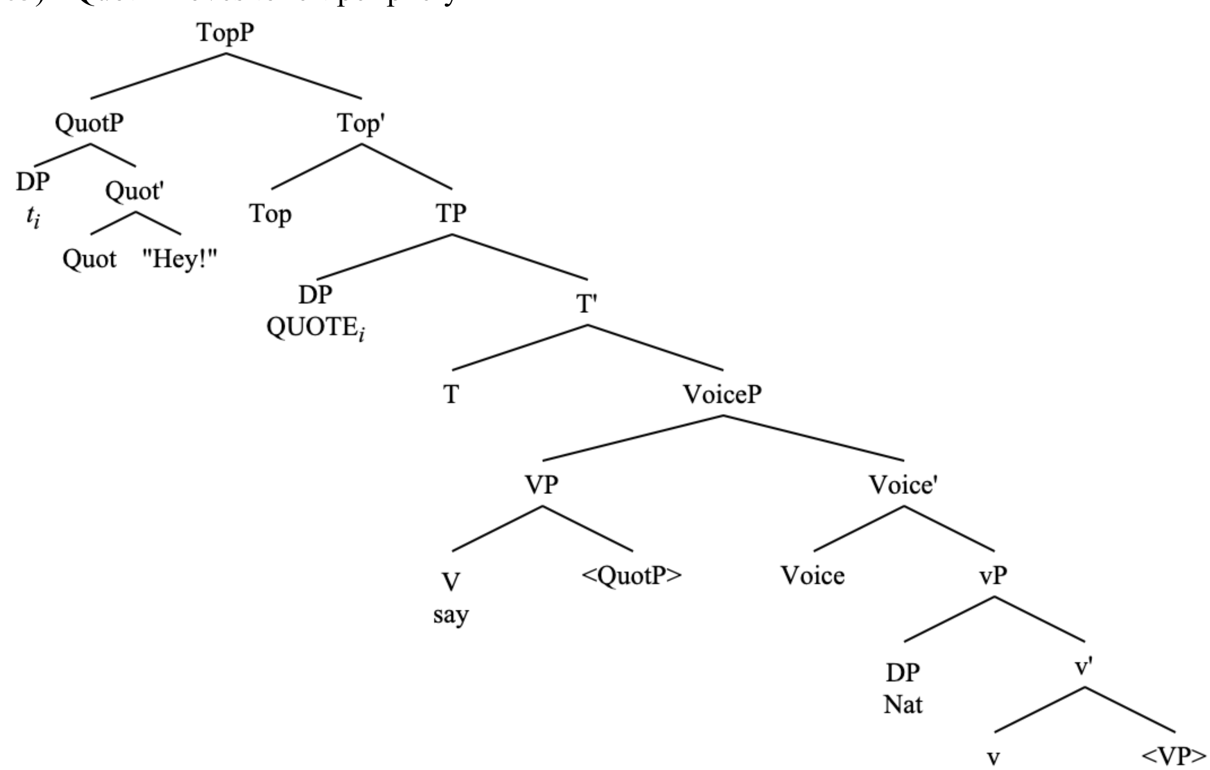
The quotative DP then moves to Spec,TP.

(382) QUOTE moves to Spec,TP



At this point T^0 may either stop probing (and spell out default verbal agreement), or it may probe again to do defective circumvention, which would cause verbal agreement to track the postverbal subject (unless that is prevented by a conflicting feature on the lower goal). The last step is that the QuotP moves to an information-structural projection in the left periphery such as TopP. This movement licenses the null subject in Spec,TP and derives the pragmatic properties of QI.

(383) QuotP moves to left periphery



This derivation is very similar to that of LI. As such, the two constructions are derivable from the same technology of smuggling via VoiceP and defective circumvention. Furthermore, QI also has a HNPS variant (whose derivation is identical to the HNPS derivation of LI with unergatives). See the following evidence.

(384) a. “D’oh!” said (*drunk) the man (drunk)
b. “D’oh!” said (drunk) the man from the bar who I had never met before (?drunk)

(385) a. Fire!” cried (*to get the crowd’s attention) Taylor (to get the crowd’s attention)
b. “Fire!” cried (to get the crowd’s attention) the only observant woman in the entire room (to get the crowd’s attention)

This is further supported by the same contrast with modals and auxiliaries seen with LI.

(386) a. ??/*“My preciousss” was saying Gollum
b. ??/* “Don’t tempt me, Frodo” has said Gandalf
c. ??/*“What the fuck?” would think my mom
d. ??/*“I don’t know, can you?” may ask the teacher

(387) a. “My precioussss” was saying the slimy creature who lives in the Misty Mountains
b. “Don’t tempt me, Frodo!” has said the only gray wizard to ever visit the Shire
c. “What the fuck?” would think anyone who reads my diary without permission
d. “I don’t know, can you?” may ask the condescending teacher who likes trouble

I do not show it here, but the facts on HNPS in QI are identically explainable with movement of the AGENT to Spec,FocP before smuggling takes place.

Given the undeniable and numerous similarities between QI and LI, it is now possible to discuss how to handle the TC in both cases.

3.5.2 The Transitivity Constraint

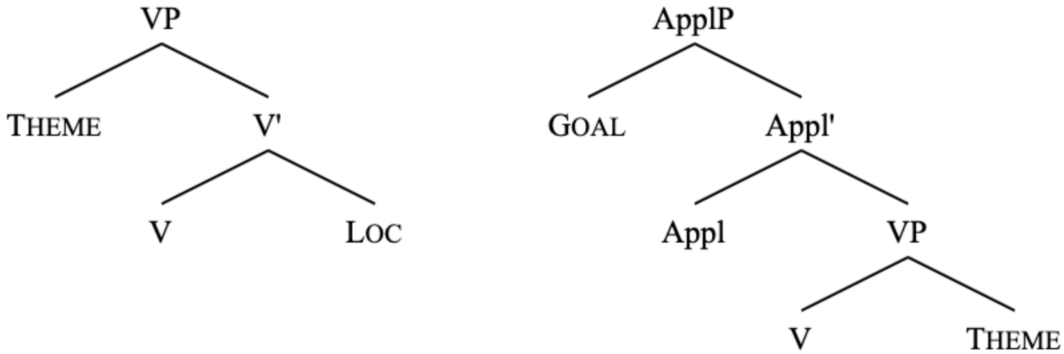
The TC is a restriction that forbids LI and QI from taking place when there is more than one internal argument in these derivations: a THEME and a LOC in the case of LI, and a GOAL and a THEME in the case of QI.

(388) a. Cristie walked the dog into the room
b. *Into the room walked the dog Cristie

(389) a. Gustavo asked Daniel, “What’s up?”
b. *“What’s up?” asked Daniel Gustavo

I assume the following structures for the projection of these arguments, using an applicative head to introduce the GOAL (Pylkkänen 2008). The left structure shows the projection of a THEME together with a LOC, and the right structure shows a GOAL together with a THEME.

(390) Projection of more than one internal argument with locatives and quotes



As such, the higher arguments serve as interveners for accessing a deficient lower argument. The higher internal argument blocks the inversion of the defective element (the LOC in LI, the quote THEME in QI). This is shown for (388b) in (391), and for (389b) in (390).

(391) [T^0 … [$_{\text{VoiceP}}$ [$_{\text{VP}}$ [$_{\text{DP}}$ the dog [$_{\text{V'}}$ [[$_{\text{PP}}$ *into the room*]]$_i$ … [$_{\text{vP}}$ [$_{\text{DP}}$ *Cristie*] … [t_i]]]]

(392) [T^0 … [$_{\text{VoiceP}}$ [$_{\text{ApplP}}$ [$_{\text{DP}}$ *Mary*] [$_{\text{VP}}$ [$_{\text{QuotP}}$ *"Beans!"*]]]$_i$ … [$_{\text{vP}}$ [$_{\text{DP}}$ *John*] … [t_i]]]]

Given the discussion in the end of section 3.2.3 on the purpose of these smuggling derivations with locative and quotative predicates, which is to move the deficient element to a position where it is a viable candidate for A-movement, these structures fail. There is no way in these structures to A-move the deficient arguments, and agreement with the thematic subject is not possible at all.

As such, the higher internal argument must first be moved out of the way in order for the smuggled VP to allow the deficient internal argument to become visible for A-movement.

One way to do this is with dative shift with quotative GOALS, going from a double object construction to having a prepositional dative (Collins 2021; 2024). With a prepositional dative, which involves the movement of the entire VP over the Goal argument, QI is then possible.

(393) a. Gustavo asked "What's up?" to Daniel
b. "What's up?" asked Gustavo to Daniel

QI becomes possible because, in the prepositional dative configuration (Collins 2024:ch 9), the VP moves above the GOAL to become a viable target for smuggling to VoiceP without the GOAL, which is impossible in the double object configuration (389).

Dative shift is not available for the ditransitive LI configurations as they do not involve a GOAL argument. This does not mean, however, that the TC in LI configurations is completely insurmountable.

The THEME in ditransitive locative constructions can be extraposed via HNPS. We can displace the THEME to outside of the VP with HNPS.

(394) McGovern ushered into the room all manner of Americans who had previously been marginalized politically because of who they were or the ideas with which they identified.

While (388b) is still very bad, the following is not so bad.

(395) Into the room ushered McGovern all manner of Americans who had previously been marginalized politically because of who they were or the ideas with which they identified.

Just as dative shift allows the VP to move above the GOAL in LI, HNPS allows the VP to move above the THEME in ditransitive LI constructions, since HNPS involves the movement of the THEME to a low FocP that the VP then can be moved around. Both of these “shifts” allow the VP to be moved over the higher internal argument. For a formal implementation of HNPS and dative shift involving smuggling, see Collins (2024c) for HNPS, and Collins (2021; 2024a; 2024b) for dative shift.

The TC (which should perhaps be more accurately named the Ditransitive Constraint) holds, then, because the position of the THEME in LI and the GOAL in QI block the deficient element from being smuggled above the thematic subject. The TC, then, is not a restriction on the projection of certain thematic arguments in these derivations at all; rather, it is a restriction on derivations that are not able to make the deficient internal argument a viable goal for A-movement. This explanation of the TC adds to the ways in which QI and LI are similar.

3.6 Summary

In this chapter, I have shown that LI is best analyzed as a voice phenomenon involving smuggling. There are empirical and theoretical justifications for this analysis.

Several properties of LI constructions indicate clearly that these constructions involve A-movement of the locative. My primary piece of evidence for this is the agreement and Case assignment data that is distinct from non-inverted locative sentences. Additionally, I present evidence in the form of obviation of weak crossover, the failure to license parasitic gaps, and interaction with raising predicates.

The mechanism of inversion via smuggling driven by VoiceP sufficiently accounts for the data on LI, including the thematic structure of LI verbs, the agreement facts in LI, and the syntax of HNPS. This analysis is empirically supported by the distribution of VP-internal constituents such as verb particles and control clauses in LI constructions, which indicate VP movement over the external argument.

This analysis is subsequently extended to QI constructions, which structurally resemble LI derivations in many key ways, showing that these two constructions are derivable by the same mechanism of inversion via smuggling via VoiceP, which allows for the agreement alternation of defective circumvention. Furthermore, these constructions show the same properties of A-movement of the fronted element. I have thus shown a technical implementation of smuggling via VoiceP across a variety of related constructions, showing that this formulation of VoiceP accurately accounts for locative inversion and quotative inversion.

In the following chapter, I show that this mechanism of inversion as voice as smuggling accurately accounts for other types of inversion constructions as well: namely existential ‘there’ constructions (which I show are really a subtype of LI) and predicate inversion, all of which exhibit the same basic properties including the agreement alternations, PCC effects, and VP phrasal movement. I further show that both existential inversion and predicate inversion have distinct information structural properties which I derive in the same way that I derive postverbal subject movement in unaccusative LI derivations.

4 Existential 'there' sentences as inversion

In this chapter, I extend the analysis of LI and QI constructions as inversion phenomena involving smuggling via VoiceP in Chapter 3 to existential constructions involving the so-called expletive 'there'. This 'there', which I treat as a locative, is crucially not deictic, and is instead existential. This is the same 'there' element that I have described in section 3.2.2. I distinguish two sub-classes of existentials in English: one involving unaccusative verbs (396a), and another involving the copula (396b).

(396) a. There came a soft knock at the front-door
b. There was a splendid supper for everyone

I show in this chapter that the unaccusative existential clauses are functionally the same derivation as LI with unaccusative verbs (with the caveat that the locative element is *existential*, not deictic, and syntactically *weak* (Cardinaletti & Starke 1999; Tortora 2014a)), in that they both involve movement of the THEME to Spec,FocP and inversion of the VP via VoiceP. I detail the featural and semantic differences of standard locative PPs and the deficient PP containing existential 'there', effectively explaining the restrictions on the distribution of the existential 'there' (i.e., it cannot appear as an adjunct, it cannot appear in object positions, it can only appear with unaccusative verbs).

This analysis explains the similarities observed between existential sentences and inversion sentences such as LI and QI; namely, that existential sentences show the same agreement alternations (397) and PCC effects (398), and the same restriction on the distribution of VP-internal elements such as particles (399).

(397) a. There {appears/appear} several solutions
b. There {was/were} many problems

(398) a. There {arrives/*arrive} me
b. There {was/*were} you

(399) There popped (up) a warning message (*up)

Furthermore, I detail an analysis of predicate inversion (henceforth PI) constructions, namely specificational copular sentences which are an inverted variant of predicational copular structures (Higgins 1976; Moro 1997; Heycock 2012; den Dikken 2017; Hartmann & Heycock 2020; Gravely et al. 2024; Krapp López & Gravely 2024; a.o), showing that existential sentences containing the 'there' subject with a copula are an inverted predicate structure (Moro 1997) also derived via inversion via smuggling via VoiceP.

First, I present evidence that 'there' cannot be an expletive, and that the idea of 'there' as a syntactic expletive is not useful (section 4.1). I then defend the idea of 'there' as a thematic argument (section 4.1.2). Next, I detail the analysis of EI with unaccusative verbs involving smuggling via VoiceP (section 4.2). I then take a slight detour to frame PI in the theory of inversion presented in this thesis (section 4.3), before casting EI with the copula under this theory as well (section 4.4).

4.1 There is no (expletive) 'there' there

The idea of an expletive as a semantically vacuous syntactic placeholder goes back to Chomsky (1981) (perhaps even earlier) and is a standard assumption in the field today (Chomsky 1981; 1995; Lasnik 2001; Epstein & Seely 2002; Richards 2002; Svenonius 2002; McGinnis & Richards 2005; Bever 2009). The word 'expletive' suggests something that is merely a placeholder. This expletive account of existential 'there' contains two core assumptions that I argue against here.

(400) *Traditional expletive account of existential 'there'*[39]
a. Existential 'there' is semantically vacuous
b. Existential 'there' is externally merged in Spec,TP

The main empirical support against a high-origin account of existential 'there' comes in the form of agreement optionality. 'There' can control verbal phi-agreement. Given the conceptualization of Agree presented in Chapter 2, the fact that existential 'there' can control verbal phi-agreement means that it must originate below Spec,TP.

(401) a. There {has/have} to be an explanation
b. There {was/were} no more seats available

(402) a. There {was/*were} you
b. There {is/*am} me

Furthermore, in dialects such as African American English, the existential argument 'it' obligatorily controls phi-agreement in all cases (Green 2002), and yet, given that the use of this existential 'it' appears in exactly the same contexts as existential 'there', there is no obvious reason to assume that it should be base-generated in a different position than existential 'there'.

(403) a. It was a lot of things going on in this lesson (Green 2002:81)
b. It was seventy in the family that went down to Israel

(404) It's a lot of people backstage (Green 2002:82)

I discuss the facts on agreement with existential 'there' in English further on in the analysis of existential inversion, but the data in (401) clearly show that existential 'there' is externally merged low enough to serve as a suitable goal for the phi-agreement probe on T[40].
Existential 'there' is unambiguously locative in form (Kayne 2005; 2010; 2018), which informs the analysis presented in this chapter. Given these facts, I present an analysis of existential 'there' constructions in which I reject the claims of (400) and motivate this 'there' as a thematic argument.

If existential 'there' were a true expletive, what would prevent it from appearing as the subject of most unergative verbs in English? On this, Larson (1989:20) says that "if there is indeed unselected, then we expect no lexical restrictions on the predicates with which it may co-occur."

[39] See example (14) in Kayne (2016:3) for a similar explanation.

He goes on to say that “expletive there appears only with a specific semantic class of intransitive predicates”. In other words, existential ‘there’ does not appear with all predicates.

(405) *There walked a man

The distribution of ‘there’ seems even more puzzling considering that it is not even possible with all unaccusative verbs (without the presence of a locative PP).

(406) a. *There melted a ton of ice
b. *There burned a candle

It is not possible to restrict the presence of existential ‘there’ to all intransitives[41], or even to all unaccusatives. This stands out as a major problem for an account of ‘there’ as a true expletive, a problem which is remedied by treating existential ‘there’ as a thematic argument. As a thematic argument, the distribution of existential ‘there’ is governed by the selectional properties of certain verb classes, the specifics of which I detail further on.

This discussion relates to an observation on word order in Romance from Goodall (2001) following Pinto (1994), Adger (1996), and Zubizaretta (1998), in which verbs that select null locative arguments and have only one non-locative overt argument in Italian and Spanish allow for postverbal subjects, but those that do not select locative arguments do not.

(407) a. E arrivato Gigi Italian (Goodall 2001:204)
is arrived Gigi
‘There has arrived Gigi’
b. Ha telefonato Gigi
has called Gigi
‘There has called Gigi’

(408) a. *Ha riso Gigi Italian (Goodall 2001:205)
has laughed Gigi
int: ‘There has laughed Gigi’
b. *Hanno starnutito tre leoni
have sneezed three lions
int: ‘There have sneezed three lions’

While ‘call’, ‘laugh’, and ‘sneeze’ may all be unergative in Italian (as evidenced by the ‘have’ auxiliary instead of the ‘be’ auxiliary seen with ‘arrive’), only ‘call’ in (407b) allows for postverbal subjects.

For Goodall, this is because ‘call’ is the only one of these verbs that selects a locative argument. The postverbal subject is then permitted because the null locative argument is able to satisfy the EPP. Again, the issue of whether or not a locative argument is present in these structures boils

[41] Although, if existential ‘there’ is an argument, which I assert that it is, then these verbs as they are used in these sentences are not truly intransitive, as they have more than one thematic argument.

down to lexical selection (Tortora 2014a:57-59), although this point does not handle the issue of selection of specifically the existential locative argument. In other words, lexical selection of a locative argument does not explain the following paradigm.

(409) a. At the station arrived a man
b. Into the room walked Robert

(410) a. There arrived a man
b. *There walked Robert

In order to explain this paradigm, I turn to the semantics of the existential argument.

4.1.1 Semantic vacuity

I now argue against the first assumption in (400) that existential 'there' is semantically vacuous, showing that this is not an accurate characterization. In order to do this, I discuss the definiteness effect, a complication to the idea that the postverbal DP in so-called expletive-associate constructions raise to the position of 'there' at LF (Chomsky 1991; 1993; 1995), and the fact that sentences with an existential 'there' subject do have a distinct semantics (Kimball 1973; Belvin & den Dikken 1997; Tortora 2014a; a.o).

4.1.1.1 The definiteness effect

The definiteness effect is a well-known and documented crosslinguistic phenomenon in which definite DPs are restricted in certain positions, namely as postverbal subjects in sentences which contain a so-called expletive pronoun as the preverbal subject (Milsark 1977; 1979; Safir 1982; 1987; Szabolcsi 1986a; b; Heim 1987; Diesing 1992; Abbott 1993; Rigau 1993; 1997; Moro 1997; Keenan 2003; Sheehan 2010; 2016; Fischer & Kupsich 2016; Kayne 2016; 2018, Agulló 2023; a.o). Take the following contrast in English, in which the (b) example is not acceptable without the appropriate pragmatic context.

(411) a. There appeared a ghost out of thin air
b. ???There appeared Casper out of thin air

The definiteness effect is notably not present when there is no existential 'there' subject.

(412) a. A ghost appeared out of thin air
b. Casper appeared out of thin air

It is also not the case that every postverbal THEME is subject to the definiteness effect.

(413) a. I saw a ghost
b. I saw Casper

The definiteness effect does not even hold of all postverbal subject themes of verbs of motion. See the following examples of LI.

(414) a. All of a sudden, under the archway appeared a ghost
b. All of a sudden, under the archway appeared Casper

These examples show that the definiteness effect cannot be a property of inversion constructions, either. The definiteness effect, therefore, is a property – at least in English – of postverbal DPs only in sentences which contain existential 'there' as a subject (which is well-known).

The definiteness effect is also able to be circumvented in a variety of contexts, such as list readings, and as the answer to certain questions.

(415) Q: Which ghosts appeared out of thin air?
A: There appeared Casper, Obi-Wan, Moaning Myrtle, Bloody Mary, and the Ghost of Christmas Past, all out of thin air

(416) Q: Who was at the party?
A: There was John, there was Bill, there was Mary, there was Susan, there was me…

(417) Q: Have we dealt with all of the issues?
A: Well, there is still the problem of the budget to contend with.

This is all very surprising of an element that has been purported to have no semantic contribution, as the definiteness effect looks to be a semantic restriction since it cannot be predictably derived from the syntactic structure, since it seems to hinge on the presence of the existential 'there', and since it can be obviated in certain contexts.

I do not pursue a syntactic, Case-based explanation of the definiteness effect here (Safir 1982; Belletti 1988; Belletti & Bianchi 2016; a.o), as such an explanation would carry the stipulation that the underlying structures of (a) and (b) sentences of the following examples be distinct in some way that allows for the postverbal subjects of the (a) sentences to either not receive Case (Safir 1982) or to receive some abstract Case such as partitive (Belletti 1988).

(418) a. There appeared a ghost out of thin air
b. There appeared Casper out of thin air (pragmatically restricted)

(419) a. There was a man
b. There was John (pragmatically restricted)

(420) a. There was an issue
b. There was the issue of the budget (pragmatically restricted)

Rather, I conclude that the explanation of the definiteness effect lies in the semantic/pragmatic context of the utterance (Milsark 1977; 1979; Szabolcsi 1986a; b), which allows us to maintain the fact that there is no evidence aside from the definiteness of the postverbal subject that points toward the (a) and (b) examples above having any difference in their syntactic structures.

The definiteness effect of postverbal DPs holds in most sentences involving existential 'there' because definite DPs presuppose the existence and uniqueness of the entities to which they refer (Russell 1905; Šimík 2014), which creates a complication for the focus properties of the postverbal DP position in this construction, as well as the semantics of existential 'there' itself (to be presented further on). Thus, to say 'There is John' without any additional pragmatic context creates the issue of being tautological and thereby semantically infelicitous. Additional pragmatic context is needed, then, for definite DPs to be compatible with the semantics of the existential argument and the focus properties of the postverbal subject.

It must be the case that this restriction is tied to both the focus position of the postverbal DP as well as the semantics of existential 'there', as the focus position of the postverbal DP in inversion constructions with HNPS or unaccusative verbs is not enough to trigger the definiteness effect. Note that the following examples are fine without any pragmatic setup.

(421) a. At the table sat Erin
b. Into the room walked ME!
c. "Hey!" shouted drunk the man who everybody was sick of listening to

Interestingly, the definiteness effect is parameterized (e.g., it is much stronger in Spanish than in Catalan (Ordóñez p.c.), which suggests that, crosslinguistically, there may be a difference in the focus properties of the existential construction or even the semantics of the existential argument, but there is no evidence to believe in Case-based definiteness effect in English.

This discussion of the definiteness effect as a semantic/pragmatic restriction shows that existential 'there' cannot be semantically vacuous as its presence is responsible for the semantic restriction of the definiteness effect. That is, the definiteness effect reduces to the semantics of existential 'there' (which I expound upon further on) and the focus properties of the postverbal subject. These two properties together are responsible for the definiteness effect.

4.1.1.2 Against LF raising

Another argument against the semantic vacuity of existential 'there' comes in the rejection of the idea that the postverbal subject of existential 'there' sentences raise to the position of the 'there' at LF (Chomsky 1991; 1993; 1995). Under this view – introduced in Chomsky (1991) – the postverbal subject (the associate) of existential 'there' sentences (expletive-associate constructions) raises at LF to the position of 'there', where it is interpreted as the structural subject of the clause. This happens, allegedly, because 'there' is semantically vacuous. Under close scrutiny, however, this claim does not seem to hold up. See the first piece of evidence against this from den Dikken (1995), who shows that a postverbal subject cannot bind a reflexive anaphor that precedes it, even when 'there' is the clausal subject.

(422) a. Some applicants$_i$ seem to each other$_i$ to be eligible for the job (Den Dikken 1995:348)
b. *There seem to each other$_i$ to be some applicants$_i$ eligible for the job

Furthermore, as is noted in several sources (Lebeaux 1990; Fox 1999; Lechner 2015), a postverbal DP a 'There seems to be…' construction may not take scope over the verb 'seem', which is surprising under an account of LF raising to the position of 'there'.

(423) a. There seems to be a unicorn in the garden ('seem' $\gg \exists$)
b. A unicorn seems to be in the garden ($\exists \gg$ 'seem')

If the associate does not need to raise to the position of 'there' at LF in order for the clause to have something that is interpreted as a subject (because every clause needs something to be interpreted as a subject (call this some LF-equivalent notion of the EPP)) this opens up the possibility that 'there' itself is interpreted as the structural subject of the clause, which is only possible if 'there' has a semantic value.

We also have to contend with the fact that sentences with existential 'there' as the subject simply do have a different interpretation from their 'there'-less counterparts. See the following.

(424) a. A train arrived
b. There arrived a train

(425) a. Two cats were eating the tuna
b. There were two cats eating the tuna

The difference between the (a) and (b) examples here is that the (b) examples have an interpretation that is notably lacking from the (a) ones, which is the *existential reading*[42] (Keenan 1987; 2003; Diesing 1992; Moro 1997; 2006; Hartmann 2008; a.o). The existential reading is the assertion that a given argument (the postverbal subject DP) exists in some contextual specification of the utterance, which typically means that the given argument whose existence is being asserted is new information (Szabolcsi 1986a; b; 1992; Kiss 2023). This is the only possible reading of the (b) examples above, which is not necessarily the case of the (a) examples. Furthermore, Kimball (1979:265) refers to this interpretation as one of "coming into the perceptual field of the speaker". Because of the fact that there is a distinct reading of sentences with an existential 'there' subject when compared to those without, it seems very unlikely that this element should be semantically vacuous.

Before giving a semantic formulation of existential 'there', I point out one more issue with the semantically vacuous expletive account of existential 'there'. Take the following example.

(426) There is a man

Assuming that the copula is semantically vacuous (Pollard & Sag 1994; Heycock & Kroch 1998; Partee 1998) (or, at the very least, denotes an identity function (Heim & Kratzer 1998)), then there is a glaring issue with (426) under the view that 'there' is semantically vacuous: namely, it does not form a proposition; it cannot be true or false. Take another sentence that does not form a complete proposition, which is ungrammatical (in the vast majority of contexts).

[42] Another terminology for a reading associated with the (b) sentences in (424)-(425) is the *presentational reading* (e.g., in Belvin & den Dikken 1997). In the aforementioned source it is not clear if the authors intend for the presentational reading to be distinct from the existential reading, though some authors claim a unique syntax and semantics of presentational sentences (Wood & Zanuttini 2023). I assume here that, in this context, the presentational reading and the existential reading are more or less equivalent notions.

(427) *A man is

The expletive account of ‘there’ does not do anything to explain why (426) does not have the same issues of well-formedness as (427). It also cannot be ignored that (426) is not interpreted as (427), but rather as something like ‘A man exists in the world’ (‘world’ here being contextually restricted as needed). This only makes sense if ‘there’ has a semantics, which I explain now.

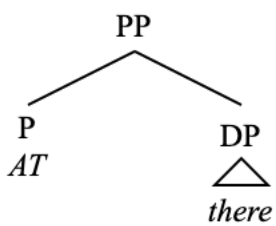
4.1.1.3 Semantics of existential ‘there’

Given that existential ‘there’ has a semantics which contributes to the meaning of the utterances in which it appears, I propose that ‘there’ refers to the universe $\mathfrak{U}$, which can be contextually restricted as needed[43]. See the following semantic denotation of ‘there’.

(428) $[\![\text{there}]\!] = \mathfrak{U}$

This cannot be all, though. Otherwise (426) would carry the interpretation that ‘a man’ is equal to the contextually specified universe. It cannot be overlooked that ‘there’ (and elements like it) have a locative component to their interpretation (Freeze 1992) and therefore should have a locative component reflected in their syntactic structure. For this reason, I analyze ‘there’ as being introduced by a silent preposition AT (Collins 2007; Myler 2013; Schoenmakers & Storment 2021), making existential ‘there’ a PP, the underlying structure of which (to be explained in greater detail later) I present below in (429).

(429) Existential ‘there’ as a PP



In this structure, the silent preposition AT has the following semantic denotation.

(430) $[\![\text{AT}]\!] = \lambda P \lambda x.x \in P$

The entire PP, then, has the following semantic denotation.

(431) $[\![\text{AT there}]\!] = \lambda x.x \in \mathfrak{U}$

This correctly derives the semantics of the sentence in (426). It has the following interpretation.

(432) A man exists in the (contextually restricted) universe

With this semantic definition of existential ‘there’ in mind, it is now possible to talk about its role as a thematic argument, and the structure of sentences which contain it as an argument. Furthermore, given the presence of this null P^0, we can formally derive the restriction on the

[43] This contextual restriction can happen in the discourse, or in the utterance (e.g., ‘There is a man in the garden’).

distribution of the existential ‘there’ subject.

4.1.2 The weak locative

Given that existential ‘there’ has a locative interpretation, I analyze the PP [AT there] as a thematic argument which receives the theta-role LOCATION (LOC). This analysis also tracks with the fact that ‘there’ is locative in form (Kayne 2010; 2018) and has been analyzed elsewhere as a locative argument (Tortora 2014a), given that it appears only with verbs which select a locative argument. This analysis explains the point discussed above that existential ‘there’ is lexically selected by certain verbs as it is a thematic argument, and it takes into the consideration the observation that existentials are a sub-class of locatives (Freeze 1992).

Tortora’s (2014a) account of the existential ‘there’ subject in English frames this element as a *weak locative*, meaning that it is a kind of locative pronominal that is syntactically weak in the sense of Cardinaletti & Starke (1996; 1999). The weakness of this element accounts for its deficient semantics and the fact that it is not licensed in its base position, and is thereby only licensed as a subject. Tortora remains agnostic about the syntactic category of ‘there’, but she is clear that it must be featurally deficient in order to derive its syntactic and semantic weakness.

Given that I have described existential ‘there’ as being the complement of a null P ‘AT’, I believe that the story of the featural deficiency of existential there can be clarified and expanded upon to explain why existential ‘there’ only appears as a subject (not as an object or an adjunct), and why it only appears as the subject of a subset of unaccusative verbs (when it is not accompanied by a deictic PP). Under the view presented here, both the DP ‘there’ and the null PP that contains it are featurally deficient in some way: the phi-featural deficiency of the DP allows for the agreement alternation with defective circumvention that is outlined in Chapter 2, and the featural deficiency of the PP explains the distribution of existential ‘there’ as a subject. I focus now on the latter.

The existential PP headed by null ‘AT’ introducing the DP ‘there’ is not deictic. As opposed to describing the spatial or temporal orientation of some DP with respect to some established ground (deixis), it is asserting that an element exists. As such, this existential PP is not deictic[44].

A standard deictic PP (such as those which contain the subject of LI sentences explored in Chapter 3) has a head P^0 which bears a deictic feature [Deix]. This null ‘AT’ of the existential PP lacks such a feature. This is parallel to the aspect of Tortora’s analysis in which existential ‘there’ lacks a [±speaker] feature.

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[44] There are other non-deictic preposition-like elements whose purpose seems only to be introducing certain arguments. Collins (2024a) refers to these as argument prepositions, which he analyses as being underlyingly KPs: that is, their purpose is to Case-license a given argument. This is notably not what is going on with null ‘AT’, which is crucially unable to license arguments. So, while other non-deictic prepositional elements may be more accurately characterized as KPs, this null ‘AT’ is a true instance of a featurally deficient PP. In this section I explore the consequences of this claim.

(433) Deictic vs non-deictic PP

That is, deictic PPs bear a [Deix] feature, which is notably lacking from the PP that introduces existential ‘there’. The deictic feature of ‘into the room’ licenses the arguments of the PP in their base positions; there is no need for these arguments to move out of the PP. The deficient PP, on the other hand, which lacks a deictic feature, is not able to license its arguments in their base positions. The arguments of the non-deictic PP must evacuate their base positions, which is exactly the case for existential ‘there’. This account of the deficiency of the existential PP makes additional sense with the conjecture that the [Deix] feature is connected to some abstract Case licensing of the PP-internal arguments, which I leave open as a possibility without getting into the weeds of the issue of PP-internal Case assignment.

Because of the deficiency of the existential PP, the PP-internal argument ‘there’ must move to an A-position in which it is licensed. This explains the following contrast between deictic ‘there’ and non-deictic ‘there’.

(434) a. Madeline arrived there (deictic)
b. There (deictic) arrived Madeline

(435) a. *Madeline arrived there (existential)
b. There (existential) arrived Madeline

The properties of the deictic ‘there’ of (434) of course also track with deictic PPs, while only existential ‘there’ is not licensed in its base position.

(436) a. Madeline walked into the room
b. Into the room walked Madeline

The fact that existential ‘there’ must be licensed only via A-movement is reflected in the fact that deictic ‘there’ and full deictic PPs may function as fronted focused elements, whereas existential ‘there’ cannot.

(437) a. There (deictic), Madeline arrived
b. Into the room, Madeline walked
c. *There (existential), Madeline arrived

The only way for existential ‘there’ to be licensed is for it to A-move to Spec,TP, where it undergoes Agree with T^0 and receives nominative Case. This need of ‘there’ to be licensed outside of the deficient PP is the reason why existential ‘there’ only appears as a subject, and not in the object position where VP-internal Loc arguments normally appear (even though that is where it is base-generated). The only way to accomplish this is via inversion as smuggling facilitated by the projection VoiceP, a claim which I show further on to have empirical support.

The featural deficiency of the existential PP also explains the selectional restriction on existential 'there'; namely, that the existential PP does not show up as an adjunct, and is only able to appear as an argument of certain unaccusative verbs of motion.

While a non-deictic element 'there' (which has the same universe-denoting semantics as the existential 'there' which appears as the argument of the deficient existential PP) may appear as the subject of agentive verbs, doing so is only possible when that 'there' is part of a deictic PP. See the following contrast.

(438) a. There walked into the room a man i've never seen before
b. Into the room there walked Minnie

(439) a. *There walked a man I've never seen before
b. *There walked Minnie

The 'there' subject of (438) is only able to appear alongside agentive verbs when it is base-generated inside of a non-deficient, deictic PP (see (433)). Thus, the selectional restriction of existential 'there' is not a property of the DP 'there' itself, but of the deficient existential PP.

The deficient, non-deictic PP containing existential 'there' is only selected by certain unaccusative verbs of motion that select a locative argument. Tortora (2014a) calls these verbs of inherently-directed motion (ultimately from Levin 1993; Levin & Rappaport Hovav 1995). That is, there is a certain class of predicates which are able to select existential 'there' as an argument (so long as that 'there' is not contained with in a deictic PP). This is better said as follows:

(440) *Restriction on the selection of existential 'there'*
There is a only a certain class of predicates which are able to select the deficient existential PP 'AT there' as an argument. These are unaccusative verbs of inherently-directed motion

Common to these verbs is that they obligatorily select a LOC argument, and that they are not able to select an AGENT. Interestingly, these two phenomena seem to pattern together. Those unaccusative verbs which do not obligatorily select a locative argument (and are thus incompatible with existential 'there') are the same unaccusative verbs which participate in causativity alternations (and thus, at times, select an agent).

(441) a. *There melted some ice
b. *There burned a candle
c. *There hung a picture

(442) a. Juan melted some ice
b. I burned a candle
c. Robert hung a picture

The deficient PP which introduces existential ‘there’ is also not compatible with agentive verbs of motion.

(443) a. *There walked Marty
b. *There ran Janet
c. *There swam Patrick

While the fact that existential ‘there’ is lexically selected only by certain verbs has been well-documented (Larson 1989; Tortora 2014a; a.o), few analyses have attempted to capture the fact that the existential PP ‘AT there’ is incompatible with any verb which may project an AGENT. Deal (2009) offers an explanation of this fact, claiming that ‘there’ is generated in Spec,vP and is thus in complementary distribution with an AGENT. This analysis loses out on the fact that existential ‘there’ is ultimately locative (though not deictic) in nature, and should therefore be base-generated as a locative argument. It also creates the issue of having a (what Deal claims to be) non-thematic non-argument base-generated in the position of a thematic AGENT argument, which is a major issue for the Argument Criterion.

The generalization is that the deficient, non-deictic PP ‘AT there’ is only selected by verbs of inherently-directed motion and the copula. It is further incompatible with verbs that may select an AGENT. This restriction, which requires that the existential PP be selected only by a certain subset of unaccusative verbs, also explains why existential ‘there’ cannot appear as an adjunct, as adjuncts are not selected (at least, not in the way that arguments are).

(444) a. We ate lunch there (deictic) with our family
b. *We ate lunch there (existential) with our family

Only deictic PPs are licensed as spatial or temporal adjuncts. The deficiency of this existential PP explains several restrictions on the distribution of existential ‘there’, both in terms of positions in which it is licensed and also of verbs that select it as an argument.

4.1.2.1 Deficient movement and anti-locality

Since existential ‘there’ must move out of the deficient PP in order to be licensed, it must first move to a position where it can be extracted from the PP.

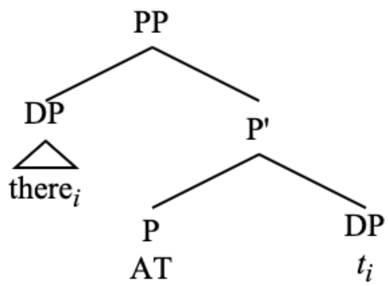
The complement position of a PP is not a licit position in which to be a target for A-movement (or any other A-probing operation such as Agree). If it were, given the analysis of LI in Chapter 3, we may expect verbal agreement in LI constructions to be able to track the internal argument of a deictic PP, which is notably impossible.

(445) a. *Into the rooms walk Wendhy
b. *Through the doors run Edu

Given the opaqueness of PPs in this way (i.e., it is a phase), it must be the case that the ‘there’ argument of the existential PP first moves to the PP-edge, where it is visible for A-probing (van Riemsdijk 1978).

According to Collins (2007), light nouns move to the edge of the PP for similar reasons of licensing that the ‘there’ complement of the deficient existential PP finds itself having to contend with. Under a Collins (2007) framework, this looks something like the following.

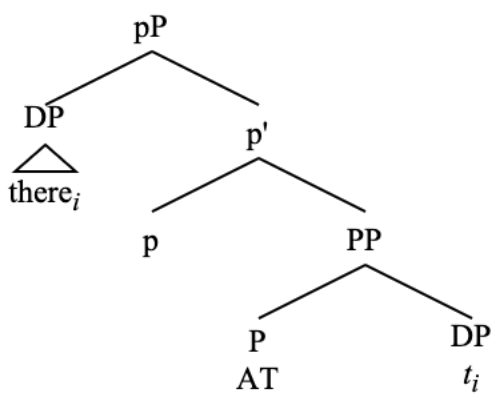
(446) Movement to Spec,PP



This structure has a problem, though, as this movement is a violation of anti-locality, which I have argued is a major issue at previous points of this thesis. From here, there are at least two paths forward. First, we could say that anti-locality is only a major issue in the verbal clausal domain, which upholds all prior claims and also allows for the structure in (446) to be licit. Second, we could maintain that anti-locality is a serious issue for any constituent structure and accordingly amend the structure in (446) with this in mind.

Pursuing the second path, there would need to be a kind of functional extended pP projection (McInnerney 2022) (a shell, so to speak) to allow for the movement of ‘there’ to the phase edge.

(447) Movement to phase edge with an pP shell



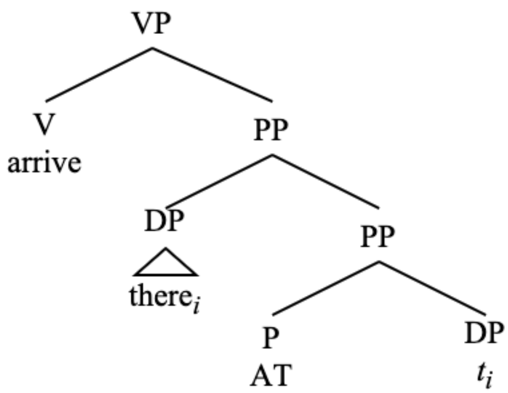
There is, however, also a discussion to be had on the nature of anti-locality and whether or not it should apply here in the same way that it does regarding the selection of a verbal projection by VoiceP. Recall from the previous chapter that anti-locality would arise with VoiceP selecting its complement for movement to its specifier because the relevant feature [V] that Voice^0 searches for movement to its specifier is the same feature that caused the complement of Voice^0 to be selected by it. The complement of Voice^0 is thus inactive to the search for an element to move to Spec,VoiceP.

This process is notably different than from what is going on with the movement of ‘there’ to

Spec,PP in (446). Namely, this movement is not driven by any category-selectional feature on P^0, but rather by the necessity of the DP 'there' to be licensed by some higher projection, since it cannot be licensed by the non-deictic deficient PP. In other words, anti-locality effects arise from selectional features on the relevant head, while the movement of 'there' to Spec,PP is necessary due to the featural deficiency of 'there' and is not driven by some featural specification of the head p^P. All of this is to say that the structure of existential 'there' in (446) is just fine for the current purposes.

As a LOCATION, the PP containing existential 'there' is externally merged as the VP-internal argument in the complement position of VP, just as any other locative argument.

(448) LOC argument of VP



With this structure in mind, it is now possible to explore the analysis of existential sentences as inversion. Before getting fully into the analysis, I briefly go over further motivation for the fact that existential 'there' originates low, in the LOC argument position.

4.1.2.2 Low origin of 'there'

As a LOC argument, existential 'there' originates low, as the complement of V^0. This, in part, can be assumed based on the fact that other LOC arguments originate in the same position. As I argued that existential 'there' is not semantically vacuous, I also argue that it is not externally merged in Spec,TP, which is the second part of the definition of expletive 'there' presented in (400). Refuting this claim is important to state in no unclear terms that existential 'there' is a LOCATION.

The claim that existential 'there' is externally merged in a position lower than Spec,TP is not an uncommon claim throughout the literature. It seems to be more of a common stance than the idea that it has semantic value. A large volume of work treats existential 'there' as a kind of predicate (PredP) (Hoekstra & Mulder 1990; Freeze 1992; Moro 1997; 2000; Belvin & den Dikken 1997; Bjorkman 2015; a.o), an argument on its own (Vikner 1995; Deal 2009; Tortora 2014a), or some kind of subconstituent of the associate (Sabel 2000; Kayne 2016).

I maintain that existential 'there' is a locative argument that can function as a predicate (complement of PredP) in copular sentences and small clauses, as I demonstrate further on. Thus, for unaccusative EI sentences, the existential PP is base-generated as a locative argument (inside VP), and, for copular EI sentences, the existential PP is base-generated as the complement of

PredP. In both of these cases, the existential PP is base-generated *below* the thematic subject, and therefore any surface order in which 'there' and the verb appear before the thematic subject must be derived via inversion.

4.2 English existentials as inversion via VoiceP

English existential inversion (EI) refers to any sentence in which the existential 'there' externally merged as part of the deficient, non-deictic PP 'AT there' appears as the argument in Spec,TP of a clause. Before presenting an analysis of these sentences, it is important to keep in mind that there are demonstrably two different types of EI sentences. This split concerns the difference between sentences in which existential 'there' appears as the structural subject of an unaccusative verb of inherently directed motion, and sentences in which it appears as the structural subject of a copular clause (Hartmann 2008; a.o). Aside from the type of verbal element appearing in these sentences, they have demonstrable differences, as well as significant similarity. The similarities, I argue, are a result of the mechanism of inversion via smuggling to Spec,VoiceP that derives both types of EI sentence, although the differences are great enough to warrant a separate analysis for each clause type.

4.2.1 Word order

I first turn to the case of EI with unaccusative verbs of inherently directed motion.

(449) a. There arrived a man
b. There appeared a solution
c. There a stood a tower of solid rock

Here, I show the same tests concerning word order involving particles, control clauses, and secondary predication as I did with LI and QI sentences to demonstrate that EI sentences also involve movement of the VP over the postverbal THEME subject.

First, the movement of the verb over the THEME does not strand particles; the particle must move with the verb.

(450) a. There came (up) a serious problem (*up)
b. There popped (up) an ad (*up)
c. There rushed (in) a pack of wild dogs (*in)

Control clauses (VP complements) must also precede the THEME[45].

(451) a. There managed (to appear) a solution (*to appear)
b. There stopped (being) any smoking sections on airplanes (*being)

These facts with particles and control clauses are identical to those shown with LI and QI in the previous chapter. These facts diagnose VP movement over the THEME.

Secondary depictive predicates may either precede or follow the THEME argument. This is consistent with the data on LI with unaccusative verbs from Chapter 3, which makes sense given

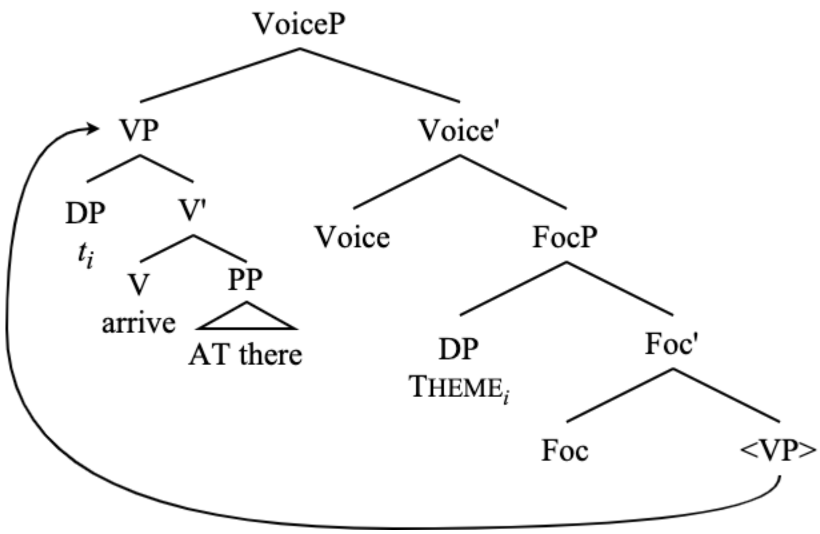
[45] This fact also sets these English inversion constructions apart from V-2 constructions in Germanic.

that these are also unaccusative predicates involved in LI constructions.

(452) a. There arrived (drunk) a man (drunk)
b. There appeared (barefoot) a child (barefoot)

These facts, keeping in mind the obligatory movement of the THEME to Spec,FocP when there is smuggling, and the mechanism of smuggling to Spec,VoiceP, present us with the following familiar structure for EI with unaccusative verbs.

(453) Smuggling of VP to Spec,VoiceP



This is the same derivation as LI with unaccusative verbs, the only difference being that LI involves a deictic PP, while EI involves the non-deictic existential PP, which makes EI obligatory, while LI is optional. This is because the PP-internal argument of the deficient PP is not licensed in its base position.

(454) a. *A man arrived there (existential)
b. There arrived a man

(455) a. A man arrived at the station
b. At the station arrived a man

The obligatory movement of the THEME to Spec,FocP in EI sentences also explains the information-structural properties of the existential construction (and its "presentational" reading); namely, that this construction is used to assert the existence of the postverbal subject as new information in a specific context (which is responsible for the definiteness effect). This connection of inversion as smuggling to a specific information-structural reading further solidifies the role of VoiceP in enabling further possibilities from argument structure to LF.

4.2.2 Agreement alternations

With structure in (453) in mind, I present the data on phi-agreement with existential unaccusative sentences, showing that it is once again identical to the other inversion constructions outlined thus far and that it fits into the general paradigm of defective circumvention in English presented in Chapter 2. Given this discussion, it is possible to be less ambiguous about the structures given for English existential sentences in Chapter 2.

4.2.2.1 Agreement optionality with 3pl arguments

Verbal agreement in unaccusative EI sentences with 'there' as a preverbal subject and a third person plural postverbal subject DP may optionally surface as either singular or plural. In line with the theory of Agree presented earlier, this is because the phi-probe on T^0 may optionally enter into an Agree relation with just the phi-deficient existential argument 'there', which lacks a number feature (Chomsky 1981; 2001; Béjar 2003:62; Deal 2009; Tsiakmakis & Espinal 2022; a.o), or with both 'there' and the postverbal subject DP.

Understanding that plural verbal agreement in these examples is the prescriptive norm, and that all of the examples with singular verbal agreement are understood to also be grammatical with plural agreement, I present examples from the Internet showing singular verbal agreement with plural postverbal subjects. I show this with 'appear' (456), 'arise' (457), 'arrive' (458), 'come' (459), and a few others (460). Despite being reported as ungrammatical in the literature (Rochemont 1978; Schütze 1999; Deal 2009), these sentences are extremely well-attested online.

(456) a. Another snap, and there appears two glasses. A pop of the bottle later, and he silently pours the king a mere half-glass. (Twitter)
b. Hi Blake, what was the source of the part-article you've sourced? There appears many mistakes and exaggerations. Just interested, cheers Phil (Twitter)
c. My relationship with the future from whatever frame I occupy always appears defined in the past direction and undefined in the future direction where there appears many (un)equally possible solutions to where I will be in future frames (Twitter)
d. So there appears two routes (Twitter)
e. There appears two runners coming up @TheValley in Cox Plate. (Twitter)
f. Oh there appears many more options then! (Twitter)
g. In one particular Word .doc (Word 97-2003) in my Word 2007, there appears several very small words which are underscored with a red squiggly line (Blog)
h. And in the last 10 minutes of the film, suddenly there appears people wearing robes, smeared with "blood", chasing after the "paranormal" crew.. (Amazon review)
i. With the sun there appears people who shed their shoes and walk barefoot (Instagram)
j. Initially, there seems to be no problems, but some time after the honeymoon phase ends, realities begin to settle in, there appears problems (LinkedIn)
k. There appears problems with some medical Colleges allowing doctors with alleged conflict of interest onto such committees (Facebook)

(457) a. The dialectic is first born here. Out of one, there arises two. (Twitter)
b. there arises two channels of 'knowing,' being & phenomena for every entity. (Twitter)
c. Throughout all walks of life, there arises several instances where a need for a vehicle to showcase and amplify information presents itself, and in our modern day society PowerPoint takes the cake (Blog)
d. On top of this thing, there arises these round hollow things that have this dark-colored liquid in them (Blog)
e. Usually when I forget I'm in a body and where I am, who I am, there arises these

sensations that I'm jumping off a cliff and my stomach drops, but I remain non-judging. (Reddit)
f. Point is, when a resource is so scarce and limited by the physical world and the only way for a firm to get bigger is to have more of it, there arises issues from the get-go (Twitter)
g. that could be anybody, not necessarily Jesus because there arises Sons of Man in many generations that oversee the transition from one era to another (Blog)
h. Just when you think most of the damage is done there arises those who blend their narrow theology and narrow political ideology into yet another poisonous stew. (Blog)
i. From this core of vision and respect, he argues that there arises 7 primary traditional values (Newhouse 2000)
j. When your child crosses the line and begins testing boundaries and breaking rules, there arises opportunities for a parent/caregiver to set or re-set safe and appropriate boundaries (Blog)

(458) a. First harvest takes place between March and April in which there arrives two leaves and a bud in the bushes. (Blog)
b. So when all seems lost there arrives two options: Soraya can be married to a second cousin willing to take them all out of Kabul or Parvana can cut her long hair and pose as a boy to become the "man of the house." (Blog)
c. Between the winter thaw and the summer sun there arrives two annual influxes of new area residents. (Blog)
d. There arrives many times in our lives when we require information to accomplish one or more deeds (Facebook)
e. For every international conference, framework, and legal finding ruling in women's favour; there arrives different forms of backlash, opposition, and critique from across the political spectrum. (Davies & Harman 2020)
f. The slave class acquires separateness only as fast as there arrives some restrictions on the powers of the owners (Mee & Hammerton 2008)
g. And there arrives, dozens and dozens , if not hundreds, of these bent 3ft x 1 1/4 " tent pegs , made from Re-bar , for the Big Top (Blog)
h. But then there arrives three unknown slayers with unknown intentions… (Apple TV)
i. Just then there arrives four of Steve's cowboys, for whom he telephoned at the same time that he telephoned the club (IMDb)
j. every 10 days, the computer is capable of doing 10^8 tasks, and there arrives 7×10^7 time-critical tasks and 3×10^7 background tasks. (Blog)

(459) a. THERE COMES MANY TIMES IN YOUR LIFE WHERE U FEEL THAT A SITUATION IS HOPELESS & WITHOUT RESOLUTION BUT KNOW IF U PUT YOUR BELIEF IN GOD & CONTINUE YOUR EFFORTS WHILE U LEAVE WHAT IMMEDIATELY CANNOT BE HANDLED IN HIS HANDS KNOW THAT OUR LORD WILL NEVER LEAVE U IN DISTRESS (Twitter)
b. The boys are fucked and there comes several female members of the cast with super powers to save them, every finale (Twitter)
c. There comes several times in a girl's life where she craves chips, queso, guac and salsa (Twitter)

d. There comes those moments in Life when we must self-encourage. (Twitter)
e. Just when you think you're connecting with your male coworker there comes two more male coworkers to talk about Aaron Rodgers for thirty minutes (Twitter)
f. And I celebrate those moments, but then as the journey continues on, there comes these — these valleys where it doesn't feel like we have arrived (Podcast) (Audio ~26:00)
g. There comes millions of unsolveable problems when companies like that get privatized, its enough to drive a tea totaller to beer because its cheaper than water (Blog)
h. & there comes bitches trynna ruin the fun "Ur entire ass is out pull ur skirt down" LEMME BE A HOE IN PEACE!!!!!!!! (Twitter)
i. With any product launch, there comes months of preparing from legal, finance, technology, marketing strategies, marketing roll outs and much more (Blog)
j. There comes seasons in life for plenty (Blog)

(460) a. Downstairs at his Berkeley home there hangs pictures on a brick mantel and chimney of those who've died (Blog)
b. In the Healing Garden of the Witch, there grows plants with a spirit, with a purpose (Instagram)
c. What I mean when I say there lives two wolves inside me (TikTok)
d. Well there goes the nurses, doctors, engineers, scientists, researchers, teachers... (Blog)
e. But then there pops up these horrible ones, he's trying to rape you (Interview)
f. Low and behold, there pops up these pictures that look very close to mine. (Blog)
g. In my experience there exists two subclasses within polyamory: "Boardgames" and "Drugs" (Twitter)
h. And what is the manner by which there occurs two distinct populations of the ribosomes in both the perinuclear region as well as along the perimeter of the cells? (Welch & Suhan 1986)
i. there emerges new things to be said, ways of expression and hints of a pathway to a new future!! (LinkedIn)
j. In this case, there results two systems, one a shift of finite type, the other not, with isomorphic unital ordered groups (Boyle & Handelman 2006)

Clearly, these data are very well-attested and this alternation is possible for many English speakers. This is the exact same agreement alternation described in Chapter 2 for English defective circumvention, and, as such, this agreement alternation is explainable in the exact same way as the agreement alternation in LI sentences detailed in Chapter 3.

4.2.2.2 PCC effects

As further evidence that agreement works the same way in these EI sentences as it does for LI and QI, take the PCC effects that can be seen with postverbal [Part]-bearing DPs in EI sentences with unaccusative verbs.

Nominative first person forms are not licensed in this position; only the accusative forms are.

(461) a. There appeared {me/*I}
b. There arrived {us/*we}

For second person pronouns, which do not make Case distinctions in English, verbal agreement must surface as third singular.

(462) a. There {comes/*come} you
b. There {appears/*appear} y'all

This is true of the first person pronominals as well.

(463) a. There {appears/*appear} me
b. There {appears/*appear} us

Accusative third person pronouns in existential 'there' constructions are presented as ungrammatical in some sources (Deal 2009; Kang 2019) and grammatical elsewhere (Citko 2014:38). Citko (2014) explicitly deems nominative pronouns in these contexts ungrammatical, and the two aforementioned papers that state that accusative Case is ungrammatical do not mention nominative Case at all. Examples with both nominative and accusative Case are both poorly attested online. A small survey of native English speakers reveals that nominative Case on postverbal third person pronouns in the existential 'there' construction is worse than accusative Case. Additionally, the agreement alternation with the third person plural pronominal 'them' is still possible. My own judgments align with (464)-(465) as well.

(464) a. There arrived them
b. There{arrive/arrives} them
c. *There {arrive/arrives/arrived} they

(465) a. There {arrives/arrived} him
b. *There {arrives/arrived} he

Again, this is not a ban on agreement with accusative subjects in all cases. See (464b), as well as the following examples.

(466) a. There appear "them" in contrast to "us" (Blog)
b. But then there come them (Tumblr)
c. If there exist them now, I apologize but I haven't seen them (Forum)

To summarize, the postverbal subject DPs of EI sentences with unaccusative verbs do not ever receive nominative Case. When the postverbal subject DP is third person, verbal agreement may either track the postverbal thematic subject or existential 'there', which is deficient for number. When the postverbal DP is first or second person, verbal agreement may only track the existential argument. This is because the [Part] feature is incompatible with a probe that has already agreed with a third person goal.

4.2.3 FocP and modals

Another way that unaccusative EI and LI function similarly is with respect to some of the HNPS diagnostics presented in Chapter 3. EI sentences naturally allow for modals and auxiliaries.

(467) a. There will come a soft rain
b. There has arrived a train
c. There could appear a solution

This is similar to the discussion of modals and auxiliaries with unaccusative LI verbs, and can be contrasted with types of inversion that do not necessarily involve movement of the postverbal subject to Spec,FocP, such as unergative LI, or QI.

(468) a. *Into the room will walk Martha
b. *Through the doors has pushed María
c. *In can walk Rafael

Thus, it is the obligatory FocP of EI sentences that allows for modals and auxiliaries to be present in this kind of inversion construction (see the formal implementation of this in section 3.4.2).

The compatibility of EI with modals extends to EI with copular verbs as well.

(469) a. There will be blood
b. There has been significant progress
c. There can be no more mistakes

The obligatory FocP is present in all cases of EI, then. This variety of EI involving the copula is derivable through the same mechanism of inversion via smuggling (via VoiceP) as EI with unaccusative verbs, the main difference being that, in the case of copular EI, the existential PP is a predicate, not an argument of the verb.

4.3 Predicate Inversion

Before discussing the other subclass of EI involving copular verbs, it is necessary to clarify an analysis of predicate inversion (PI) constructions. Here I make reference to two specific kinds of copular sentences: predicational and specificational (Higgins 1976; den Dikken 2017)[46]. See the

[46] There are many other types of copular sentences, which I do not consider relevant for this discussion on PI (den Dikken 2017). For instance, I ignore predicate inversion construction in which the fronted element is adjectival. Verbal agreement here always tracks the postverbal subject, regardless of its features. Pronouns always appear nominative as well.

vii. a. Cold {are/*is} the nights without you
b. Comfy and cozy {are/*is} {we/*us}
c. Absorbent and yellow and porous is {he/*him}

This suggests that, unlike with specificational copula sentences and other types of inversion, agreement in adjectival predicate inversion construction either takes place before inversion, can only happen with the postverbal subject, or both. Evidence with word order shows that this may be a V-2 construction, and not "true" inversion.

viii. a. Dark have {my dreams} been {my dreams} of late
b. Long have {the winter nights} been {the winter nights} without you

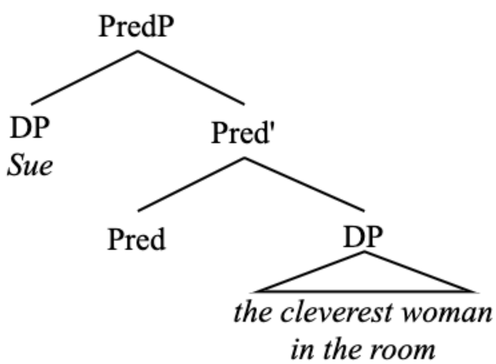
following examples from Heycock (2012:209).

(470) a. Su is the cleverest woman in the room (predicational)
b. The cleverest woman in the room is Su (specificational)

The specificational order is known to be derived from the predicational order via inversion (Moro 1997; Heycock 2012; Hartmann & Heycock 2020; Gravely et al. 2024; Krapp López & Gravely 2024; a.o). According to Heycock (2012), the specificational structure is distinguished from the predicational structure in that the first noun phrase does not refer straightforwardly to an individual, while the second does. In other words, the first DP is attributive, and the second is referential. In other words, the structural subject of the specificational clause in (b) denotes a predicate with a variable, and the postverbal DP denotes a value for that variable. As such, when I refer to PI in this chapter, I am referring specifically specificational copular clauses.

This is the inverse of the predicational structure, in which the first DP is referential, and the second is attributive. In (470), (b) is the inverted structure of (a), and (a) is the non-inverted variant of (b). In a predicational copula construction, the DP which appears as the structural subject of the clause is the external argument, and the postverbal DP is the predicative DP: the internal argument of a PredP (Bowers 1993).

(471) Predicational copular structure



From this point, extraction of the external argument of the PredP to Spec,TP is straightforward. T, merged above PredP, targets the closest goal for Agree and movement to its Spec. This goal is the external argument 'Sue'.

4.3.1 Specificational copular clauses

What happens with the inverse (specificational) sentence is more complicated. A functional projection merged directly above PredP that can select the predicative internal argument DP directly for movement above the external argument DP (Moro 1997; Krapp López & Gravely

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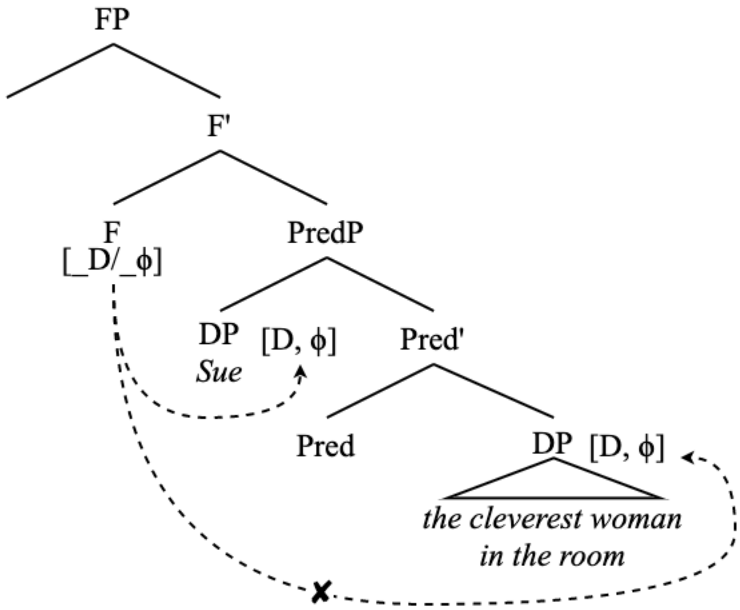
This word order alternation is notably not available with PI clauses.

ix. My favorite thing about this experience has {*the encouragement I've received} been {the encourage I've received}

This fact holds of the other inversion constructions shown here as well – the postverbal subject must follow *all* verbal elements in the clause. The fact that this does not hold of adjectival predicate inversion shows that it may not be "true" inversion (that is, inversion via smuggling via VoiceP).

2024) does not seem to work for issues of minimality, at least not if this functional projection is targeting a goal for movement based on a feature such as [D] or [ϕ].

(472) Failed selection of the predicative DP

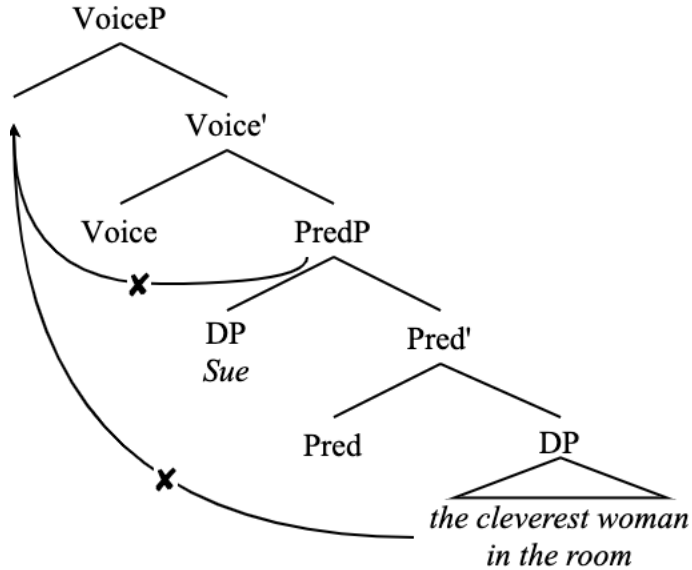


In this structure, the external argument DP is always closer to the probe, and thus the selection of the internal argument DP would violate minimality. This issue could be circumvented, perhaps, if this FP is selecting something for movement that bears a feature that the external argument DP lacks, say some kind of nominal [Pred] feature, but that does not seem to take us in the right direction as Krapp López & Gravely (2024) explicitly indicate that this functional projection can select *either* the external or internal argument DP[47].

Additionally, VoiceP gets us nowhere with this structure, as it currently stands. Given that VoiceP is restricted to only moving verbal projections to its specifier, the only option for that movement – PredP – would be an anti-locality violation (bearing in mind the discussion on anti-locality and selectional features above). Furthermore, if VoiceP could select PredP for movement to its specifier, this would effectively do nothing to raise the internal argument DP over the external argument DP as the external argument DP would still asymmetrically c-command the internal argument DP inside the moved PredP. If VoiceP were somehow able to also select a nominal for movement to its specifier, then the issues of minimality and the nominal-selecting FP from the previous example come up again (and it has already been established that VoiceP moves verbal projections to its specifier). As such, VoiceP is not sufficient to derive specificational copular sentences, at least not yet.

[47] Again, perhaps equidistance is a consideration here that I am leaving out, but there is good reason to leave it out. First, there is no way around the fact that the external argument DP does asymmetrically c-command the internal argument DP, and is thereby closer to any goal merged above the two. Equidistance is an unnecessary complication to any formal implementation of Search, and, as I will show here, it is not necessary (see also my discussion of equidistance in the introduction of this thesis.

(473) No viable desirable candidate for movement to Spec,VoiceP



It seems, then, that we have reached an impasse regarding predicate inversion. Fortunately, this can be remedied by further examining the information structural properties of the specificational copular construction.

4.3.2 Focus movement

Another diagnostic for the specificational copular structure is focus. In specificational copular structures, the postverbal DP (the external argument to the PredP) necessarily bears contrastive focus (Higgins 1976; Akmajian 1979; Declerck 1988; Mikkelsen 2005; den Dikken 2017).

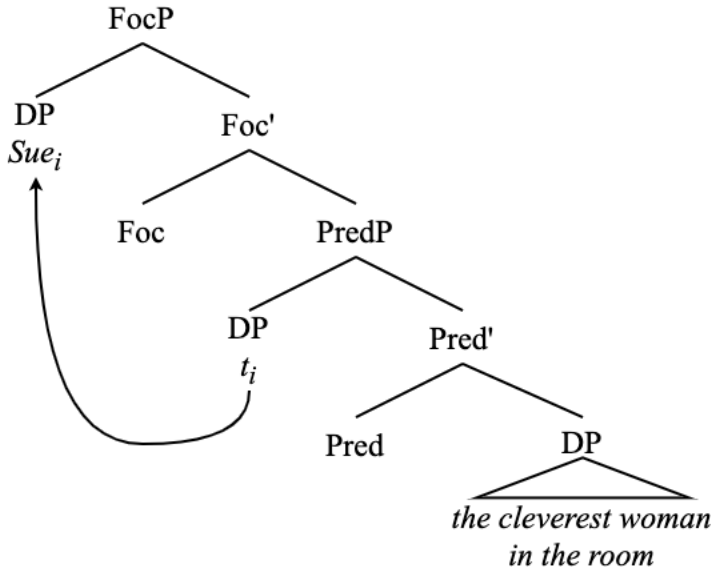
(474) a. The cleverest woman in the room is SUE (not Mary)
b. The president is TRUMP (not Biden)
c. The person who planned your surprise party was ME (not anyone else)

This is not necessarily true of the external argument DP in non-inverted predicational structures. The following sentences may have a reading with contrastive focus on the structural subject, but it is not a necessary property of the sentences.

(475) a. Sue is the cleverest person in the room
b. Trump is the president
c. I was the person who planned your surprise party

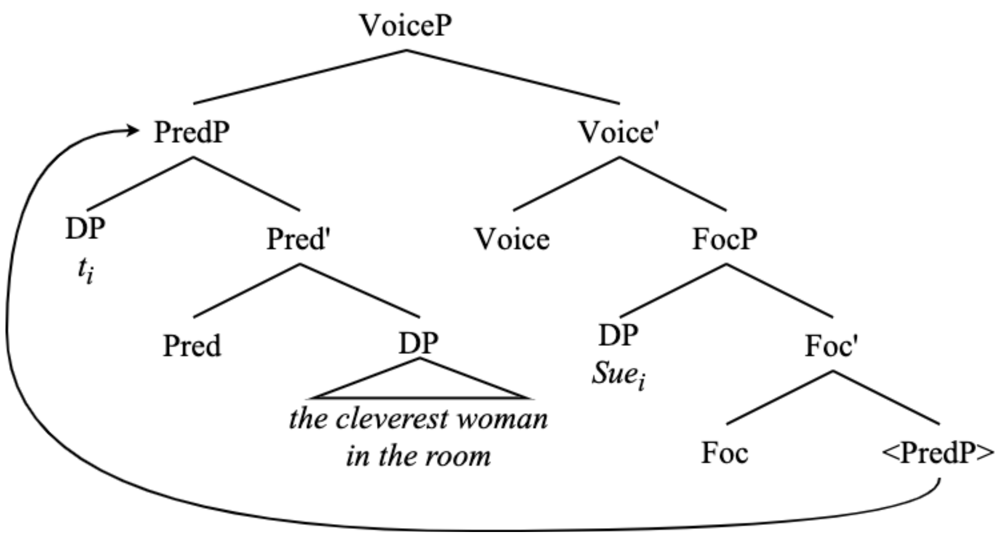
The sentences in (475) do not have a focus reading of the postverbal DP, which is the predicative DP, not the external argument. It is possible for sentrnces like (475) to have the postverbal DP focused, but, given the ambiguity of copular clauses (Kripke 1972; Higgins 1976; Mikkelsen 2005; den Dikken 2017), these sentences are more accurately characterized as specificational. As such, the focus property of the postverbal DP is characteristic of PI sentences (just as it is characteristic of EI sentences).

(476) Movement of the external argument DP to Spec,FocP



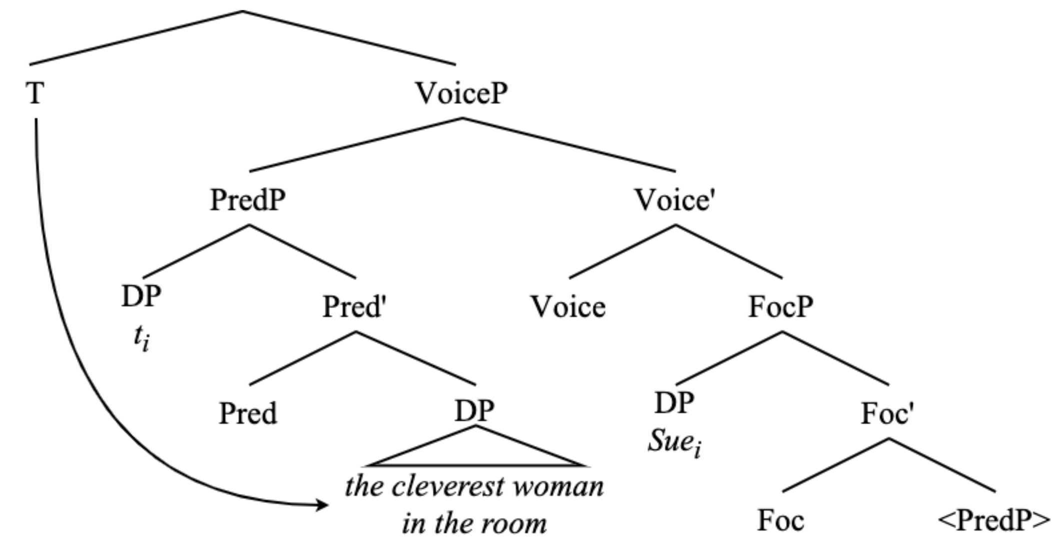
With this movement in mind, let us reconsider what VoiceP is able to do without violating minimality or locality conditions.

(477) Movement of PredP to Spec,VoiceP



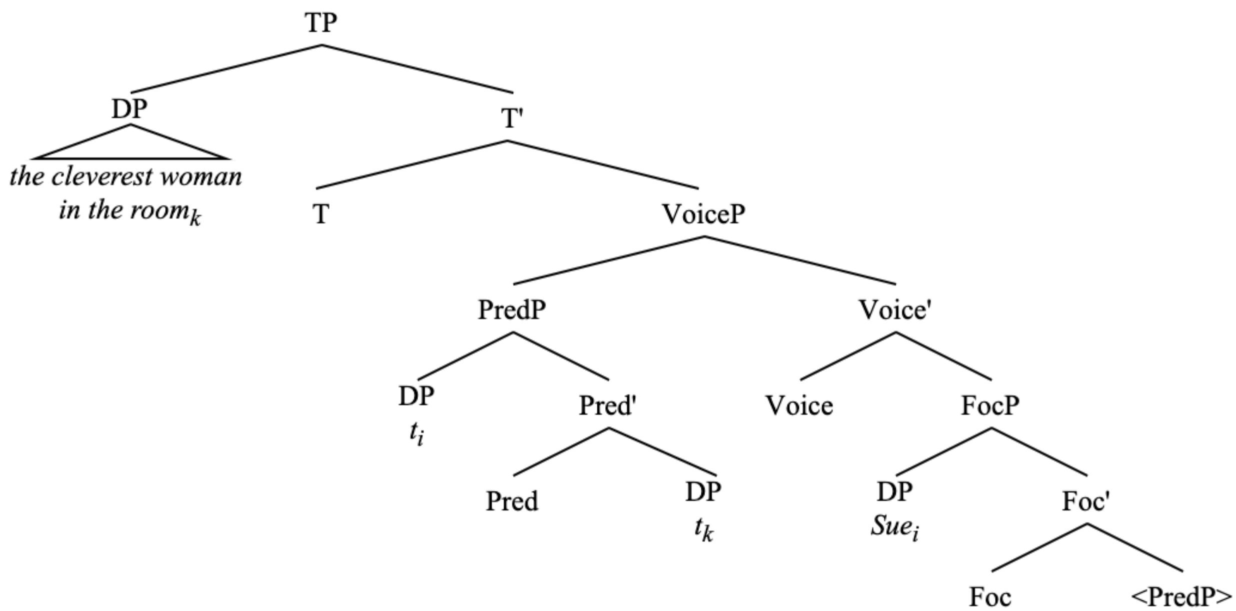
Here, VoiceP is able to select a verbal projection which is not its complement – PredP – for movement to its specifier. Since the external argument DP moved out of the PredP to Spec,FocP, the way is now cleared for the internal argument DP to move to the position of the structural subject. The internal argument DP is the element that T encounters during its search for a phi-bearing in goal, keeping in mind the DFS parsing algorithm of closeness without c-command

(478) T probes, finds VoiceP



As a result of the phi-probe on T finding the internal argument DP, this DP moves to Spec,TP and functions as the structural subject.

(479) Specificational copular clause derived via PI



The presence of the FocP, then, allows for inversion via smuggling to derive the grammatical variant of the sentence, in addition to explaining the semantics of specificational copular clauses. Furthermore, the FocP once again allows for the presence of modals. For this reason, modals, auxiliaries, and negation are all perfectly grammatical in PI.

(480) a. The president will be Trump (not Harris)
b. The president should be Harris (not Trump)
c. The president isn't Biden (it's Trump)

As evidence that the entire projection containing the verbal element and the internal argument is smuggled above the external argument DP, see the fact that control clauses must precede the postverbal DP.

(481) a. The cleverest woman in the room seems (to be) Sue (*to be)
b. The president stopped (being) Biden (*being)

Again, these facts set this construction apart from something that could simply involve head movement of the verb above the postverbal subject.

Secondary depictive predicates behave somewhat differently from the other constructions involving FocP here, though. It is not the case that smuggling over FocP causes depictives to be able to appear to the left of the postverbal DP. Note that secondary depictives must follow the postverbal DP in predicational clauses

(482) Sue is still (*drunk) the cleverest woman in the room (drunk)

And this is still true of the specificational PI clause.

(483) The cleverest woman in the room is still (*drunk) Sue (drunk)

It must be the case, then, that these depictivies are adjoined higher than PredP, perhaps to FocP, which makes sense given that secondary depictive predicates are not the kind of adjuncts that can directly adjoin to a PredP like this (Ernst 2001; Harley & Jung 2015; *cf.* Bruening 2018)

4.3.3 The copula

I do not include the copular verb 'be' in these trees as it is not extremely consequential for the analysis of the internal argument DP's movement to Spec,TP presented here. Additionally, there is some disagreement on whether the copular verb originates as the head of PredP (Bowers 1993) or above the PredP as a realization of tense (Moro 1997). I believe that this issue merits its own discussion before we can include the copula in these structures.

In previous work I claimed that the copula in the Kalahari-Khoe language Tshila originates as the head of a copular phrase (CopP) that is the complement of the PredP (Storment 2025). I did this because Tshila predicate nominal sentences have a copula verb *ʔè* which is unique to predicate nominals (484). Other kinds of predication relation verbal elements are distinct. For example, the locative copula is *hã̋ã̄* (485), and the morpheme that introduces adjectival predicates is *hì̃* (486).

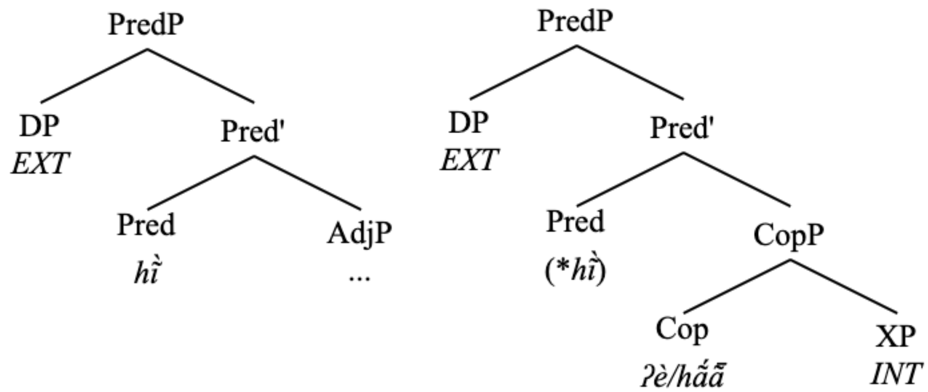
(484) kí ǀχ'ūn̄ è Thabo b -è **ʔè**
1SG name NOM Thabo PGN.M-NOM **COP**
'My name is Thabo

(485) ʔābā **è** ŋǀã̋ũ̏ **hã̋ã̄**
dog **NOM** there **COP.LOC**
'The dog is there'

(486) kí ŋúū è !áò **hȉ̃**
1SG house NOM tall **PRED**
'My house is tall'

As such, I took *hȉ̃* to be the head of a PredP and *ʔè* and *hấ̃ã* to be CopP heads, with only the lowest of whichever of these heads is projected able to be realized at once.

(487) Tshila adjectival PredP vs. PredP with CopP complement

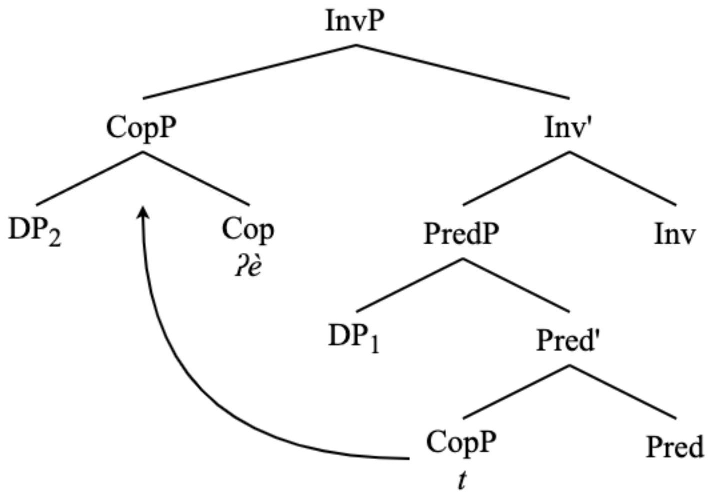


The issue of seeing three different copular/predicative morphemes is not present in English as English uses the same copula in all three contexts presented in Tshila in (484)- (486).

(488) a. My name **is** JD
b. The dog **is** over there
c. My house **is** tall

As such, there is no immediate reason to believe that these copulas are the projections of separate heads. Furthermore, the inclusion of a low FocP in the specificational predicate structure allows for smuggling via VoiceP to work without the need for another verbal projection layer such as CopP. Without FocP and with CopP, inversion via smuggling looked as follows (keeping in mind that I referred to VoiceP as InvP in that paper).

(489) Smuggling in Tshila with CopP



Revising the analysis of Tshila predicate nominal structures in Storment (2025) to include FocP and thereby not need CopP, I would say that the three different predicative morphemes in (484)-(486) are the realizations of three PredPs each with distinct selectional features: Pred_{N}P, Pred_{Loc}P, and Pred_{Adj}P, respectively. Under this view, it would not make sense for these copular elements to be an overt realization of tense (as in Moro 1997) as T^0 plays no direct role in the selection of the complement of PredP.
Assuming, then, that some universality in the way that predicative copulas are base-generated across languages is preferred, then I would favor an analysis of English in which the predicative copula is the overt realization of the PredP head, and I shall indicate such in this chapter.

4.3.4 Agreement and Predicate Inversion

When the inverted structural subject in a specificational copular clause is phi-deficient, then the same Agree phenomenon regarding defective circumvention and PCC effects can be observed in PI constructions (Partee 1998; Heycock 2012, fn. 3; Krapp Loópez & Gravely 2024).

Normally, in PI involving two DPs, the DPs match in number features due to the semantic relation that holds between the two DPs involved in the construction. When there is a mismatch in these features, though, and when there is not a phi-deficient element involved, phi-agreement on the copula tracks the structural subject.

(490) a. My cats {are/*is} my reason for getting up in the morning (predicational)
b. My best friend {is/*are} my cats (specificational)

When the inverted specificational subject is phi-deficient, however, there appears optionality. This optionality is derived via defective circumvention.

(491) My reason for getting up in the morning {is/are} my cats

Singular agreement with third person plural postverbal DPs in specificational copular clauses seems much more prescriptively permissible than singular agreement with third person plural postverbal DPs in other inversion constructions (LI and QI). In the presence of a phi-deficient structural subject such as a wh-cleft (Percus 1997; Berg 1998; Heycock 2012; Lee 2016; Krapp

López & Gravely 2024), this optionality can be observed. I present fewer of these examples than I have for other inversion constructions because it seems more prescriptively permissible and because it has been described elsewhere in the literature. See the following examples from social media.

(492) a. I buy him way too many toys and I swear what he loves most are these 99 cent mice (TikTok)
b. What he loves most is those meetings with the audience members (The Oakland Press)

(493) a. The thing I love most is those relationships I've enjoyed with people I've connected with directly and, in some cases, indirectly through MSU (Michigan State University)
b. This Narciso Rodruguez ensemble is edgy and cool but the thing I love most are those gloves (Blog)

With the non-inverted (predicational) word order, this alternation is not possible.

(494) a. These 99 cent mice {are/*is} what he loves most
b. Those relationships {are/*is} the thing I love most

PI sentences also show the same PCC effects as other inversion constructions. Postverbal DPs which are first or second person may not control phi-agreement, and they cannot receive nominative Case.

(495) a. Since what she loves most is me, wouldn't that be really vain? (Fanfiction)
b. She loves her tennis ball, rolling in the grass and watching out the front door. What she loves most is YOU! (Pet website)
c. Let's just suppose that what he loves most is us, his family (Raschka 2011)
d. I love art and music, but what I love most is y'all! (Blog)
e. Technically everything is as sound as you can imagine, but what I love most is him, his atmosphere (Business review)

Once again, this is not a ban on agreement with postverbal accusative DPs.

(496) a. What their families really need are them (Blog)
b. All I miss are them (Instagram)
c. The demographic who does it the most are them (Forum)
d. What excites me the most are them! (Instagram)
e. The one that caught my eyes the post are them (Reddit)
f. The thing that I like most are them and their songs (Instagram)
g. The only thing getting in the way are them (Blog)
h. Now all I need are them! (Blog)
i. When I walk to the store, all I see are them (Wiki)

The same also applies to specificational copular structures with a locative predicate, both in terms of defective circumvention and of PCC effects.

(497) a. In the box {was/were} three magic rings
b. In the box {is/*am} {me/*I}
c. In the box {was/*were} you

Indeed, locative predicates form a superset which contains the existential predicate 'there', the analysis of which is presented in the following section.

4.4 Copular EI as Predicate Inversion

I now extend this analysis of PI to copular EI sentences involving a copula, such as the following.

(498) There is a problem

Using diagnostics such as word order, Agree, and information structure, I show that these sentences have essentially the same derivation as the specificational copular clause, the difference being that the inverted predicative internal argument is the existential locative 'there'.

First, briefly concerning word order, EI sentences are identical to specificational PI sentences in that control clauses must precede the postverbal DP.

(499) a. There seems (to be) a solution (*to be)
b. There stopped (being) any smoking on flights (*being)

Again, this sets these sentences apart from V-2 constructions and also shows that they behave exactly like every other inversion construction detailed thus far in this thesis in this regard.

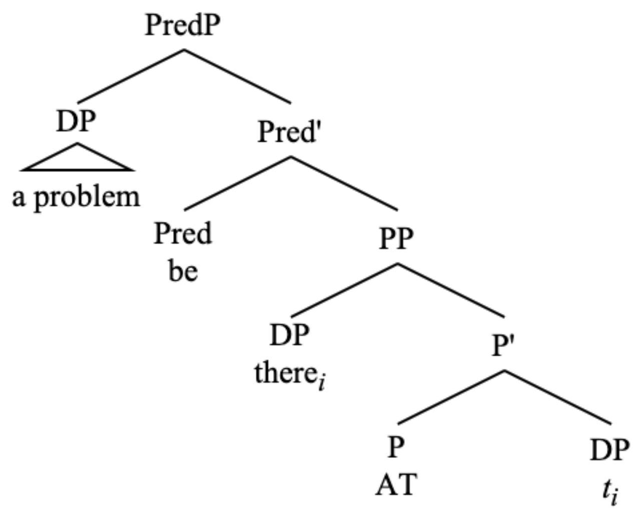
The same is also true of EI that is true of specificational PI regarding secondary depictive predicates, that they must follow the postverbal DP.

(500) a. There was (*drunk off his ass) a man (drunk off his ass)
b. Then there was (*drunk off his ass) John (drunk off his ass)

This is once again explained by the fact that secondary depictive predicates cannot adjoin directly to PredP and instead are adjoined somewhere such as FocP in these structures (i.e., this fact on secondary predication is further evidence that we should have a FocP in EI sentences).

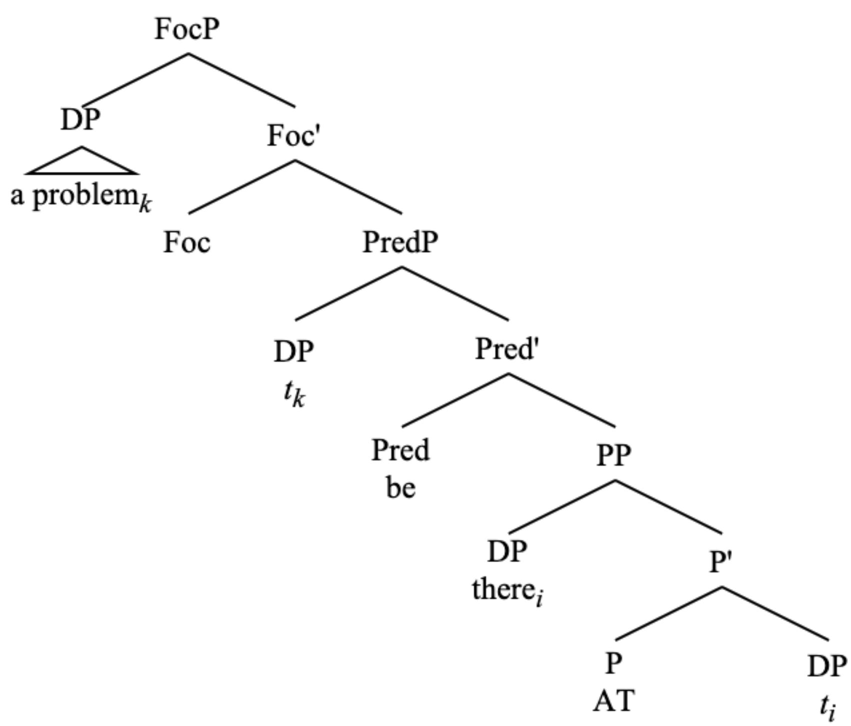
As such, I present the following underlying structure for copular EI sentences, using the example in (498).

(501) Underlying structure of copular EI



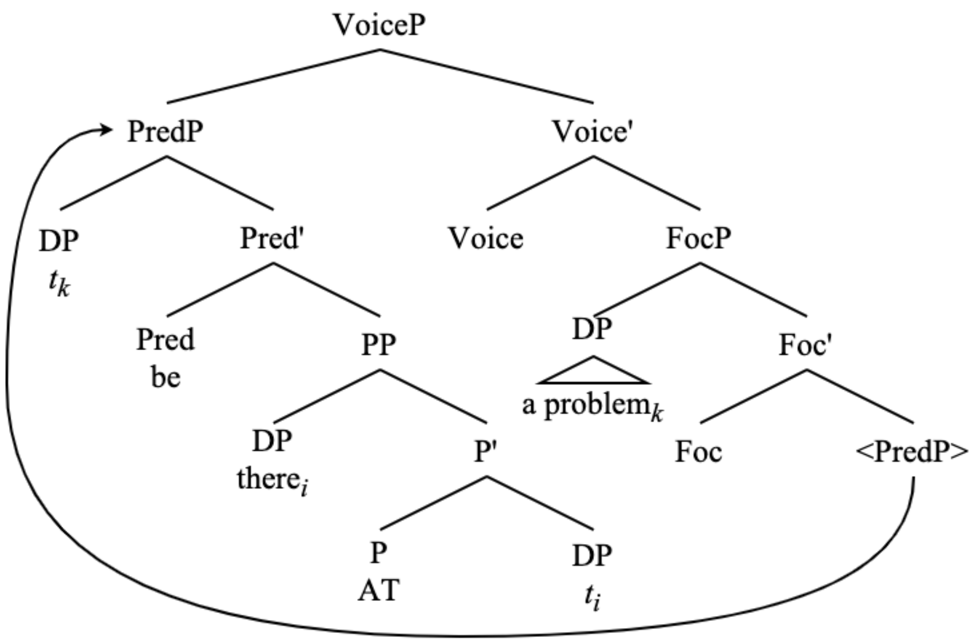
Just as with PI and unaccusative EI, the external argument in Spec,PredP moves to Spec,FocP.

(502) External argument moves to Spec,FocP



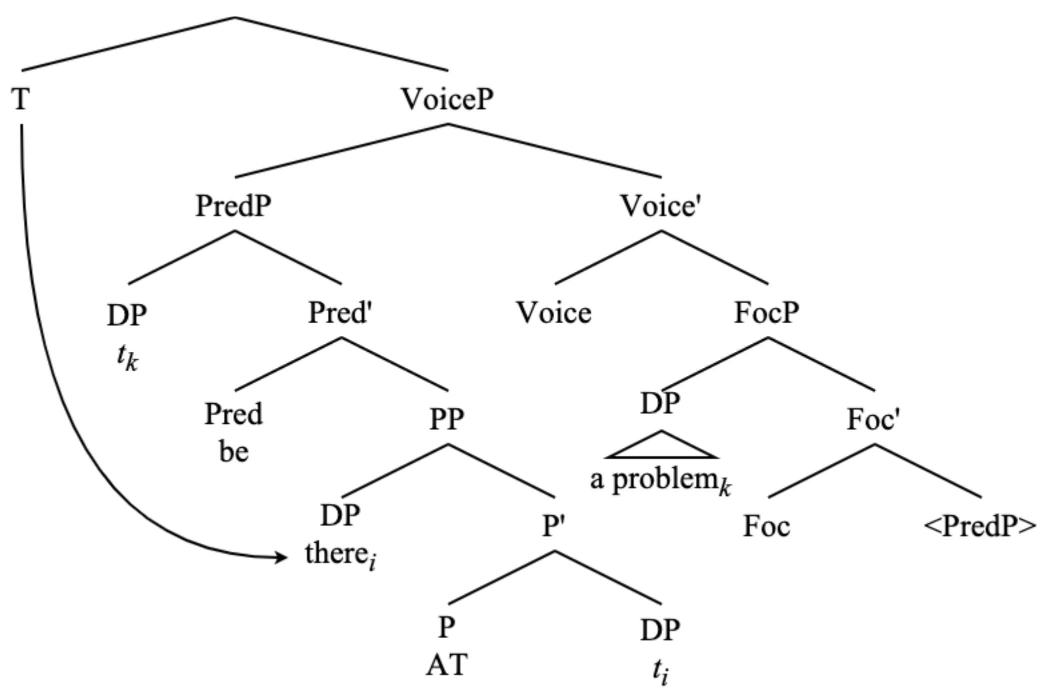
From here, VoiceP is merged above FocP, and PredP is attracted to Spec,VoiceP.

(503) PredP moves to Spec,VoiceP



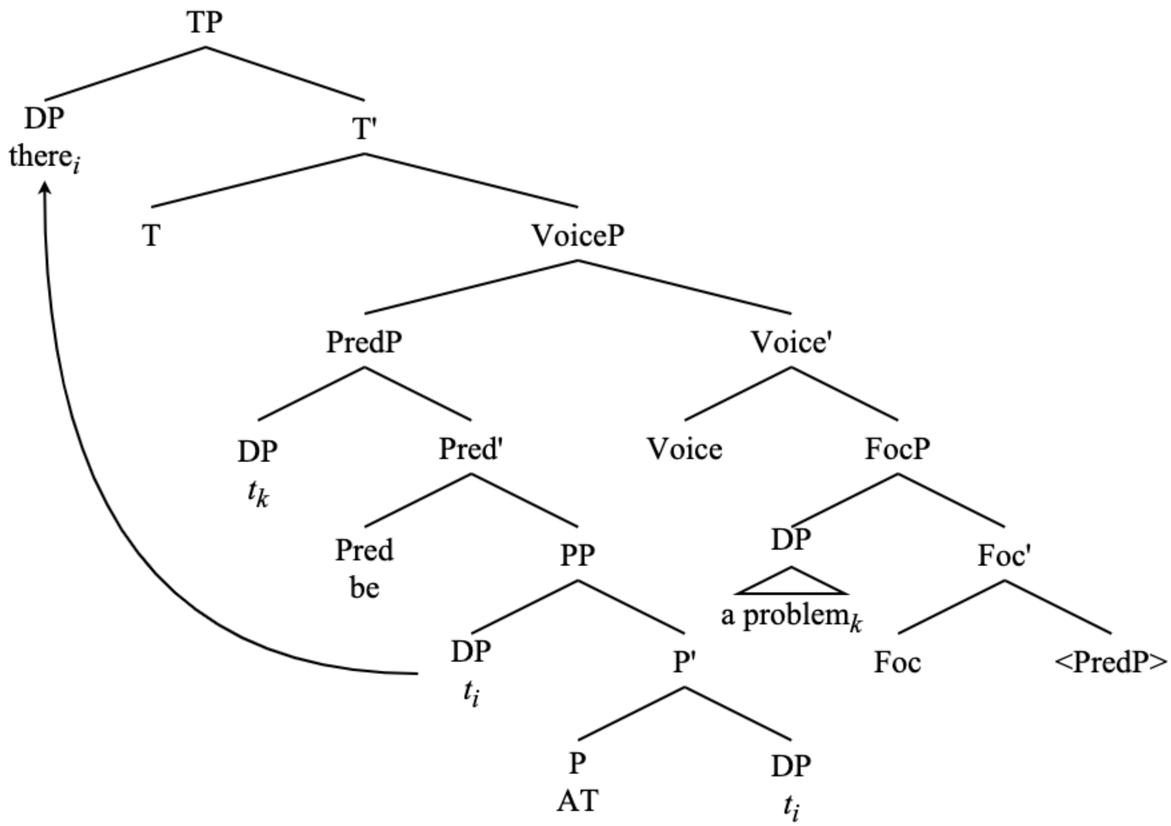
At this point, when TP is merged above VoiceP, T^0 probes and first encounters the phi-deficient DP 'there' in Spec,PP, given the stipulations on closeness without c-command and DFS in Chapter 3.

(504) T^0 finds DP 'there'



Given that the existential DP 'there' is phi-deficient, at this point, T^0 may optionally probe again to undergo defective circumvention, which creates the agreement optionality presented in Chapter 2 and in the following section. In either case, though, the DP 'there' moves to Spec,TP.

(505) ‘There’ moves to Spec,TP



In Spec,TP, the DP ‘there’ is licensed via undergoing Agree and receiving Case, something that it was unable to do in its base position inside the deficient existential PP.

The fact that ‘there’ must undergo this movement in order to be licensed makes this inversion with an existential PP predicate obligatory. This is a marked difference between copular EI and other specificational PI sentences for which the uninverted variant of the sentence is also grammatical.

(506) a. Sue is the cleverest woman in the room
b. The cleverest woman in the room is Sue

(507) a. *A problem is there (existential)
b. There is a problem

This structure also allows for the agreement optionality and PCC effects presented in Chapter 2. I present further discussion of these agreement facts in the following section, keeping in mind the derivation presented above.

4.4.1 Agreement

For many English speakers, with a third person plural postverbal DP, the copular verb in EI constructions in English can optionally show third singular or third person plural agreement.

(508) a. There have to be exceptions to the rule
b. There has to be exceptions to the rule

While the (a) example is certainly the prescriptively correct one, and despite the agreement pattern in the (b) example being reported as ungrammatical in the literature (Deal 2009) both are well-attested.

Here I present Internet examples showing that the singular agreement with a third person plural postverbal DP is productive across all tenses and several structures of plural postverbal DP, showing that this agreement alternation is a general property of EI sentences.

I ignore contraction because everyone generally finds those okay. That is, all examples with "there is" okay are also okay with "there's". I take the contracted form to be syntactically equivalent to the uncontracted form because, despite claims to the contrary (MacKenzie 2013), there is no evidence to assume that these two allomorphs behave differently with respect to syntactic agreement. I also assume that 'there is [PL]' sentence also has a grammatical 'there are [PL]' counterpart. This agreement alternation is possible regardless of tense. Indeed, the fact that this alternation is possible in the past tense alone demonstrates that this must not be a by-product of contraction.

(509) a. There is two teams playing (Twitter)
b. There was cats in the garden but they didn't wanna be friends (Twitter)
c. There has been problems using foreign machines so why do we continue using it??? (Twitter)

Moving on to the syntactic properties of the postverbal DP, this agreement pattern is available with bare plural NPs (510), plural DPs with a definite determiner (511), plural DPs with demonstrative determiners (512), plural DPs with quantifiers (513)-(515), and plural DPs with numerals (516).

(510) a. They don't choose though, a choice implies that there is options (Twitter)
b. There was seats bitch (Twitter)

(511) a. There is the results of the referendum… (Twitter)
b. There was the cats doing it in the car wash and the drunk homeless guys sitting by the dumpster (Twitter)

(512) a. on holiday and im ngl i HATEEEE it here there is these lil beetle bug things everywhere (Twitter)
b. It was a spaceship. And there was these things, these killer clowns, and they shot popcorn at us! We barely got away! (Twitter)
c. The cowards won't get him to step down.........and then there is those prisons concerns hovering. Flee to friendly dictatorships? (Twitter)
d. And there was those aliens on Jupiter's moon Europa……. (Twitter)

(513) a. Lucky there is no dogs in that yard (Twitter)
b. I never said there was no countries! I said there is no cities! (Twitter)

(514) a. There is some movies I don't want to watch (Twitter)
b. And there was some people doubting Tim at the beginning of the window…. Shame on you (Twitter)

(515) a. Paper. Kinda old school but find it super cool that there is both options (Twitter)
b. There was both options available mate (Twitter)

(516) a. How you lose to one guy when there is seven of y'all (Twitter)
b. forgot there was four quarters in basketball, how embarrassing (Twitter)

In addition to defective circumvention, EI sentences also show PCC effects, just like the other inversion constructions. This has been described to a very limited extent.

(517) There {is/*am} only me (Richards 2004:159)

The inability of agreement to track the postverbal DP here is not result of the DP having accusative Case. Nominative first and second person pronouns are not permitted at all, and they cannot be agreed with here.

(518) a. *There {was/were} we
b. There {was/*were} us

(519) There {was/*were} you

Third person pronominals in the postverbal position may not receive nominative Case, but they can optionally be agreed with (which we can see in the case of third person plurals).

(520) a. There was {him/*he}
b. There was {her/*she}

And, once again, this is not a ban on agreement with postverbal accusative DPs.

(521) a. There are them that cannot sing, and them that cannot weep (The Crucible)
b. There are them and then there's #ME ! (Instagram)
c. There were them and there was us; and he never ever saw it any differently (Interview)

The fact that the accusative pronouns can optionally control agreement here suggests that Case does not feed phi-agreement, and that phi-agreement is not enough on its own to assign nominative Case in English; it has to come from a Spec-Head configuration (see the discussion on Case in Chapter 2). Additionally, these agreement facts are consistent with the other inversion constructions presented thus far under the theory of Agree presented in this thesis.

4.4.2 Modals and auxiliaries

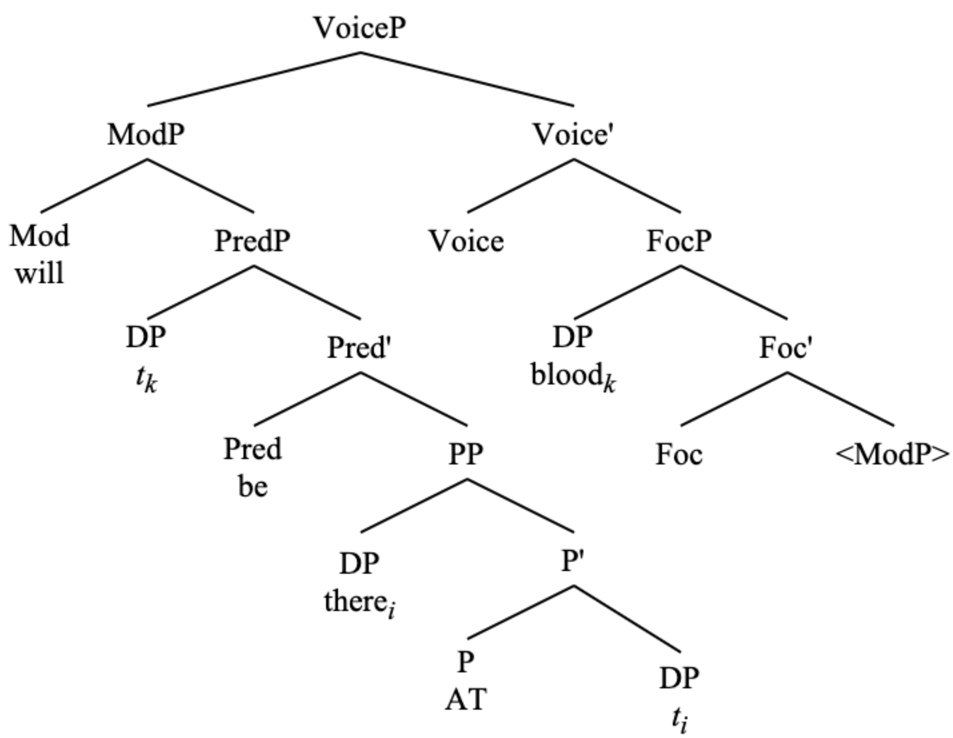
Another way in which EI sentences are similar to PI – and a way that sets them apart from other inversion constructions such as LI and QI – are how grammatical modals and auxiliaries are in EI sentences. This is somewhat surprising given how marked modals have been in other inversion constructions.

(522) a. There will be a man
b. There should be a man
c. There has been a man

The availability of modals and auxiliaries here also follows from what makes modals and auxiliaries available in specificational PI sentences and in other inversion constructions with obligatory movement of the thematic subject to Spec,FocP. Take the following example.

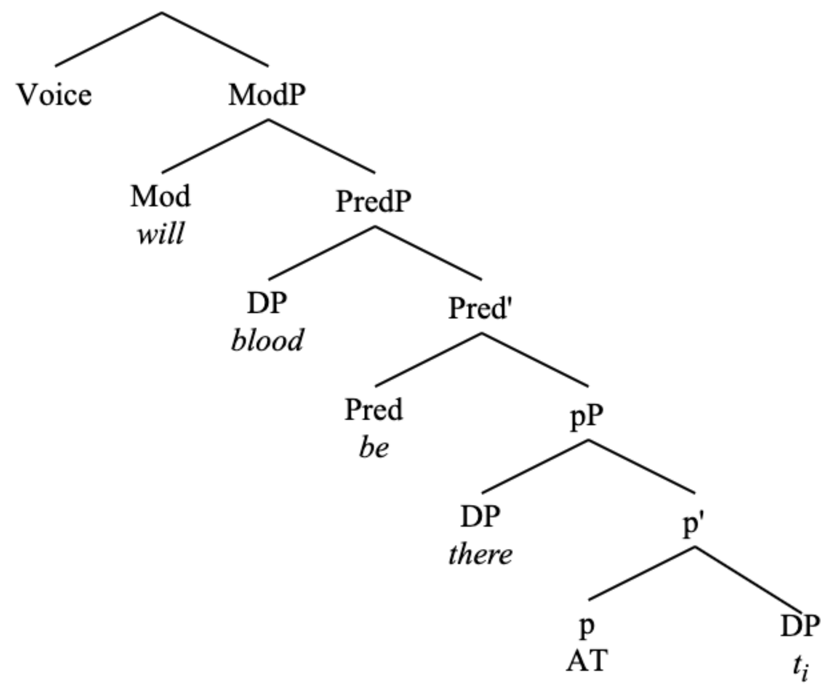
(523) There will be blood

(524) Smuggling of ModP to Spec,VoiceP



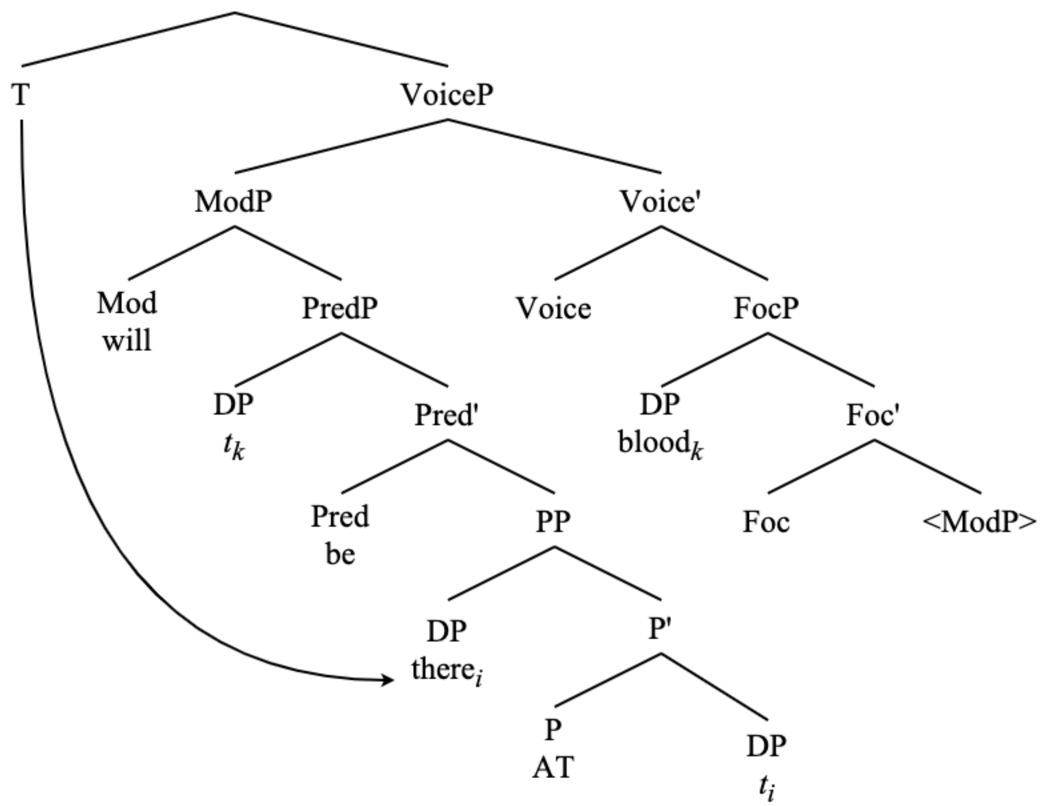
Without the FocP where it is, movement of both the modal and the copula (as well as existential 'there') above the external argument is not possible. Without FocP, the verbal projection that is targeted for movement to Spec,VoiceP would have to be PredP as it is the closest viable target

(525) EI without FocP



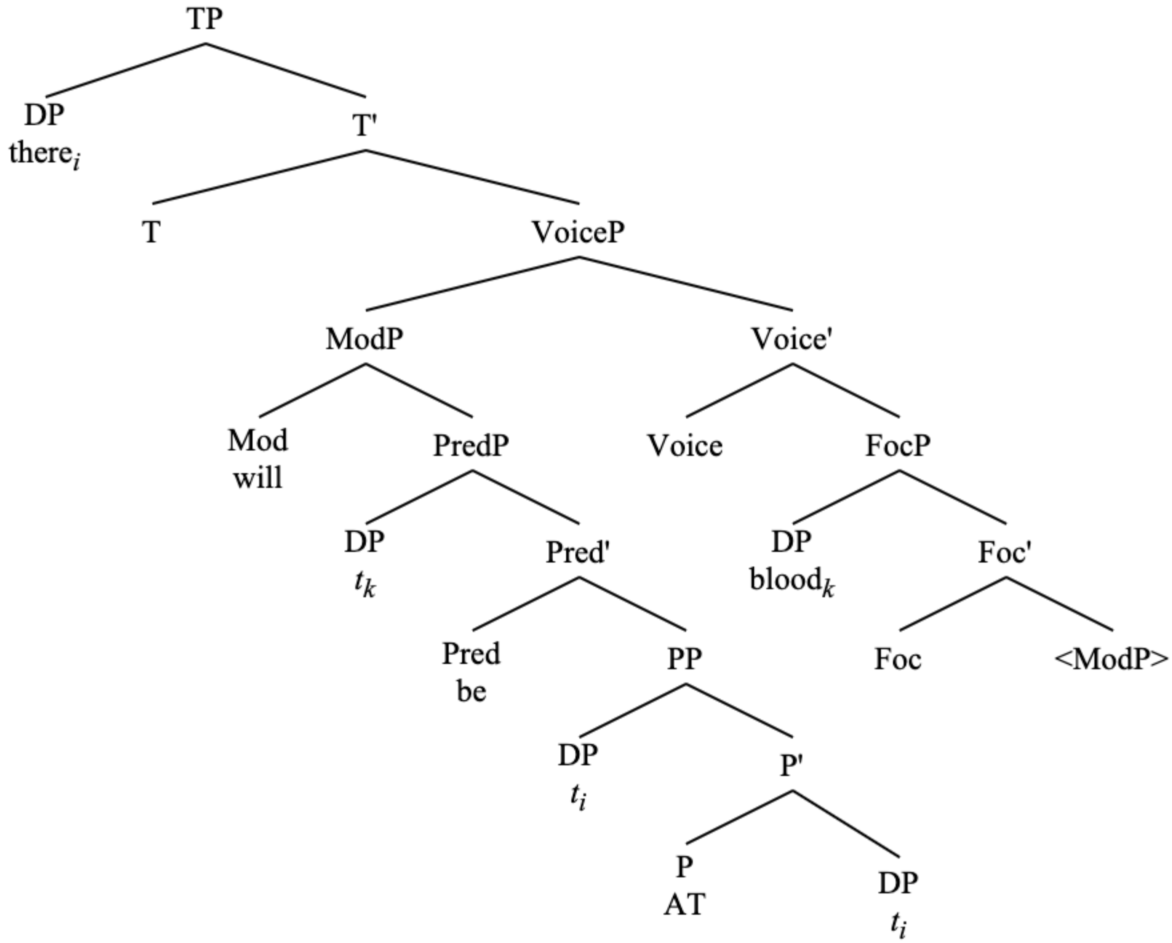
In this structure, the external argument 'blood' would be selected for movement to Spec,TP, and, as such, would not allow movement of 'there' outside of the deficient PP to be licensed. As such, the FocP is necessary in this derivation not only to derive the information-structural properties of the external argument but also to allow for smuggling to facilitate the licensing of the existential 'there' DP, which cannot be licensed in its base position.

(526) DP ‘there’ is selected for movement to Spec,TP after smuggling



At this point there is once again the option for defective circumvention, but ‘there’ nonetheless moves to Spec,TP.

(527) Movement of ‘there’ to Spec,TP in ‘There will be blood’



The presence of the low FocP to which the external argument DP moves is crucial for deriving the structure of EI with all its considerations, including agreement, and the presence of modals and auxiliaries.

4.4.3 FocP and information structure

Another consideration is that the postverbal DP in EI sentences is necessarily focused; it is always new information. This seems crucial to the existential/presentational reading.

(528) Q: Who was at the party?
A: There was John…

(529) Q: Was John at the party?
A: #There was {John/him}…

EI sentences are, by definition, the assertion of the existence of some entity (the postverbal subject DP which originates as the external argument of PredP). This is only compatible with a focus reading of this DP. As such, the information structural properties of this construction follow from the obligatory movement of the external argument to Spec,FocP (which is also necessary for smuggling).

At this point, a dichotomy emerges from the inversion constructions surveyed thus far. There are those constructions which necessarily involve movement of the thematic subject to Spec,FocP (PI, EI, and unaccusative LI), and those which do not necessarily involve FocP, but which may have one in contexts such as HNPS (QI and unergative LI).

(530) Distribution of FocP across inversion constructions

Inversion	FocP required?
Quotative (QI)	*
Locative (LI)[48]	*
Specificational predicate (PI)	✓
Existential (EI)	✓

This observation about the necessity of FocP to derive the inversion seems to track with another observation: whether or not these constructions are heavily restricted to a specific context or register.

QI and LI are very contextually restricted in English. They typically do not occur outside of narrative contexts. QI especially seems most restricted to a literary register. EI and specificational PI, on the other hand, do not seem to be contextually restricted in this way at all. Examples of the latter two cases are far more attested in the real world and in the literature than are QI and LI.

Given that EI and specificational PI obligatorily involve movement of the thematic subject to Spec,FocP in their derivation, I relate this contextual restrictedness to the lack of an obligatory FocP in the derivations of QI and LI. This is because the FocP in EI and PI forces there to be a focus interpretation of the postverbal subject DP. As such, the information-structural properties of these sentences follow naturally from their syntactic structures. Since QI and unergative LI sentences do not have this obligatory FocP, their interpretation is more restricted to a specific context in which their information-structural properties can be computed pragmatically.

It is not the case, however, that QI and LI never involve a low FocP. Both allow for HNPS, which involves a low FocP.

(531) a. *"What the fuck?!" would say Deane
b. "What the fuck?!" would say anyone who reads my diary without permission

(532) a. "D'oh!" said (*drunk) Homer (drunk)
b. "D'oh!" said (drunk) the man from Springfield with three kids and a wife named Marge (*drunk)

[48] While it may be true that one variety of LI – that which involves unaccusatives – does obligatorily involve movement of the thematic subject to Spec,FocP – the fact that this is not the case with unergative LI sentences show that movement of the thematic subject to Spec,FocP is not a necessary condition of LI constructions as a whole.

(533) a. "Fire!" yelled (*to get the crowd's attention) Ed (to get the crowd's attention)
b. "Fire!" yelled (to get the crowd's attention) the anxious man who was desperate to find a solution (*to get the crowd's attention)

It has been my experience that, in many cases, HNPS helps QI and LI sentences to sound more natural, which could be a by-product of the FocP contextually "freeing up" the utterances, so to speak.

These observations demonstrate the intimate connection between voice alternations and information structure, and the syntactic representation of such with smuggling via VoiceP and movement of thematic subjects to FocP, which further solidifies the conception of voice constructions as a way to mediate between argument structure and LF mappings.

This is not to say, of course, that VoiceP itself must bear an information structure feature; see numerous arguments against this in section 3.2. Rather, VoiceP facilitates (and is facilitated by) relations between arguments and information-structural projections such as FocP and TopP. As opposed to being associated with a specific informational feature itself, VoiceP's role is simply to facilitate the movement of arguments via smuggling and influence the way that arguments are mapped onto A-positions, which has a subsequent effect on information structure.

4.5 Summary

In this chapter, I have presented an analysis of existential 'there' in which it is a DP introduced inside a deficient PP headed by a null preposition that does not project a deictic feature. For this reason, existential 'there' is only licensed if it raises out of its base position to Spec,TP. Its base position inside the PP is as a VP-internal argument that receives the theta-role LOCATION.

Existential 'there' has the semantic contribution of denoting that an event takes place in some contextually restricted universe $\mathfrak{U}$, which creates the existential reading of the sentences in which it appears. Additionally, existential 'there' is base-generated low, as the LOC complement of V^0. Evidence that it is base-generated below the position in which it appears is the fact that it can control phi-agreement.

As such, any sentence containing an existential 'there' subject is so because the LOC argument moves over any other argument in the clause in order to undergo A-movement to Spec,TP. These existential sentences are derived via inversion in the same way that the other inversion constructions presented in this thesis are: smuggling via VoiceP. The main difference between EI and the other inversion constructions analyzed in this thesis is that the non-inverted counterpart of EI is not grammatical, which is not the case for the other inversion constructions.

(534) a. John walked into the room
b. Into the room walked John

(535) a. Chris said "Hey!"
b. "Hey!" said Chris

(536) a. Paco is the chair
b. The chair is Paco

(537) a. *A solution to the problem was there (existential)
b. There was a solution to the problem

This asymmetry is due to the featural deficiency of the PP into which existential 'there' is externally merged, which causes 'there' to not be licensed in its base position.

The analyses of EI and PI presented in this chapter also explain the focus properties of the postverbal thematic subject in these constructions, the syntactic representation of which is necessary to allow for smuggling via VoiceP. The four English constructions which display the agreement alternation and multiple subject PCC effects with agreement are then unified as smuggling constructions, which involve movement of the VP to a position above the thematic subject, which is empirically supported by the data concerning the distribution of particles and control clauses in these constructions.

In this way, defective circumvention and inversion via smuggling are intimately connected[49] in these constructions, as it is inversion via smuggling that allows for defective circumvention configurations in the first place. All the constructions presented in this thesis – locative inversion, quotative inversion, predicate inversion, and existential inversion – similarly involve agreement optionality via defective circumvention (and its associated PCC effects) brought about by inversion as smuggling via VoiceP.

[49] Although, it is not the case that defective circumvention and inversion via smuggling only occur in the presence of each other. For example, the dative subject constructions in Icelandic and Spanish shown in Chapter 2 show agreement optionality via defective circumvention, but there is no reason to believe that those constructions involve inversion via smuggling. Similarly, passives involve inversion via smuggling (Collins 2005), but for the most part do not show the agreement alternations that are a signature of defective circumvention.

5 Conclusion

In this dissertation, I have presented a previously unconsidered paradigm of verbal agreement in multiple subject inversion constructions: namely, locative inversion, quotative inversion, predicate inversion, and existential inversion. I have shown that these constructions are unified in their ability to optionally have verbal phi-agreement track either a phi-deficient preverbal subject or a non-deficient postverbal subject.

These agreement alternations are characterized as defective circumvention, which involves Agreeing with a deficient goal, then conditionally Agreeing past the deficient goal to a non-deficient goal, given that the non-deficient goal's features are compatible with the specifications of the probe. Defective circumvention manifests as optional agreement with a third person postverbal DP argument in the constructions surveyed in this thesis, as well as the emergence of weak PCC effects in English with [Part]-bearing postverbal subjects.

I have additionally shown that there is a difference between English and Spanish in these defective circumvention configurations which comes about due to the relativized specifications on the probe which I have ultimately connected to the null subject parameter. Whereas English has weak PCC effects with [Part]-bearing postverbal subjects, Spanish has obligatory agreement with [Part]-bearing postverbal subjects due to the presence of a [_Part] feature on T^0 in Spanish, but not in English. This is summarized in the following tables.

(538) Postverbal subjects in English, where T^0 does not search for [Part]

Phi-features	*Verbal agreement*	*Case*
1st person	Impossible	Accusative
2nd person		
3rd person	Optional	

(539) Postverbal subjects in Spanish, where T^0 does search for [Part]

Phi-features	*Verbal agreement*	*Case*
1st person	Obligatory	Nominative
2nd person		
3rd person	Optional	

How do these agreement configurations come about, given that the defective element that appears in the preverbal position is an internal argument that is externally merged in a position below the argument that surfaces as a postverbal subject in these constructions? Given the minimal assumptions underlying the formulations of Agree and Argument Structure, and given that the fronted constituents in inversion constructions involve A-movement, which is evident not just from these facts on Agree and Case assignment but also from facts presented throughout this thesis concerning scope and binding, weak crossover, licensing of parasitic gaps, and raising, a way is needed to make an internal argument accessible over an external argument to a phi-probe without resorting to stipulative complicative technology such as equidistance.

Enter: smuggling, which I have demonstrated is at the core of these inversion constructions (and other voice alternations) in English that show defective circumvention. Smuggling of a phi-deficient element to a position where it is in the search domain of T^0 and crucially higher than another non-defective goal is what allows for defective circumvention and PCC effects to arise.

Smuggling is motivated by a functional projection VoiceP. Inversion as smuggling involves two applications of internal merge: one of a verbal projection, and another of an argument within the verbal constituent to an A-position. That is, smuggling facilitates new A-dependencies.

Voice mediates between the positions into which arguments are externally merged (Argument structure) and the mapping of those arguments onto specific grammatical and information-structural roles at LF. In other words, smuggling occurs in these derivations in order to facilitate the promotion of some arguments over others. This can be seen transparently in passives, which involve the promotion of an internal argument, as well as inversion constructions, which is a novel contribution of this thesis: that these English inversion constructions should be characterized as voice alternations.

VoiceP triggers VP movement over the external argument (just as it does in other voice constructions such as passives (Collins 2005)), which I have shown happens consistently across these different inversion constructions via the distribution of VP-internal elements such as particles and control clauses.

As such, my thesis unifies these disparate constructions involving postverbal subjects under a single mechanism of inversion via smuggling that facilitates defective circumvention, which ultimately is a reflex of the fact that smuggling allows for A-movement of an internal argument that would be otherwise inaccessible to the phi-probe on T^0. These constructions differ in the kind of predicates involved, the elements that appear as structural subjects, and the specific information structural properties of each construction, though the analysis presented here allows for the technology of inversion via VoiceP and defective circumvention to derive all these constructions and their complex agreement data despite their differences.

I present the inversion quadrilateral from the introduction again here.

(540) English inversion quadrilateral

	Necessary FocP -	Necessary FocP +
Inverted LOC -	Quotative	Predicate
Inverted LOC +	Locative	Existential

We can now gaze upon this representation with newfound understanding. Both LI and EI involve the fronting of a locative – a deictic PP for LI, and a non-deictic existential PP for EI. In both of these cases, a PP-internal DP moves out of the PP to function as the structural subject in

Spec,TP, and, in LI, the PP is able to raise above Spec,TP in order to license an empty category in that position.

QI and specificational copular PI sentences both involve the fronting of a non-locative element. In the case of QI, this is a quote, and, for PI, it is a specificational DP. A null DP is licensed as the structural subject in Spec,TP for QI in the same way that it is for LI.

As for the necessary FocP, the FocP to which the external argument subject moves is necessary to derive the syntactic structure of both PI and EI, and this necessity also corresponds to the focus reading of the postverbal subject DPs in both of those constructions. FocP is also necessary in unaccusative LI derivations, though that is not a property of all LI sentences. QI and LI may involve movement to FocP in the case of HNPS (regardless of the thematic status of the predicate), but, once again, it is not necessary to derive these constructions in the way that it is in EI and PI. It is also the case that the semantic status of the FocPs may vary.

Thus, the differences between the four different English inversion constructions analyzed in this thesis, which are all derived via smuggling via VoiceP and allow for defective circumvention, come down to distinctions in the types of predicates and internal arguments involved in the inversion derivations, as well as the information-structural properties of the arguments in the derivations (as it has been established that the information-structural properties of each inversion construction are not identical (Quesada & Skopeteas 2010)).

5.1 Core theoretical contributions

In this dissertation, I have derived complex agreement interaction effects involving English and Spanish postverbal subjects without resorting to theoretical complications such as equidistance. Additionally, I do not complicate the theory of Agree by moving it outside of the narrow syntax. The formulation of Agree that I have presented in Chapter 2 and implemented throughout the thesis effectively reduces to Minimal Search in the syntax, and as such does not rely on postsyntactic operations such as feature impoverishment or vocabulary insertion to derive the data. Theoretically, this is advantageous, as it does not require positing that Agree must cross multiple transfer domains, and is thus in conformation with SMT. I have shown here that complex agreement interaction effects involving multiple goals and a single probe can indeed be derived squarely in the narrow syntax.

The formulation of Agree as an operation driven by local Search presented in this dissertation, along with the minimalist assumptions on the uniform projection of thematic arguments (argument structure), present somewhat of a conundrum for the relevant inversion sentences, since these sentences uniformly involve an internal argument appearing in an A-position above a (more) external argument. Given the understood uniformity of the projection of arguments, as well as the locality and strict minimality of this formulation of Agree, a mechanism is needed to affect the accessibility of arguments for A-operations.

In this way, the theoretical considerations on Agree and argument structure force us into a situation of needing a device such as smuggling, which is able to move arguments over one another and make arguments accessible to A-probes that would normally be blocked by an intervener. Additionally, the VP-smuggling which I show is characteristic of voice alternations is empirically supported by the distribution of VP-internal elements.

Voice as a smuggling phenomenon is a major theoretical contribution of my work here, as is the definition of voice that I present here which is not invariably tied to the projection of the external argument. I have shown that the goal of voice is to manipulate the positions of arguments for mapping onto A-positions and subsequent mappings onto LF. This is seen with the structural and informational promotion and/or demotion of certain arguments in the syntax and also in the pragmatics, which is observed uniformly in passives, middles, antipassives, dative shift, and, of course, inversion constructions. This definition of voice allows us to reconsider what kind of constructions empirically count as voice alternations, which ultimately opens the door for a deeper understanding of what the purpose is of voice alternations in the syntax. I have also shown that this understanding of voice as smuggling is advantageous for explaining the syntax of phenomena such as Heavy-NP Shift and the Transitivity Constraint.

Another major theoretical contribution of this thesis is how I treat the classification of existential 'there'. I have shown that existential 'there' is a thematic internal argument which receives the theta-role LOCATION and originates as an internal argument of VP. The major consequence of this analysis is that it forces sentences containing existential 'there' subjects to be analyzed as inversion constructions, which is empirically supported by the fact that existential 'there' sentences display the same empirical properties as the other inversion constructions with respect to defective circumvention and VP movement. This treatment of existential 'there' categorically rejects the idea that this element is a syntactic expletive which is inserted late in the derivation and has no semantic content. Furthermore, this treatment of existential 'there' as thematic opens the door to reconsider if the notion of a syntactic expletive is one that is useful at all (Bolinger 1973; Levin & Krejci 2019; Mathurin 2020; Greeson 2024, a.o).

The final core theoretical and empirical contribution that I have made in this thesis is that colloquial, nonstandard, prescriptively marginal linguistic data is a completely valid object of scientific inquiry, and the treatment of it as such allows us to make crucial observations about the structure and combinatorics of natural language that would otherwise be missed. Indeed, the key data presented in this dissertation is mostly colloquial and prescriptively "incorrect", and has thus been barely noted in the literature before, though it informs our understanding of Agree, argument structure, voice, and the syntax of optionality in crucial ways. I show in this thesis that these data are valid, and that collecting written data from the Internet is a valid way to investigate natural language. I look forward to seeing the ways in which this methodology is used to push the boundaries of syntactic inquiry.

5.2 Reflections on methodology

Here I have presented a systematic survey of a colloquial, nonstandard agreement alternation. In doing this, I make heavy use of data collected from the Internet (from both spoken and written sources). In doing this, I discuss prior research that has made use of such methodology, as well as steps that must be taken to verify the data as valid for native speakers of the language in question.

This work is very innovative in its approach to Internet data and what counts as a valid grammatical utterance; however, this approach to collecting data is somewhat new and therefore worthy of discussion. As others have noted (e.g., Fernández-Serrano 2022), Internet data is subject to certain considerations that are not present elsewhere, such as the increased likelihood of "noisy" data, the possibility that an utterance may not be original to an author, and increased difficulty verifying the identity (and thus the native-speakerhood) of the author of an utterance. I have taken steps to decrease the issues that these considerations may pose to the authenticity and validity of the data, such as verifying Internet examples with native speakers of the relevant language, filtering out examples that are full of typos, have multiple occurrences, and are likely not from native speakers.

I believe that any study that uses Internet data would benefit from simply having considerably more of it. Moving beyond Internet searches, it would be advantageous to use more robust methods of data collection such as NLP to collect more relevant utterances. As this relates to this dissertation specifically, more data would allow us to run numbers to observe phenomena such as frequency effects; i.e., what is the overall ratio of 'There is problems'-type sentences to 'There are problems'-type sentences? Given that the Internet is such a vast place, and given that these data often need additional verification, it would be difficult to process incredibly large amounts of the relevant data, but doing so would allow deeper understanding of the relevant phenomena.

Additionally, studies such as this one would benefit from experimental verification such as performing large-scale grammaticality judgment tasks with native speakers. While the data presented in this thesis were verified with native speakers, a large-scale grammaticality survey would allow us to see trends across large numbers of speakers, and perhaps even to collect demographic data on which speakers have the relevant agreement alternations.

This is all to say that this study would be augmented by having more data; although, as this is mostly about the theory of how to derive the relevant data, the mere fact of their existence is enough for these purposes. In future work, more data collected via NLP and large-scale grammaticality surveys would be very useful to make more granular generalizations about the nature of these constructions and their agreement alternations, which would ultimately deepen the theoretical inquiry of this dissertation.

5.3 Outstanding questions

There are a several open questions that the work in this dissertation raises. By "open", I mean that the answer is not immediately clear based on the analyses and discussion that takes place throughout. Here, I raise a couple of these questions, and provide insight where I am able.

The first outstanding question I address here has to do with the difference between inversion and passives. Why are these constructions different? Why would a language need to make use of inversion when it has the option to passivize a sentence? Why would a language need to make use of passives when it has the option to invert a sentence? Given that I analyze both of these constructions as voice alternations that promote an internal argument over a (more) external argument via smuggling, it may not be immediately clear why a language should need more than one option to do this. Succinctly put: why are passives and inversion constructions different?

To this, I would first say that it is not clear to me why a language should only have one strategy of "doing something". It is true that both inversion and passives involve a displacement of the canonical order of arguments, and that both constructions do this via A-movement which is facilitated by smuggling. It is also true that both passives and inversion involve an altered information structure when compared with their non-passivized/non-inverted counterparts. None of these facts, however, serve as any reason that we should predict these constructions to not be distinct from one another. Empirically, they are distinct, and that must be accounted for.

The domains of passives and inversion are demonstrably distinct, both in terms of the kinds of sentences they can affect as well as their information-structural properties. As I show now, the licensing potential of each type of voice alternation is distinct – at least in English and languages like it – which further explains the potential necessity of both types of voice alternation.

Passives and inversion have a different distribution. To start, clauses containing an unaccusative verb or a copula cannot be passivized in English.

(541) a. *The train was arrived
b. *The station was arrived at by a train
c. *At the station was arrived by a train
d. *There was arrived by a train

(542) a. *The president was been by Trump
b. *Trump was been by the president
c. *There was been by a problem

This immediately rules out unaccusative LI sentences, all EI sentences, and all PI sentences as constructions which could have been derived via passivization. This is simply because passive derivations seem to require an AGENT argument, which is lacking in all of these sentences.

Turning to QI and unergative LI, which do involve an AGENT, it looks to be the case that the sentences which participate in QI and LI are not really valid kernels for undergoing passivization, either.

Starting with unergative LI sentences, passivization of the PP locative argument that appears preverbally in LI constructions is not possible. The only option for passivization that resembles this is a pseudopassive, which crucially involves the fronting of the internal argument of the PP, which is not the case for LI.

(543) a. Into the room walked Robert
b. *Into the room was walked by Robert
c. The room was walked into by Robert

Pseudopassives involve the movement of the PP-internal argument to Spec,TP, whereas my analysis of LI demonstrates that an argument from the edge of the PP is moved into Spec,TP while the entire PP moves into a topic position. As such, the purpose (so to speak) of the pseudopassive is to promote the internal argument of a PP to the subject position, whereas the goal of LI is to promote the PP into a topic position. This difference further shows that the domains of passivization and inversion are distinct.

As for QI, examples of passivized quotes are marginal at best.

(544) a. ??"Leave me alone!" was said by Paula
b. ??"I'm not a brat!" was said by me
c. ??"Would you get out of here?" was said by Squidward

Examples of passivized quotes are attested online, although these quotes are usually embedded inside of a visible DP.

(545) a. Did I forget to mention that the 'I love you' was said by the boy to the girl before they got into a relationship!? (Blog)
b. An "I love you" was said by Adam in the first season (Fandom Wiki)
c. Wouldn't it be awesome if the "HEY!" was said by Harry Carrey? (Forum)
d. A chipper, "Hey," was said by both Amethyst and {Name} (Fanfiction)
e. SA's slogan, "Vote it Down," was chanted by the crowd several times (Blog)

This DP structure is notably impossible with QI. It is very ungrammatical.

(546) a. *The "I love you" said the boy
b. *An "I love you" said Adam in the first season
c. *The "Hey!" said Harry Carrey
d. *A chipper "Hey" said both Amethyst and {Name}
e. *SA's slogan "Vote it Down" chanted the crowd

This suggests that the quotes that undergo passivization – including those in (544), which are marginal, albeit attested – are actually embedded inside of a DP, which is distinct from the QuotP analysis of the internal structure of quotes in QI derivations presented in section 3.5.1.3.

As such, passives and inversion are in complementary distribution in English. Predicates that undergo inversion as detailed in this thesis are not the same predicates that undergo passivization. Furthermore, the information-structural properties of each are distinct.

In passives, the passive morphology is there to license the external argument (Collins 2005; 2024a). The external argument in passives is never a potential goal for phi-agreement with T^0 because there is no possibility of probing past the non-deficient arguments that move into Spec,TP in passives. In inversion, no such licensing is needed because of the possibility of defective circumvention (brought about by the feature deficiency of the inverted argument) and the potential of the postverbal subject to move into a low FocP. This relates to a discussion at the end of section 3.2.3.

This account of the difference between passives and inversion may be more complicated for languages like Algonquian and Bantu which are freer in the types of predicates that can do inversion, but, for English, it is clear that the distribution of passives and inversion are not an issue for this analysis. Passives and inversion are distinct in their structure, capabilities for licensing arguments, and information-structural properties.

The second outstanding question that I raise here has to do with the diachronic future of grammars that display the kind of optionality seen with agreement in these inversion constructions. Certainly, this kind of optionality is not the typological norm, despite being found in a variety of constructions crosslinguistically. This question is not something that I have the means or background to answer in the way that I have done for the previous outstanding question, but it is worth mentioning nonetheless.

Do we expect instability with these kinds of optionality in grammar? Has this influenced the tendency to analyze such phenomena as changes-in-progress of some kind? As I have shown in section 2.6.2, this very agreement alternation with a postverbal subject in English has been attested for centuries, but it is not clear if this kind of optionality is something that one should expect to disappear in time, or if it can be a stable part of any given grammar indefinitely.

If the former is true, perhaps there more innovative grammars which have moved away from this optionality, which could be the case with existential ‘it’ sentences in AAE which do not display the kinds of agreement alternations seen with existential ‘there’ sentences in other varieties of English.

This question of how languages handle optionality – and how to account for it in a minimalist framework – is undoubtedly an interesting avenue of exploration for future work, though I do not provide an answer here. The scope of this dissertation has been to show how a single set of grammatical operations can produce more than one felicitous output in the case of verbal agreement with multiple subjects. Whether or not human language shows a diachronic tendency to move toward or away from these kinds of optionalities will be an interesting question to address in the future.

5.4 Extensions

There are many natural extensions of the work that I have done in this dissertation. The first and perhaps most obvious extension is the application of defective circumvention to the cases of agreement optionality in non-English languages shown in the beginning of Chapter 2, repeated here.

(547) Nos encanta{-Ø/ -n} las películas de terror (Spanish, Fernández-Serrano 2022)
us.DAT love {-3sg/-3pl} the films of terror
‘We love horror movies’

(548) Henni {líkaði/ líkuðu} þeir (Icelandic, Sigurðsson & Holmberg 2008)
her.DAT {like.3sg/like.3pl} they.NOM
‘She likes them’

(549) Nógvum kvinnum {dámar/ dáma} mannfólk (Faroese, Jónsson 2009)
many.DAT women.DAT {like.3sg/like.3pl} men.ACC
við eitt sindur av búki
with a bit of belly
“Many women like slightly fat men’

(550) Hi ha {-Ø/ -n} tres cadires (Catalan, Rigau 2005)
there have{-3sg/-3pl} three chairs
'There is/are three chairs}

(551) Hub{-o/ -ieron} dos hombres en la fiesta (Spanish, Rodríguez-Mondoñedo 2006)
had {-3sg/-3pl} two men in the party
'There {was/were} two men at the party

(552) Fech {-ou/ -aram} muitas fábricas (Portuguese, Costa 2000)
closed{-3.sg/-3.pl} many factories
'Many factories closed'

(553) Lleg {-ó/ -aron} los colectivos (Spanish, Ordóñez p.c)
arrived{-3sg/-3pl} the buses
'The buses arrived'

As I use defective circumvention and its restrictions to derive weak PCC effects in English multiple subject constructions, this theory of Agree has applications to other PCC effects across languages. PCC effects with multiple objects in Romance (Bonet 1991; Ordóñez 2002), which show essentially the same restriction as the multiple subject constructions discussed in this thesis. In multiple object clitic constructions in Spanish and Catalan, a first or second person accusative direct object clitic is not possible in the environment of a dative clitic. Given the analysis of dative DPs as involving a shell that is phi-deficient (see section 2.1), and given that dative GOALS are externally merged in a position above THEME objects, it follows that the PCC effects in Romance clitic object constructions can be uniformly analyzed in the same way as the agreement restrictions in English multiple subject constructions. I plan to pursue an analysis of these PCC effects in future work. See similar treatments of PCC effects with multiple object clitics in Basque in Preminger (2014); Coon & Keine (2021).

I believe that the analyses of voice as smuggling can – and must – be extended to other voice phenomena. There exists smuggling treatments of passives (Collins 2005; 2024a) and dative shift (Collins 2021; 2024a), but the theory of voice set forth in this dissertation predicts that smuggling should be involved in other voice alternations as well. As such, constructions such as middles, antipassives, and causatives should be systematically investigated for their compatibility with smuggling, especially given that they show the same restrictions on the distribution of VP-internal elements such as particles (see section 1.2.3). Again, I consider this a very promising avenue for future work.

Defective circumvention presents a clear treatment of optionality within the narrow syntax. As such, its capabilities to derive other points of optionality in grammar should be taken seriously. Take the following example of yes-no question formation in multiple modal constructions in Ozark English (my native variety of English).

(554) You might could go tomorrow

(555) a. Could you might go tomorrow?

b. *Might you could go tomorrow?
c. Might could you go tomorrow?

Here, there are two acceptable outputs for yes-no question formation. One involving the movement of the lower modal, and another involving the movement of both modals. To my knowledge, this exact alternation has not been worked on at the time of writing this.

We have here an alternation between a syntactic operation being done optionally to one element or to two elements. In this way, the alternation with multiple modals here is the exact same kind of optionality shown with verbal agreement in English inversion constructions, which involve a phi-probe optionally undergoing Agree with more than one goal. Given that movement is driven by Agree (section 2.2.3), these kinds of optionality in movement of one element or two could feasibly be derived via the same technology of defective circumvention. Defective circumvention may in this way prove to be very helpful in explaining cases of optionality of the application of some syntactic operation.

Another here has to do with superiority violations (Chomsky 1973). Smuggling is used to circumvent superiority effects that arise between two arguments that stand in an asymmetrical c-command relation with one another. Smuggling allows an argument to become accessible to A-probing that would otherwise be blocked by a higher, intervening argument. Smuggling in these voice alternations like inversion facilitates the derivation of what would otherwise be a superiority violation (targeting a goal past an intervener). Naturally, one wonders about the applications of smuggling to obviate other kinds of superiority effects. Take English multiple wh-question formation (Pesetsky 1987; Goodall 2015), in which wh-elements must typically obey superiority with respect to their base positions.

(556) a. Who wrote what?
b. *What did who write?

(557) a. I wonder who bought what (Goodall 2015:1)
b. *I wonder what who bought

There are, however, situations in which the superiority effects can be obviated, such as D-linking (Pesetsky 1987).

(558) a. Which student wrote which paper?
b. Which paper did which student write?

(559) a. I wonder which man bought which car (Goodall 2015:1)
b. I wonder which car which man bought

These data are reminiscent of the inversion constructions discussed in the thesis to the degree that arguments must normally obey superiority with respect to A-movement.

(560) a. Drew ate beans
b. *Beans ate Drew

However, these effects can be obviated in certain circumstances, such as inversion with predicates that take a phi-deficient internal argument.

(561) a. Drew walked into the restaurant
b. Into the restaurant walked Drew

Or, alternatively, with passivization.

(562) a. Drew ate beans
b. Beans were eaten by Drew

The superiority effects with these arguments are circumvented via inversion as smuggling via VoiceP. The question here that is relevant for the multiple wh-question data is whether or not the superiority effects shown with multiple wh-elements can be circumvented with some kind of A-bar inversion via smuggling. If smuggling is sufficient to minimally derive argument inversion, perhaps it is also sufficient to derive these alternations with D-linking that look to be a kind of wh-inversion.

Similarly, inversion as smuggling has applications to inverse voice constructions in a variety of languages including Algonquian languages (Oxford 2023) and Bantu languages (Shlonsky 2024), where agreement obligatorily tracks the fronted constituent.

(563) a. ni- wa:pam -ikw (Ojibwe inverse voice, Oxford 2023:321)
1- see- -INV
'She sees me'
b. ni- wa:pam -a:
1- see -3.OBJ
'I see her'

(564) a. Ici-ya ci- tula mw- ana (Luguru subj-obj. inv., Shlonsky 2024:439)
7- pot SM7- broke 1- child
b. Imw-ana ka- tula ici-ya
1- child SM1-broke 7- pot
'The child broke the pot'

This framework also has potential to explain inversion in which agreement obligatorily does not track the features of the fronted constituent and the verb agrees entirely or partially with the postverbal thematic subject, such as in Russian or Arabic, respectively.

(565) a. Ètu knigu čitaet Ivan (Russian, Bailyn 2004:4)
this.ACC book.ACC reads Ivan.NOM
b. Ivan čitaet ètu knigu
Ivan reads this.ACC book.ACC
'Ivan reads this book'

(566) a. qadim{-a/ *-uu} al- ʔawlaadu (Arabic, Harbert & Bahloul 2002:45)
came {-3MSG/*-3MPL} the-boys.3MPL
b. ʔal-ʔawlaadu qadim{-uu/ *-a}
the-boys.3MPL came {-3MPL/*-3MSG}
'The boys came'

These inversion constructions display crucial differences in agreement from the constructions analyzed in this thesis, but still may ultimately be analyzed as inversion via smuggling given that the languages parametrically differ with respect to agreement, which we have already seen with the differences between English and Spanish.

The theoretical proposals of this dissertation are relevant not just for agreement in multiple subject constructions and the syntax of inversion, but have implications for a wide range of empirical phenomena and theoretical questions. Defective circumvention and inversion via smuggling have a wide range of technical applications and predictive power, and as such have numerous extensions beyond the analyses presented here of agreement interaction effects in inversion constructions that are derived via A-movement facilitated by smuggling.

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